

FORM 3

COMBINED SCIENCE 4003

NEW CURRICULUM NOTES

BIOLOGY SECTION

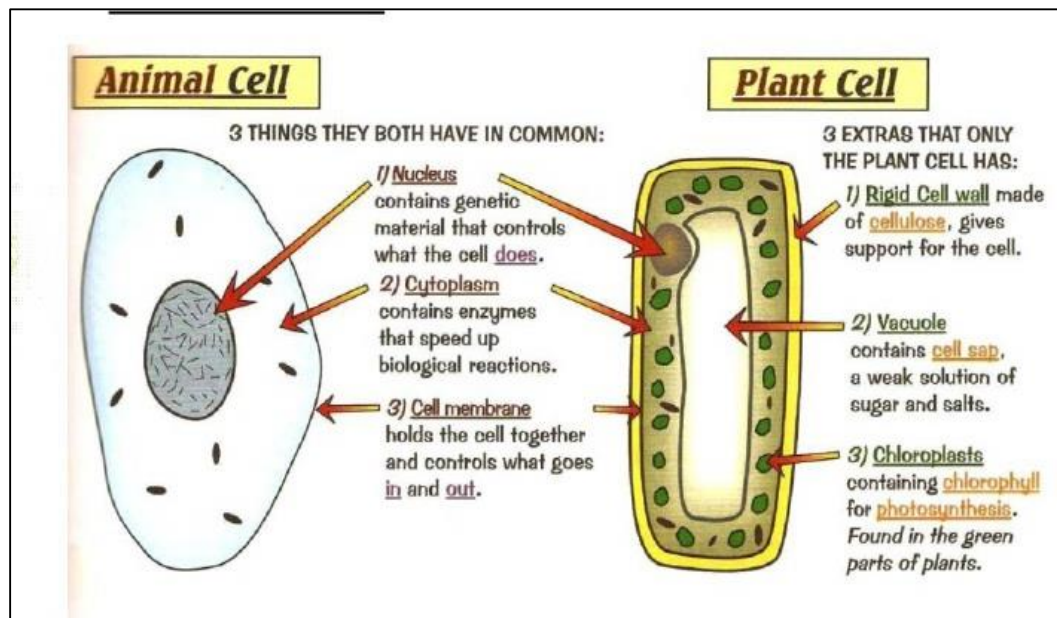
CELLS

A Cell is a basic unit of the structure and function of all living organisms.

The cells cannot be seen by naked eyes; instead microscopes are needed to telescope our view of these structures.

Cell structures:

The structure of an animal and plant cell



A Comparison between a plant and animal cells:

Plant cells	Animal cells
Have a cellulose wall covering the cell membrane	Don't have cell wall
Have a cell membrane	Have a cell membrane
Have cytoplasm	Have cytoplasm
Have a nucleus	Have a nucleus
Often have chloroplasts with chlorophyll in them	Chloroplasts absent in animal cells
Often have large vacuoles containing cell sap	No vacuoles
Often have starch grains	Only have glycogen granules present sometimes

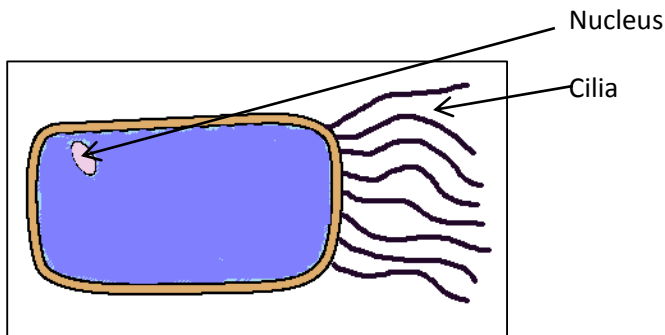
Often have a regular shape

Often irregular in shape

Specialised cells:

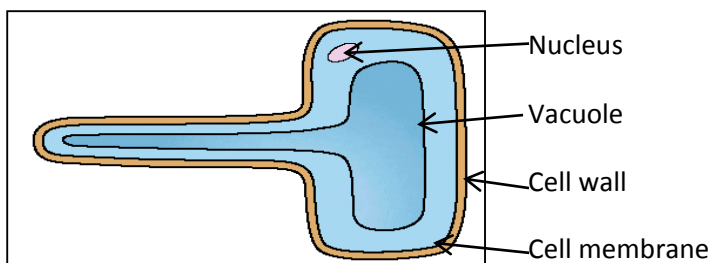
A specialised cell is designed to do a particular job.

Ciliated cell: found in the trachea and bronchi, moves the mucus towards the throat. They are adapted by the tiny hair like projections called cilia which sweeps the contaminated mucus upwards. The mucus is secreted by goblet cells which are next to ciliated cells.

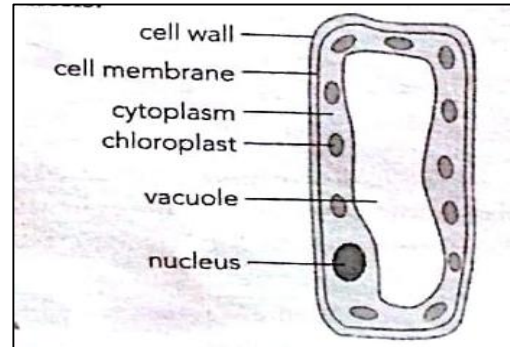


Memory cell : found in the blood, keep antibodies ready to kill pathogens that have affected you more than once.

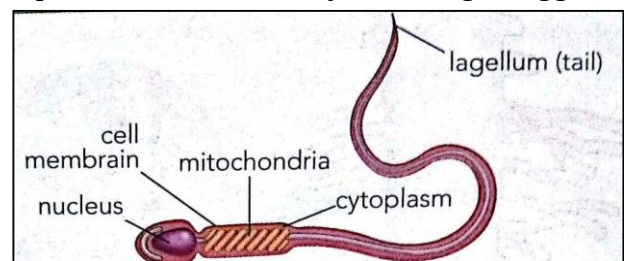
Root hair cell : found at the end of the plant roots are responsible for the absorption of minerals and water. They are adapted by 3 ways. One, they have an extension that increases the surface area for more water intake. Two, they have a large number of mitochondria for respiration to become more active. Three a concentrated vacuole to help absorbing water by osmosis.



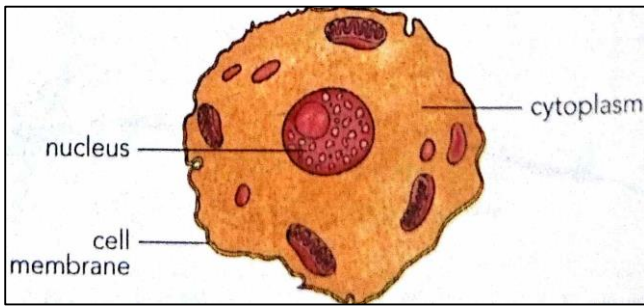
Palisade mesophyll cell : found beneath the epidermis of a leaf are specialised at photosynthesis.



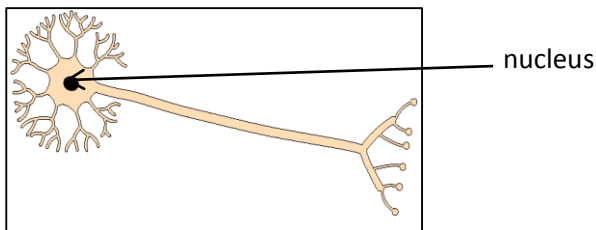
Sperm cells: are male sex cells. The front of the cell is oval shaped and contains a nucleus which carries genetic information. There is a tip, called an acrosome, which secretes enzymes to digest the cells around an egg and the egg membrane. Behind this is a mid-piece which is packed with mitochondria to provide energy for movement. The tail moves with a whip-like action enabling the sperm to swim. Their function is reproduction, achieved by fertilising an egg cell.



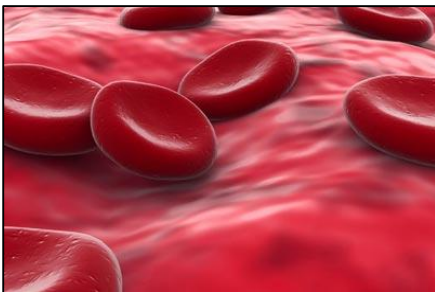
Egg cells (ova, singular: ovum): are larger than sperm cells and are spherical. They have a large amount of cytoplasm, containing yolk droplets made up of protein and fat. The nucleus carries genetic information. The function of the egg cell is reproduction.



Nerve cell : found throughout the bodies of all organisms are responsible for the transmission of electrical nerve impulses

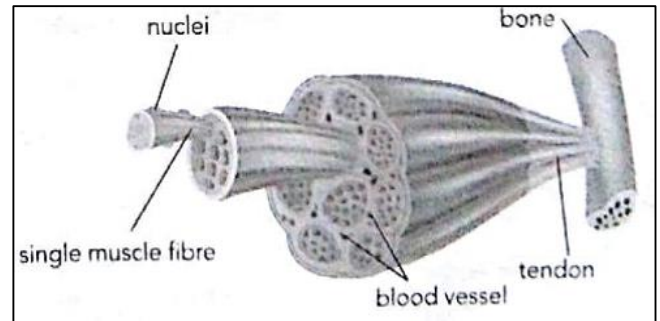


Red Blood Cell : found throughout in the blood of mammals and specialise at the transport of oxygen using the red pigment haemoglobin. Do not have a nucleus. They are adapted by four ways: They have a biconcave disc shape that gives it a large surface area to carry more oxygen. They contain a chemical called haemoglobin that combines with oxygen and carbon dioxide. They have no nucleus to carry more oxygen and carbon dioxide. They are tiny enough to squeeze through capillaries.

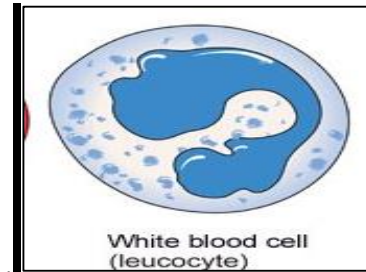


Muscle cells: They are cells found in muscles in animals, they contract and relax together to move the organisms.

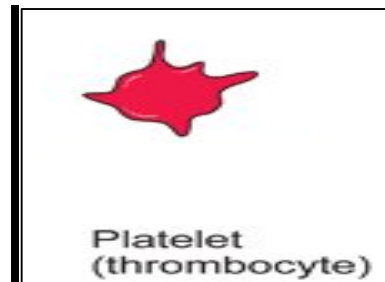
Their function is to contract to support and move the body. They are adapted by two ways, First, Is that they are made of contractile filament to help in contraction. Second is it contains lots of mitochondria to supply the cell with energy.



White blood cells: attack bacteria



Platelets: help clotting

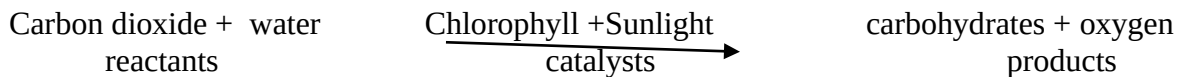


NUTRITION

- Green plants are the only living organisms able to manufacture their own food in the form of carbohydrates.
- They are called food producers of the world.
- Animals rely on green plants to make food that produces energy.

Photosynthesis

- It is whereby green plants manufacture their own food using carbon dioxide, water and light as a source of energy.



The equation shows that carbon dioxide, water and sunlight are raw materials and carbohydrates and oxygen are the end products.

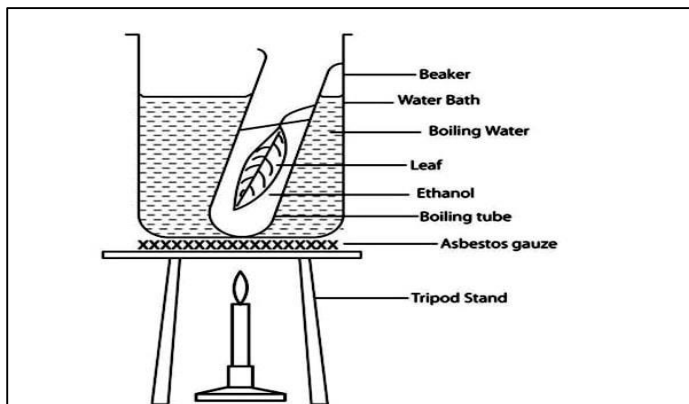
Carbohydrate is a group of foods that can be in two types (sugars and starches).

Experiment1

Testing a leaf for starch

Materials: iodine solution, dropper, alcohol, test tube, burner, stand, white tile, green leaf, beaker.

Method



- i. Dip a leaf in water in a beaker to kill cells, stop chemical reactions and to soften it.
- ii. Place a leaf in alcohol to remove chlorophyll.

NB. Alcohol is highly flammable. Put the test tube with alcohol in beaker of boiling water.

- iii. Dip the leaf in hot water to soften it again.
- iv. Place the leaf on a white tile and cover it with iodine solution using a dropper.

Observations

When the leaf is boiled in alcohol, it becomes decolourised.

The colour enables colour changes between iodine and starch to be more visible.

Alcohol makes the leaf brittle and hard and can be softened by placing it in hot water.

Results

The leaf turns to blue-black when starch is present. The leaf turns yellowish-brown when there is no starch.

Conclusion

Plants can make their own food.

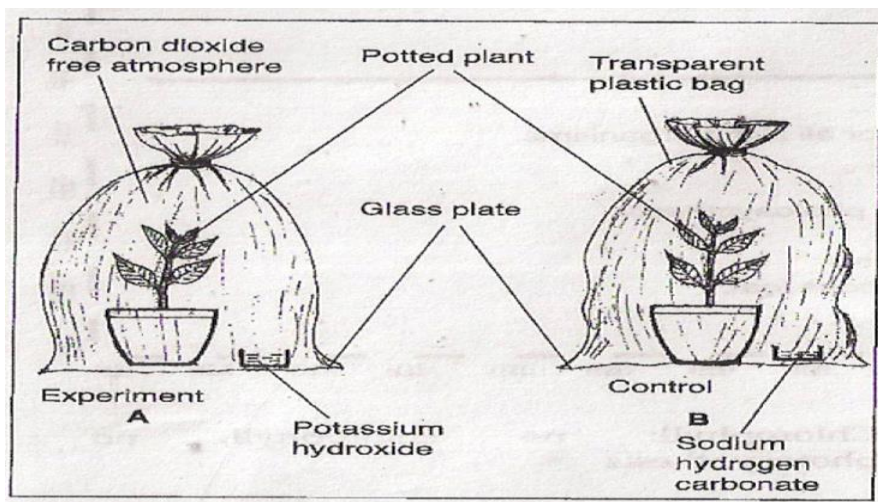
Experiment 2

Is carbon dioxide necessary for photosynthesis?

Materials

Two potted plants, soda lime, sodium hydrogen carbonate, starch test kit, polythene bags.

Method



- De-starch the leaf.
- Cover the plants in a polythene bags for two days, keeping them in a lighted place.
- Insert an inverted lid containing soda lime or calcium carbonate on the surface of the pot for plant A. This absorbs carbon dioxide.
- Cover the plant again and leave plant B untreated and keep the plants for 2-3 days.
- Perform the starch test.
- Record the results.

Observation

Soda lime absorbs carbon dioxide.

The control has all the conditions necessary for photosynthesis.

Results

Plant A does not photosynthesize since carbon dioxide is being absorbed.

Plants B makes food because carbon dioxide is produced and used by the plant.

Conclusion

Carbon dioxide is required for photosynthesis to take place.

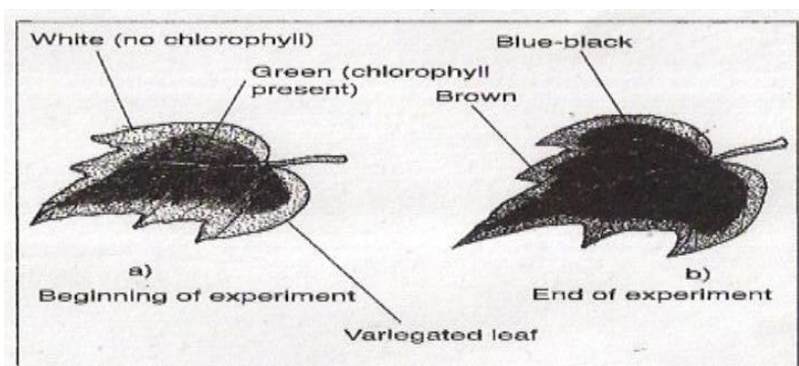
Experiment 3

Is chlorophyll essential for photosynthesis?

Materials

Variegated plant, starch test kit,

Method:



1. Starch is difficult to remove chlorophyll from a leaf without killing it so a leaf which contains chlorophyll only in patches is used.
2. Test the variegated leaf for starch.

Observations

When dropping iodine solution, only the green part changes to blue-black, but the white part remains brown.

Conclusion

Chlorophyll is necessary for photosynthesis.

Experiment 4

To investigate the leaf for sunlight

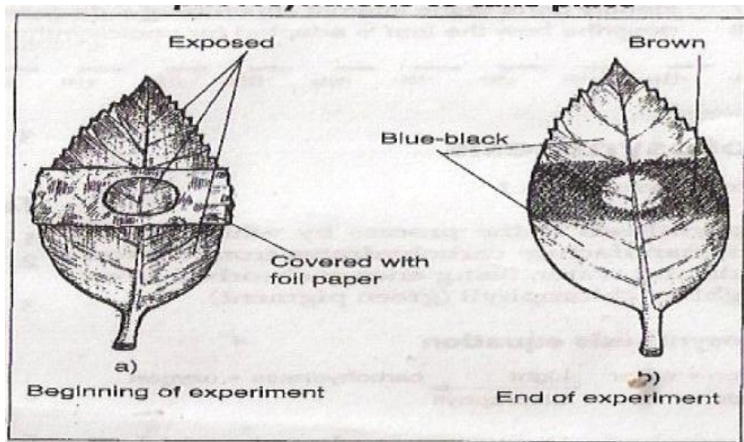
Materials

Aluminum foils, green plant, iodine solution,

Method

- De starch the potted plant by placing it in darkness for at least 24 hours.

- Take a small piece of aluminium foil or cardboard and cut simple shape out on its middle and cover leaf on the plant with the foil, ensuring the cut out side is uppermost.
- Leave the plant in sunlight for 4-6 hours.
- Remove the leaf covered by the foil and test it for starch.



Results

When testing for starch, the uncovered part will turn blue- black (starch present).

The covered part turns brown / no colour change (absent of starch).

Conclusion

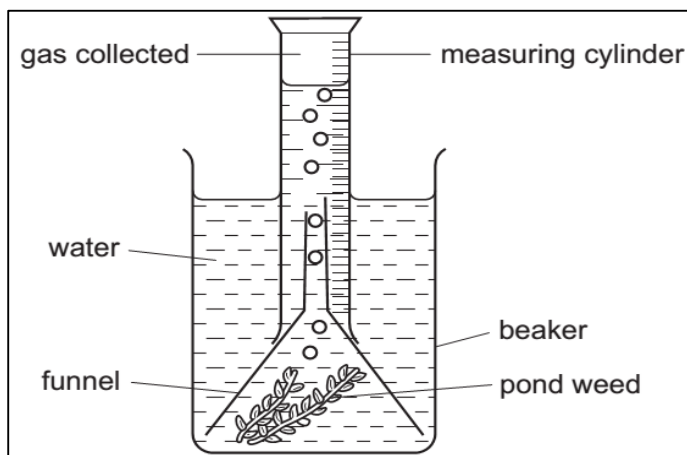
Sunlight is essential for photosynthesis.

Experiment 5

Is oxygen produced during photosynthesis?

Materials: leaf (elodea), iodine solution, test tube, beaker, funnel, water, glowing splint, stand

Method



- i. Place a water weed in a beaker with water.
- ii. Fill the test tube with water and invert it over the inverted funnel.
- iii. Place the apparatus near a window so it receives sunlight.
- iv. Leave the apparatus until the test tube is full of gas.

Observations

Gas collects in the test tube. Remove the test tube by lifting it upwards so that the gas remains in it. Test the gas in the test tube with a glowing splint.

Results; The splint burst into a flame.

Conclusion

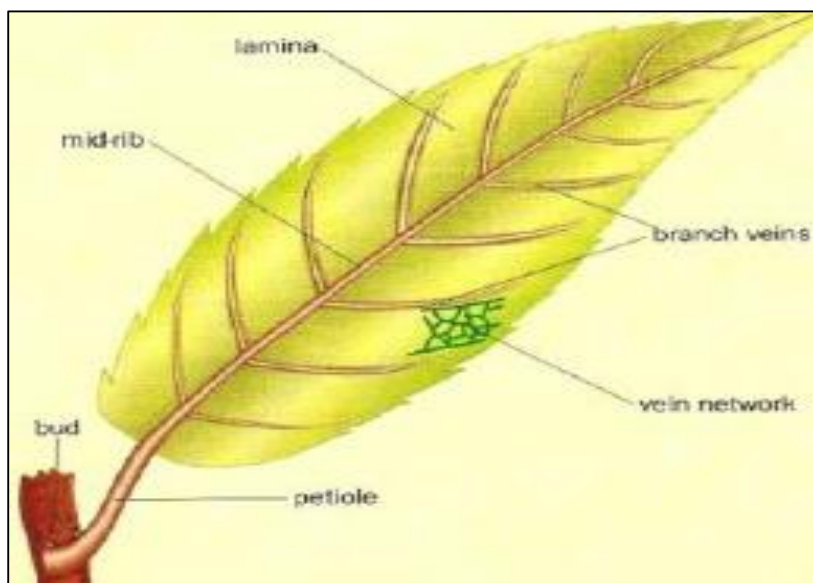
Oxygen is produced during photosynthesis.

Factors that Affect the Rate of Photosynthesis.

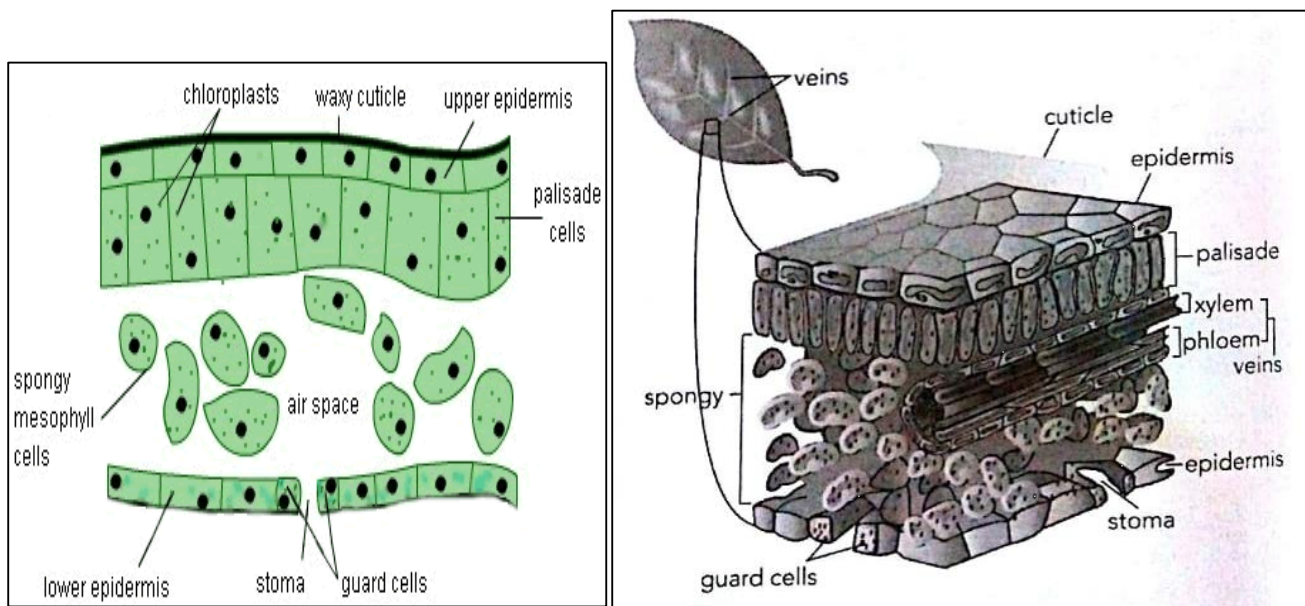
There are 4 factors affecting the rate of photosynthesis,

1. Amount of carbon dioxide – increased carbon dioxide increases rate of glucose production which speeds up photosynthesis
2. Light intensity- light energy is trapped by chlorophyll to split water molecules & therefore increasing light energy will speed up the rate of photosynthesis
3. Temperature – temperature affects rates of chemical reactions. Low temp makes enzymes inactive & slows down photosynthesis while very high temp will kill the enzymes stopping photosynthesis
4. Amount of water available – adequate amounts of water are needed & any shortage will slow down photosynthesis

The External Leaf Structure



Structure of a Leaf



Functions of the Parts

- ✚ Cuticle layer:- It is a waxy layer on top of epidermis. It is a waterproof and therefore reduces the loss of water from the leaves and also prevents entry of pathogens.
- ✚ Palisade layer/mesophyll- : main organ of food production. The cells contain a lot of chloroplast.
- ✚ Upper epidermis below the cuticle and forms a protective covering.
- ✚ Lower epidermis at the lower surface and contains stomata for gaseous exchange.
- ✚ Spongy layer :- have got a lot of air spaces and is important for gaseous exchange.
- ✚ Vascular bundle:- contains the xylem and phloem vessels.
 - Xylem vessels -carries water required for photosynthesis.
 - Phloem vessels- carry away soluble food to other parts of the plant.
- ✚ Stomata are small pores on the lower epidermis for gaseous exchange during photosynthesis, respiration and transpiration.

Adaptations of a Leaf for Photosynthesis

- ✚ Large surface area to trap light.
- ✚ Thin leaf to allow light through and a short diffusion for carbon dioxide.
- ✚ Presents of stomata on the underneath of the leaf to allow gaseous exchange.
- ✚ Spongy layer which creates air spaces to allow carbon dioxide to enter through.
- ✚ Chlorophyll to attract or trap sunlight.
- ✚ Venation (veins) to carry water into the leaf and soluble food out of the leaf.
- ✚ Many stomata on the lower surface for gaseous exchange and water vapour.
- ✚ The cuticle is transparent to allow easy passage of light.

End products and their fate

- Starch is manufactured.
- Oxygen is given out or produced. Oxygen passes out by diffusion into the surrounding air, Some of the oxygen is used by plants during respiration.

- Carbohydrates or starch is transported to other parts of plant. They are transported as sugars as these molecules are soluble.
- Carbohydrate are used for energy or stored as starch which is insoluble.
- Cellulose is also a structural compound from carbohydrates used to make plant cell walls and give plant support. It makes up the fibred found in plant such as carrots, sugarcane and mangoes.

Question 2

Why is photosynthesis important? (4)

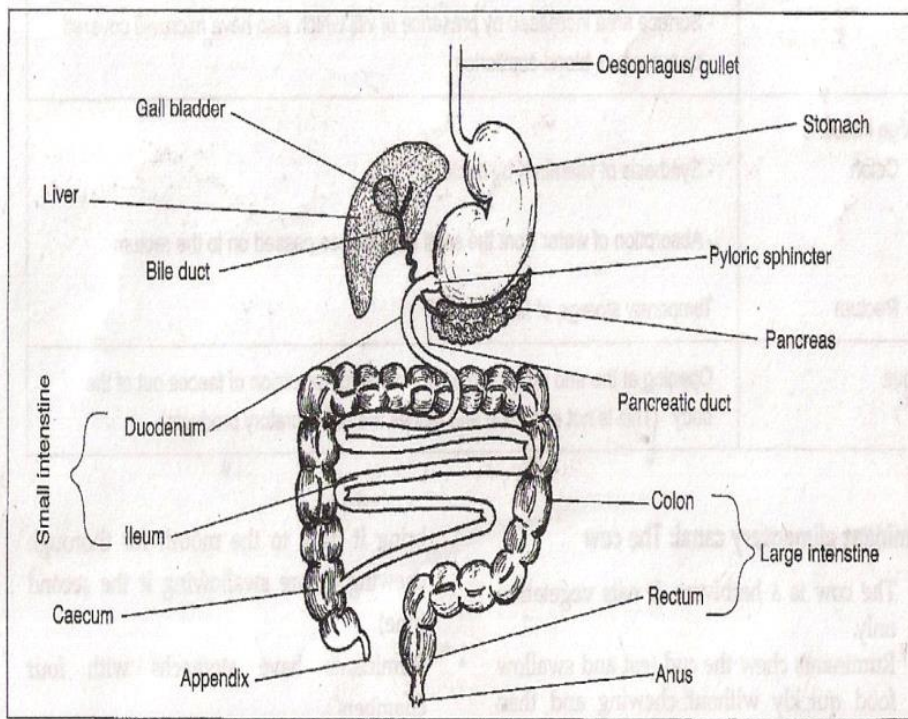
1. It produces carbohydrates which are food for animals. The plants themselves also manufacture their own food. This is why plants are called Producers when looking at food chains and food webs.
2. Photosynthesis produces oxygen which is used by plants and animals during respiration.
3. Plants use carbon dioxide during photosynthesis which helps to reduce global warming.
4. Photosynthesis converts light energy into chemical energy. This chemical energy can be used for many processes in our everyday life e.g. we burn firewood to get heat.

ANIMAL NUTRITION

Alimentary system

Is made up of many different organs. These organs together, help with the processing of food.

The alimentary canal

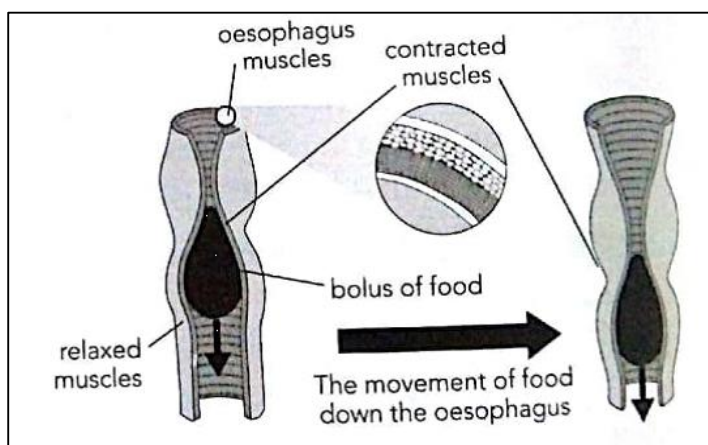


- ✚ Ingestion is a process of taking in food. In the mouth, food is broken down into smaller pieces by the teeth (physical digestion).
- ✚ Food is mixed with saliva which contains enzyme salivary amylase. It breaks starch into maltose (chemical digestion).
- ✚ From there food passes the oesophagus to the stomach. In the stomach, it is mixed with gastric juice containing gastric enzymes.
- ✚ Food remains for some hours and is then churned by movement of stomach walls (physical digestion).
- ✚ Food then passes into the duodenum where many digestive enzymes act on it and food is broken into smaller particles.(chemical digestion).
- ✚ Food travels into small intestines (ileum) where most of digested food is absorbed into blood streams.
- ✚ The remains pass into the colon (large intestines), and the rest enters into the rectum and is egested as waste material through the anus.

Parts of the digestive system

Mouth – food enters the mouth (ingestion) where it is chewed and mixed with saliva. Starch is digested in the mouth by action of enzyme called salivary amylase. The tongue rolls food into balls and pushes the food down.

Gullet/oesophagus – when food is pushed, it is forced to move slowly through this passage to the stomach. This process is called peristalsis (contraction and relaxation of muscles).

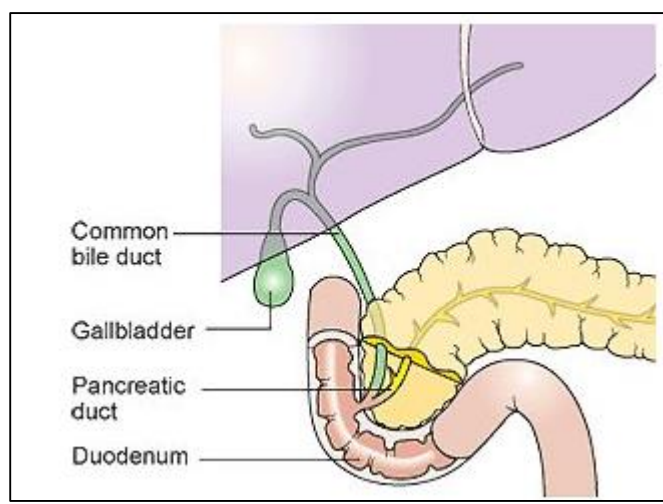


Stomach – food is kept here for up to 5 hours and chemical digestion takes place. Food is chewed into churning by action of stomach juices. Stomach-walls secrete gastric juice containing enzymes renin and pepsin. Renin assists in coagulating milk thus, preventing it passing too rapidly through the canal before it can be digested. Renin is important to children who frequently consume milk. Pepsin breaks down protein into peptones. Hydrochloric acid provides the correct pH for gastric enzymes to act.

Small intestines – food enters the small intestines from the stomach. This is where digestion is completed. The gall bladder produces bile to digest proteins and other carbohydrates. The ileum joins the large intestines

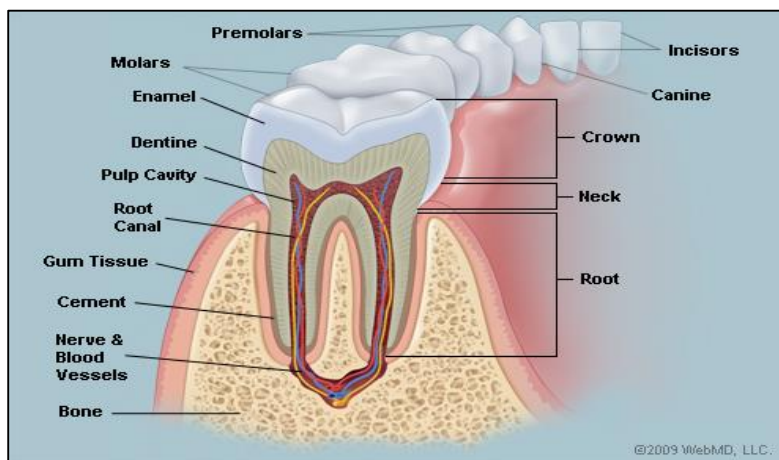
Large intestines /colon – this is for absorption of water which enables wastes (faeces) to pass out via the rectum to the anus where egestion

Duodenum- undigested food is converted into maltose. Proteins are changed into peptones. Fats are changed into fatty acids and glycerol. The digestion of fat is aided by bile which is made in the liver, stored in the gall bladder and carried to the duodenum along the bile duct. Peptones from the stomach are changed into amino acids and amino acids are small enough to be absorbed through the ileum. Sugars are converted into glucose which is absorbed through the intestinal wall.



Types of teeth and functions

Teeth are used for cutting and chewing food.



Identify types of teeth and explain their functions (8)

The teeth are used to bite, tear, crush and grind food into smaller pieces. There are four main types of human teeth. Each type is different in structure and performs a specific function:

1. Incisors (*i*): Four front teeth in the upper jaw and four front teeth in the lower jaw. They are chisel-shaped and are used for biting and cutting food.
2. Canines (*c*): These are single teeth in each half jaw next to the incisors. They are pointed and are used for gripping and tearing food.
3. Premolars (*pm*): There are two behind the canines on the upper and lower jaws, thus totalling eight. Together with the molars, they form the cheek-teeth and have flattened surfaces for crushing and grinding food.
4. Molars (*m*): There are six molars found at the back of the upper jaws and six in the lower jaws. They have flattened surfaces for crushing and grinding food.

Causes of dental decay/dental caries

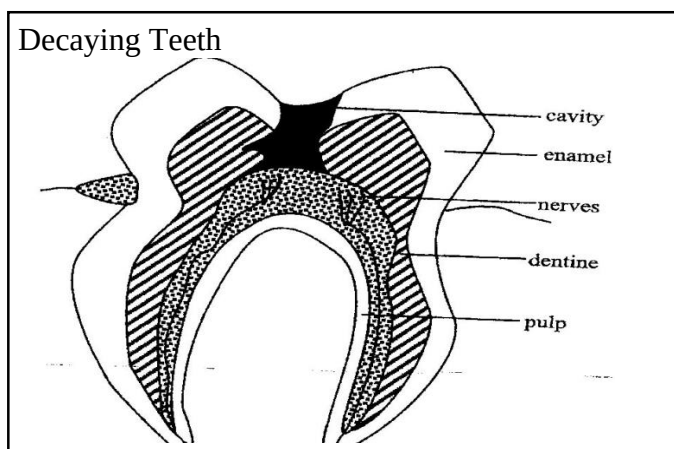
Caused by bacteria and plaque.

Plaque- It is a thin layer of food remains on the tooth surface which contains bacteria.

Stages in Tooth Decay

1. Damage to the enamel results in no pain being felt.
2. Dentine damage results in extreme cold/heat causing a pain sensation.
3. Opening into the pulp results in severe toothache as a result of exposure of the blood capillaries and nerve endings. Tooth extraction is usually recommended at this stage.

N.B- Action of bacteria on sugary food remains has been associated with dental decay.



Caring For Teeth

- ✓ Regular brushing.
- ✓ eat a balanced diet.
- ✓ Use a tooth brush .
- ✓ brush after meal.
- ✓ brush gently for 1-3 minutes.
- ✓ visit the dentist regularly
- ✓ eat fibrous foods.
- ✓ brush in circular motion.

Question 1

- a) Give the difference between non-ruminant and ruminant animals. (6)
- b) What are the functions of the hepatic portal vein and the liver? (6)

Solution

a)

Ruminants e.g. cow	Non ruminants e.g. rabbit
Have four stomach chambers	Have one stomach
Chew the cud	Do not chew the cud
Cellulose digestion in the rumen	Cellulose digestion in the caecum

Digestion

- Is the breaking down of insoluble food particles into soluble ones.

Mechanical digestion

- Is the physical breakdown of food in the alimentary canal. It starts in the mouth by the action of teeth.
- In the stomach, food is squeezed, churned and mixed with gastric juices. Involuntarily muscles in the stomach bring this automatic action about.
- Peristaltic movements in alimentary canal (successive squeezing movements in the walls of the alimentary canal)
- Anti-peristalsis in ruminants allows the cow to chew and this is physical.
- The strong contraction of the rectal muscles helps to expel the undigested matter.

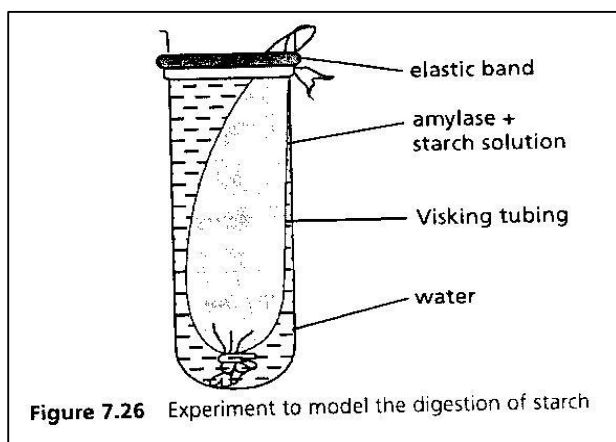
Chemical digestion

- Is the breakdown of food into smaller soluble molecules so that they can be absorbed into the blood stream. Enzymes bring about chemical digestion.
- Enzymes are produced either in the walls of the gut or in glands. They are catalysts which speed up chemical digestion.

NB: In the caecum (of rabbits) and rumen (of cattle) the bacteria secretes cellulose that changes cellulose to glucose.

Modelling the action of amylase on starch

- Collect a 15 cm length of visking tubing which has been softened in water.
- Tie one end tightly. Use a syringe to introduce 2% starch solution into the visking tubing, to about two thirds full.
- Add 2 cm³ of 5% amylase solution (or saliva if it is permissible).
- Pinch the top of the visking tubing to keep it closed, before carefully mixing its contents by squeezing the tubing.
- Rinse the outside of the visking tubing thoroughly with tap water, then place it in a boiling tube, trapping the top of the tubing with an elastic band.
- Add enough distilled water to cover the visking tubing.
- Test a small sample of the distilled water and the contents of the Visking tubing for starch and reducing sugar, using iodine solution and Benedict's solution.
- Place the boiling tube in a beaker of water or a water bath at 37 °C.
- After 20 minutes, use clean teat pipettes to remove a sample of the water surrounding the Visking tubing and from inside the visking tubing.
- Test some of each sample for starch, using iodine solution, and for reducing sugar, using Benedict's solution.
- Also test some of the original starch solution for reducing sugar, to make sure it is not contaminated with glucose.



Results

- At the start of the investigation the distilled water tests negative for starch (stays brown) and reducing sugar (stays turquoise). The contents of the visking tubing are positive for starch (blue-black), but negative for reducing sugars (stays turquoise).
- After 20 minutes, the contents of the visking tubing are yellow/brown with iodine solution, but turn orange or brick red with Benedict's solution. The water sample stays yellow/brown with iodine solution, but turns orange or brick red with Benedict's solution.

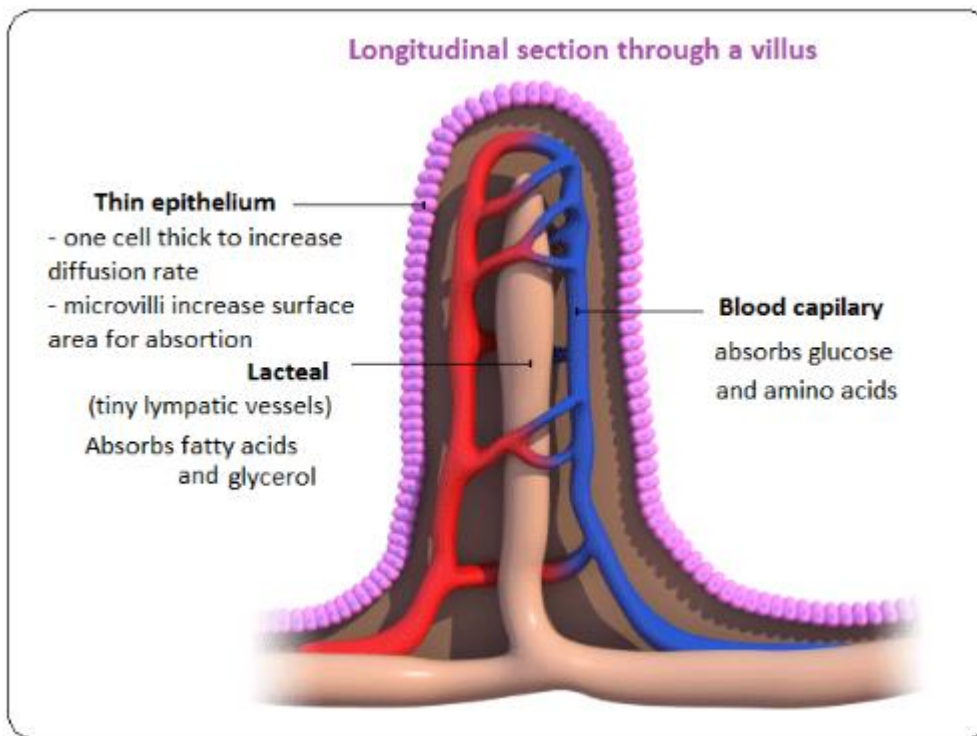
Interpretation

- The amylase digests the starch in the visking tubing, producing reducing sugar.
- The complete digestion of starch results in a negative colour change with iodine solution.
- The presence of reducing sugar (maltose or glucose) causes the Benedict's solution to turn orange or brick red.
- The reducing sugar molecules can diffuse through the visking tubing into the surrounding water, so the water gives a positive result with Benedict's solution.
- Starch is a large molecule, so it cannot diffuse through the tubing: the water gives a negative result with iodine solution.

This model can be used to represent digestion in the gut. The starch solution and amylase are the contents of the mouth or duodenum. The visking tubing represents the duodenum wall and the distilled water represents the bloodstream, into which the products of digestion are absorbed.

Absorption

- Soluble food molecules are the end products of digestion. This happens mainly in the small intestines or ileum. Water and alcohol can be observed directly through the walls of the stomach. The ileum is very long in most mammals and its inner wall with its folds and villi greatly increase the surface area available for absorption.
- The walls of villi are one cell thick and this enables simple food molecules to pass through.
- Inside each villus are capillaries with blood which collect and carry food molecules from the small intestines.
- Glucose and amino acids are absorbed by diffusion from an area of their high concentration in the ileum to one of their low concentration in the villus.
- Inorganic salts, vitamins and water are also absorbed through the villi in the ileum
- Fatty acids and glycerol pass into the blood stream via the lacteal ducts of the villi and the lymphatic system.



The role of the liver

- Glucose, amino acids and fatty acids are carried to the liver in the blood via a system of capillaries and veins to the large hepatic portal vein. This connects the blood system surrounding the ileum to the liver.

NB: The principal area for the absorption of nutrients is the ileum/small intestines.

Assimilation

- It looks at the fate of the products of digestion that have been absorbed into the body of an animal.
- The carbohydrates and amino acids are taken to the liver through the hepatic portal vein.
- Glucose is stored in the liver or in muscles as glycogen or fats
- Excess amino acids are taken back to the liver where they are de-amined to remove nitrogen which is made into urea. The urea is then taken to the kidneys where it is removed as urine.
- Fatty acids and glycerol are burnt to provide energy to the body especially when glucose is in short supply.
- Some fatty acids are made into cell membranes.
- Excess fatty acids are converted to fats which insulate the body providing warmth, and protect organs such as the heart and the intestines.

Food Tests

- It is important to know how much nutrients they contain. This would be useful to nutritionist and those who are health conscious.

Experiments

Materials

Benedict's solution /Clinistix, iodine solution, Biuret's solution / Albustix, ethanol, paper, a variety of foods to be tested, burners, suitable containers.

Testing for starch

1. Drop a few drops of iodine solution on the food.
2. Or add a food sample to a test tube containing an iodine solution

Results

The iodine solution turns blue black if starch is present.

Testing for Glucose

Method

1. Place about 2 ml of food solution in a test tube and add some Benedict's solution
2. Heat the test tubes in a water bath.
3. Observe the colour change from green/orange to red/brown/ brick red.

Results

The Benedict's solution changes from green/orange to brick red if the food contains sugar/glucose

Protein

1. Add a few drops of Biuret solution to the food sample.
2. Add a food sample to a test tube containing Biuret solution

Results

If the food sample contains proteins the solution will turn blue/ a darker shade of blue.

Fats

1. Place the food sample (about 2ml) into a test tube.
2. Add ethanol and shake thoroughly.

Results

An emulsion forms if fat or oil is present.

NB. Alternatively a small amount of the food can be smeared onto a piece of paper. And left to dry for a couple of minutes, the remaining stain is then observed. If a translucent stain forms then fat is present in the food sample being tested.

Qn 1. Describe food tests using Clinistix and Albustix. [6]

RESPIRATORY SYSTEMS

The system is responsible for gaseous exchange and breathing so that oxygen is taken in and carbon dioxide expelled out. All organisms need oxygen from the air. The oxygen is taken into cells so that glucose can be broken down (respiration).

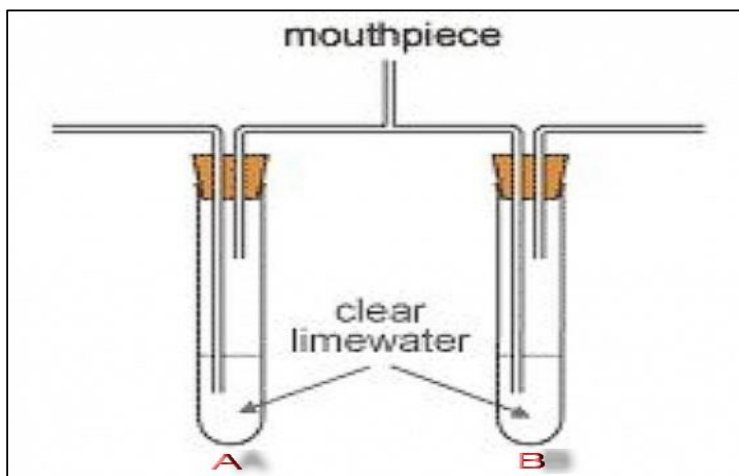
The respiratory system consists of the lungs, organs inside the head and of the chest.

Experiment 1 (a)

Comparing inhaled and exhaled air

Apparatus: carbon dioxide indicator, two test tubes, mouth piece,

Method



- ✚ Set up the apparatus.
- ✚ Breathe in and out of the mouth piece while closing pipe A and B to help the movement of air.
- ✚ Observe any colour change.

Observations

Lime water in A remains clear and in B turns milky.

Results

Lime water changes colour to milky showing the presence of carbon dioxide.

Conclusion

There is more carbon dioxide in B than in A.

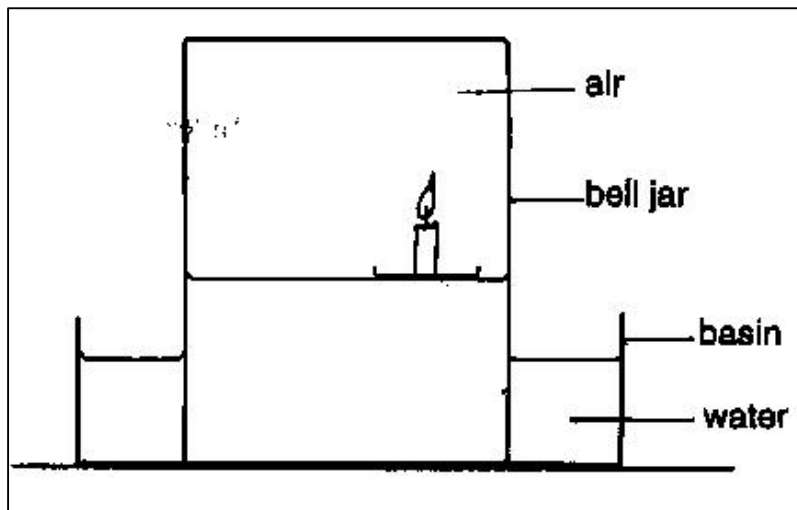
Experiment 2 (b)

Aim: comparing inhaled and exhaled air

Materials

Large transparent container e.g. bell jar, candle, tin lid, water, large basin, rubber tubing , ruler.

Method



- Place the bell jar over a lighted candle floating in a basin of water.
- Measure how far the water rises up after the candle has been extinguished.
- Repeat this with exhaled air after holding your breath for 30 mins, this will fill the container with exhaled air.
- Place a candle in the container and measure the water displacement again.

Results and explanation

- Water displacement in the first instance when natural air is used is greater due to more oxygen which is then used up by the candle.
- Oxygen combines with carbon to form carbon dioxide. The candle eventually goes out as oxygen runs out and the container is filled with carbon dioxide.
- There is less displacement in the second instance as exhaled air contains less oxygen.

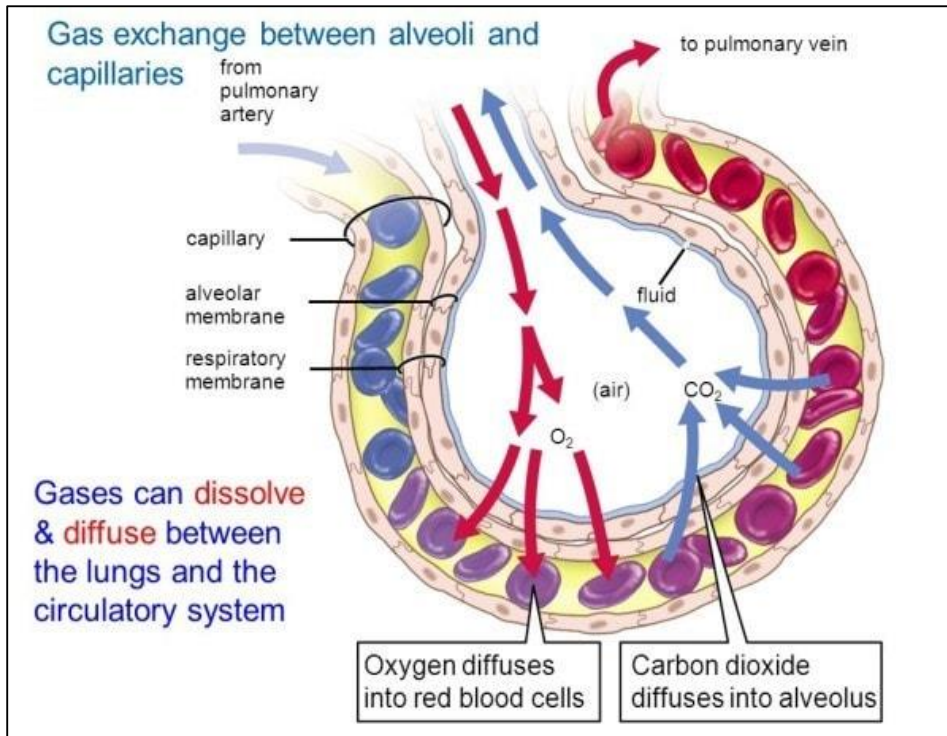
Conclusion

Inhaled air contains more oxygen than exhaled air.

Differences between Inhaled and Exhaled Air

	Inhaled	Exhaled
Oxygen	21%	16%
Carbon dioxide	0,03%	4,1%
Water vapour	Small percentage	A larger percentage
Temperature	Lower than exhaled	Higher than inhaled
Nitrogen	78%	78%

The Alveoli or Air Sacs



- The alveoli are air sacs where gaseous exchange takes place.
- They are good gaseous exchange surfaces because there are many and their total surface area is large enough intake of oxygen and expulsion of carbon dioxide.
- They are moist, thin walled to allow gaseous exchange and re surrounded by blood vessels.
- Capillaries surrounding the alveoli contain less oxygen than the air sacs.
- Oxygen therefore diffuses from the alveoli into the capillaries.
- It is then carried by blood to all parts of the body to respiring cells.
- The blood in the capillaries has more carbon dioxide than the blood in the air sacs, so by diffusion carbon dioxide moves into the air sacs and expelled through the bronchioles.

TRANSPORT SYSTEMS

Transpiration

Question 1 – define transpiration. (2)

It is the loss of water from the plant through the leaves and stem. Water is lost from the plant mainly through the stomata on the leaves.

Importance of Transpiration

- For continuous absorption of water. The continuous flow of water from the roots up the stems and out of the leaves is called transpiration stream.
- For the plant to take up dissolved mineral nutrients/ions.
- It has a cooling effect on the plant due to evaporation.

However, when plants lose more water than they can absorb from the soil wilting occurs

i.e. wilting occurs when plant leaves lose excessive water. Permanent wilting leads to plant death.

Experiment 1

Do plants lose water?

Apparatus: one potted plant, plastic bag, rubber band, blue anhydrous cobalt chloride paper.

Method

- ✚ Place a transparent plastic bag over a potted plant.
- ✚ Leave the set up for about 20 minutes.
- ✚ Observe the results, if there is a liquid, drop it on blue cobalt chloride paper to see if it is water.

Observations and results

Droplets of liquid are seen inside the plastic bag.

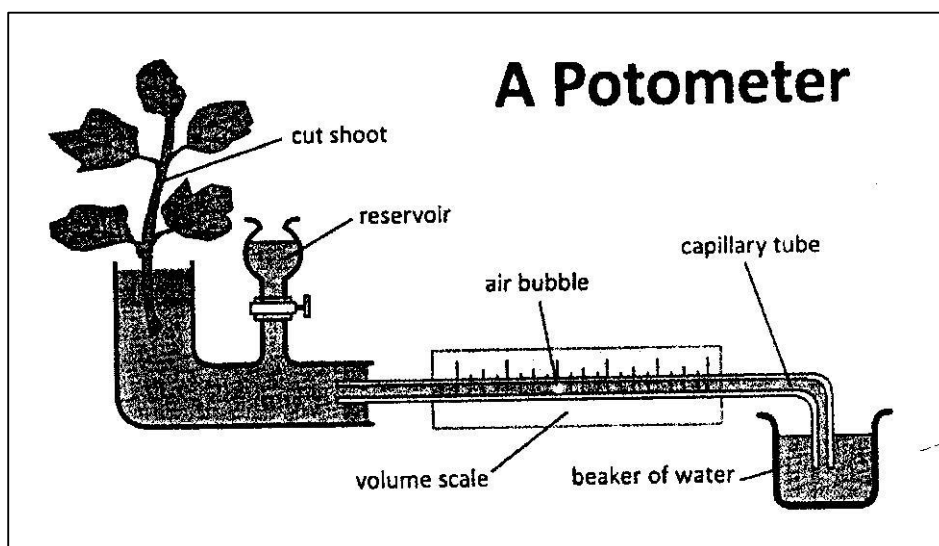
Blue cobalt chloride paper changes to pink i.e. the positive test of water.

Experiment 2

Investigating factors affecting the rate of transpiration

The apparatus shown below is called a potometer. It is used to measure the rate of transpiration.

Method



- Cut a woody shoot and immediately place the cut edge of the shoot under water.
- Immerse the potometer completely under water in a big bowl or sink with the tap under the reservoir open. Make sure the potometer is completely filled with water and that there are no air bubbles.
- Put the cut stalk of your leafy shoot into the water in the sink and cut off the last centimetre of the stem obliquely under water.
- With the potometer and the stalk of the shoot still under water, fit the stalk into the rubber bung. It must fit very tightly.
- Close the tap under the reservoir and remove the shoot and potometer from the water. Clamp in position if necessary and keep in a still, lit position.
- Any air bubbles in the capillary tube should be expelled by opening the tap to let water run into the potometer bottle from the reservoir.
- At this stage if the apparatus is not air tight, water will be seen oozing out from the sides of the stopper. If necessary smear some Vaseline around the rubber stopper to prevent any leakage.
- Prepare a results table.
- Record the distance the air bubble travelled in mm every minute for at least 10mins
- Calculate and record the distance travelled in mm per min.
- Repeat at least twice and calculate the average distance.

The conditions can now be changed in one of the following ways:

- Move the apparatus into sunlight or under a fluorescent lamp.
- Blow air past the shoot with an electric fan or merely fan it with an exercise book.
- Cover the shoot with a plastic bag.
- After each change of conditions, take three more readings of the rate of uptake and notice whether they represent an increase or a decrease in the rate of transpiration.

Results

- An increase in light intensity should make the stomata open and allow more rapid transpiration.
- Moving air should increase the rate of evaporation and, therefore, the rate of uptake.
- The plastic bag will cause a rise in humidity round the leaves and suppress transpiration.

Limitations of the potometer

- Not all the water taken up will be transpired; some will be used in photosynthesis; some may be absorbed by cells to increase their turgor. However, these quantities are very small compared with the volume of water transpired and they can be disregarded.
- The rate of uptake of a cut shoot may not reflect the rate in the intact plant. If the root system were present, it might offer resistance to the flow of water or it could be helping the flow by means of its root pressure.

Factors affecting Transpiration (Summary)

The rate of transpiration is affected by environmental factors: light, heat, air and wind.

1. Temperature – when temperatures increases the rate of transpiration increases as evaporation of water on leaves becomes rapid. When temperature is low the rate is also low.
2. Humidity – when the atmosphere is saturated with water vapour the rate of transpiration will be reduced. The rate will be low when the atmosphere is dry.
3. Light intensity – transpiration is fast when there is more light as this would encourage the opening of stomata.
4. Surface area – plant with broad leaves lose more water than those with small, thin leaves.
5. Air movement – water is lost faster on a leaf exposed to the wind, as more water is swept from the surface of the leaf thus transpiration is rapid.
6. Stomata – the greater the number of stomata on the leaf surface, the greater the rate of transpiration.

Question 2

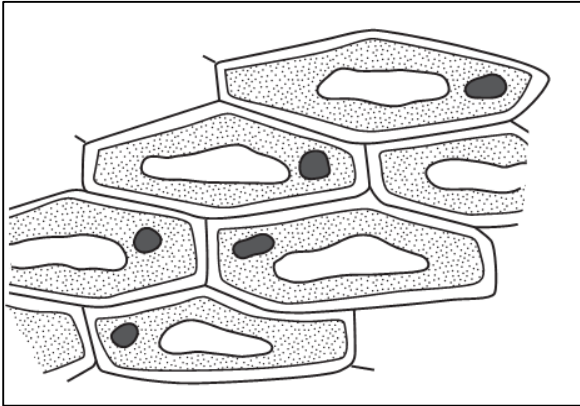
With the aid of labeled diagram, describe an experiment to illustrate that water is given off during transpiration. (6)

Turgidity and Plasmolysis

Turgidity

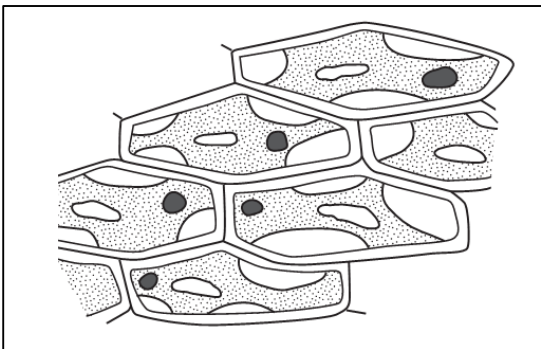
Turgid means cell is completely filled with water and cannot absorb more water.

- When plant cells are surrounded by water, there is a higher concentration of water molecules outside compared to inside the cells.
- Water moves into the cells by osmosis.
- The cells become firm and hard.
- We say the cells are turgid.
- The cellulose cell wall prevents the cells from bursting.



Plasmolysis

- Plasmolysis /flaccid means cell has lose water and become soft
- When plant cells are surrounded by concentrated sugar solution they lose water by osmosis.
- Water molecules move from the plant cells into the sugar solution where they are at a lower concentration.
- The cell walls sag inwards, the cells become soft.
- We say the cells are flaccid.



Osmosis and Diffusion

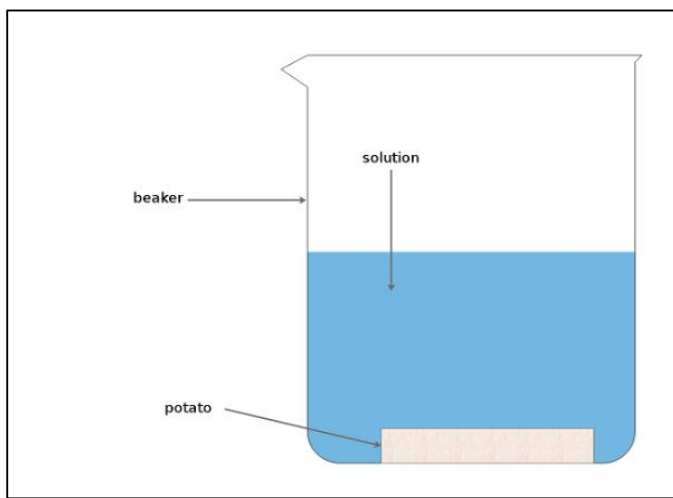
Diffusion is the movement of particles from a region of high concentration to low concentration through a concentration gradient. Diffusion only stops when the concentration of particles in region A is equal to that in region B, I.e. when there is equilibrium.

Osmosis is the movement of water particles from a region of high concentration to low concentration through a semi-permeable membrane. This type of membrane usually refers to animal and plant cell membranes.

Experiment 1

Demonstrating turgidity and plasmolysis

Method



- Cut three pieces of peeled potato of about 5cm x 1cm.
- Place the in distilled water until they are required.
- Prepare three solutions and place in three different containers as follows:
 - A - distilled water.
 - B - 5% solution by dissolving 10g of sugar in 90g of water.
 - C – 10% solution by dissolving 10g in 90g of water.

- Record the size of potato.

Results

- ✓ In A, there the potato increases in size due to osmosis.
- ✓ In B there is no discernible change in size because of more or less equal water concentration without and within the potato piece hence negligible change in size.
- ✓ In C, the salt solution increases and the potato becomes soft due to less water concentration outside the potato.

Transport System in humans

Blood circulatory

- Animals require a system to carry simple food molecules, oxygen, water other nutrients and also unwanted substances which should be removed from body cells.
- The blood provides a fluid carrying medium which makes up a circulatory system. In mammals, the system is facilitated by a pump (the heart) to create the force to push these substances through tubes or blood vessels.

The Blood

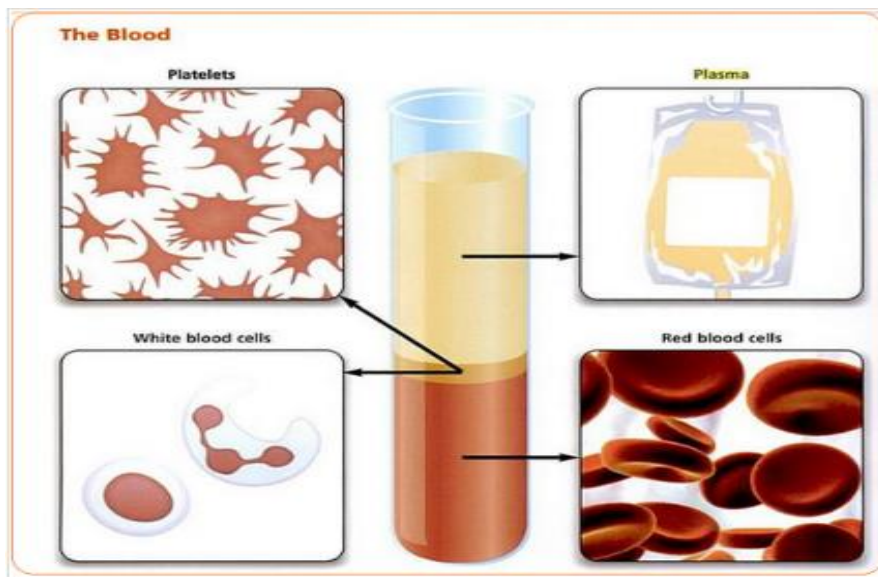
The blood is made up of two major components

1. Blood Plasma, and
2. Blood Cells (red blood cells, white blood cells and blood platelets)

Activity: Allowing blood to stand

If blood is allowed to stand without clotting, it separates out into 4 components: plasma, red blood cells, white blood cells and **platelets**.

The **plasma** and **red** blood cells play an important role in the **transportation** of substances, around the body. **White blood cells** and **platelets** are part of the body's **immune system**



Blood Plasma

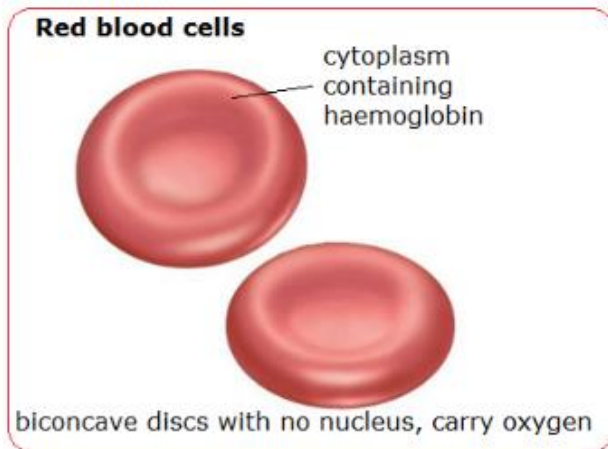
Is the liquid part of blood which carries all the other components of the blood around the body. 55% of the blood is plasma. This straw-coloured liquid contains water with many important dissolved substances which must be carried around the body. Most materials are carried by the blood plasma either in solution or suspension, except for oxygen which is carried by the red blood cells.

Plasma transports:

- blood cells (**red blood cells, white blood cells and platelets**)
- soluble **nutrients** e.g. **glucose and amino acids** (products of digestion) from the small intestine to the organs

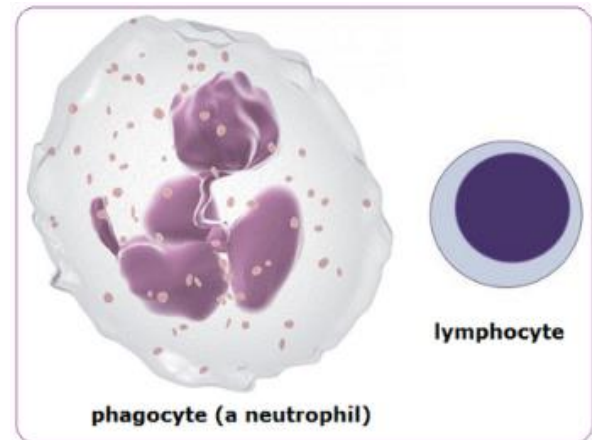
- **amino acids** (plasma acts as a pool for amino acids for these cannot be stored in the body)
- **Plasma proteins** that are important in blood **clotting** (e.g. fibrinogen).
- **Carbon dioxide** (waste gas produced by respiration in cells) from the organs to lungs
- Other wastes of digestion (e.g. **urea**) from the liver to the kidneys.
- **Antibodies** and **antitoxins produced by white blood cells**
- **Hormones**
- **Heat** from the liver and muscles to all parts of the body.

1. Red blood cells



- They are made in the **bone marrow** of some bones, including ribs, vertebrae and some limb bones.
- **Transport Oxygen** from lungs to all respiring tissues.
- Contain **haemoglobin**, a red iron-containing pigment which can **carry Oxygen**. In the lungs, Haemoglobin combines with Oxygen to form oxy-haemoglobin. In other organs, oxy-haemoglobin splits up into Haemoglobin and Oxygen.
- Red blood cells have **no nucleus**.
- Have a special **biconcave disc** shape which increases the surface area and makes the diffusion of oxygen into & out of the cell easier.
- Old red blood cells are broken down in the liver, spleen and bone marrow. Some of the **iron** from the Haemoglobin is stored, and used for making new Haemoglobin, some of it is turned into bile which is used during digestion.

2. White blood cells



- These are made in the **bone marrow** and in the **lymph nodes**.
- They have a **nucleus**, and shape of nucleus depends on type as shown in the diagram above.
- Can **move around** and squeeze out through the walls of blood capillaries into all parts of the body to fight off infection.
- There are two main different kinds of white blood cells and all have the function of **fighting pathogens** (disease-causing bacteria and viruses) and to clear up any dead body cells in your body:

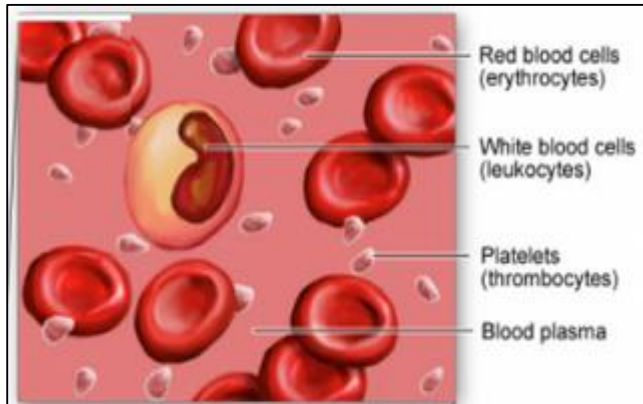
a. Phagocytes

- Have a lobed nuclei and **granular cytoplasm**.
- Can move out of capillaries to the site of an infection.
- Remove any microorganisms that invade the body and might cause infection, **engulf** (ingest) and kill them by **digesting** them.

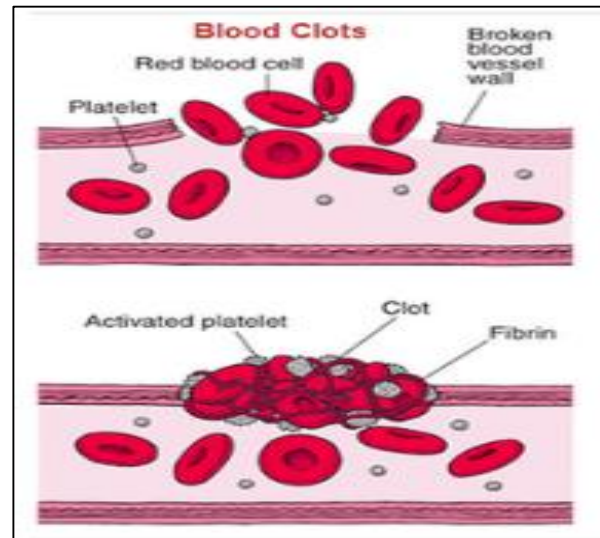
b. Lymphocytes

- produce antibodies to fight bacteria and foreign materials.
- **Have large nuclei**
- Responsible for **immunity**

3. Platelets.



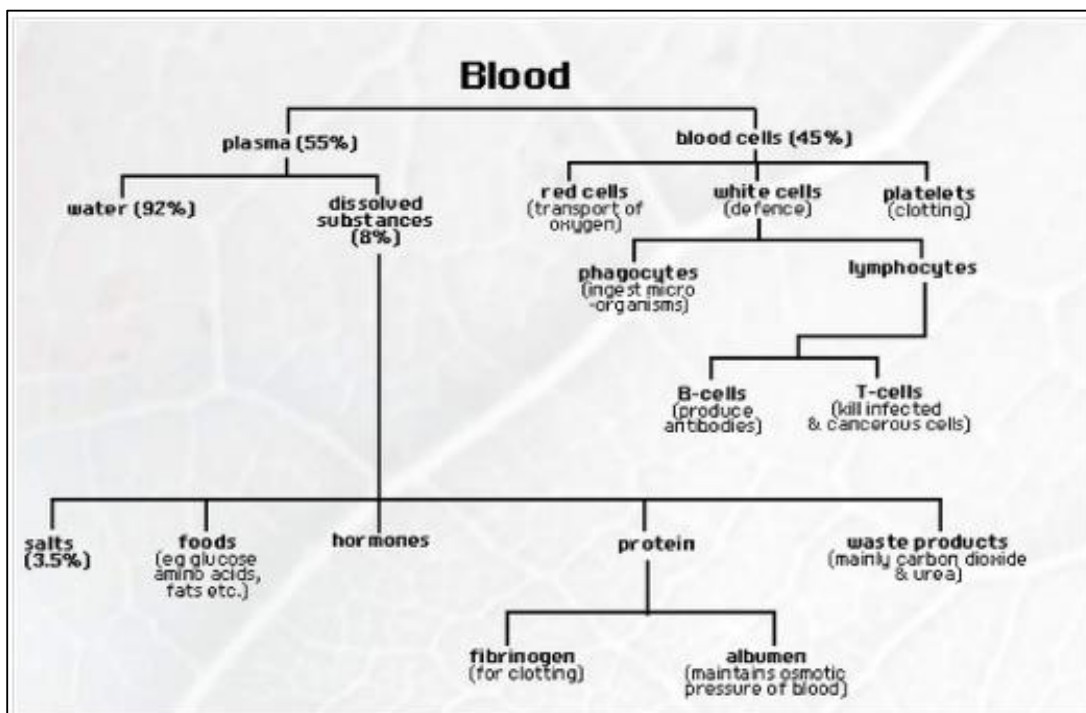
Platelets are responsible for blood clotting. When an injury occurs, fibrinogen in the blood interacts with platelets to form insoluble fibrin. This forms a net over the wound and the net traps blood cells plugging the wound and stopping the bleeding.



Necessity for blood clotting

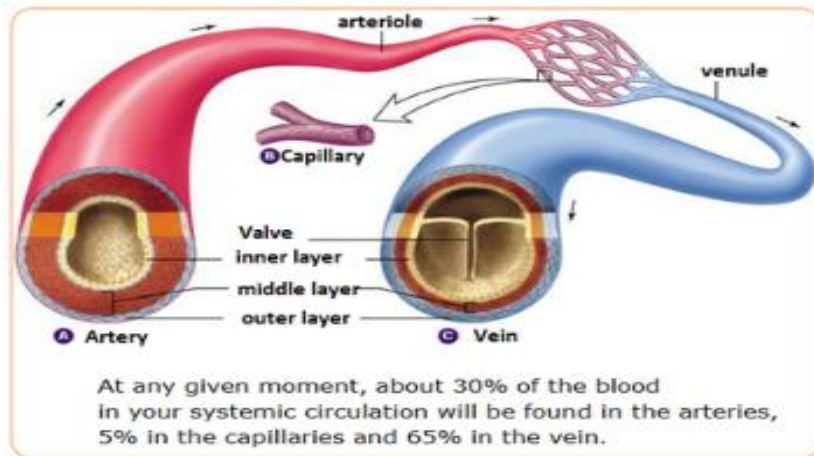
- Prevent excessive blood loss from the body when there is a damage of the blood vessel.
- Maintain the blood pressure.
- Prevent the entry of microorganism and foreign particles into the body.
- Promote wound healing.

Summary of the Composition of the blood.



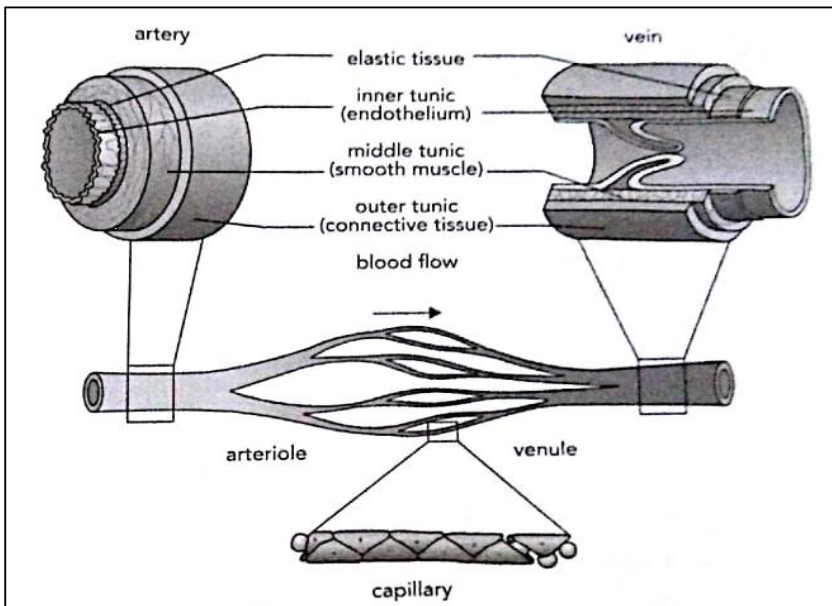
Blood Vessels.

These are the tubes through which blood flows around the body. There are 3 main kinds of blood vessels – arteries, veins and capillaries. Blood leaves the heart through arteries. These break up into smaller arterioles which divide again and again, and eventually form very tiny vessels called capillaries. The **capillaries** gradually join up with one another to form large vessels called veins. **Veins** carry blood towards the heart.



Plan of the main blood vessels in the human body

The comparison of blood vessels structure and functions



The transfer of materials between capillaries and tissue fluid

As blood enters capillaries from arterioles (small arteries), it slows down. This allows **substances** in the plasma, as well as **Oxygen** from red blood cells, to **diffuse** through the capillary wall into the

surrounding tissues (the capillary wall is thin and permeable).

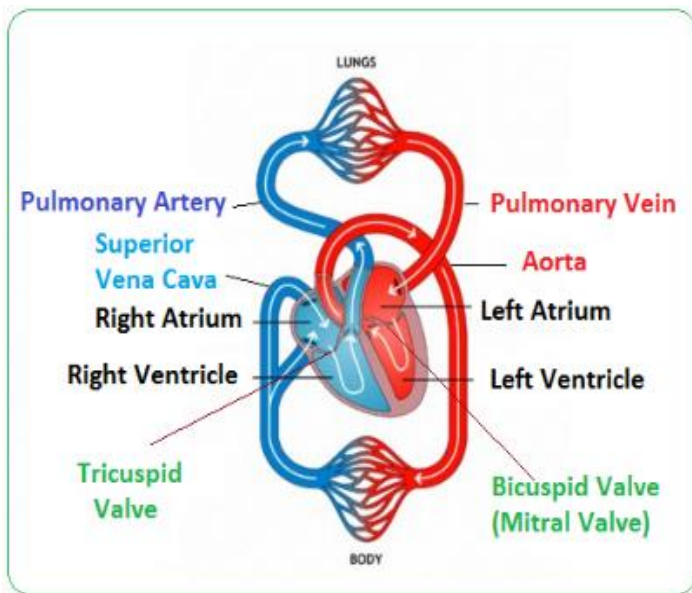
Liquid in the **plasma** also passes out. This forms **tissue fluid**, bathing the cells. **Waste products** from the cells, e.g. Carbon dioxide, **diffuse** back through the capillary walls into the plasma. Some of the tissue fluid also passes back.

Diffusion is responsible for the transfer of materials between capillaries and tissue fluid.

	Arteries	Capillaries	Veins
Function	Carry blood away from the heart at high pressure	-Supply all cells with their requirements -Take away waste products	Return blood to the heart at low pressure
Structure of wall	- Thick , strong -Contain muscles , elastic fibres and fibrous tissue	Very thin , only one cell thick	- Thin -Mainly fibrous tissue -Contain far less muscle and elastic tissue than arteries
Lumen	- Narrow -Varies with heartbeat (increases as a pulse of blood passes through)	- Very narrow -Just wide enough for a red blood cell to pass through	Wide
Valves	(-)	(-)	(+) Prevent backflow
How structure fits function	-Strength and elasticity needed to withstand the pulsing of the blood, prevent bursting and maintain pressure wave -Helps to maintain high blood pressure , preventing blood flowing backwards	- No need for strong walls, as most of the blood pressure has been lost -Thin walls and narrow lumen bring blood into close contact with body tissue, allowing diffusion of materials between capillary and surrounding tissues. -White blood cells can squeeze between cells of the wall	- No need for strong walls, as most of the blood pressure has been lost - Wide lumen offers less resistance to blood flow

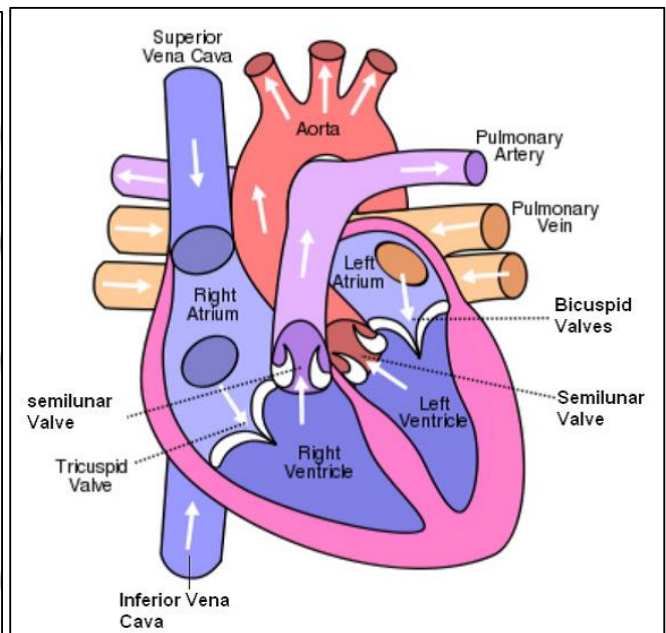
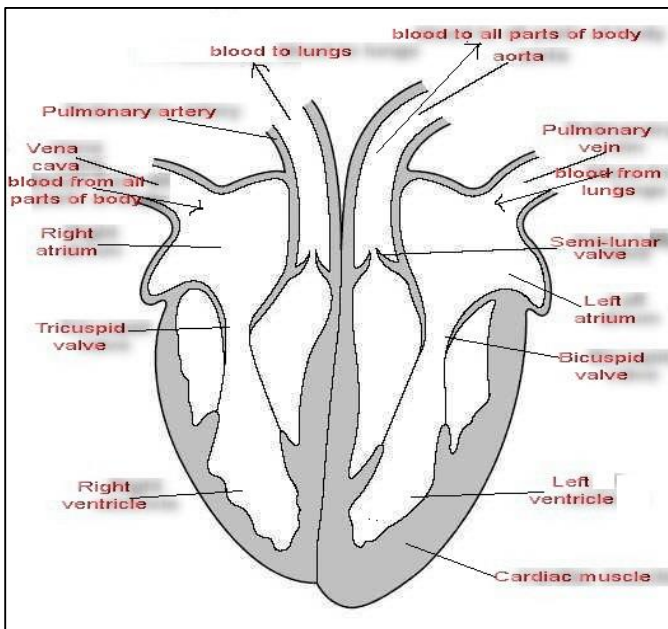
The Circulatory System

- It consists of the heart and blood vessels and these valves to make sure that there is a one way flow.
- Mammals have got a double circulation or pump action. The double action keeps the blood high enough to force the blood through the vessels.



The Heart

Blood from the body enters the right atrium through the vena cava (main vein). The right atrium contracts and forces blood into the right ventricle. The right ventricle contracts and the tricuspid valve closes. Why? Deoxygenated blood is pumped into lungs through the pulmonary artery. In the lungs blood picks oxygen and gets rid of carbon dioxide. Oxygenated blood moves to the left side of the heart and enters the heart through the pulmonary vein, then into the left atrium and the left ventricle through the bicuspid valve. Blood is forced into the aorta to all parts of the body.



Parts of the Heart

- Pulmonary Artery: carries deoxygenated blood from the heart to the lungs.
- Vena Cava :- a large vein returning blood from the body to the right atrium of the heart (deoxygenated blood).
- Right Artrium: -thin walled chamber that receives the deoxygenated blood from the body tissues.
- Tricuspid valve -: these are three flaps to prevent back flow of blood.
- Aorta :- the main artery from the heart to the body. It carries oxygenated blood to the rest of the body.
- Pulmonary vein: -carries oxygenated blood from the lungs to the heart.
- Left atrium:- upper chamber of the heart receiving blood from veins.
- Blood entering the left side of the heart carries oxygen from the lungs.
- Blood entering the right side comes from the body and is then pumped to the lungs to absorb more oxygen.

NB. 1) Blood on the left side of the heart is oxygenated and that on the right side is de-oxygenated.

2) Muscular walls of the left ventricle are much thicker than those on the right because the blood has to reach all parts of the body.

Other main blood vessels in the body

Hepatic Artery	Carries oxygenated blood to the liver
Hepatic Portal Vein	Carries de-oxygenated blood and products of digestion from the small intestines.
Hepatic Vein	Carries de-oxygenated blood from the liver.

PLANT REPRODUCTION

Reproduction

Is a feature of all living organisms and shows continuity of life. Is a process whereby organisms produce off springs of their kind.

Types of Reproduction

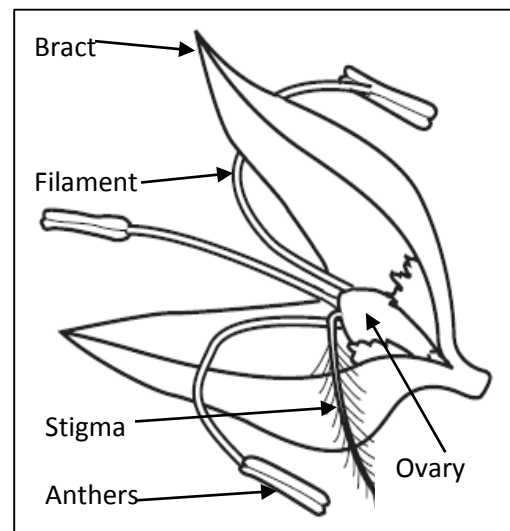
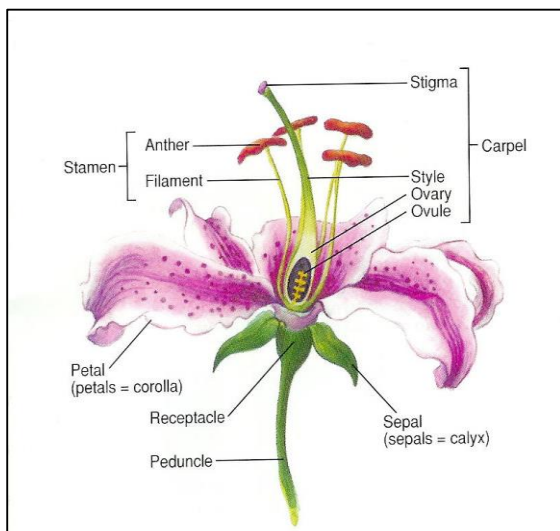
- a) Sexual reproduction-it involves the fusion of male and female sex cells or gametes.
- b) Asexual reproduction-is where by organisms reproduce from multiplication of a single cell e.g. bacteria and amoeba.

In plants, this is named vegetative reproduction (apart of the plant grows and give rise to a new independent plant).

Sexual Reproduction in Plants

The main reproduction organ is the flower and produces fruits and seeds.

Structure of a Flower



- On the diagram, label the filament and style. (2)

Functions of Parts of a Flower

Stigma	Where pollen grains are deposited, produces sticky substance to ensure pollen grains get stuck
receptacle	Where the floral parts of the flower grow and are usually swollen.
Style	Hold the stigma and is a path way for pollen tube to ovules.
Ovary	Produces ovules and develops into fruit.

Ovule	Contains female sex cells and develops into seeds after fertilization.
Petals	Attract insects for pollination and the corolla tube directs insects to the nectaries.
Sepals	Protect the flower when still in bud.
Anther	Male sex cells (produces pollen grains).
Filament	Hold the anther

Features of Wind and Insect Pollinated Flower

Wind pollinated flower	Insect pollinated flower
has small dull petals	large bright petals
do not produce nectar	produce scent
have long stamens which hang out of the flower	produces nectar
produce plenty of pollen	have short stamens
have small, smooth , light and dry pollen	they do not produce much pollen
Have hairy stigma hanging out of the flower	have big, rough, sticky pollen
	have short style with a sticky stigma

Qn 1 a. Define pollination. [2]

b. State types of pollination and the agents of pollination. [4]

Pollination:

Pollination is the transfer of pollen grains from the anthers of a flower to the stigma of a flower.

Methods of Pollination

There are two methods of pollination,

1. insect pollination and
2. wind pollination.

Some flowers pollinate by insects while others pollinate by wind and wind and insects can also be called the agents of pollination.

Insect pollination

Insect pollinating flowers have special attractive features like brightly colored petals, attractive scents and sugary nectar. These features' aim is to attract insects like bees to come and collect their nectar. While an insect is collecting the nectar, its body will touch the anthers. The pollen grains of insect pollinating flowers have hooks and spikes all over them in order to stick to the bodies of the insect that touch it. When this happens, the pollen grain sticks to the body of the insect. When the insect moves on to another flower to collect its nectar, the pollen grain falls off the insect onto the stigma of the flower, thus insect pollination took place.

Wind Pollination

Wind pollinating flowers however look very different from insect pollinating ones. This is because they do not need attractive features such as bright colors and scents and nectar to attract insects. Instead, their petals are green or brown with their anthers and stigmas hanging outside the flower to be exposed to the wind. When wind is strong enough, it will blow the pollen grains off the anther and carry it along. At some point the wind will get weaker and the pollen grain will land, if it lands on the stigma of a flower of the same species then the pollination was successful. The surface of pollen grains of wind pollinated plants is smooth because it does not need spikes to catch on insects.

Types of Pollination

There are two type of pollination

1. Self Pollination and
2. Cross Pollination:

Self pollination - The transfer of a pollen grain from the anther to the stigma of the same flower or the stigma of a flower of the same plant.

Advantages of self pollination

1. Good characteristics can be maintained in the offspring because there is no variation.
2. Chances of success for self pollination are much greater especially were plant population is small.

Cross Pollination - The transfer of a pollen grain from the anthers of one plant to the stigma of a flower on another plant of the same species.

Self pollination is considered sexual reproduction because it involves the fusion of two gametes together even though they might come from the same flower or plant.

Why cross pollination is considered desirable

1. It causes variation in plants which allows plants to adapt to changing environments. This prevents the extinction of plant species.
2. It allows the breeding of better varieties by combining plants with desired characteristics.

How are plants adapted to promote cross pollination

1. Unisex plants – some plants have female plants and male plants and no plant can pollinate itself e.g. paw paw plant
2. Stigmas and anthers of plants in certain plants mature at different times. The stigma matures earlier than the anthers and can only receive pollen from another plant.

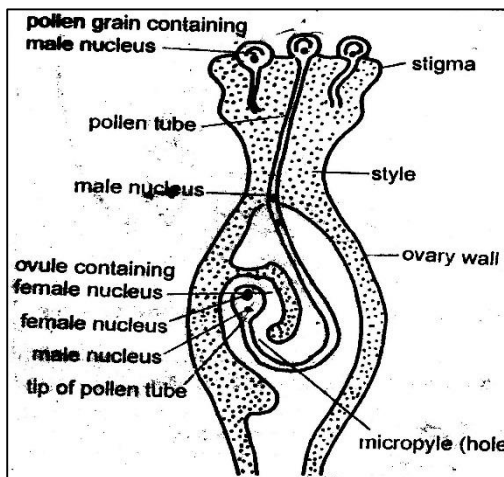
3. Anthers which are lower than stigmas. This ensures that insects touch the stigma on their way in with pollen from another plant. On their way out, they crawl out using the petals carrying pollen from this flower to another plant.
4. Self sterilisation – some plants produce chemicals on the stigma which do not allow pollen from the same plant to develop.

Disadvantages of Cross pollination

Refer to advantages of self pollination.

Fertilisation

Is the fusion or joining of male and female sex cells to form a zygote.



Occurrence

- Fertilization occurs after pollination.
- When a pollen lands on the stigma, it grows a pollen tube(s)
- The pollen tube grows down through the style absorbing a fluid from the style.
- On reaching the ovary, the pollen tube grows towards one of the ovules and enters through a hole (micropyle).
- The male nucleus fuses with the female nucleus to form a zygote.

After Fertilisation

- The zygote develops into an embryo.
- The ovule becomes the seeds and the ovary develops into fruit.

The zygote divides to form a mass of food material called endosperm.

Germination

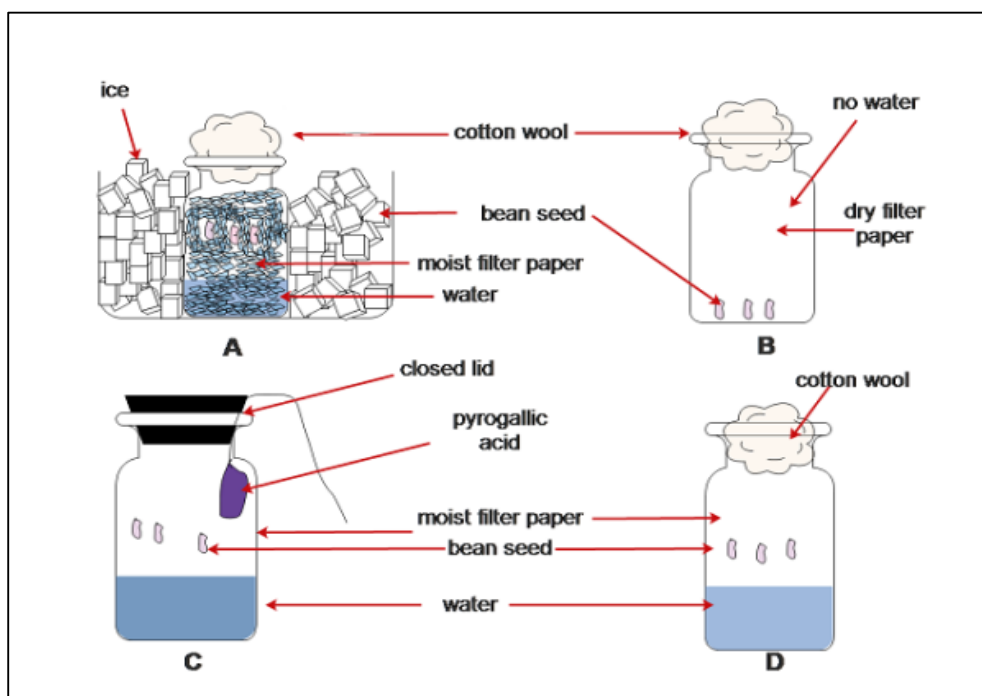
It is the beginning of growth in a plant or a process by which a seed becomes a seedling.

Experiment 1

Investigating conditions necessary for Germination

Materials: maize seeds, containers, cotton wool, filter paper, pyrogalllic acid.

Method



- Take four batches with ten seeds in each.
- Put them into containers as shown above.
- Container A has water, oxygen, low temperature.
- Container B has oxygen, suitable temperature but no water.
- Container C has water, suitable temperature but no oxygen(pyrogallous acid absorbs oxygen).
- Container D is the control.

Results

- ✓ Very few seeds in container A germinated.
- ✓ The dry seeds in container B did not germinate.
- ✓ Seeds in container C did not germinate at all.
- ✓ Most of seeds in D germinated.

Conclusion

- Temperature, moisture and oxygen at optimum are necessary for germination.
- Light is not necessary for germination.

Question1: Explain the main functions of temperature, moisture and oxygen during germination. (6)

Answer:

Condition	Purpose for seed germination
Suitable temperature	Increases enzymes activity which speeds up biological reactions and development of the seed.
Moisture (water)	Necessary for chemical reactions. The medium for dissolving food substances.
Oxygen	Needed for respiration. Oxygen is used to burn the endosperm and /cotyledon (food sources) so that energy is released for the plumule to shoot out of the soil and the radicle to develop into the root system.

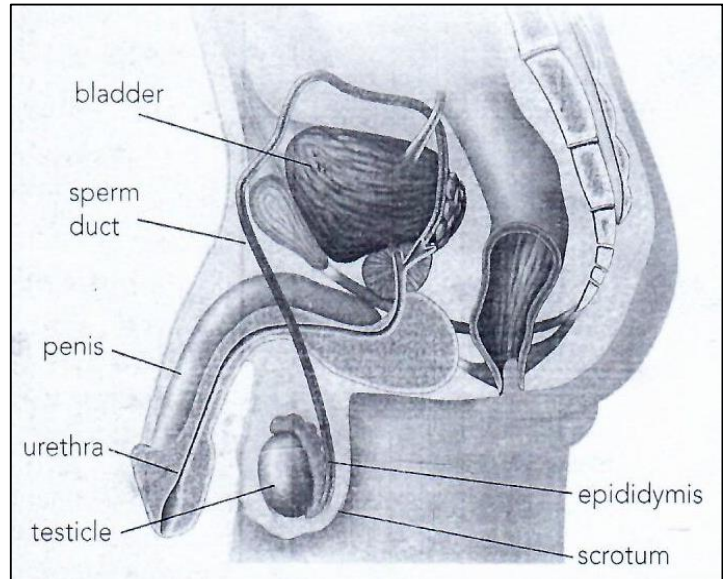
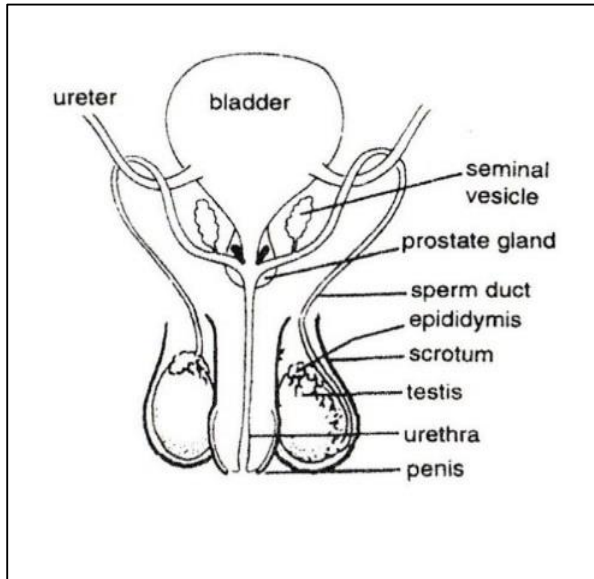
Percentage Germination

Percentage germination= $\frac{\text{Number of seeds that germinated}}{\text{Number of seeds that were planted}} \times 100\%$

Reproductive systems (Human)

Males and females become sexually mature at puberty stage when their bodies undergo changes in appearance and behavior. Reproduction occurs when male and female sex cells fuse (fertilisation). There is internal and external fertilization. The organs and mechanism for reproduction are adapted to ensure successful role (fertilization).

Male Reproductive System

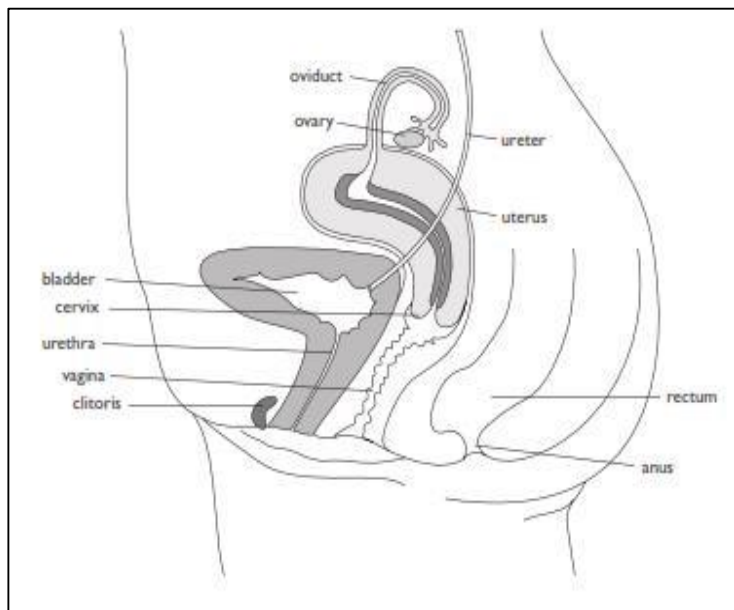
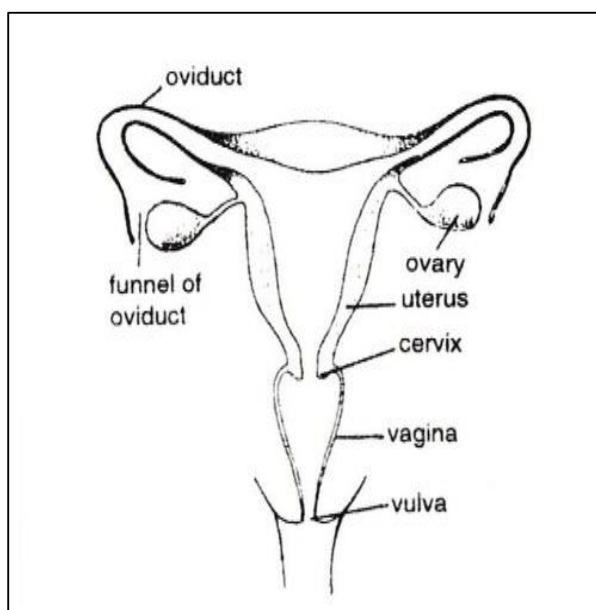


➤ The male reproductive system is also located under the stomach. It consists of the following parts:

- *.**Testes**: a male human has two testicles. A testes is a male gland which produces sperms and the male sex hormone testosterone,
 - *.**Scrotum**: it is the sac which contains the testicles,
 - *.**Sperm Ducts (Vas deferens)**: They are two muscular tubes, each connected to a testis. They carry the sperms from the testis to the urethra,
 - *.**Prostate Gland**: It secretes a nutritive fluid to the sperms to form a mixture called semen,
 - *.**Urethra**: It is a tube inside the penis which is the pathway of semen and urine out of the body,
 - *.**Penis**: It is the male sex organ which ejaculates semen into the vagina during sexual intercourse,
 - *.**Epididymis**: coiled tubes in which sperms are stored,
 - *.**Seminal vesicle**: it is another gland like the prostate gland. It also secretes nutritive fluids for sperms to feed from and swim in forming semen.
- The **testicles** are inside the scrotum which hangs outside the body. This is because testicles work best at a temperature below that of the body.

The Female Reproductive System:

- The female reproductive system is located under the stomach. It consists of the following parts:
- *.**The Ovaries**: There are two ovaries, one on each side. They contain follicles where eggs are produced,
- *.**Oviducts (Fallopian Tube)**: They are two tubes, one on each side connected to an ovary. They are where fertilization occurs and they provide a pathway for the eggs to travel to the uterus by sweeping them by cilia on its walls,
- *.**Uterus (Womb)**: Where the fetus develops,
- *.**Cervix**: A muscular tissue which separates the vagina from the uterus,
- *.**Vagina**: it receives the male penis during sexual intercourse.

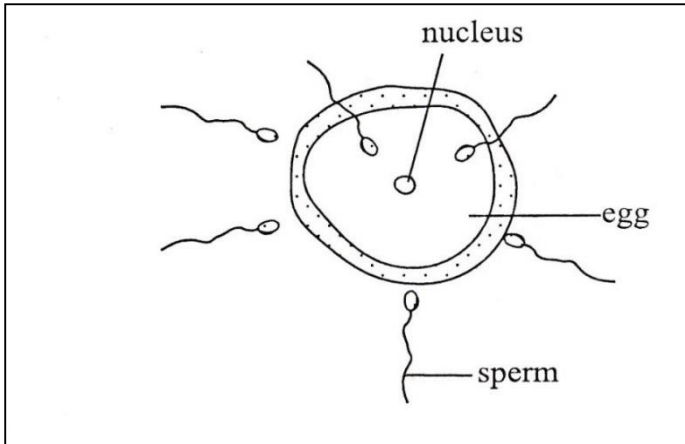


Sex Cells

Ova	Sperm
Small spherical cell with a nucleus, cytoplasm and thick membrane.	Single cell, with oval shaped head.
Larger than sperm-this is because it needs space to store nutrients on which the embryo feeds on before it reaches the uterus.	Smaller than an ovum and in larger quantities to increase the chance of successful fertilisation. Has small middle part and a tail.
Does not move by itself – they are swept to the uterus by cilia in the walls of the oviduct.	Can swim to meet the ovum because it has a large number of mitochondria to release lots of energy to be used in swimming.
Once a month – puberty to menopause.	Million produced.

QN 1 Draw and label the sex cells [6]

Fertilisation



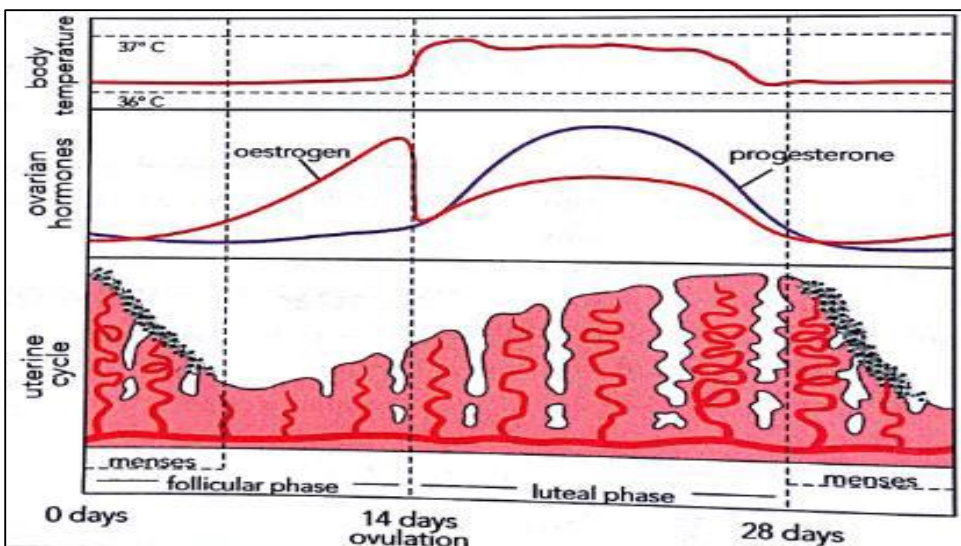
Define the term fertilisation. (2)

During the formation of the gametes the chromosome number is reduced half so that the normal number is retained (46) at fertilisation.

This refers to the fusion of the male and female gamete nuclei to form a zygote; it takes place in the oviduct or fallopian tube.

- Sperm swim through the cervix into the uterus by wriggling movements of their tail.
- They pass through the uterus to the oviduct, one of the sperm may bump into it and stick to its surface.
- The sperm then enters the cytoplasm of the ovum.
- Then the nucleus fuses with the female nucleus.
- A single ejaculation may contain about 5 hundred million sperms, only one sperm fertilizes the ovum.
- When one sperm enters the ovum, it forms a barrier such that other sperms will not enter and will die after fertilization.
- Before fertilization the released ovum can survive for about 24 hours and a sperm for 2 to 3 days.

Menstrual Cycle



Day 1-5:

- The menstrual cycle begins on the first day of bleeding. The thickened spongy layer of the endometrium (uterus lining) breaks down and passes through the vagina. During menstruation, another ovum starts to mature in one of the ovaries

Day 6-14

- The maturing ovum releases the hormone oestrogen that stimulates the endometrium in the uterus to form a new layer of spongy tissue. On day 14 the mature ovum is released from the ovary into the fallopian tube. This is called **ovulation**.

Day 15-28

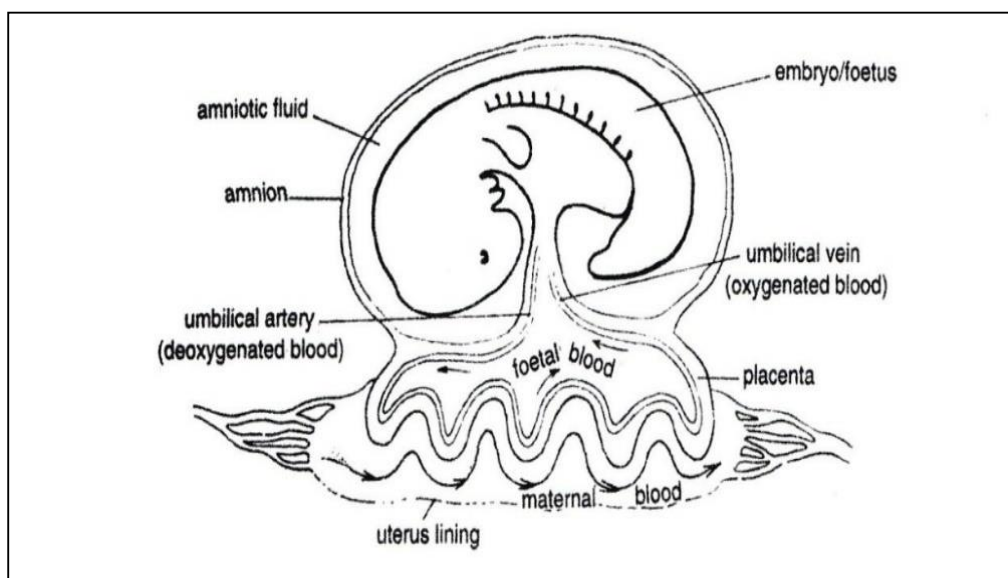
- A yellowish body called the corpus luteum develops in the ovary where the ovum has been released. The corpus luteum produces the hormone progesterone that stimulates the tissue of the endometrium to thicken to prepare for the possible implantation of a fertilized ovum. The ovum passes down the fallopian tube to the uterus. If there are sperms, fertilization can occur. If the ovum is not fertilized it passes through the uterus to the vagina and out of the body. The corpus luteum degenerates and stops producing progesterone. With no progesterone present, the endometrium breaks down again, restarting the menstrual cycle.

Causes of Infertility

- Low sperm count.
- Poor quality of sperm.
- Physical conditions.
- Cancer.
- Damage by sexually transmitted diseases (STDs).
- Blockage of the oviducts or sperm ducts.

Development of the Foetus / Embryo (Pregnancy)

-After fertilisation, a zygote is formed in the oviduct or fallopian tube. The embryo then implants itself into the lining of the uterus and begins to develop on placenta. At about eight weeks after fertilisation, the embryo begins to develop features such as a heart, head and limbs, at this stage is called fetus (foetus). The normal gestation period is about 40 weeks.



Functions of The Placenta, Umbilical Cord And Amnion.

1. Placenta - allows the diffusion of substances such as oxygen, glucose, amino-acids, salts and antibodies from the mother's blood to the fetus.
-Also waste substances like carbon dioxide and urea are excreted together with the mother's waste substances. NB: The placenta is selectively permeable, not all substances in the mother's blood pass to the fetus. Alcohol and nicotine do pass through the placenta and may affect development of the fetus.
2. Umbilical Cord – attaches the fetus to the placenta and carries the umbilical vein (carries oxygenated blood) and umbilical artery (carries deoxygenated blood).
3. Amnion fluid - is a fluid filled sac that supports the fetus and protects it from physical knocks and mechanical damage.

1. What is pregnancy? (3)
2. Describe the functions of the following organs:
 - a) Placenta.
 - b) Amnion / water sac.
 - c) Umbilical cord. (9)
 - d) State any four problems associated with pregnancy. (4)
3. What are the effects of smoking to pregnant women? (4)

HEALTH AND DISEASES

Sexually Transmitted Diseases:

- Sexually transmitted diseases or STD's for short are caused by viruses or bacterium passing from one partner or another during sexual intercourse.
- The most spread STD's are gonorrhea and HIV (also known as AIDS).

Gonorrhea

- Gonorrhea is caused by a bacterium that lives and breeds in damp conditions at 37°C called *Neisseria gonorrhoeae* (also known as *Gonococci/ Gonococcus*).
- It lives in the reproductive system of the carrier person.
- The bacteria could be easily transferred to the partner during sexual intercourse.
- Both males and females can be infected
- Infections usually occur in the urethra, cervix, anal canal and the conjunctiva of the eye.
- The rate of gonorrhea acquisition for males is 35% after a single exposure and this figure rises to 75% after multiple sexual contact with the same infected person
- Gonorrhea can also lead to sterility through infecting the prostate gland in males and fallopian tube in females.

Symptoms of gonorrhea

- In men the symptoms are very clear.
- They include sores on the penis, discharges and burning sensation during urination.
- In women however, the symptoms are less noticeable because women's reproductive organs are inside her body.
- They include discharges which the woman won't be able to differentiate between it and the normal vaginal discharges.
- Women may suffer from swelling of the vulva and abnormal menstrual bleeding
- If a pregnant woman catches the disease, it is likely that the baby will catch it too during giving birth.
- In both sexes untreated gonorrhea may spread to other body parts including the heart valves, the brain and joints
- When it reaches the joints, it causes severe arthritis

Treatment of Gonorrhea

- The disease is easily cured by a course of antibiotics such as penicillin and tetracycline although some strains have become resistant to treatment
- A previously cured infection does not confer immunity. In other words, gonorrhea can be contracted again

Syphilis

- It is a bacterial infection caused by *treponema pallidum* and is passed on during sexual intercourse/ during childbirth from mother to child
- The bacteria enter the body through cuts in the epithelium and by penetrating into mucus membranes
- In a pregnant woman the bacteria cross through the placenta to infect the foetus

➤ Babies with congenital syphilis are usually very weak and ill, living only a few hours after birth

Signs and symptoms of syphilis

➤ The disease develops in three stages

➤ During the first stage a lump appears on the penis, vagina or cervix

This turns into an ulcer that disappears after about six weeks. This stage may pass unnoticed.

➤ In the second stage starting six to eight weeks later, the infected person develops a mild fever and a rash or sores around the genitals, anus, mouth and eyes. The lymph glands especially in the neck, may also swell.

➤ The third stage can occur as many as ten or more years after infection. The bacteria will have invaded most parts of the body and they will affect many tissues and organs. This is the most destructive stage as the bacteria destroy nerves, cause heart disease, blindness, sometimes insanity and eventually death.

Treatment of Syphilis

➤ Can be treated with penicillin and other antibiotics, but only during the early stages.

➤ Once the disease has reached the third stage it is difficult to cure as the lesions may have already caused permanent damage to organs

Chancroid (venereal ulcers)

➤ Is an STI caused by the bacterium *haemophilus ducreyi*

➤ The bacteria attack the tissue around the external areas of the reproductive organs of both men and women

➤ An open sore referred to as a chancroid or venereal ulcer, appears on the outside of the genitals.

➤ The ulcer can bleed or produce a contagious fluid that can spread bacteria during sexual intercourse

➤ Chancroid can be acquired through touching an open sore as well as through sexual intercourse

Signs and symptoms of Chancroid

➤ Small bumps appear on the penis or the labia three to five days after sexual intercourse or infection

➤ In men this develops into open sores on the penis and scrotum

➤ In females the bumps turn into ulcers that cause a painful burning sensation during urination or bowel movements.

➤ The sores usually heal quickly but they may persist for months if left untreated

Treatment of Chancroid

➤ Is treated with antibiotics such as tetracycline and sulphanilamide

➤ Antibiotics decrease scarring after the healing of the venereal ulcers

Genital herpes

➤ The herpes simplex virus is categorized into two groups, the herpes simplex virus type 1 (HSV-1) and the herpes simplex virus type 2 (HSV-2)

➤ You can get the herpes simplex virus by having sexual intercourse with an infected person. Fluids found in herpes sores also carry the virus and can cause infection

➤ The virus enters through the skin and thereafter the infection spreads to nerves in the skin

Signs and symptoms of Genital Herpes

➤ Most people infected by the H.S virus have no signs and symptoms

- When signs occur, they are small blisters that eventually break open and become raw painful sores
- This causes itching in the infected area
- The sores later form scabs and heal within a few weeks
- Flu-like symptoms with fever and swollen lymph nodes may also appear

Treatment of Genital Herpes

- There is no cure for genital herpes but the symptoms can be prevented with treatment
- Antiviral medication reduces pain and discomfort from outbreaks of sores
- Taking daily medication can also suppress the virus and reduce the risk of infecting others

Effects of sexually transmitted infections

- STIs are a huge health and economic burden for developing countries
- Herpes and syphilis dramatically increase the chances of getting infected with HIV
- People with STIs are at 2-5% more risk of getting HIV through sexual intercourse
- Syphilis in pregnant women is linked with a low birth weight of babies, premature birth and stillbirth
- Gonorrhea causes pelvic floor infections and infertility in women
- Controlling sexually transmitted infections
- In Zimbabwe, the Ministry of Health is running health awareness campaigns to educate and inform the public on the effects of sexually transmitted diseases
- Programs promoting condoms, vaccines and screening can help reduce the incidence of STIs
- The mass treatment of vulnerable groups (such as pregnant women) and high-risk groups (such as sex workers) helps to reduce numbers of new infections
- The best way to prevent STIs is abstinence, in other words, not to have sex
- If a person is sexually active, it is important to practice safe sex, by always using a barrier such as latex condoms, during sexual intercourse
- This is not a guarantee against STIs, but condoms can significantly reduce the risk the risk of STIs and HIV
- Limiting the number of sexual partners reduces a person's risk to get STIs
- A sexually active person should get regular tests for STIs
- The guidelines are twice a year for gonorrhea and once a year for syphilis
- Sexually active women should have annual Pap smear to detect cervical cancer

QN. How are these sexually transmitted diseases transmitted?(5)

Other communicable diseases

- A communicable disease is a disease that can be passed on from person to person, or in some other way
- Infections are caused by pathogens such as viruses, bacteria or protozoa entering the body and causing illness
- The table below shows some examples of pathogens commonly found in Zimbabwe and the diseases they cause

PATHOGEN TYPES	INFECTIOUS DISEASE
Bacteria	Cholera, typhoid fever, tuberculosis
Protists	Malaria, sleeping sickness
viruses	Ebola virus disease, HIV/AIDS, polio, influenza, chicken pox

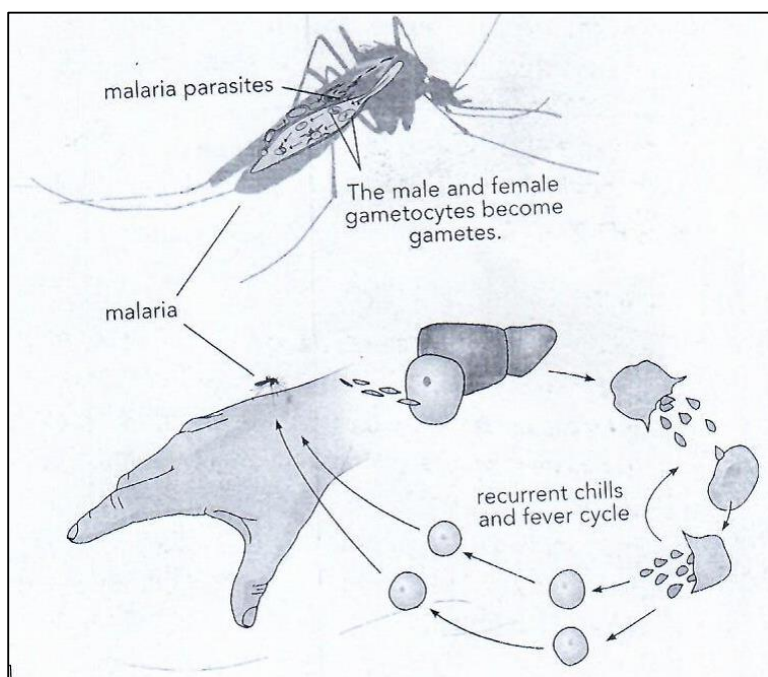
➤ Infectious diseases' spread can be reduced by good health habits such as washing hands, cleaning and disinfecting surfaces, covering your mouth and nose when sneezing, not sharing personal items and staying at home when ill

Malaria

- Is caused by a plasmodium, a parasitic protozoan, found in the salivary glands of the female mosquito of the anopheles genus
- The anopheles mosquito is a vector, that is, an insect carrying a parasite to its next host
- The malaria parasite completes its life cycle partly in the female mosquito and partly in humans

The life cycle of the malarial parasite

- When the anopheles female mosquito (first vector) bites a human, the malaria parasites are injected into the bloodstream
- The malaria parasites travel in the blood to the liver and invade the liver cells
- The malaria parasite reproduces asexually in the liver cells, forming thousands of parasites
- The parasites are released from the liver into the blood, where they enter the red blood cells
- In the red blood cells parasites reproduce asexually
- The parasites burst the red blood cells causing cycles of violent feverish attacks of malaria
- The released parasites then infect new red blood cells
- Some of the parasites become male and female gametocytes
- When a mosquito (second vector) bites an infected human, it sucks up the blood and the gametocytes pass into the mosquito's intestine
- Inside the mosquito, the male and female gametocytes become gametes
- Fertilization takes place inside the mosquito and the parasites mature
- Mature parasites finally leave the mosquito's intestine and migrate to its salivary glands



Signs and symptoms of malaria

- The symptoms appear about two weeks after infection
- These symptoms include high fever, shivers and sweats, headaches, vomiting and an enlarged spleen
- In some cases the malaria parasite attacks the brain and the patient can fall into a coma and die

Controlling and treating malaria

- There is no vaccine for malaria but the disease can be prevented by taking preventative drugs such as quinine and chloroquine when visiting a malaria prone area
- Other preventative drugs are paludrine, deltaprim, malasone and malaquine
- High rates of malaria occur in the lowveld of Zimbabwe, in the Zambezi escarpment and in Mashonaland Central.
- Other ways of preventing the disease are to sleep under mosquito nets, spray mosquito repellents and to wear long-sleeved clothes from early evening when mosquitos become active
- The best protection against malaria is to avoid being bitten especially for little children who cannot be given antimalarial drugs
- Stagnant waters should be drained to destroy breeding areas
- Spraying oil on the surface of stagnant water such as drains suffocates and kills mosquito larvae
- In some rural areas, the poison DDT is used to spray houses to kill mosquitos
- Biologically anopheles mosquitos can be controlled using toxins produced by bacteria referred to as *Bacillus thuringiensis* and by stocking ponds with fish that feed on mosquito larvae
- Cutting down tall grasses to destroy their habitat, breeding places and food

Typhoid

- Is caused by the bacterium *salmonella typhosus* and is common in most parts of the world but Southern Asia, Africa, Latin America, India and Pakistan are high risk areas
- The bacteria spread through contaminated food and water

- The bacteria enter the small intestine and the bloodstream of a person that has come into contact with contaminated food or water
- In the blood, the bacteria are carried by the white blood cells to the liver, spleen, gall bladder and bone marrow where they multiply very quickly

Signs and symptoms of typhoid

- These include high fevers, weakness, stomach pains, headaches, a poor appetite and a rash (rose spots) on the abdomen and chest
- The first signs of infection are high temperature, stomach pains and diarrhea
- Typhoid can be diagnosed by testing a stool sample or blood sample

Controlling and treating Typhoid

- Treatment with antibiotics usually leads to patients recovering fully within 7-10 days
- There are vaccines for people travelling to high risk areas
- Other methods of control are putting good sanitation in place, giving people access to clean water, ensuring waste disposal and protecting food supplies from contamination
- Personal hygiene is critical to avoid contracting typhoid
- People with typhoid should not handle or prepare food

Ebola virus disease (EVD)

- EVD is caused by the deadly Ebola virus that is transmitted to humans through close contact with the blood, secretions and organs of animals such as chimpanzees, gorillas or fruit bats that are infected
- This contact happens mostly when people hunt bush meat in the rain forests
- Human-to-human transmission of the Ebola virus takes place when a person touches the broken skin of an infected person or comes into contact with infected body fluids such as blood, saliva, sweat, vomit, semen and breast milk
- Touching a person who has died from EVD or even touching contaminated bedding or clothing may lead to infection
- A person is infectious as long as one's blood contains the virus

Signs and symptoms of EVD

- The incubation period of EVD is 2-21 days after infection
- The first symptoms of EVD are headaches, high fever, weakness, muscle pain and a sore throat
- The virus destroys all tissues and organs of the body, except for the skeletal muscles and bones
- The patient will suffer from a high temperature, vomiting, diarrhea and a rash about 10 days after the appearance of the first symptoms
- The rash is caused by blood clots that weaken blood flow and the spots get bigger as the disease progresses
- By day 11, there will be bruising, brain damage and external bleeding from the eyes, nose, mouth and anus
- By day 12-16, there is massive internal bleeding of the lungs, brain, liver, intestines, kidneys, testicles, and breast tissue
- The person loses consciousness, will have seizures and will eventually die

Controlling and treating EVD

- There is no cure for EVD so that treatment is limited to supportive therapy
- Very ill patients require intensive supportive care: balancing the patient's fluids and electrolytes, maintaining their oxygen levels and blood pressure and treating complicating infections early
- The Ebola virus kills about 65% of its victims
- To protect yourself from getting EVD, make sure you wash your hands with soap and water or with an alcohol-based disinfectant
- People should cover their nose and mouth when coughing and sneezing, using a handkerchief or tissue
- Avoid contact with blood and bodily fluids
- Do not handle clothes, bedding, needles, and medical equipment that may have been contaminated with an infected person's blood or body fluids
- Do not touch someone at a funeral who has died of EVD

Cholera

- Is a bacterial disease caused by some strains of *Vibrio cholerae*
- It is mostly waterborne and is common in areas with poor sanitation and poor water purification facilities
- The disease also spreads through food that is contaminated by human or animal faeces, by human carriers, flies and rodents

Signs and symptoms of cholera

- A cholera patient will suffer from watery diarrhea (rice water) and vomiting
- This can lead to dehydration and, if fluids and electrolytes are not replaced, the patient will go into coma and die

Controlling and treating cholera

- Can be treated by administering tetracycline and chloramphenicol to the patient
- There are vaccinations against cholera
- To prevent the spread of cholera, good food hygiene is essential
- Proper use of toilet facilities is vital such that faeces do not end up in water sources; e.g. building Blair pit toilets in rural areas
- Washing of hands with soap and water after using the toilet is necessary
- Having enough treated or boiled clean water for drinking and cooking

Lifestyle diseases and substance abuse

- These are diseases caused by a person's habits or the way people live their lives
- Common examples are alcohol and substance (drug) abuse, smoking, a lack of physical exercise and unhealthy eating

Effects of tobacco smoking on health

- Tobacco smoke contains over 4000 harmful and poisonous substances
- Among these tar and carbon monoxide are known carcinogens, i.e. cancer-causing substances

Effects of tobacco smoke on human health

Chemicals in tobacco smoke	Properties of the chemical	Effects on the body
Nicotine	➤ Addictive ➤ Causes release of the hormone adrenaline ➤ Makes blood clot easily	➤ Increase in heartbeat and blood pressure ➤ Increased risk of blood clots in blood vessels
Carbon monoxide	➤ Combines with haemoglobin to form carboxyhaemoglobin-reduces oxygen transport efficiency of red blood cells ➤ Increases the rate of fatty deposits on the inner arterial walls ➤ Damages the lining of blood vessels	➤ Death if concentrations in the air are increased by 1% ➤ Causes coronary heart diseases ➤ Increased risk of atherosclerosis ➤ Increased risk of blood clotting in the arteries
Tar	➤ Is carcinogenic (contains cancer causing chemicals) ➤ Paralyzes cilia lining the air passages	➤ Blockage in the air sacs and reduction in gas exchange efficiency ➤ Dust particles trapped in the mucus lining the airways cannot be removed
Irritants (e.g. hydrogen cyanide, acrolein, formaldehyde)	➤ Paralyzes cilia lining the air passages	➤ Increased risk of chronic bronchitis and emphysema

- The cilia of the epithelium are paralysed. Mucus and dust particles cannot be removed
- The airways become blocked, making breathing difficult
- The person has to cough persistently to clear his airways in order to breathe. This increases risk of getting lung infections

Emphysema

- Tar irritates alveoli walls causing persistent and violent coughing which may lead to emphysema.
- The signs of emphysema are as follows
 - Violent coughing breaks the partition walls between the air sacs leaving large spaces or blebs
 - The blebs fill up with mucus and thus surface area for gaseous exchange decreases
 - The lungs become inflated with air
 - The lungs lose their elasticity

- Breathing becomes difficult. The person wheezes and suffers severe breathlessness
- Carbon dioxide also collects in the blebs, eventually leading to suffocation

Coronary heart disease

- Carbon monoxide is absorbed from the lungs into the blood stream where it combines with haemoglobin forming carboxyhaemoglobin in the red blood cells
- With carbon monoxide attached to the haemoglobin, less oxygen is carried to the cells for respiration
- The body increases its breathing and heart rates because of the oxygen shortage
- Therefore, the heart works harder increasing chances of heart attacks
- New-born babies whose mothers smoked while pregnant usually have lower birth weights and often have breathing problems
- Passive smoking is also dangerous, i.e. inhaling smoke from a cigarette smoker

Excessive alcohol consumption

- Alcohol is a depressant
- It can enter the brain and change the way you think and feel
- It affects the proper functioning of the nervous system
- When under the influence of alcohol, pain messages are impaired and it is difficult to coordinate body movements
- It can make the user feel happy and carefree and have impaired judgement especially in dangerous situations such as driving
- Alcohol results in reduced self-control and reduces reaction time
- The long-term use of alcohol causes liver cirrhosis or liver damage
- The liver is severely scarred so that it shrinks and hardens
- The hardened tissue reduces blood flow through the liver, leading to poor liver function.
- This, in turn, affects many body functions
- Alcohol, being a depressant, can lead to addiction, depression, memory loss, aggression and anxiety
- Alcohol abuse leads to people losing their inhibitions so that they behave impulsively
- It can even lead to actions such as self-harm and suicide
- Alcoholism is also linked to mental conditions such as having hallucinations and believing that you are being persecuted
- Alcohol has social implications such as neglecting families financially, emotionally and morally, leaving members being destitute
- Other social issues are that the alcoholics are often on sick leave more than the non-alcoholics
- Social immorality can result in unacceptable behaviour such as prostitution and crime
- Alcohol drinkers are often unaware of their problems
- Alcohol abuse may lead to unemployment which in turn may lead to more drinking out of hopelessness
- A developing foetus is affected by the drinking of the mother during pregnancy
- The first four months of pregnancy are important for brain development
- The usage of alcohol during pregnancy can make the baby addicted to alcohol

- Some babies are born brain- damaged and will have learning disabilities
- This condition is known as Foetal Alcohol Syndrome (FAS)

Effects of use of mandrax and cannabis/mbanje

- Alteration of personality
- Loss of self-control or maniac behaviour
- Criminal tendencies like violence or murder
- Increased chances of HIV infection
- Addiction
- Can cause blood poisoning

Effects of inhaling solvent e.g. glue sniffing

- Results in :
 - Hallucinations
 - Reduced self control
 - Nasal passage damage
 - Heart muscle damage

FORM 3 NOTES

PHYSICS SECTION

DATA PRESENTATION

- Scientists carry out some observations and experiments and in the process, record the data in form of tables, tallies, line graphs, pie charts etc.

Pie Chart

- Is a diagram used to represent data in a circle form for example a class of learners which is grouped into gender i.e. males and females.
- Also can be used to show marks got by few pupils in a maths test.

How to construct a pie chart

Table 1 below shows average rainfall of a region over six months

Months	Average rainfall (mm)
January	40
February	50
March	45
April	35
May	20
June	10

- Calculate the percentage of your data relative to the total. E.g. the percentage for June = average rainfall in June divided by total rainfall over six months.
- Calculate the angle which corresponds to the percentage out of 360° e.g. $5\% \times 360^\circ = 18^\circ$
- Draw a circle to represent the total quantity and find the central point.
- Draw a vertical line from the centre to the top of the circumference.
- Draw in segments measuring them using a protractor starting with the largest segment.
- Different shading may be used.
- Label the pie chart either by numbering or writing in some words.

The interpretation and explanation of the pie

Pie charts are visually appealing as they are easy to understand.

Line Graphs

- A line graph is used to show some relationships between two sets of results / comparing two sets of quantities that depend on each other e.g. the size of a family and the food required for that family.
- It may be used to show changes over a period of time for example growth rate of maize broilers over a period of one month or changes in temperature of water being heated at given time intervals.

How to draw a line graph

- Draw the x- axis and the y-axis.
- Label the scale using variables (the scale which we are controlling should be the x-axis and the one which we cannot control should be the y-axis).
- Select a suitable scale both axis.
- Mark points using a dot.
- Join the points using a line which must pass through all the marked points.

Bar and line graphs give more information than pie charts hence they are more useful.

Question 1

Exercise 16.2 page 128. (5)

Question 2

Use data on table 1 on page 1, draw a bar graph. (4)

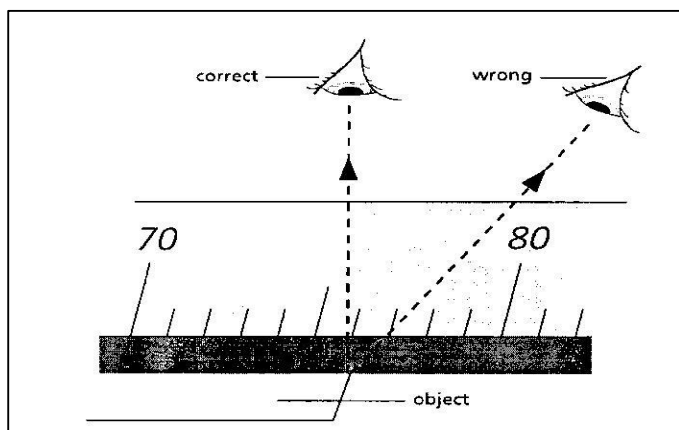
MEASUREMENT

Physics experiments involve the measurement of a variety of quantities, and a great deal of effort goes into making these measurements as accurate as possible.

- Before a measurement can be made, a standard or **unit and appropriate instrument** must be chosen.
- Three quantities measured in physics include, **length, mass, voltage, current and time.**
- The SI (System International Units) system is a set of metric units now used in many countries.
- It is a decimal system in which units are divided or multiplied by 10 to give smaller or larger units.

Length

- Is the distance between two points.
- The SI unit for length is the metre (m).
- A **rule** (ruler) is rigid piece of wood, plastic or perspex with markings.
- It is used to measure length for distances between 1mm and 1metre;
- Your eye must be directly over the mark on the scale to avoid parallax errors.



- $1\text{cm} = 10\text{mm}$; $1\text{m} = 100\text{cm}$; $1\text{km} = 1000\text{m}$; $1\text{km} = 100\,000\text{ cm}$.

Convert

210 cm to m; 33mm to cm; 96 m to km and 840m to km.

Activity: 17.1 Step Ahead f3 page 130

Measuring tape

- Is a flexible piece of material with markings similar to those on the ruler.
- It is used to measure objects that are not flat or straight such as the circumference of a pipe.

Voltage

- Voltage is the difference in electric potential between two points or objects. Voltage is measured in **volts (V)** and is measured by a **voltmeter** (connected in parallel). If a cell has 1 Volt, it delivers 1 Joule of energy to each coulomb of charge (J/C).
- Voltage = Energy / Charge
- Volts = Joules / Coulomb
- $V = E / C$

Measuring Electricity / Current

- Electric current is the flow of electricity in a complete circuit. Or flow of electrons through a conducting circuit.
- An ammeter measures the strength of an electrical current, that is, how many charges move through a circuit per second.
- Measured in Amperes usually abbreviated as Amps / (A).
- The ammeter is connected in series in a circuit.
- State the formula for current. [1]

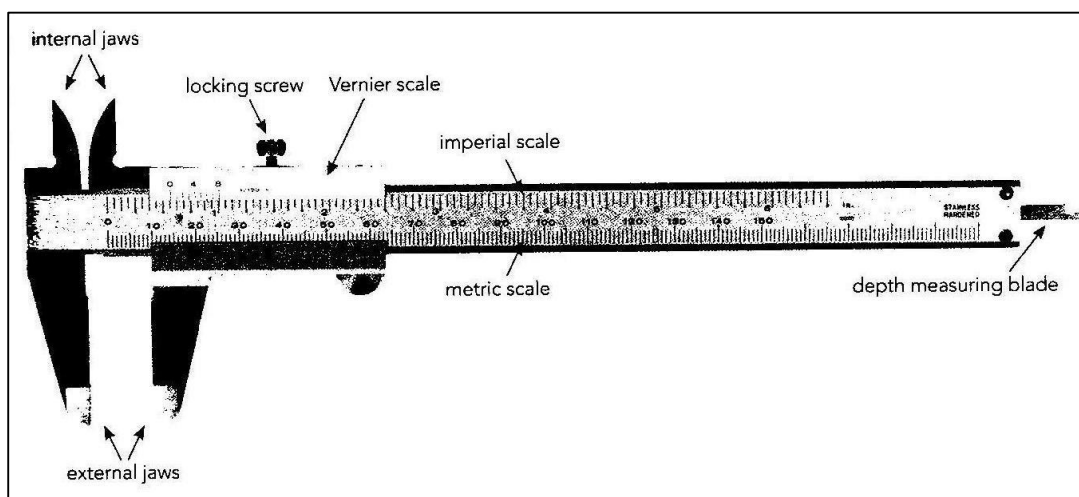
Activity 17.3 Step Ahead f3 page 131

Vernier Calliper

- Is a tool used to take highly accurate measurements.
- It has two scales, the main scale with markings similar to those on a ruler and a vernier scale with markings in 0.1mm increments.
- This measures diameter of very small objects to 0.1mm.

How to use Vernier callipers

- Close the jaws to allow for zero errors.
- Open the jaws slowly to fit the object firmly.
- To take a reading, first look at the main scale and then from the Vernier scale
- Record the measurement.



Activity: 17.2 Step Ahead f3 page 131

Density

- This is the **mass per unit volume** of a substance. The SI unit of density is the **grams per cubic metre**.

$$\text{Density} = \frac{\text{mass}}{\text{volume}}$$

- To convert a density from g/cm³, normally the most suitable unit for the size of sample we use, to kg/m³, we multiply by 10 to the power of 3.
- Using the symbols ρ (rho) for density, m for mass and V for volume, the expression for density is

$$\rho = \frac{m}{V}$$

Rearranging the expression gives

$$M = V \times \rho \text{ and } V = \frac{m}{\rho}$$

To determine the density of a liquid

- Weigh a measuring cylinder on a balance.
- Fill the measuring cylinder with some liquid and weigh.
- Subtract the two measurements of mass from one another i.e. the change in mass is the mass of the liquid and the volume is shown on the scale, then use the formula:

$$\rho = \frac{m}{V}$$

Activity: 17.5 Step Ahead f3 page 132

Density of irregular objects

- Weigh an irregular object and record its mass.
- Pour water in a measuring cylinder and take a reading.
- Place an object in the cylinder and measure new level of water.

- Subtract the first reading from the second reading to get the volume of the irregular object.
- Divide the mass by the volume to get its density.

Activity: 17.6 Step Ahead f3 page 133

Force

- Is a push or a pull. It can cause a body at rest to move, or if the body is already moving it can change its speed or direction of motion. A force can also change a body's shape or size.
- The unit of force is the **newton** (N).

Force of friction

- Is force that opposes motion.
- It produces heat, stops moving objects and causes wearing away of materials which are in contact.
- Gives balance to bodies,

- Qn 1. State any four ways of reducing friction. [4]
 2. State the principle of moments. [2]
 3. Explain the turning effects of force. [2]

Weight

- is the amount of gravitational force that acts on an object, measured in Newtons, and given by the formula:

Weight = mass $\times$ acceleration due to gravity / $W = mg$.

- The weight of a body can be measured by hanging it on a spring balance marked in Newtons.
- Weights (and hence masses) may be compared using a balance.

Mass

- Is the amount of matter in a substance.
- Is measured using a balance.
- Mass depends only on the inertia of an object.
- The more the inertia, the more the mass it has.
- A more massive object has great tendencies of resisting change in its state of motion.
- E.g. if two bricks are made of different substances, one can tell which brick is made of foam or cement by pushing the bricks and change its motion.
- The one with least resistance has least inertia, mass and thus made of foam.

Centre of Mass

- Is a point where the mass of an object is centred.

Momentum

- Is a useful quantity to consider when bodies are involved in collisions and explosions.
- It is defined as the mass in motion. That is, all objects have mass so if all moving, have momentum.
- Is measured in kilogram metre per second (kg m/s) or newton second (N. s).
- Momentum = mass $\times$ velocity.

- A 2 kg mass moving at 10 m/s has momentum 20 kg m/s, the same as the momentum of a 5 kg mass moving at 4 m/s.

Inertia

- Newton's first law states that an object at rest stays at rest and an object in motion stays in motion with the same speed and direction unless acted on by an unbalanced force.
- Objects tend to keep on doing what they are doing.
- This natural tendency of objects to resist change in their states of motion is called **inertia**. **e.g.** The reluctance of a stationary object to move can be shown by placing a large coin on a piece of card on your finger. If the card is flicked sharply the coin stays where it is while the card flies off.

Newton's first law of motion

- Movement in uniform motion is a state of all objects.
- An object in uniform motion will remain in uniform motion unless something changes its motion.
- Thus, if you roll a ball, it will eventually slow down and stop due to friction (external force).

Summary: ***A body stays at rest, or if moving it continues to move with uniform velocity, unless an external force makes it behave differently.***

Newton's second law

- It shows that the acceleration a is
 1. directly proportional to the applied force F for a fixed mass, i.e., $a \propto F$, and
 2. inversely proportional to the mass m for a fixed force, i.e., $a \propto 1/m$.

Combining the results into one equation, we get $a \propto F/m$ or $F \propto ma$

$$F = kma$$

where k is the constant of proportionality.

One newton is defined as the force which gives a mass of 1 kg an acceleration of 1 m/s², i.e., 1 N = 1 kg m/s².

So if $m = 1$ kg and $a = 1$ m/s², then $F = 1$ N.

Substituting in $F = kma$, we get $k = 1$ and so we can write $F = ma$

Example 1: A force of 10N is applied to the body of mass 15kg. Calculate the acceleration.

Solution: $F = ma$
 $F = 10$ N
 $M = 15$ kg
 $10N = 15a \div 15$ both sides
 $a = 0.6666$ therefore $a = 0.67\text{m/s}^2$

Experiment

Aim: Using a ticker timer to measure time

Materials: a ticker timer, ticker tape, stop watch or clock, a machine trolley or wind up/pull back toy car (optional)

Procedure

1. Thread a short length of ticker tape through the ticker timer. If there is a carbon paper disc, make sure the tape goes underneath the disc.
2. Turn the ticker timer on for a few seconds. It vibrates rapidly and hits the top of the carbon paper. It makes a lot of dots on the tape, at regular intervals.
3. Remove the tape from the ticker timer. If the tape didn't move when the ticker timer was switched on, then all the dots will be in the same place.
4. Thread a longer piece of ticker tape, about 1 metre long, through the ticker timer. Switch the ticker timer on. Pull the tape slowly through the ticker timer.
5. Check the tape to see if you can see each individual dot, with a space between. We can say that each dot-to-dot space stands for a 'tick' of time.
6. Thread another 1 metre piece of tape through the ticker timer.
7. You need a 'start' signal and a 'stop' signal. These could be hand-claps by one of your group or by your teacher. They should be just a few seconds apart. Pull the tape slowly and switch the ticker timer on at the start signal. Switch it off at the stop signal.
8. Count the number of dot-to-dot spaces between the start and the stop. That is the time between the signals, measured in 'ticks'.
9. Use a fresh piece of tape, and a stopwatch or stopclock. Pull the tape through the ticker timer for 3 seconds. Find out how many 'ticks' there are in 3 seconds. Find out how many there are in 1 second. Work out the time in seconds that is the same as 1 tick.

Results

Once you have determined how many dots are made in 1 second (or 3 seconds), you can measure the time by dividing the number of dots by how many are made in 1 second.

MACHINES

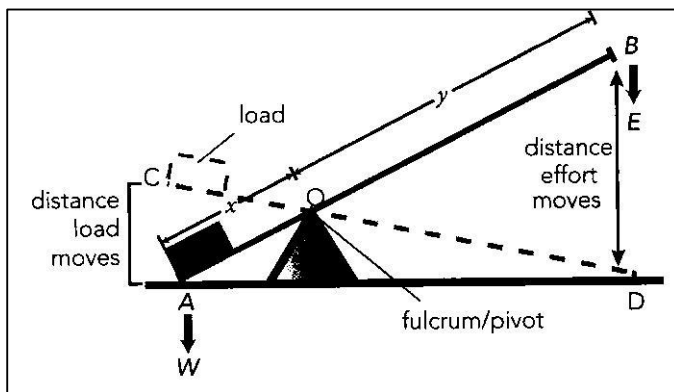
- A machine is mechanical system that makes hard work easier to do e.g. , bottle opener, pliers, levers, pulleys, and gears and incline planes.
- The principles of machines are : Mechanical Advantage (MA), Velocity Ratio (VR) and Efficiency.

1. Levers

-Levers are simple machines

-It is a bar or string tool used for lifting or moving bearing stiff objects. Levers have got a pivot/fulcrum where the lever turns, the point where the load is placed and another end where effort is applied e.g. wheelbarrows, crowbar, hammer, e.t.c.

- Size of effort depends on distance of load and effort from the pivot.
- The greater the distance of effort from the pivot, the smaller the size of the effort e.g. borehole can be made more efficient by putting a longer handle i.e. increasing the distance of the effort from the pivot.



Mechanical advantage

- Is the ratio of the load lifted to the effort applied. A machine makes big jobs lighter and thus an advantage, called Mechanical Advantage.

$$\text{Mechanical advantage (MA)} = \frac{\text{load (L)}}{\text{Effort (E)}}$$

E.g. if effort of 40N is used to lift 80N of soya beans. Calculate the MA

$$\text{MA} = L/E = 80\text{N}/40\text{N} = 2 \text{ (MA)}$$

- The lever has in this case enable the effort to raise a load twice as large.
- Thus a lever moved twice the distance of the load.

Velocity Ratio (VR)

- Is the ratio of the distance moved by the effort to the distance of the load.

$$\text{VR} = \frac{\text{Distance moved by effort (BD)}}{\text{Distance moved by load during the same time (AC)}}$$

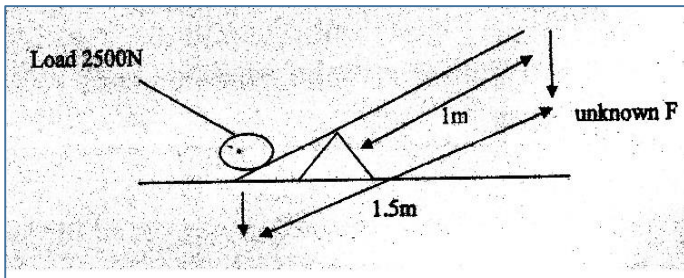
Work done

Is the amount of force multiplied by the distance moved in the direction of force.

Work done (joules) = force (newtons) x distance (meters)

Example:

A lever is used to lift a 2500N rock as shown below:



Calculate

- Unknown force F
- MA
- VR
- Efficiency (8)

Efficiency

- Is the ability of machine to do work satisfactorily. It can be reduced by friction or the weight of the machine hence no machine is 100% efficient.

Efficiency = $\frac{\text{work obtained from machine}}{\text{work put into machine}} \times 100\%$

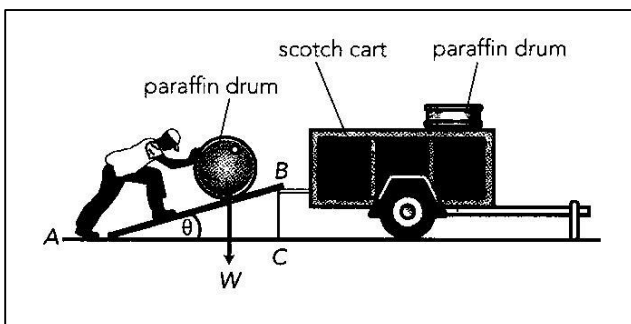
Or $\frac{\text{load} \times \text{distance moved by load}}{\text{effort} \times \text{distance moved by effort}} \times 100\%$

2. Inclined planes

-An inclined plane is a sloping which is less than a right angle (90°)

-It is a machine used to lift heavy loads using a smaller effort but the load move through a great distance.

-The smaller the angle of the inclined plane the smaller the effort applied. It is easier to load a lorry by pushing up an inclined plane than lift the load vertically.



$M.A = \frac{\text{load}}{\text{Effort}}$

$V.R = \frac{\text{distance moved by Effort}}{\text{distance moved by load}}$

$$= \frac{AB}{CB} = \frac{1}{\sin \theta}$$

3. Pulleys

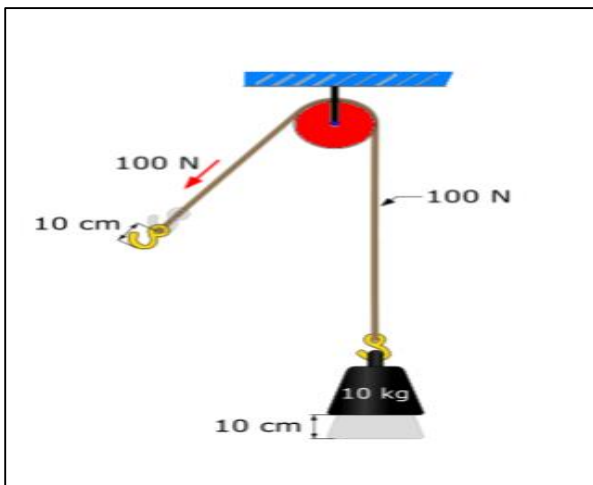
-A pulley is wheel with a groove through which a rope passes. Two or more pulleys on the same axle is a pulley block.

-Pulleys are used to lift heavy loads. They change the direction in which the effort is applied.

Types of pulleys

1. Single fixed pulley

– enables lifting or load by applying a downward force. if there is no friction, it will have a MA and VR of one but because of friction the effort is always slightly bigger than the load.



a. $M.A = \frac{\text{load}}{\text{effort}}$

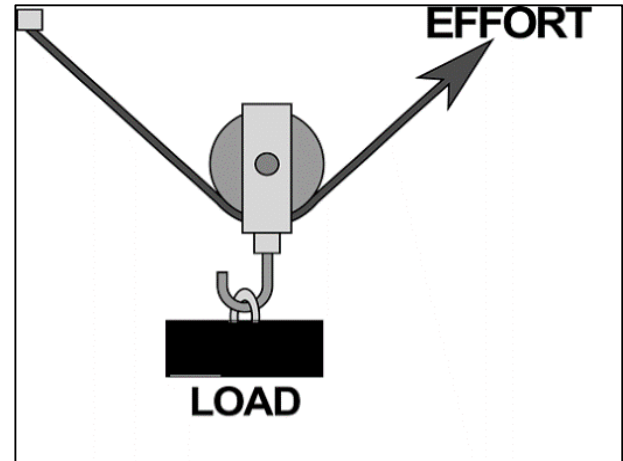
b. $V.R = 1$

c. $\text{EFFICIENCY} = \frac{M.A}{V.R} \times 100 \%$

2. Single movable pulley

-The effort applied is two times the load since two parts of ropes supports one load, if there is no friction ,VR and MA would be 2 but in practice is always slightly less due to friction.

$V.R = \text{No. Of movable pulleys} \times 2 = 2$



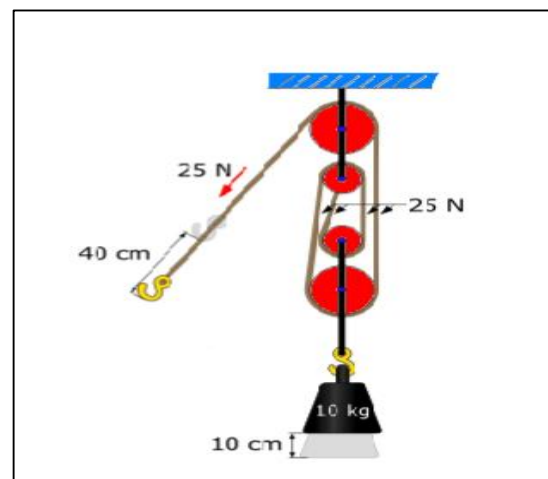
3. Block and tackle

- Normally used in crane s, it consists of two blocks each with one or move pulleys mounted on the same axle.
- Each fixed pulley simply changes the direction in which the effort is applied.
- Each pulley increases the MA of the system by 2 (provided friction is negligible and light parts are used for the lower block).
- This means the more the pulleys; the easier it is to lift a load. A block and tackle combines fixed and movable pulleys.

NB: In pulleys

-Efficiency increase as the load increases.

-Efficiency is reduced by friction in strings and weight of movable pulleys.



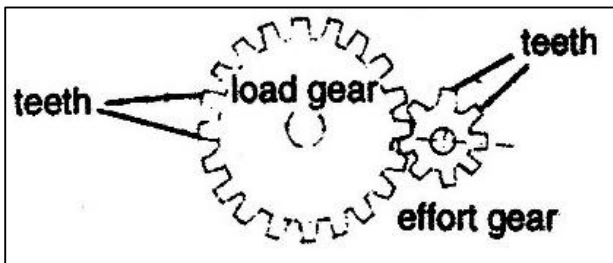
$$M.A = \frac{\text{load}}{\text{effort}}$$

V.R = Number of pulleys in the system

$$\text{Efficiency} = \frac{M.A}{V.R} \times 100\%$$

4. Gears

- Are systems of wheels with interlocking teeth so that one wheel drives another.
- Bigger gear is the load gear and the small gear is the effort gear.



- $M.A = \frac{\text{Load}}{\text{Effort}}$
- $V.R = \frac{\text{number of teeth on load}}{\text{number of teeth on effort}}$

Ways of improving efficiency

- Lubricating – with oil or grease so that the surfaces can slide more easily over each other.
- Using rollers or ball bearings so that surfaces roll rather than slide over each other.
- Using lighter parts for machines because friction increases as the weight of the machine increases.
- Making surfaces smooth – so that surfaces slide over each other easily.
- Rough surfaces have more friction.

NB: Friction is useful in brakes, for stopping moving vehicles and for balance e.g. when wearing shoes in a rough surface.

$$\text{EFFICIENCY} = \frac{M.A}{V.R} \times 100\%$$

OR $\frac{\text{No. of effort gear}}{\text{No. of teeth on driven gear (load)}} \times 100\%$

Wheel and axle

- Is a system that comprises of a big wheel attached to a small wheel that it turns.
- The direction of effort is opposite to that of load for example on a screw driver and screw, car steering wheel and the windlass.
- When the effort turns the wheel (radius R) a distance of one circumference ($2\pi R$), the load (radius r) turns a distance = $2\pi r$, the circumference of the axle.
- $V.R = \frac{\text{distance moved by effort}}{\text{Distance moved by load}}$

$$= \frac{R}{r}$$

Energy losses in machines

-In many machines, it is possible to obtain velocity ratio but not high value of mechanical advantage due to energy losses caused by friction.

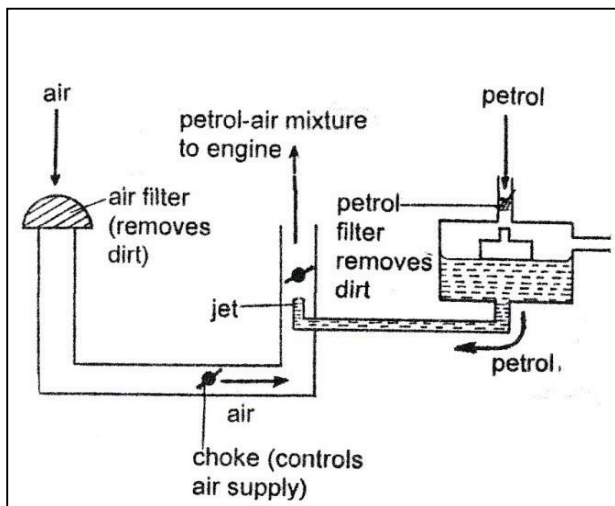
PETROL AND DIESEL ENGINES

- Fuel engines such as petrol and diesel convert chemical energy into heat energy and then into kinetic energy.

Chemical energy → Heat energy → Kinetic energy

- Both petrol and diesel engines work by internal combustion but in slightly different ways.
- Both engine shave a fuel injection system from the tank to the engine and into the system.
- Fuel injection can be direct or indirect. Direct systems spray the fuel direct into cylinders while indirect uses the carburettor.

The Carburettor



The carburettor is a connection to the engine used to filter petrol and mix it with air in desired proportions.

- ❖ Petrol is cleaned by the petrol filter to remove dirt so that the jet does not get blocked.
- ❖ The air filter filters dust and dirt from the drawn air.
- ❖ The choke controls air supply.
- ❖ The throttle accelerator controls the amount of petrol - air mixture to the engine.
- ❖ Opening the throttle allows more mixture to the engine and speeds it up.
 - Newer engines have electronic direct fuel injection instead of carburettors and fuel is delivered straight to the engine, the pressure caused by the quick expansion of the gas pushes piston down, rotating a flywheel connected to the crankshaft.

The Petrol Engine

1. Intake Stroke

- Downward moving piston sucks a mixture of air and petrol gas into the cylinder through the inlet valve.
- The exhaust valve is closed.

2. Compression Stroke

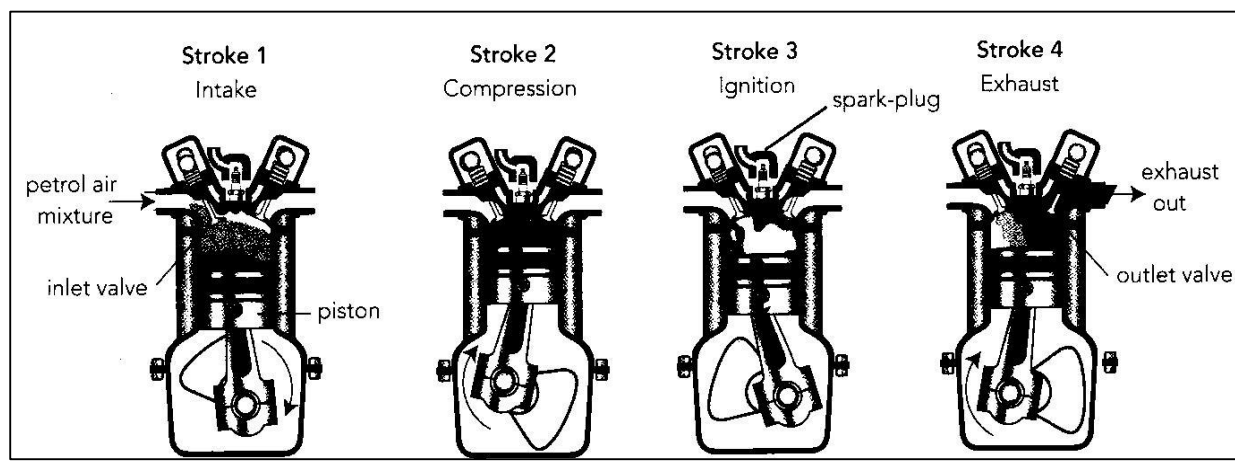
- The piston moves upwards compressing the gas mixture.
- Both valves are closed.

4. Ignition Stroke

- A spark jumps across a point in the spark plug exploding or igniting the petrol air mixture. This forces the piston to move downwards, rotating a flywheel connected to the crankshaft.
- Thus, the rotation used to drive the wheels of the car.
- Both valves are closed.

4.Exhaust Stroke

- The piston moves upwards pushing the exhaust gases out through the exhaust valve.
- The inlet valve is closed while the exhaust valve is opened.
- And , as the piston moves down, it pulls more of the fuel air mixture in to restart the cycle.



The Diesel Engine

1. Induction stroke

- Air is drawn into the cylinder through the inlet valve.

2. Compression stroke

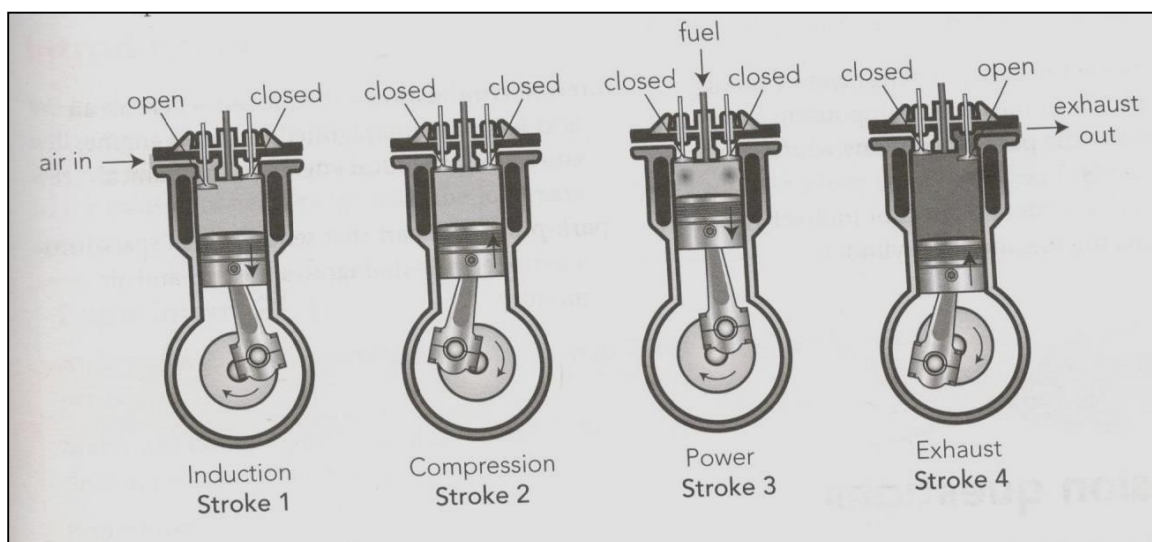
- The piston moves upward, compressing the air so that so that it heats up.

3. Power or ignition

- Fuel is injected into the hot, compressed air through a central fuel injection valve, making it spontaneously ignite.
- The pressure due to expansion of gas pushes the piston, rotating a flywheel that is connected to the crankshaft. This drives the wheels of the car.

4. Exhaust stroke

- The piston moves upward again in the cylinder and pushes out the gas through the exhaust valve into the exhaust system of the vehicle.
- As the piston moves down, it pulls more fuel air mixture in to restart the cycle.



State the difference between petrol and diesel engines (6)

The Crankshaft

- It is driven by the pistons to transfer energy to the wheels.
- It is the main rotating shaft running the length of the engine.
- It is supported by main bearings
- As the piston move up and down, the connecting rods move the crankshaft around.
- The turning motion of the crankshaft is transmitted to the transmission and to the driving wheels.



ENERGY AND HEAT TRANSFER

Heat Transfer

Three methods include conduction, convection and radiation.

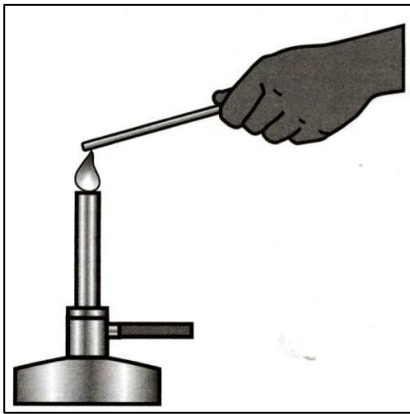
Heat Transfer in Solids – Conduction

Experiment 1a.

To investigate the rate of heat transfer in different materials.

Materials: Bunsen burner or candle wax, rods, of different materials such as iron, steel, carbon.

Procedure



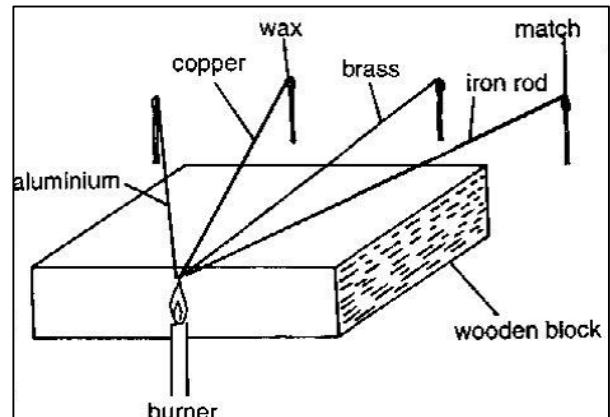
1. Hold the different rods with one end in the burner.
2. Remove the rod the moment you can feel it getting warm.
3. Sort the materials in order of how quickly heat was transferred from the flame to the end of the rod.

Results

Metals conducts heat better than non – metals.

Experiment 1b.

Comparing the rate of heat conduction



- After some time, the candle wax will melt and the match stick will fall in the order; copper, brass, aluminium, and iron.
- Therefore, copper conducts heat faster and iron the least.

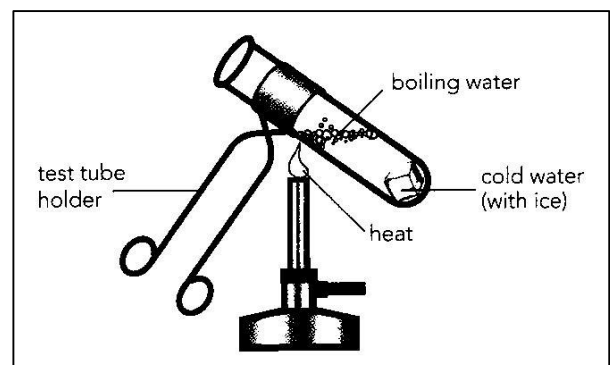
Heat Transfer in Liquids (convection)

Experiment 2

To determine how well water conducts heat Step Ahead f3 page 154

Materials: a test tube, test tube holder, a bunsen burner, water, ice,

Procedure



1. Put a small block of ice into the test tube and add water until it is about three – quarters full.
2. Heat the water at the top as shown.
3. Use your hand to feel the temperature at the bottom of the test tube.

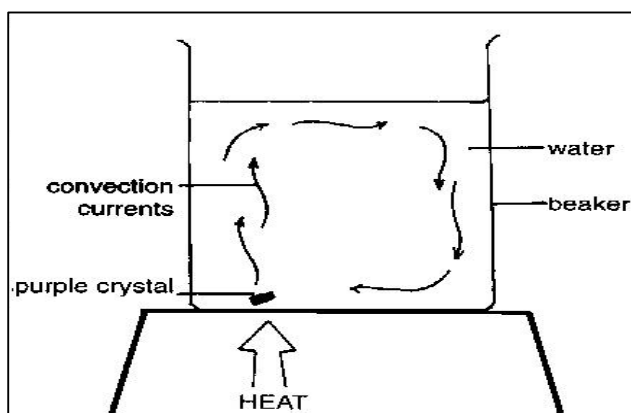
Results

- The water at the top boiled while at the bottom remained cold.
- Heat does not conduct well through water.
- However, if we heat water at the bottom of the test tube, heat is quickly transferred throughout the liquid due to convection.

Experiment 3

To demonstrate convection currents in water.

Method



- Fill a beaker with water to about three quarters.
- Drop a crystal of potassium permanganate to the bottom edge of the beaker.
- Heat the water around the crystal.

Results

When a liquid is heated, it expands and its density decreases. The liquid nearest heat source is less dense so it rises taking heat in it. The purple colour makes stream of water visible. The cooler, denser liquid moves downwards to heat source where it is warmed and then rises due to convection current.

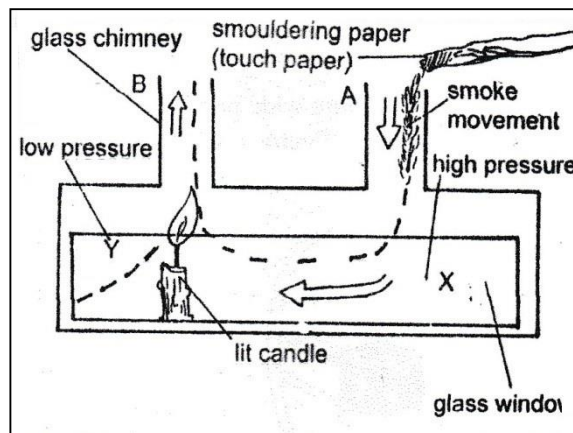
Heat Transfer in Gases (convection)

Experiment 4

To demonstrate convection currents in gases.

Apparatus: a box with two chimneys, candle, a rag or rolled paper,

Method



- Position a lit candle inside the chimney to the left as shown.
- Place the back of your hand just above left chimney for about two seconds.
- Bring the lit brown paper to chimney A.
- Watch the movement of smoke through the clear front / glass window and describe the movement of smoke.

Observations and results

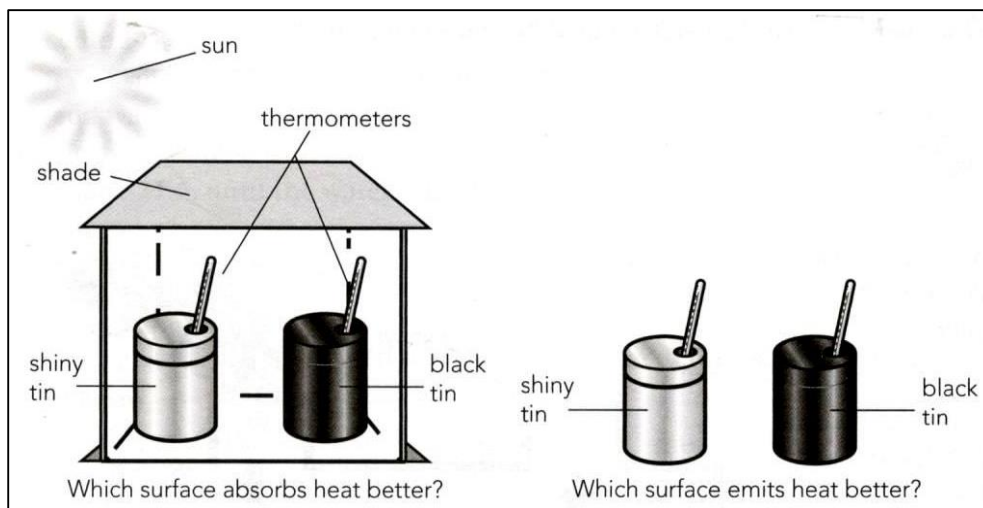
- ✓ Smoke moves in the direction shown by the arrow.
- ✓ The smoke in chimney A is dense, so it sinks. Thus, there is high pressure in region X.
- ✓ The candle at the other end of the chimney heats the air around it, causing it to expand and become less dense, rise and escape through chimney B. Thus, region Y has low pressure.
- ✓ Therefore, by pressure difference, smoke particles drift from X to Y. On reaching Y, the candle energizes the smoke which then become less dense and rise through chimney B.

Experiment 5

Aim: To investigating the effect of different surfaces on the rate heat is absorbed or emitted.

Materials: two thermometers, 2 tins with lids, (one painted black and the other white), tap water,

Procedure



1. Pour the same amounts of water into both tins.
2. Close the tins and place a thermometer in each tin.
3. Record the initial temperature of water in each tin.
4. Place the tins outside in the sun for about 20 minutes.
5. Record the temperature again.
6. Bring the tins into a shade or classroom for another 20 minutes.
7. Record your results in a table as below:

Time of measurement	Temperature of black tin	Temperature of other tin
At start		
After 20 minutes		
After 40 minutes		

Results

- Black or dull painted test tube records higher temperatures than white painted test tube.
- Black absorbs more heat than white surfaces. White reflects heat.
- Black surfaces are good emitters of heat while white are poor emitters of heat.

Heat and transparent objects

- A house with a window facing the sun get hot in summer.
- Cars get very hot when left in the sun with all windows closed
- Glasshouses get warm quickly when the sun rises.
- Thus, heat moves through transparent objects such as glass and plastic.

Application of heat emission

1. In hot deserts people wear loose white clothes so as to reflect heat.
2. Milk and petrol tanks are painted white or are shiny to avoid heat entrance into the tank.

ELECTROMAGNETISM

- There is a relationship between electricity and the force of magnets.
- The flow of current generates a small magnetic field around the wire carrying current.

Magnetic effect of an electric current

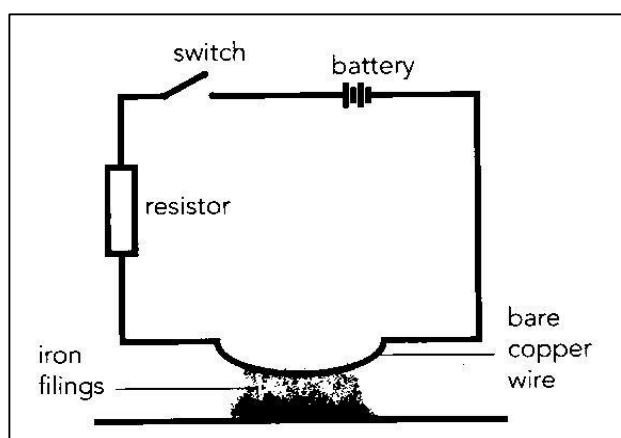
Experiment 2

Aim: To show that a wire carrying an electric current generates a magnetic field.

Materials: three 1.5 V cells, a switch, connecting wire, thin copper wire, iron filings, a 5 Ω resistor,

Procedure

1. Connect copper wires as shown.
2. Dip the copper wire into the iron filings. What happens?
3. Switch the current on and dip the copper wires into the iron filings. What happens?
4. Turn the switch off. What happens to the iron filings on the copper wire?



Results

- The electrical wire with current generates a magnet field around it.
- Iron filings are attracted to the wire.
- This type of magnetism is called electromagnetism.

Motor effect

Motors

A motor is a device which converts electrical energy to mechanical energy. Generators produce electricity by changing mechanical into electrical energy e.g. in power stations thus, a motor is the reverse of a generator.

How It Works?

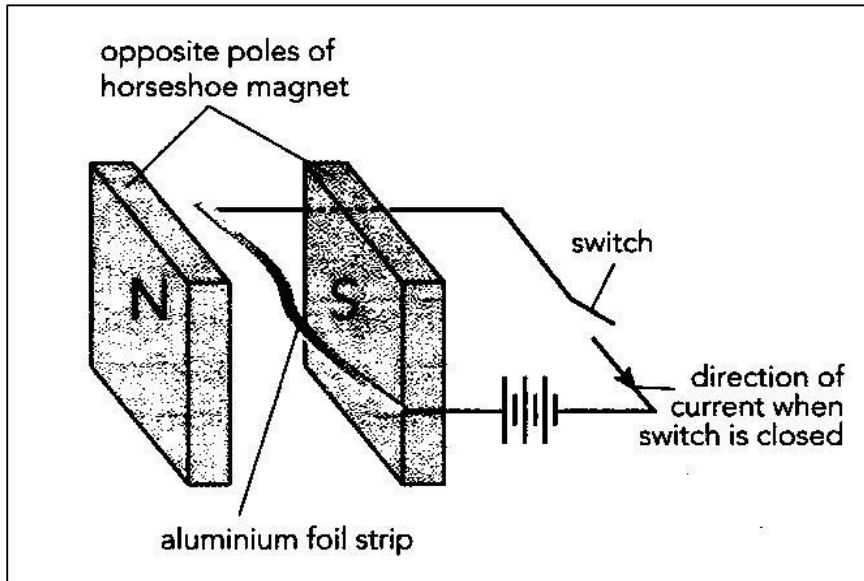
If a magnet is plunged into a coil of wires connected to an electrical meter (galvanometer), a current is induced in the coil. When the magnet is pulled out of the coil, a current is again induced, but in the opposite direction. This is due to the lines of force around the magnet cutting the wire of the coil.

Experiment: Motor effect

Aim: To demonstrate that a conductor carrying an electrical current experiences a force in the magnetic field.

Materials: a length of flexible wire, crocodile clips, 2 strong magnets / horseshoe magnet, battery with 6V or more, aluminium foil,

Procedure



1. Note: do not close the switch for as long as the circuit is a short circuit. The large current will deplete the battery quickly.
2. Close the switch and observe what happens to the aluminium foil.
3. Reverse the connection of the battery and note the direction the foil moves.

Results

- ✚ The aluminium foil moves down as current flows as shown on the diagram.
- ✚ If the terminals are reversed, it moves up.

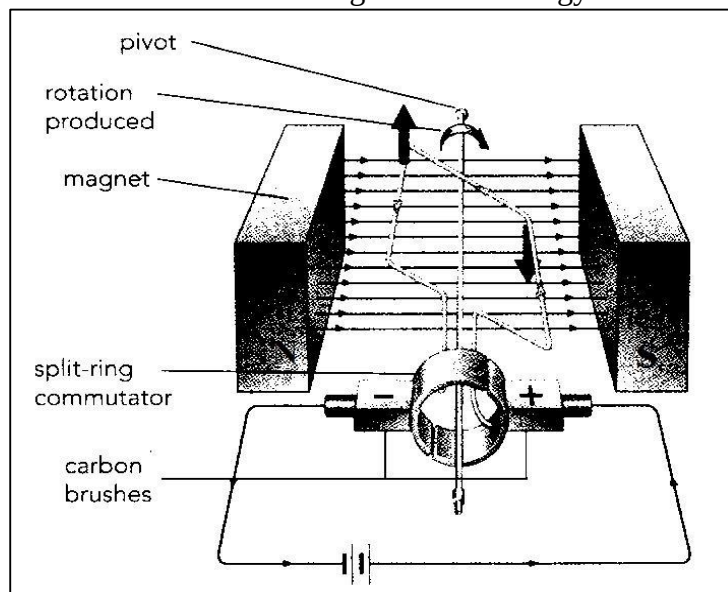
Conclusion

- ✚ A current-carrying wire's field and a magnetic field interact with one another to produce the motor effect.
- ✚ The motor effect is used in making electric motors.
- ✚ Which convert electrical energy into kinetic energy.

Direct Current (DC) motor

- A wire that carries current is 'kicked out' of a magnetic field in a direction that depends on the direction of flow of current in the wire relative to the direction of the magnetic field.

- If a wire is bent into a rectangular shape, and an electrical current flows through it, it can be placed inside a magnetic field so that one side of the wire experiences an upward force, while the other side experiences a downward force.
- If the wire is pivoted properly and current supplied is continuous, the wire will rotate.
- The current enters the coil through the right brush of the commutator, and leaves through the left brush.
- As the commutator segments exchange brushes every half cycle, the current that flows through each side of the rectangular coil changes direction hence causes rotation of the coil.
- An electric motor changes electric energy into mechanical energy.



Factors Affecting the Speed of the Electric Motor

- a) Size of electric current – increases currents and speed of the motor.
- b) Strength of magnet – the stronger the magnet, the stronger the magnetic field and the faster the motor/coil will turn.
- c) Number of turns of coil – the more the number of turns of coil, the faster the motor moves.

Generator Principle

The functioning of a DC generator is based on the generator principle:

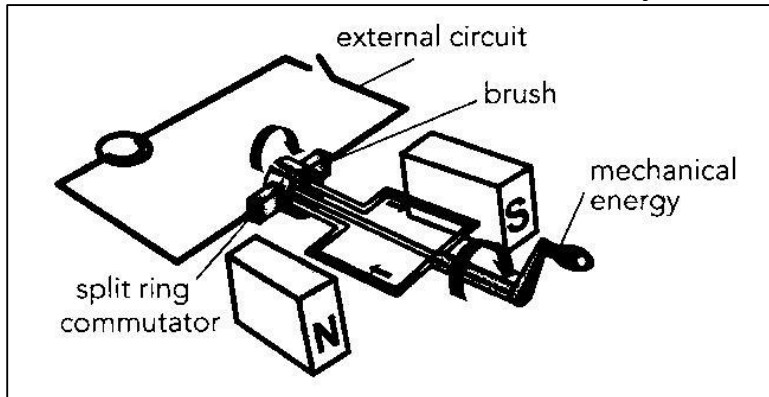
- When a conductor is placed in a changing magnetic field, or when a conductor is moved in a magnetic field, an electromotive force (emf) is induced in the conductor.
- Electromotive force is a difference in electrical potential and gives rise to electrical current.

The magnitude of electromotive force depends on -;

- a) The strength of the magnet. The stronger the magnet, the more the induced current.
- b) The number of turns of coil – The more the number of turns of the coil, the greater the induced current.
- c) The rate at which the magnet moves cuts across the coil. The faster the rate of movement, the greater the induced current.
- d) Area of the coil.

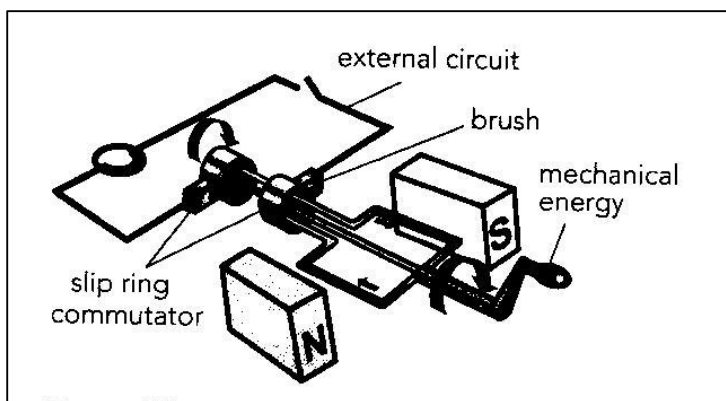
The Direct Current Generator (DC) generator

- Electric motors use electrical energy to cause movement (mechanical energy), while electrical generators produce electricity from movement.
- The **d.c.** generator is similar to the **a.c** motor except that the coil of the generator is made to rotate. The split rings commutator causes an exchange of ring segments and brushes every time the coil is vertical.
- The overall effect is that the induced current always leaves the generator from one brush and enter through the other. Therefore, one brush stays positive and the other negative
- The current in the external circuit flows in only one direction.



Alternating current (AC) generator

- A direct current generator produces direct current while an alternating current produces alternating current.
- A simple change to the circuit of DC will change the type of current generated.
- Slip ring commutator is used so that each side of the rectangular coil is always attached to the same brush.
- Thus, current changes direction with every rotation of 180° .



ELECTRICITY AND ELECTROSTATICS

Electrostatics

Is the study of static electricity.

Electric charge is due to the presence of extra electrons (- ve) or absence of electrons (+ve) charge.

Experiment 1

Static electricity

Materials: a woolen cloth or jersey, balloons, a plastic comb and a tap, string

Procedures

- Blow up 2 balloons and tie each one closed.
- Tie a long thread onto each end of the balloon.
- Give each balloon by charging it with a woolen cloth.
- Hold each balloon by the end of the thread and bring them together.
- Observe what happens.

Results

Because both balloons have like charges on their surfaces, they repel each other.

Experiment 2

Procedure

- Rub a plastic comb with wool.
- Open a tap so thin, steady stream of water comes out.
- Bring the comb near the stream of water.
- What happens?

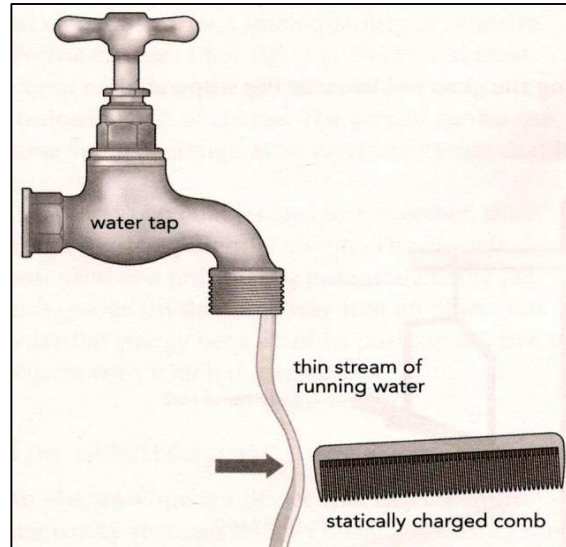
Experiment 3

Aim: To demonstrate the behaviour of like and unlike charge brought together.

Materials: 2 polythene rods, 1 glass or Perspex rod, thread, 2 copper stirrups, a retort stand, a piece of fur, a piece of silk

Results

The stream of water is attracted to the comb as shown below:



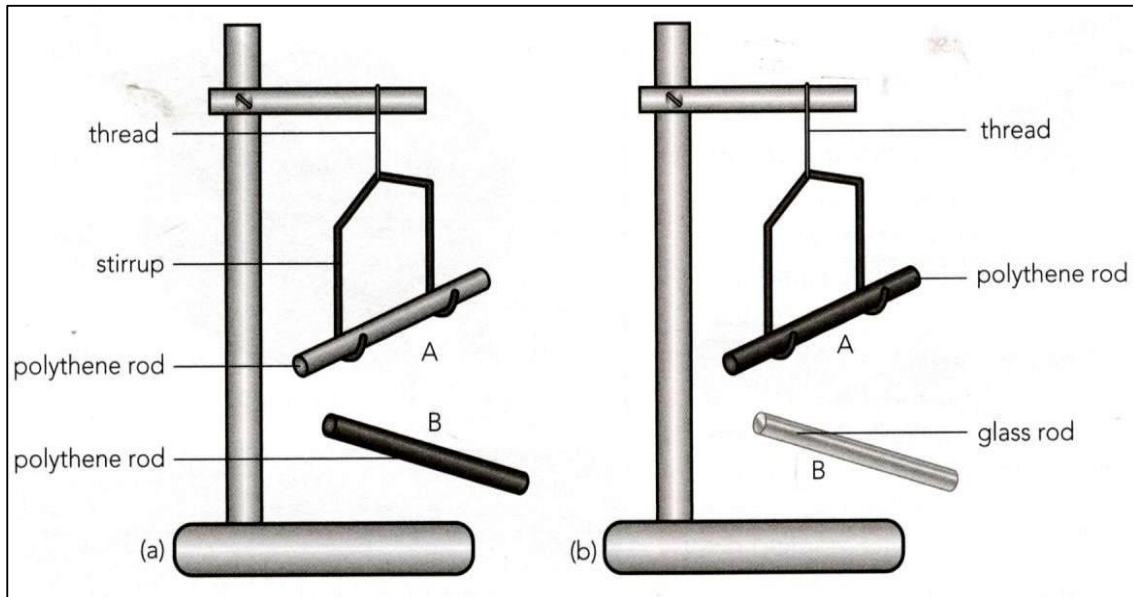
Conductors -are materials that allow electrons to move freely in them. They cannot be charged except by holding them in an insulating handle to prevent the charge passing through one's body to the earth e.g. metals and carbon.

Insulators

- Are materials in which electrons are firmly held by their atoms.
- They can be charged e.g. plastic like polythene and Perspex, nylon, rubber, our thunder clouds, etc
- As one rubs a rod, electron from it atom move into the cloth. The rod will then be short of electron hence become positively charged.
- The cloth will be negatively charged due to excess electron
- Charges generated by friction are called **triboelectricity**.

Procedure

- Set the experiment as shown below. Charge the polythene by rubbing it with fur. Place the charged rod on a stirrup suspended by a thread. Charge a second polythene rod in the same way and bring it towards the suspended rod.
- Charge a glass rod by rubbing it with silk. Now bring glass rod towards the suspended polythene rod.
- Record what happens.



Results

- Rubbed polythene is therefore said to contain a negative charge.
- When a glass/Perspex is rubbed in hair or on a woolen cloth, it loses electrons, thus gaining a positive charge.
- A rubbed Perspex/glass is therefore said to contain a positive charge. When a polythene rod is rubbed, electrons are transferred into it but when a Perspex rod is rubbed, electrons move to it.

Conclusion

- Like charges repel each other.
- Unlike charges attract each other.
- Charges exert force at a distant (the suspended rod was repelled or attracted).
- The field created by electrostatic charge is similar to the magnetic field.

Questions

- What are conductors and insulators?(4)
- Give two examples of conductors and two examples of insulators. (4)

The electroscope

- If an object with negative charges is placed near the ball of the electroscope, electrons in the electroscope plate or ball will be repelled down to the metal rod and to the metal leaves.
- Since each leaves will have a negative charge, they will repel each other and open out.

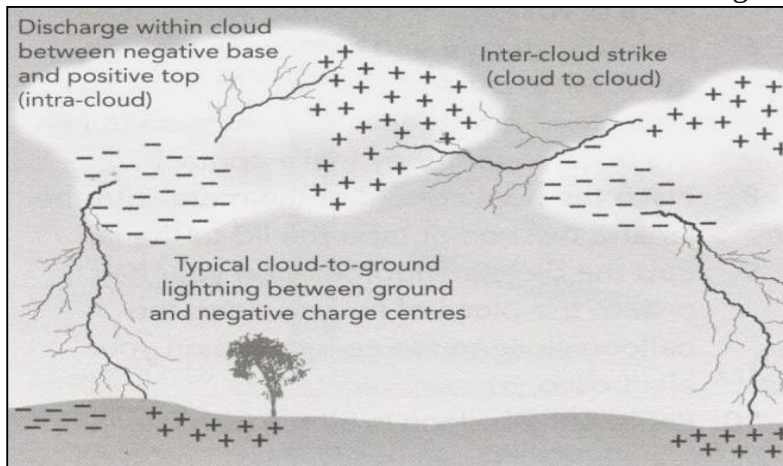
- Laboratory electroscopes detect static charges and magnitude of the charge.

Lightning

- Rapid movement of air and small frozen water droplets in a thundercloud causes the frozen water droplets to be charged.
- The smaller water droplets charge positive and rise to the top of the cloud and the large droplets become negative and occupy the bottom of the cloud.
- Charging continues as violent movement continues in the cloud.
- Very high voltage builds up between positive and negative charges within the cloud.
- The negative charges at the bottom of the thundercloud repel electrons at the surface of the ground, pushing them deeper into the ground.
- The ground becomes positively charged.

Conduction path ways

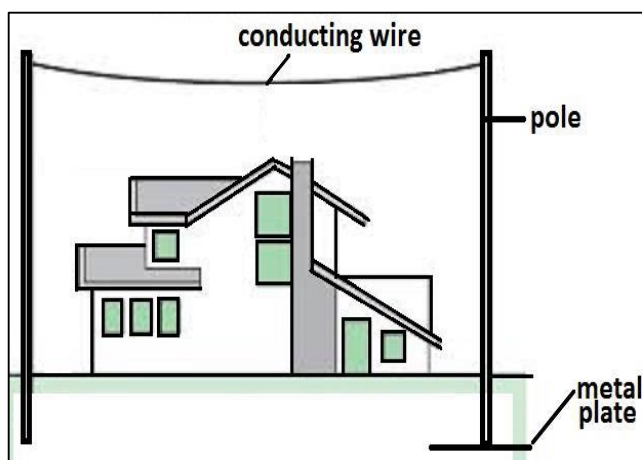
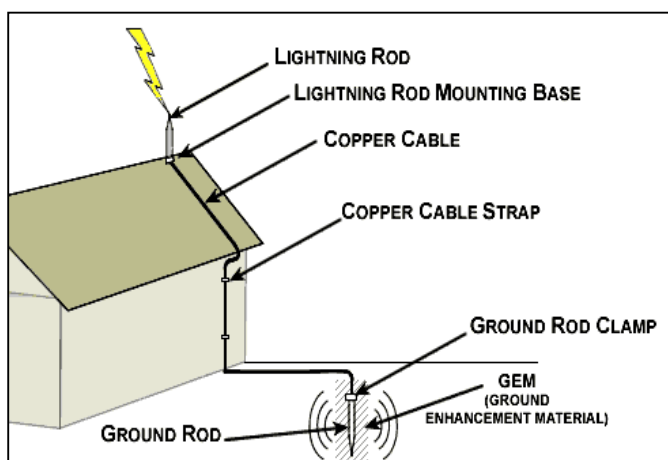
- Conduction between positive and negative charges within the same cloud.
- Conduction between two clouds shown.
- Conduction between the bottom of the cloud and the ground.



Qn 7. Describe the houses below are protected from lightning? (6)

A lightning Conductor

- Protects tall buildings by providing electrons with an easy path to the ground.
- Lightning conductors are made of a metal rod with a sharp point, conducting wires must connect a conductor to a metal rod in the ground so that lightning strikes the building, the charges will flow through the conductor and safely into the ground.
- Electricity poles and telecommunication towers act as lightning conductors.



Precautions against Lightning

- Construct a lightning conductor.
- Do not shelter under tree or isolated ones during a thunderstorm.
- Disconnect roof – top television aerials and unplug computers and radios.
- Do not use landline telephones.
- Keep away from metallic objects like fences and electrical appliances and water sources.
- It is dangerous to be near hill tops, large rocks and tall objects.
- Make sure there is at least five metres between you and other people.

OHM'S LAW AND RESISTORS

Resistance

- When applying a fixed potential difference across different components in a circuit, different values of current flows.
- This is due to that different components resist the flow of electrons by different amounts.
- Copper and aluminum offer little resistance to the flow of electric current hence used as electrical conductors.

Measuring resistance

Experiment 1

Aim: To measure resistance

Materials: 3 torch cells, a rheostat, some nichrome wire, a voltmeter, an ammeter, some conductors

Procedure

- i. Connect a circuit as shown below:
- ii. Starting with the rheostat at its minimum resistance, close the switch and record the voltmeter and ammeter readings.
- iii. Increase the resistance of the rheostat by small amount and again record the readings of voltmeter and ammeter.
- iv. Repeat step three until you have six pairs of readings.
- v. Record the results in table and calculate the ratio $\frac{V}{I}$ for each pair of reading.
- vi. Plot a graph.

Results

The ratio $\frac{V}{I}$ is a constant for a particular resistor.

Factors affecting resistance

- Length of a conducting wire and its cross – sectional area:
 - If you measure the resistance of different length of wire with same cross – sectional area, you will find that doubling the length of wires will double its resistance. Thus, resistance is directly proportional to length of wire.
 - If you measure the resistance of two pieces of wire of same length, but of different thickness, you will find that the resistance is inversely proportional to the cross – sectional area. Thus, the larger the diameter, the lower the resistance, and the thinner the wire, the higher the resistance.

Ohm's law

It gives the relationship between voltage and current. It states that current passing through a wire is proportional to the potential difference across it ends, provided temperature remains constant.

$\frac{\text{potential difference}}{\text{current}}$ = a constant, the constant is the resistance. Thus, Ohm's can be summarised as:

$$\frac{V}{I} = R$$

- Conductors that obey the ohm's law are called ohmic conductors e.g. metals and certain alloys.
- Semi – conductors such as diodes and conduction through gases do not obey ohm's law and are called non – ohmic conductors.

Limitations of Ohm's law

Under the following circumstances, Ohm's law is not obeyed by ohmic conductors.

- The temperature of the conductor changes (examples are bulbs and heater elements).
- The conductor is coiled e.g. filament of bulb.
- The conductor does not have a uniform cross – sectional area when it is under tension.
- The conductor is placed in a strong magnetic field that alters the internal structure of the metal so that its conducting capacity changes.

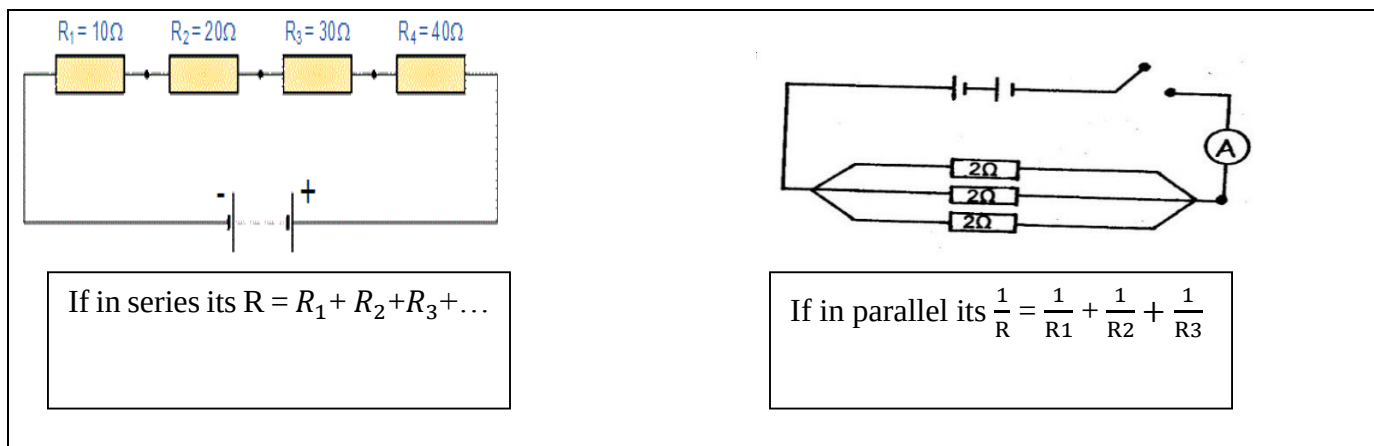
Resistors in Series and Parallel

Resistors are connected in series when they are connected end to end and the same current flow through them. Parallel connection implies that one is placed above one another and their corresponding ends joined together.

Total Resistance

This means combined resistance.

Calculate total resistance of circuits below:



Qn. State and explain the factors that affect resistance. [6]

Power

It is the rate at which electrical energy is changed to other forms of energy e.g. electrical to light energy. That is the rate of using energy. It is measured in watts or joules / second using the formula:

Power = current x volt OR Watts = Amps x Volts

Electrical energy = Voltage X Current X Time

$$E = VIt$$

[Worked examples: Step A head Combined Science Form 3 page 176]

END

FORM 3 NOTES

CHEMISTRY SECTION

SEPARATION

Simple Distillation

- Simple Distillation – separation of pure liquid from a solution by condensing vaporised liquid.

Process of Distillation:

- Solution is heated, and steam (pure vapour) is produced.
- The steam is cooled in condenser to form pure liquid.
- Liquid collected in the beaker is called distillate
- Solute remains in the flask.

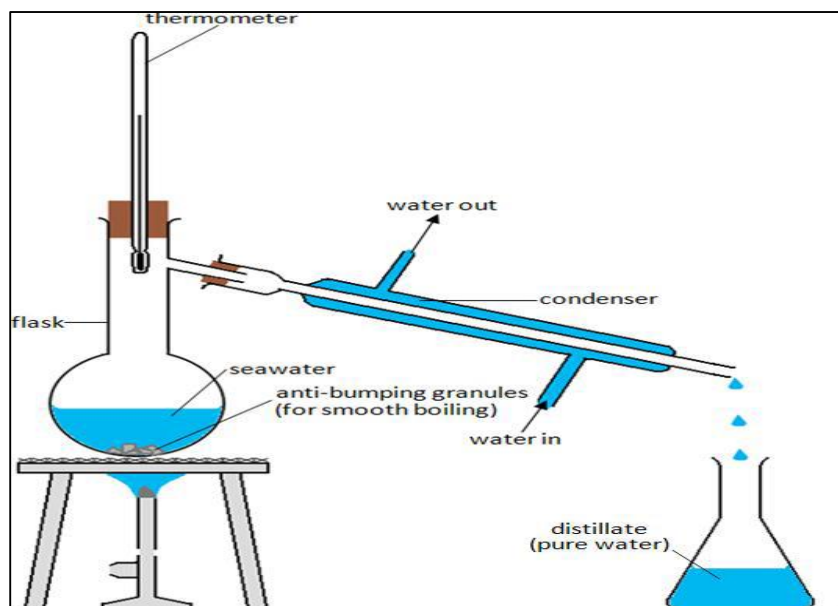
Experiment 1

Aim: To separate salt from salt water

Materials: round-bottomed flasks, a condenser, a thermometer, a beaker, a clamp, rubber tubing, burner, salt solution.

Procedure

3. Set up apparatus as shown.
4. Turn on the water tap to allow steam to be cooled in the condenser.
5. Heat the flask until all liquid evaporates.
6. Record the temperature at which steam is collected.
7. Reduce the heat to allow crystals to be formed in the flask.



Results

- At sea, steam is collected at 100 degree Celsius, but in Zimbabwe is at 96 degrees Celsius.
- Salt remains at the bottom of the flask.
- In distillation the liquid can be retained and in evaporation is lost.

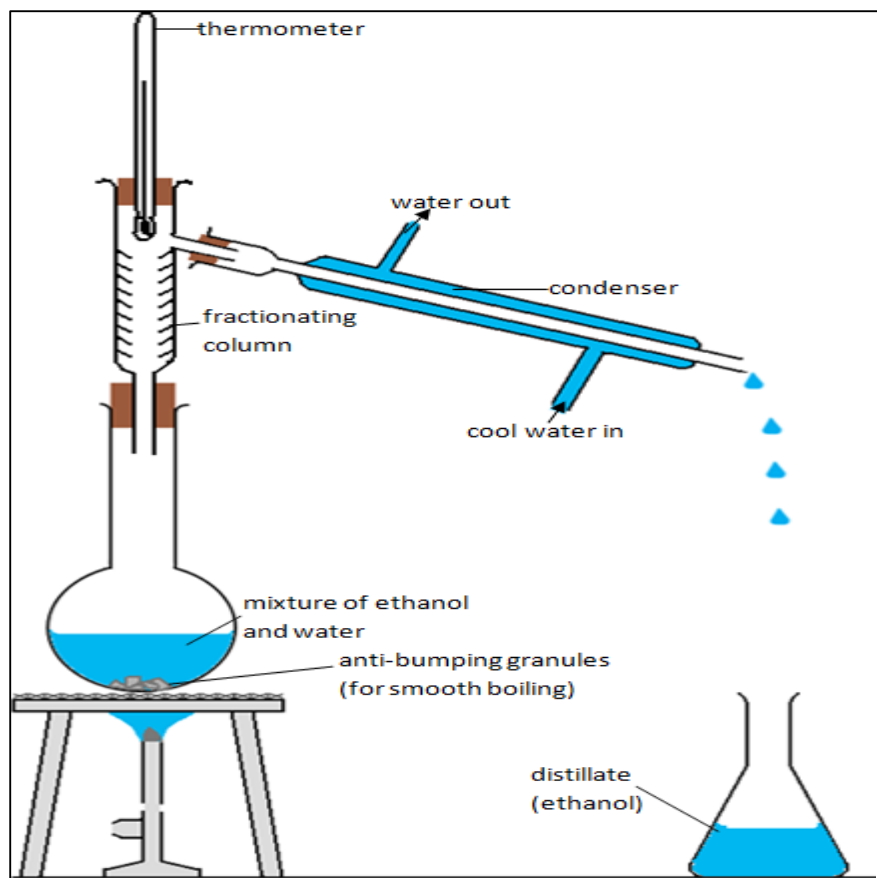
Fractional Distillation

- Fractional Distillation – separates mixture of miscible (soluble) liquids with widely differing boiling points.
- Use of fractionating column separates them.

Process of Fractional Distillation:

Aim: To Separating a mixture of ethanol and water.

- Mixture of ethanol and water is placed in flask and heated. Ethanol with lower boiling point boils and vaporises first and reach fractionating column then cools and condenses into ethanol as it passes through condenser.



Results

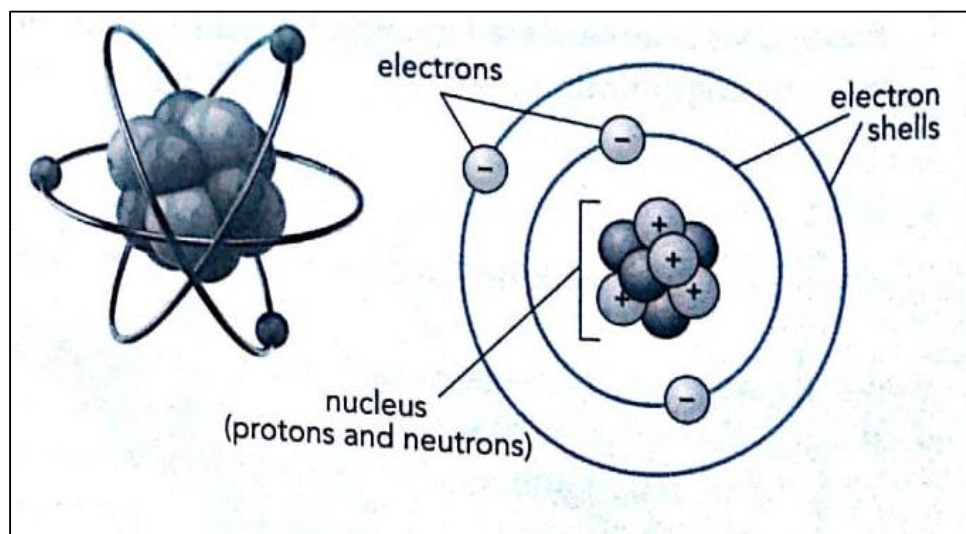
1. The boiling point of ethanol is 78, 37 degrees Celsius.
2. Ethanol is the first fraction.
3. Water boils at 100 degrees Celsius so it's above that of ethanol.
4. To make the second fraction pure, we should be distilled the same way and collected in the beaker on the other side.

Uses of fractional distillation

- Separates pure oxygen and pure nitrogen from liquefied air.
- Separates substances in petroleum (crude oil) into fractions.
- Separates alcohol to produce alcoholic drinks.

MATTER

- Dalton suggested the existence of atoms in all matter, thus matter is made up of atoms.
- An atom is the smallest unit of an element.
- Atoms cannot be divided any further or cannot be created or destroyed.
- Are identical or have same properties if are of the same substance
- Are too small and cannot be seen using the naked eye.
- Atoms consist of sub-atomic particles. These are as follows:



Electrons

- Negatively charged particles that revolve around the nucleus. They are found in shells.

Protons

These are positively charged particles found in nucleus.

Proton number for hydrogen atom = 1

Proton number for helium atom = 2

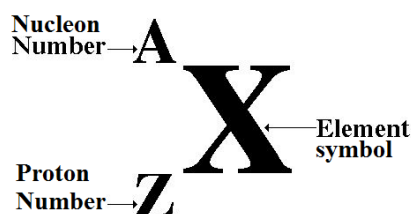
Neutrons

- These are neutral (no charge) and found in nucleus.
- The nucleus is composed of protons and neutrons.

Name the sub-atomic particles, their relative charges, masses and position in the atom

Particle	Position	Charge	Mass	Symbol
Proton	Nucleus	+1	1	p
Neutron	Nucleus	0	1	n
Electron	Atomic shells	-1	$\frac{1}{1840}$	e

Nuclide Notation



- E.g. the element carbon may be represented like this

$^{12}_6\text{C}$ Meaning carbon has 12 nucleons of which 6 are protons. C is the chemical symbol for the element.

Define relative mass / mass number

- Is the number of protons and neutrons in a nucleus (nucleons)

Define the proton number / atomic number

- Is the number of protons in the nucleus of an atom.
- The atom has no charge (electrically neutral) because the number of electrons is equal the number of protons.

Isotopes are atoms of the same element with the same atomic /proton number, but different atomic mass.

E.g. Carbon 12; 13 and Carbon 14.

Isotopes have same proton number, but have different numbers of neutrons hence their masses are different.

The fact that they have different numbers of neutrons does not make a difference in the way they react during chemical reactions. The chemical properties of an atom depends on the number of electrons and their arrangement.

1. Chlorine has two isotopes i.e. $^{35}_{17}\text{Cl}$ and $^{37}_{17}\text{Cl}$
2. Oxygen has two isotopes i.e. $^{16}_8\text{O}$ and $^{18}_8\text{O}$

3. Carbon has two isotopes i.e. $^{12}_6\text{C}$ and $^{16}_6\text{C}$

Calculation of number of neutrons

Number of neutrons = mass number – proton number

Chlorine 35 has $(35 - 17) = 18$ neutrons

Chlorine 37 has $(37 - 17) = 20$ neutrons

Oxygen 16 has $(16 - 8) = 8$ neutrons

Oxygen 18 has $(18 - 8) = 10$ neutrons

Carbon 12 has $(12 - 6) = 6$ neutrons

Carbon 16 has $(16 - 6) = 10$ neutrons

THE PERIODIC TABLE

- Scientists have developed a modern way of arranging elements by their proton numbers.
- Elements are arranged in order of increasing proton number.
- Elements with similar properties are in the same column or group.

GROUP 1: The alkali metals

Group 2: The alkaline earth metals

GROUP 17: The halogens.

GROUP 18: The noble gases

- The horizontal rows are called periods, which are numbered from 1 to 7. Between group 2 and 13, there is a group of elements called transition metals such as copper, iron, nickel, zinc and chromium.

This shows that metals are mostly found on the left and in the middle of the periodic table and most non-metals are on the far right side. There are more metals than non-metals on the table.

Periodic Table of the Elements

1

H

hydrogen

1.0079

3

Li

lithium

6.941

11

Na

sodium

22.990

19

K

potassium

39.098

37

Rb

rubidium

85.468

55

Cs

cesium

132.91

87

Fr

francium

223

2

Be

beryllium

9.0122

12

Mg

magnesium

24.305

20

Ca

calcium

40.078

38

Sr

strontium

87.62

56

Ba

barium

137.33

88

Ra

radium

226

3

Sc

scandium

44.956

21

Ti

titanium

47.867

39

Y

yttrium

88.906

57-71

La-Lu

89-103

Ac-Lr

4

V

vanadium

50.942

22

Cr

chromium

51.996

40

Mn

manganese

54.938

41

Fe

iron

55.845

42

Co

cobalt

58.933

43

Ni

nickel

58.693

44

Cu

copper

63.546

45

Zn

zinc

65.38

5

Sc

scandium

44.956

21

Ti

titanium

47.867

39

Y

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88.906

57-71

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Cu

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Zn

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65.38

13

Al

aluminum

26.982

31

Ga

gallium

69.723

49

In

indium

114.82

81

Tl

thallium

204.38

113

Uut

ununium

113

14

Si

silicon

28.086

32

Ge

germanium

72.64

50

Sn

tin

118.71

82

Pb

lead

207.2

114

Uuq

ununium

114

15

P

phosphorus

30.974

33

As

arsenic

74.922

51

Sb

antimony

121.76

83

Bi

bismuth

208.98

115

Uup

ununium

115

16

S

sulfur

32.065

34

Se

selenium

78.96

52

Te

tellurium

127.60

84

Po

polonium

209

116

Uuh

ununium

116

17

Cl

chlorine

35.453

35

Br

bromine

79.904

53

I

iodine

126.90

85

At

astatine

210

117

Uus

ununium

117

18

Ar

argon

39.948

36

Kr

krypton

83.798

54

Xe

xenon

131.29

86

Rn

radon

222

118

Uuo

ununium

118

1

H

hydrogen

1.0079

3

Li

lithium

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Na

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Mn

manganese

54.938

42

Fe

iron

55.845

43

Co

cobalt

58.933

44

Ni

nickel

58.693

45

Cu

copper

63.546

46

Zn

zinc

65.38

13

Al

aluminum

26.982

31

Ga

gallium

69.723

49

In

indium

114.82

81

Tl

thallium

204.38

113

Uut

ununium

113

14

Si

silicon

28.086

32

Ge

germanium

72.64

50

Sn

tin

118.71

82

Pb

lead

207.2

114

Uuq

ununium

114

15

P

phosphorus

30.974

33

As

arsenic

74.922

51

Sb

antimony

121.76

83

Bi

bismuth

208.98

115

Uup

ununium

115

16

S

sulfur

32.065

34

Se

selenium

78.96

52

Te

tellurium

127.60

84

Po

polonium

209

116

Uuh

ununium

116

17

Cl

chlorine

35.453

35

Br

bromine

79.904

53

I

iodine

126.90

85

At

astatine

210

117

Uus

ununium

117

18

Ar

argon

39.948

36

Kr

krypton

83.798

54

Xe

xenon

131.29

86

Rn

radon

222

118

Uuo

ununium

118

1

H

hydrogen

1.0079

3

Li

lithium

6.941

11

Na

sodium

22.990

19

K

potassium

39.098

37

Rb

rubidium

85.468

55

Cs

cesium

132.91

87

Fr

francium

223

2

Be

beryllium

9.0122

12

Mg

magnesium

24.305

20

Ca

calcium

40.078

38

Sr

strontium

87.62

56

Ba

barium

137.33

88

Ra

radium

226

3

Sc

scandium

44.956

21

Ti

titanium

47.867

39

Y

yttrium

88.906

57-71

La-Lu

89-103

Ac-Lr

4

V

vanadium

50.942

22

Cr

chromium

51.996

40

Mn

manganese

54.938

41

Fe

iron

55.845

42

Co

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V

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Cr

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51.996

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Mn

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Ni

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58.693

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Cu

copper

63.546

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Zn

zinc

65.38

10

V

vanadium

50.942

23

Cr

chromium

51.996

41

Mn

manganese

54.938

42

Fe

iron

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Co

cobalt

58.933

44

Ni

nickel

58.693

45

Cu

copper

63.546

46

Zn

zinc

65.38

11

V

vanadium

50.942

23

Cr

chromium

51.996

41

Mn

manganese

54.938

42

Fe

iron

55.845

43

Co

cobalt

58.933

44

Ni

nickel

58.693

45

Cu

copper

63.546

46

Zn

zinc

65.38

12

V

vanadium

50.942

23

Cr

chromium

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41

Mn

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Ar

argon

39.948

36

Kr

krypton

83.798

54

Xe

xenon

131.29

86</

Name the first 20 elements in the periodic table stating their symbols

1. Hydrogen	H	11. Sodium	Na
2. Helium	He	12. Magnesium	Mg
3. Lithium	Li	13. Aluminium	Al
4. Beryllium	Be	14. Silicon	Si
5. Boron	B	15. Phosphorus	P
6. Carbon	C	16. Sulphur	S
7. Nitrogen	N	17. Chlorine	Cl
8. Oxygen	O	18. Argon	Ar
9. Fluorine	F	19. Potassium	K
10. Neon	Ne	20. Calcium	Ca

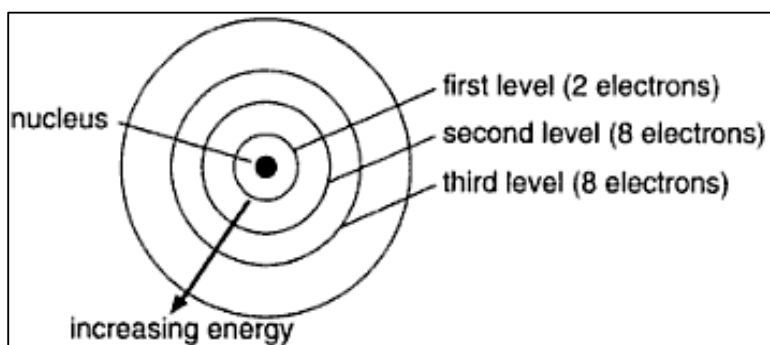
Write the electronic configuration of the first 20 elements

- Is the arrangement of electrons in an atom

Hydrogen	1	Sodium	2.8.1
Helium	2	Magnesium	2.8.2
Lithium	2.1	Aluminium	2.8.3
Beryllium	2.2	Silicon	2.8.4
Boron	2.3	Phosphorus	2.8.5
Carbon	2.4	Sulphur	2.8.6
Nitrogen	2.5	Chlorine	2.8.7
Oxygen	2.6	Argon	2.8.8
Fluorine	2.7	Potassium	2.8.8.1
Neon	2.8	Calcium	2.8.8

Electronic Configuration

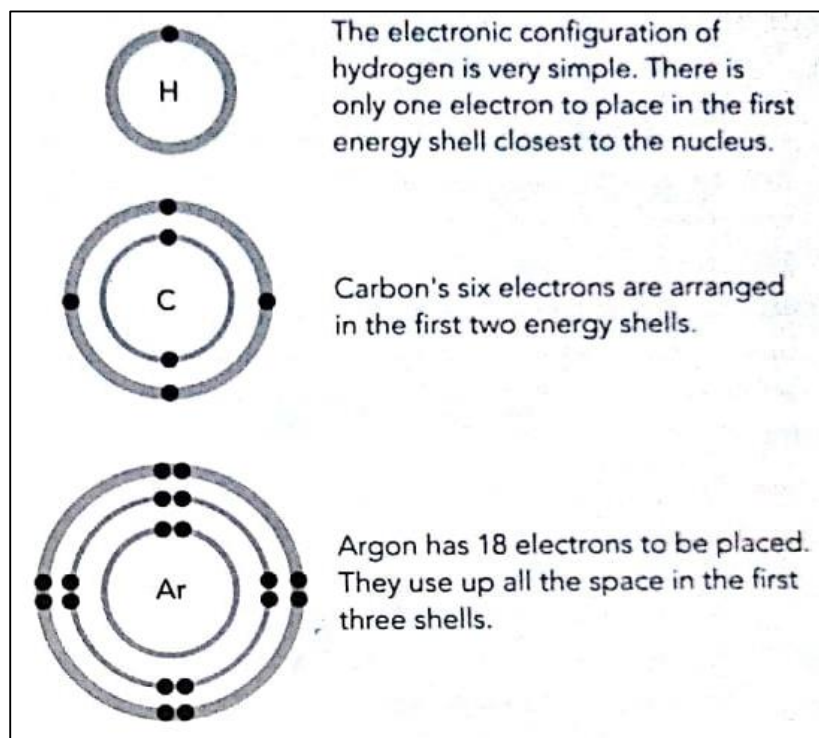
- Atoms are electrically neutral. Thus, the number of protons is equal to the number of electrons.
- Electrons are placed in orbits / series of energy shells around the nucleus. This nature of arrangement is called electronic configuration.
- Energy shells can hold a certain number of electrons. The shells become larger as they are further from the nucleus hence a larger shell can hold more electrons.
- First energy shell contains maximum 2 electrons. Second shell and so on has maximum of 8 electrons. The fourth shell can hold a maximum of 18.



To write electronic configuration: Sodium example

- ❖ Look up the atomic number in the periodic table. The number is 11.
- ❖ The number of electrons and protons are equal in neutral sodium atom.
- ❖ Arrange the electrons in shells. Fill up the lower energy shell before going to the outer one.
- ❖ In sodium, the electrons are arranged as two (2) in the first shell, eight in the second shell and one in the third. This is written as: (n,n,n) i.e. 2,8,1.

Other Configuration diagrams



Significance of the electronic configuration

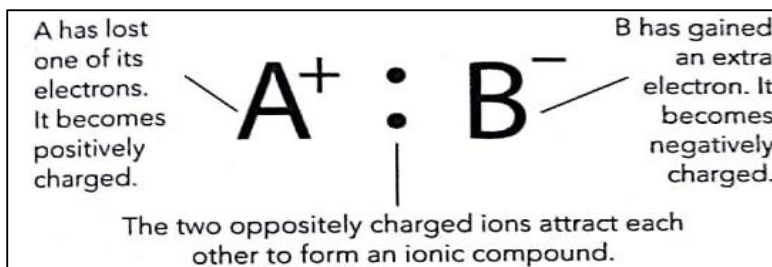
- ❖ The number of electrons in the outmost shell of an atom determines the chemical properties of an element including how it forms bonds with other elements.
- ❖ There are a group of element called noble gases eg. Helium, neon, argon, krypton, xenon and radon. All have 8 electrons in the outmost shell except helium which has only 2.
- ❖ The gases are completely **inert or nonreactive**.

Chemical Bonding

- Is the combination of elements to form compounds. A compound is a chemical composition of two or more elements.
- E.g. salt is composed of one atom of sodium and one of chlorine. Hydrogen is made up of two hydrogen atoms and one oxygen atom.
- To bond is to join and this makes a compound. When reacting, atoms will donate and receive electrons so as to form ionic compounds or will share electrons to form molecular compounds.

Ionic bonding / electrovalent bonding

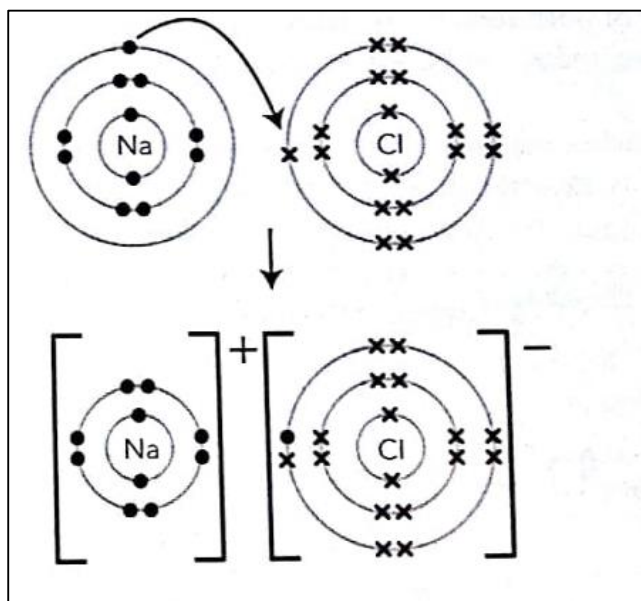
An ion is a charged atom. It is an atom which has too many or too few electrons to be neutral. Ions of opposite charges attract one another to form ionic bonds as shown:



- When atoms gain electrons and others lose electrons, there is attraction between two ions to form a compound. This is called ionic bonding.
- Metals form positive ions (**cations**) because they lose electrons while non-metals form negative ions (**anions**) because they gain electrons and become negative. The electrons that form are called valence electrons.
- **Valency** of an atom is the number of electrons which an atom should gain or lose to attain a noble gas configuration.
- Metals in group 1 have a valency of 1 because they lose 1 electron to become a cation. Group 2 have a valency of 2 and group 13 and 14 elements have a valency of 3 and 4 respectively.
- For example: $^{22}_{11}\text{Na}$ and $^{35}_{17}\text{Cl}$

Charge atom is Na^+ and is Cl^-

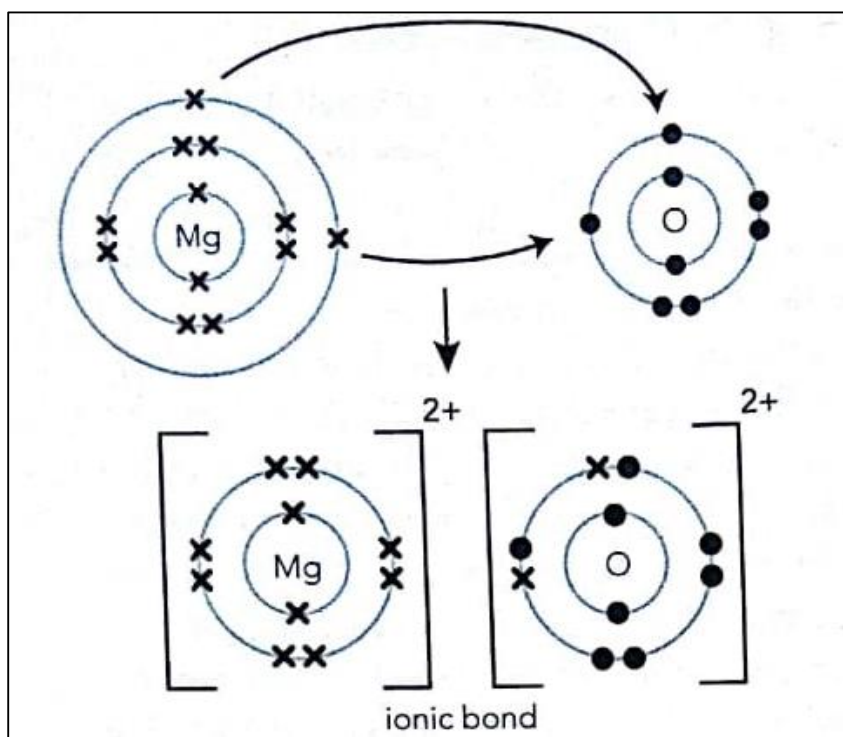
Sodium + Chlorine $\rightarrow$ Sodium chloride.



Bonding in Magnesium oxide

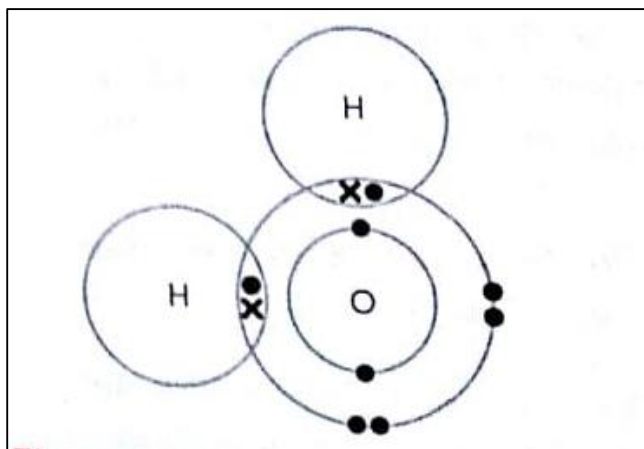
Magnesium has two electrons in its outer shell so it can become stable by losing these two valence electrons. It therefore becomes a magnesium ion with a charge of $2+$.

An oxygen atom has 8 electrons it needs two valence electrons in order for it to attain a stable outer shell hence the combination will give MgO as shown below:



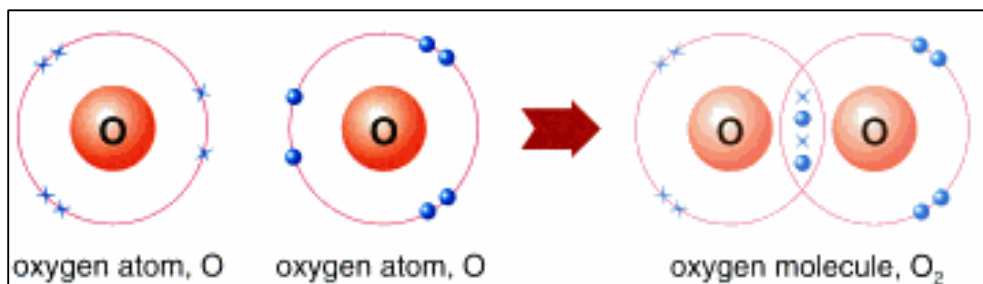
Covalent Bonding

- Involves sharing of electrons.
- When atoms combine by sharing electrons to attain a noble gas configuration.
- Example 1: $\text{H}_2 + \text{O}_2 = \text{H}_2\text{O}$.
- Oxygen has six electrons in its outermost shell. It therefore needs to gain or share another two to achieve stability.
- Hydrogen atoms each have one electron in their outermost shell(s) therefore, two hydrogens have to bond covalently with one oxygen atom to form water molecule.

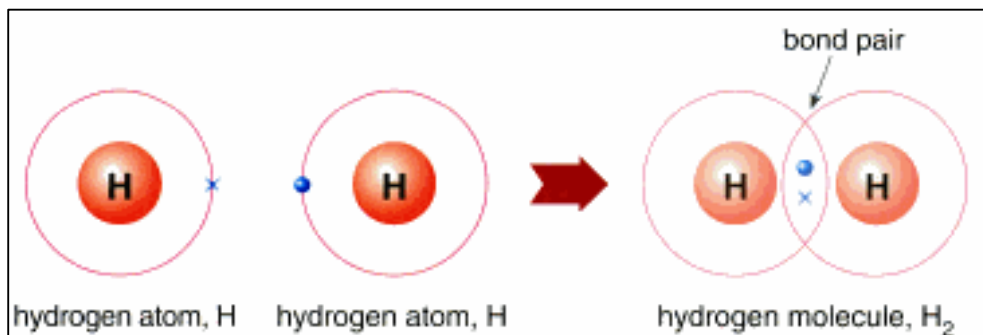


- Example 2: carbon + oxygen $\rightarrow$ carbon dioxide.

Example 3



Example 4



Properties of Covalent & Ionic Compounds

Covalent Compound

- Are usually gases and liquids.
- Does not conduct electricity in any state (because does not contain ions)
- Low melting point “less than 200°C” (because weak forces of attraction between molecules).
- Insoluble in water, Soluble in organic solvents.

Ionic Compound

- Conduct electricity when molten, Aqueous state - because it contains moving ions - does not conduct on solid as ions cannot move.
- High Melting point “More than 1000 °C” hence ionic compounds are generally solids at room temp because of strong forces of attraction between ions.
- Soluble in water

Stoichiometry and the Mole Concept

Avogadro number (L)

- Is the number of particles in one mole of an element or compound.

$$L = 6.02 \times 10^{23}$$

- A mole is atomic or molecular mass expressed in grams.
- A mole is the amount of substance which contains the same number of molecules or atoms as there are atoms in 12g of $^{12}_6C$

$$1 \text{ mol of MgO} = 24 + 16 = 40\text{g}$$

$$2 \text{ mol of HCl} = 2[1 + 35.5] = 73\text{g}$$

$$1 \text{ mol of O}_2 = 2[16] = 32\text{g}$$

$$\text{Number of moles} = \frac{\text{mass}}{\text{molecular mass}} \text{ or}$$

$$n = \frac{m}{Mr}$$

Examples

1. Express the following in moles

(a) 10g NH₃

$$1 \text{ mole (NH}_3\text{)} = 14 + (1) 3 = 17\text{g}$$

$$10\text{g NH}_3 = \frac{10}{17} = 0.59 \text{ moles}$$

(b) 5g Na

(c)

2. Calculate the mass

(a) 0.2 moles C_2H_4

$$1\text{mole } (C_2H_4) = 2(12) + 4(1) = 28\text{g}$$

$$0.2\text{moles} = 28 \times 0.2 = 5.6\text{g}$$

(b) 0.25 moles of carbon monoxide

Molecular Mass (Mr)

- Is given by the sum of atomic masses of the elements.

$$\text{Mr (NaCl)} = 23 + 35.5 = 58.5\text{g}$$

$$\text{Mr (Na}_2\text{O)} = 2(23) + 16 = 62\text{g}$$

$$\text{Mr (Na}_2\text{CO}_3)$$

$$\text{Mr (HCl)}$$

$$\text{Mr (CO}_2)$$

$$\text{Mr (NH}_3)$$

Empirical formula

- Is a formula that shows the simplest ratio combination of atoms that form up a given substance,

Example

1. 6g of Magnesium is heated in oxygen, after cooling and reweighing it is found that there is 10g of Magnesium Oxide. Find the empirical formula for Magnesium Oxide

$$\begin{array}{rclcl} \text{Ratio: Mg} & : & \text{O} & & \\ \frac{6}{24} & : & \frac{4}{16} & & \\ = 0.25 & : & 0.25 & & \\ & & (\text{divide by the smallest}) & & \\ 1 & : & 1 & & \end{array}$$

$\therefore$ **empirical formula = MgO**

2. Calculate the empirical formulae of substances which have the following compositions by mass;
43.4% Sodium, 11.3% Carbon, 45.3% Oxygen

$$\begin{array}{rclcl} \text{Ratio: Na} & : & \text{C} & : & \text{O} \\ \frac{43.4}{23} & : & \frac{11.3}{12} & : & \frac{45.3}{16} \\ = 1.89 & : & 0.942 & : & 2.83 \\ & & (\text{Divide by the smallest}) & & \end{array}$$

$$\frac{1.89}{0.942} : \frac{0.942}{0.942} : \frac{2.83}{0.942}$$

$$= 2.01 : 1 : 3$$

$\therefore$ empirical formula = Na_2CO_3

Concentration

- It is the ratio of number of moles of a solute to the volume of a solution.

$$\text{Concentration} = \frac{\text{number of moles}}{\text{volume}}$$

- Units are mol/dm^3 or g/dm^3

Examples

1. Find the number of moles of Sodium hydroxide in 25cm^3 of solution of concentration 0.1mol/dm^3

$$\begin{aligned} \text{Number of moles} &= \text{concentration} \times \text{volume} \\ &= 0.1 \times 25 \\ &= 0.0025 \end{aligned}$$

2. Find volume of solution of concentration 2 mol/dm^3 that contain 0.005 moles of hydrochloric acid.

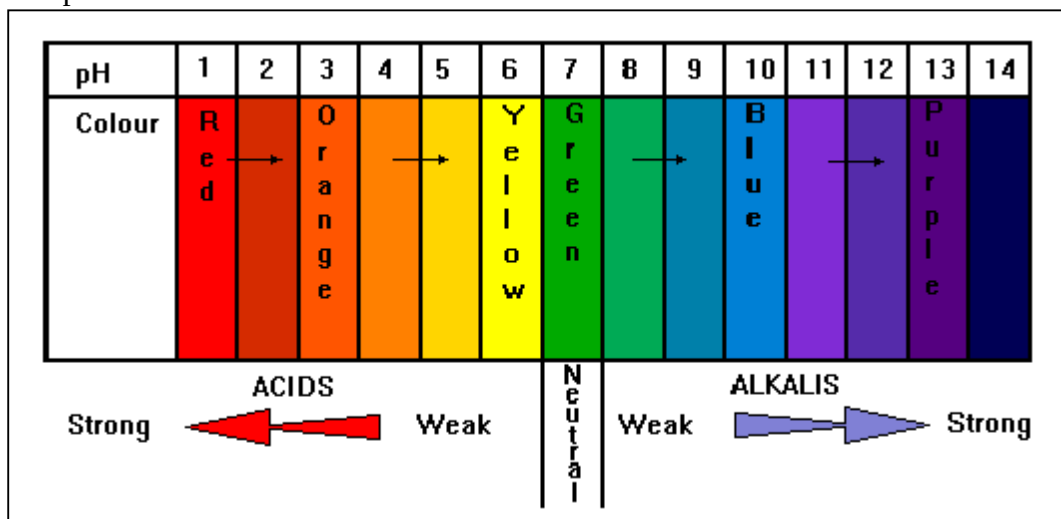
$$\begin{aligned} \text{Volume} &= \frac{\text{number of moles}}{\text{concentration}} \\ &= \frac{0.005 \times 1000}{2} \\ &= 25\text{cm}^3 \end{aligned}$$

Acid, Bases and Salts

- All substances are divided into acids, basis and salts or neutral.
- When one puts these compounds in water, they break down or dissociate and either release hydrogen ions (H^+) or hydroxides ion(OH^-)
- Acids or bases can be strong or weak and all are dangerous as they are corrosive.
- In the laboratory acids and alkali used are always diluted with water.

pH is a measure of acidity or alkaline.

The p.H Scale



- ✚ It ranges from zero to fourteen (0 – 14)
- ✚ It shows how acidic or alkaline a solution is. Acids have pH values less than 7 while alkaline have pH values greater than 7
- ✚ If a substance has a pH of 7, it is neutral e.g. pure water

Indicators

Litmus paper is an example of an indicator and the universal indicator being most important – it is a mixture of dyes that change colour gradually over a range of pH.

Indicator	Colour in acid	Colour in alkali
litmus	red	blue
methyl orange	red	yellow
phenolphthalein	colourless	red
universal	red	purple

Strong and weak acids

- When acids dissolve in water, they give away their hydrogen ions. Adding water dilutes an acid but does not make it weaker as only few hydrogen ions given away.

Strong and weak alkali

- Some alkali dissolve in water to give away their hydroxyl ions. Alkali that completely dissociate in water are strong alkali. E.g. sodium hydroxide and potassium hydroxide.
- Weak alkali such as ammonia can partially dissociate in water and does not give away their hydroxyl ions.

Experiment 1

Testing the pH of common substances

- In this activity a range of household and everyday products are tested for their pH.
- Using universal indicator paper or red and blue litmus.

Materials

White tiles, Dropping pipettes, Universal indicator paper and a colour chart, Red and blue litmus paper, a range of household substances (include fruit juices, soft drinks, vinegar, different soap solutions, floor cleaner containing ammonia and toothpaste.)

You could also include some of the dilute acids and alkalis in the kit (e.g. HCl, Ammonia solution)

NB: Acids and alkalis are corrosive, extreme care must be taken' wash with large ' amounts of water if they come in contact in eyes. Splashes on the skin, should be washed off as soon as possible. Always wipe up any small spills.

Method

- Put a small drop of one of the substances you wish to test on a white tile.
 - Tear off small pieces of blue and red litmus paper and universal indicator.
 - Drop the three pieces of paper in the substances. Record your results in a table like the one below:
- N.B. You can also carryout this experiment in test tubes, but this involves lots more washing up!

Substance	Colour of red litmus paper	Colour of blue litmus paper	Colour of Universal Indicator	Acid, alkali, or neutral	pH value
Hydrochloric acid				Strong acid	1-2
Sodium hydroxide				Strong base	12-14
Distilled water				neutral	7
Tooth paste				Weak alkali	9
Lemon juice				Acid	2.5
coffee				Weak acid	5

QN: How many acids are there?
Complete the table above.

Preparation of Acids

Non – metallic oxide + water → acid.

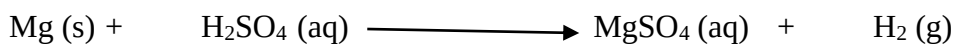
E.g. carbon dioxide + water → carbonic acid.

Sulphur dioxide + water → sulphuric acid.

Reactions of acids with metal

- Some metals react with acids while some do not react for example gold and silver do not react with acids.
- Metal react with acids to form a salt and hydrogen gas.
- metal + acid → salt + hydrogen gas.

Magnesium + sulphuric acid → magnesium sulphate + hydrogen



Experiment

Aim: To investigate the reactions of acids with metals

Materials: test tubes, dilute hydrochloric acid, sulphuric acid, nitric acid, ethanoic acid, magnesium ribbon, metal turnings of zinc, iron and copper.

Procedure

- Fill 5 test tubes - one quarter full with one of the acids.
- Drop a piece of magnesium ribbon into each test tube. Observe what happens.
- Repeat the experiment using different acids and a set of all metals.
- Record the results in a table as shown:

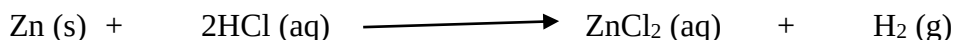
Results

- Not all metals react with acids to form hydrogen.
- Copper only reacted with nitric acid and a brown gas evolved.

Example: Magnesium + hydrochloric acid → magnesium chloride + hydrogen gas.

QN: write down the chemical equation for the above reaction. (2)

Zinc + hydrochloric acid → zinc chloride + hydrogen gas.



*Use other metals to come up with the above equation. (8)

BASES – are substances with pH above 7.

Preparation of basis

Metal oxide + water → base.

Calcium oxide + water → calcium hydroxide.

Magnesium oxide + water → magnesium hydroxide.

Some materials used in everyday are acidic and acids are identified by considering their properties as summed below:

Comparison of acids and base

Acid	Bases
Has a sour taste (never taste it).	Has a slippery feel and tastes bitter (never taste it).
Turns blue litmus paper to red.	Turns red litmus paper to blue.
Acids are corrosive	Some bases are caustic e.g. potassium hydroxide.
Reacts with a base to form salt and water.	Reacts with bases to form salt and water.
Universal indicator paper changes to a series of various colours as acidity and alkalinity of the solution changes.	

Bases

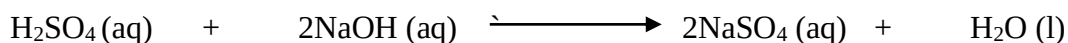
Bases can be made by dissolving metal oxides or adding water to reactive metals. Bases are chemicals that contain hydroxyl ions (OH⁻) and metal or metal like ions.

Salts

The salt we eat is sodium chloride or common salt. The term salt can mean other combination of elements too, some of which are poisonous.

The reaction between an acid and base to form salt and water is called neutralization. Acid + base → salt + water e.g. hydrochloric acid + sodium oxide → sodium chloride + water.

Qn: Write down the chemical equation for the above reaction. (2)



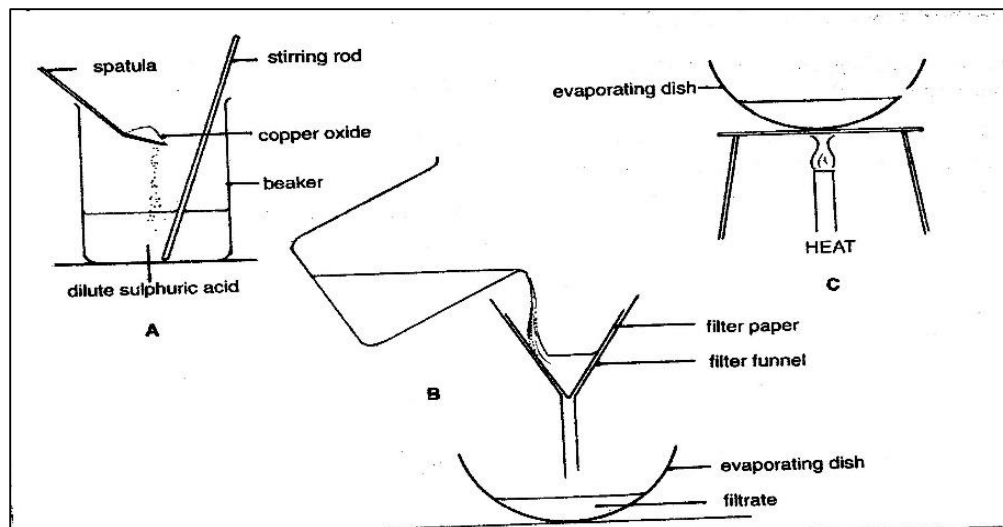
Qn: Write down the word equation for the above reaction. (2)

Experiment 2

Making salt, copper sulphate.

Materials: a beaker, stirrer, filter paper, filter funnel, evaporating dish, burner, and stand, dilute sulphuric acid, copper oxide

Method



- ✓ Warm a mixture of a small spatula load of copper oxide and 15ml of sulphuric acid in a beaker.
- ✓ Continue adding copper oxide until no more will dissolve.
- ✓ Filter the solution to remove the undissolved copper oxide.
- ✓ Heat the filtrate over a burner to remove excess water, a saturated solution will be formed.
- ✓ Leave the evaporating dish over a window sill for few days.
- ✓ Observe the results.

Results

When copper oxide is added to dilute sulphuric acid, it starts to dissolve and a blue filtrate is collected in the evaporating dish.

When the filtrate allowed to cool for some days, forms crystals of copper sulphate.

Conclusion

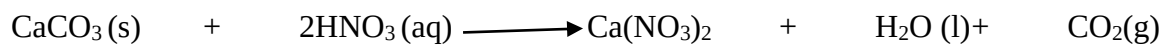
Copper oxide reacts with dilute sulphuric acid to form copper sulphate and water.

Copper oxide + dilute sulphuric acid $\rightarrow$ copper sulphate + water.

Qn: Write down the chemical equation for this reaction. (2)

Reaction of acids with carbonates

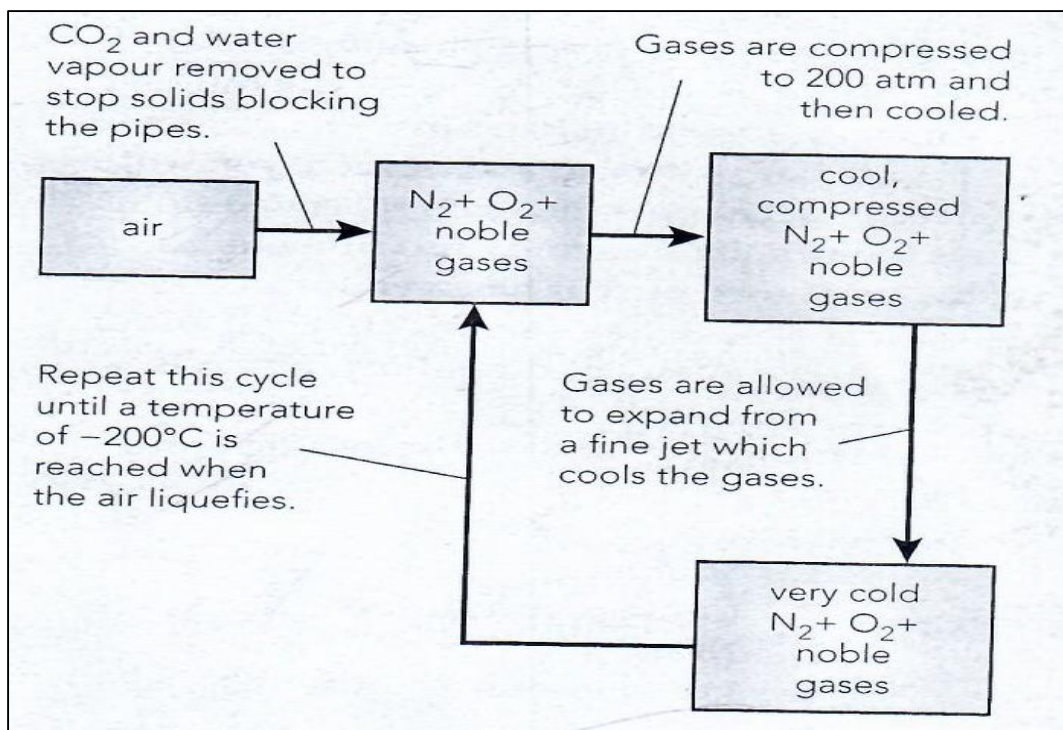
- Carbonates react with acids to give salt, water and carbon dioxide. There is fizzing or effervescence.



INDUSTRIAL PROCESSES

Production of Nitrogen and oxygen

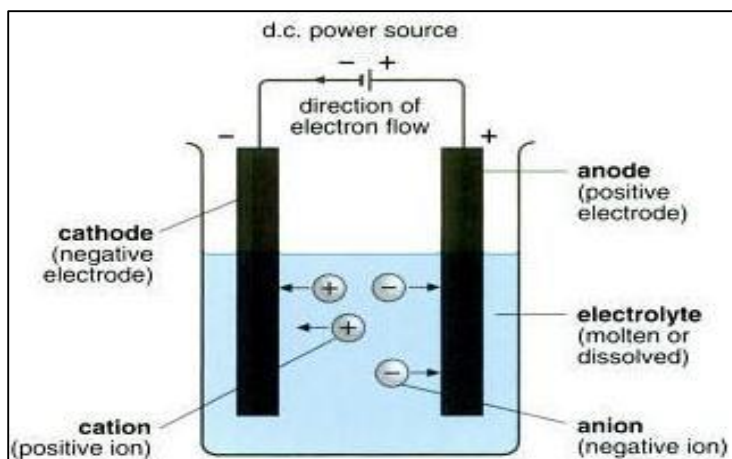
Liquefaction of air



Air is cooled to -78 degrees Celsius. Water and carbon dioxide are removed at this point since they freeze below this. The remaining air is compressed to about 150 atm. The gas becomes hot. The compressed gases are allowed to expand, thus cooling them. The compression is repeated several times so temperature drops to -200 degrees Celsius hence gases turn into liquids except helium and neon. The liquid air is reheated to -196 degrees Celsius and nitrogen evaporates and is collected at the top of the distillation. Oxygen which boils at -183 degrees Celsius remains a liquid and is collected at the bottom of the tower.

Electrolysis - is the decomposition of a compound molten or in solution, by electricity.

Components of an electrolytic cell



- Electrode is a rod or plate where electricity enters or leaves electrolyte during electrolysis. Reactions occur at electrodes.
- Electrolyte is an ionic compound which conducts electric current in molten or aqueous solution, being decomposed in the process.
- Cathode is negative electrode connected to negative terminal of d.c. source. Reduction occurs here. Cathode gains negative charge as electrons flow from the battery towards the cathode, making cathode negatively charged. This causes cation to be attracted and gains electrons to be an atom.
- Anode is positive electrode connected to positive terminal of d.c. source. Oxidation occurs here. Anode loses negative charge as electrons flow towards the battery, leaving anode positively charged. This causes anion to discharge its electrons here to replace lost electrons and also, negative charge are attracted to positive charge.
- Discharge is the removal of electrons from negative ions to form atoms or the gain of electrons of positive ions to become atoms.
- Cation is positive ion. It's attracted to cathode.
- Anion is negative ion. It's attracted to anode.

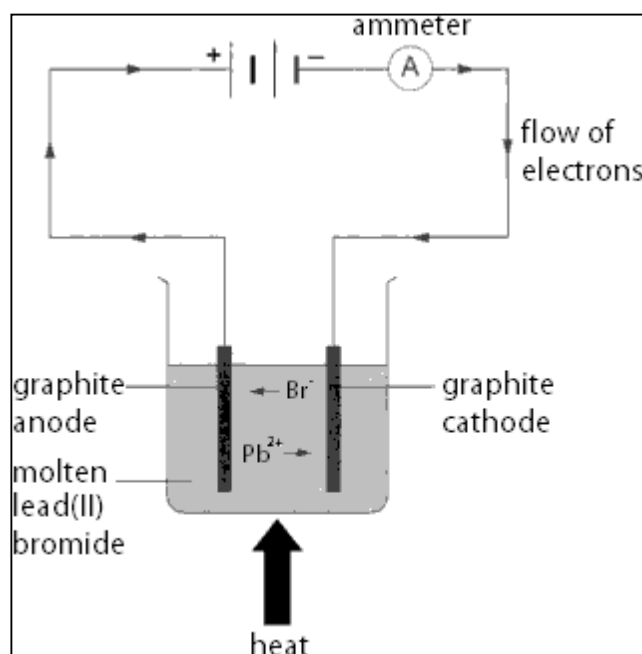
Electrolysis of Molten Compounds

- Molten/aqueous ionic compounds conduct electricity because ions free to move.
- In solid state, these ions are held in fixed position within the crystal lattice.
- Hence solid ionic compounds do not conduct electricity.

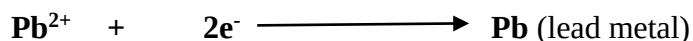
Electrolysis Of Molten Lead Bromide

Describe anode and cathode reactions for electrolysis of molten lead bromide

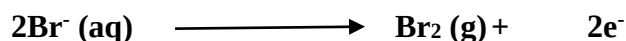
State observations for the electrolysis of molten lead bromide



- To make molten lead (II) bromide, PbBr_2 , we strongly heat the solid until it melts. To electrolyse it, pass current through the molten PbBr_2 .
- Ions Present are Pb^{2+} and Br^-
- At the cathode lead ions gains electrons to form lead atoms. The lead will form a liquid at the bottom of the reaction vessel.



- At the anode bromide ions loses electrons to form bromine atoms which combine to form molecules of bromine gas/vapour. Bromine appears as a brown gas at the anode.



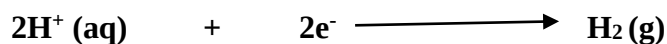
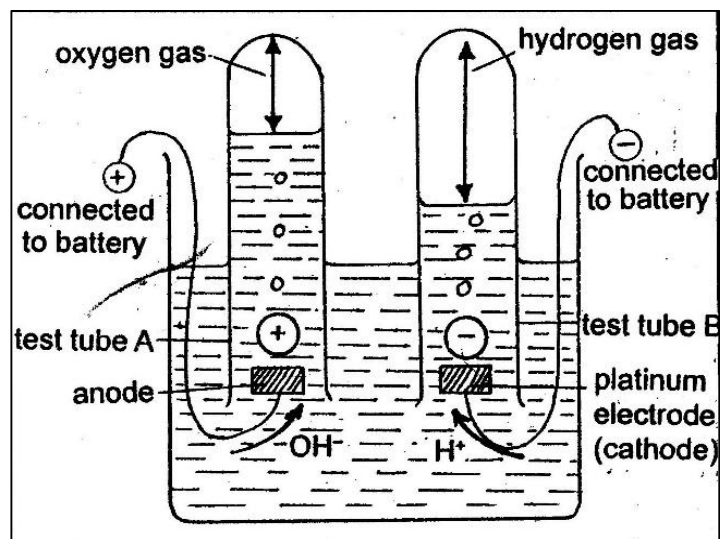
Overall Equation



Electrolysis of Water

Describe the electrolysis of water

- Inert platinum/carbon electrodes are used as electrodes.
- Water is acidified
- Water is broken into hydrogen and oxygen in the ratios 2 : 1 respectively
- While current is flowing positive H^+ ions migrate to the cathode, accept electrons to form hydrogen atoms, which then combine to form molecules of hydrogen gas.



- Negative hydroxide ions migrate to the anode and give up electrons forming water and oxygen molecules.



State products formed during electrolysis of water

- The gas at the cathode is twice the gas at the anode.
- Hydrogen is produced at the cathode. It burns with a pop sound when a burning wooden splint was brought near the mouth of the test tube.
- Oxygen is produced at the anode. It relights a glowing wooden splint.

State uses of oxygen and hydrogen

Uses of oxygen

1. Manufacture of steel
2. Cutting and welding metals
3. Medical purposes

Uses of hydrogen.

1. Manufacture of ammonia at Sable Chemicals
2. Manufacture of margarine
3. Welding

Experiment 1

Chrome Plating

Aim: To chrome plate a nickel cathode

Materials: nickel-plated electrode, lead electrode, beaker, chromic acid

Method

1. Set up an electrolytic cell using a nickel plated cathode from this experiment.
2. Place 200cm³ of chromic acid solution in a beaker.
3. Warm it up to 40°C.
4. Pass a current through the solution for about 20 minutes.
5. Switch off the current and disconnect the cathode

Observations and results

- The nickel plated cathode is covered by chrome and achieves a glossy finish.
- It does adhere well even during polishing
- Chrome shines more than nickel.

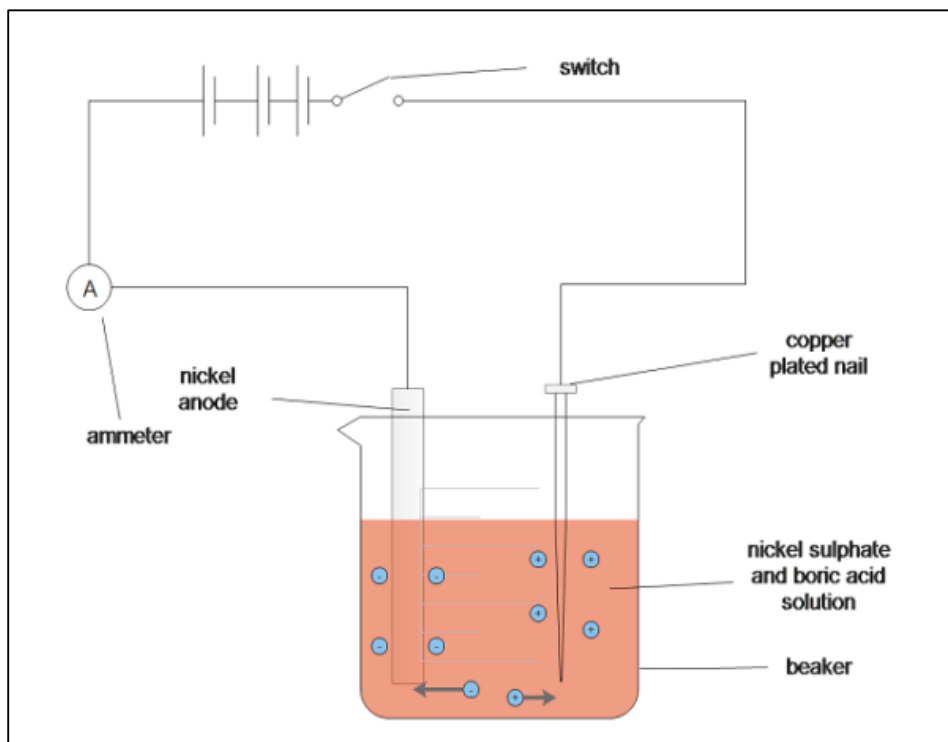
Experiment 2

Nickel plating a piece of copper sheet.

Aim: To nickel plate a piece of copper

Materials: beaker, nickel sulphate, boric acid, burner, nickel electrode, 1.5 v cell, copper electrode/copper plated item, universal indicator

Method



1. In a beaker make 20cm³ a solution using 54g nickel sulphate and 6g boric acid.
2. Warm the solution to about 70°C.
3. Test the pH and add more boric acid if it is too high (i.e. if is greater than 4).
4. Immerse a copper cathode and a nickel anode into the solution.
5. Allow a current to flow for about 20 minutes.
6. Switch off the current and disconnect the cathode.

Results and Observation

- The copper sheet is covered (plated) with nickel.
- The nickel is shiny i.e. it has a bright sheen.
- It does adhere well to the copper even during polishing.

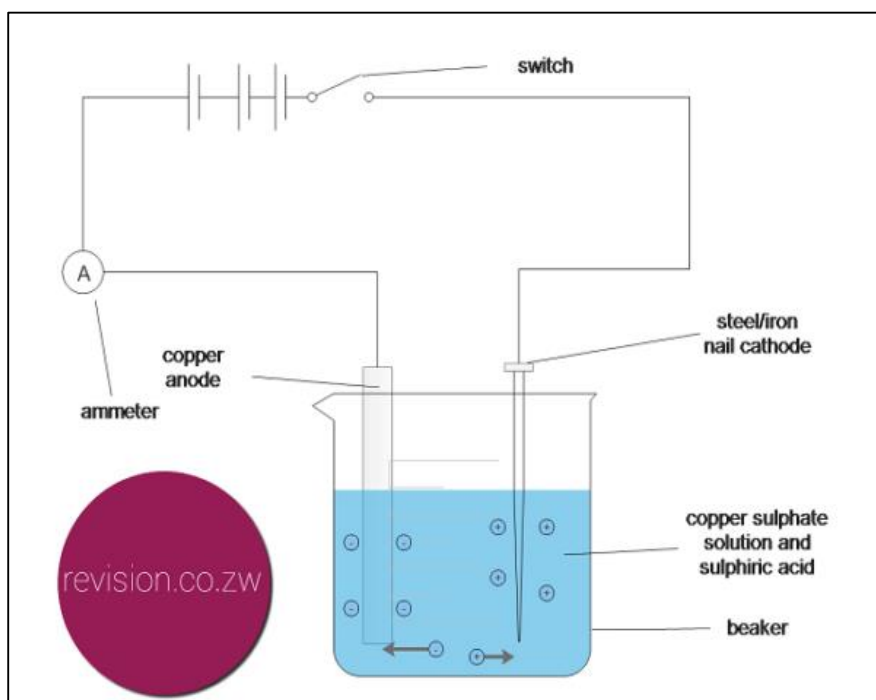
Experiment 3

Copper plating an iron nail.

Aim: Copper plating an iron nail

Materials: beaker, copper sulphate, dilute sulphuric acid, iron nail, copper electrode, ammeter, 1.5V torch cells, sandpaper

Method



1. In a beaker, make up 100 cm³ of solution containing 40g of copper sulphate.
2. Carefully add 100cm³ copper sulphuric acid.
3. Clean the iron nail using sandpaper.
4. Immerse a copper electrode and the iron nail into the solution.
5. Allow a current to flow for about 20 minutes.
6. Switch off the current and disconnect the cathode.

Results and observations

- The iron nail is coated with a thin layer of copper after the process.
- The copper coat is shiny.
- It does not adhere very well and comes off if cleaned using sandpaper.
- The copper electrode will be thinner after the process.

State reasons for electroplating materials

1. Prevent corrosion or rusting
2. Decoration

Painting- ensures that metals are not in contact with water and oxygen (rusting).

Galvanizing-is coating material (iron and steel) with zinc. What are the advantages of using zinc when galvanizing.

Assignment1

- b. Described chrome plating. (6)
- c. What are the reasons for plating metals? [4]

OXIDATION AND REDUCTION

Oxidation – is a chemical reaction in which a substance gains oxygen or loses hydrogen. For example, oxidation occurs during combustion/burning of elements e.g. carbon + oxygen $\longrightarrow$ carbon dioxide
Magnesium + Oxygen $\longrightarrow$ magnesium oxide

$S(s) + O_2(g) \rightarrow SO_2(g)$	Sulphur is oxidised to sulphur dioxide by gaining oxygen.
$H_2S(g) + Cl_2(g) \rightarrow 2HCl(g) + S(s)$	Hydrogen sulphide is oxidised to sulphur by losing hydrogen.

- In the above processes, oxygen is the oxidizing agent.
- Oxidation can also be a reaction in which hydrogen is removed. In this case, the oxidizing agent removes hydrogen. For example water is the oxidizing agent in the reaction below:-
Magnesium + water (steam) $\longrightarrow$ magnesium oxide + hydrogen gas

Reduction – is the removal of oxygen but can also be the addition of hydrogen. A reducing agent adds hydrogen or removes oxygen.

$N_2(g) + 3H_2(g) \rightarrow 2NH_3(g)$	Nitrogen is reduced to ammonia by gaining hydrogen.
$PbO(s) + H_2(g) \rightarrow H_2O(l) + Pb(s)$	Lead oxide is reduced by losing oxygen. (Hydrogen is oxidised to water by gaining oxygen.)

Reduction of Metal Oxides

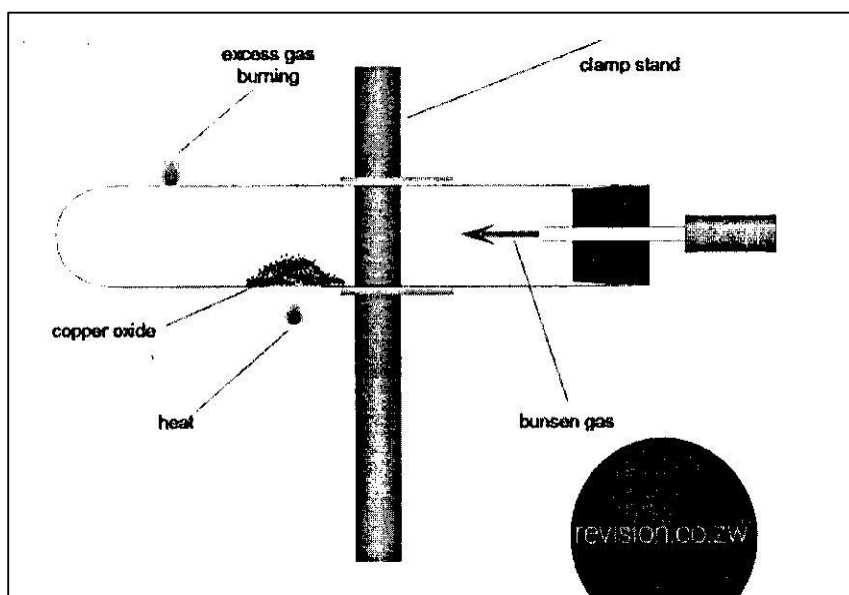
- Metals cannot be obtained by heating the oxides.
- The oxides must be heated with something to remove the oxygen.

Experiment 1

Aim: To show that hydrogen can act as a reducing agent

Materials: hard glass test-tube with a small hole near the end, copper oxide, burner, cork with a short glass tube, rubber tube

Method



- Place a spatula load of black copper oxide in the test tube.
- Set up the apparatus as shown above.
- Insert a rubber cork with a short glass tube in the test tube.
- Connect a rubber tube from a gas cylinder containing Bunsen gas to the glass tube.
- Allow a very slow stream of gas to flow through the apparatus and burn it at the small hole. Adjust the gas so that the flame is no more than 2 cm in height.
- Heat the copper oxide with a burner until there is a change in colour.
- Remove the heat.
- Turn off the Bunsen gas and allow the test tube to cool.

NB Bunsen gas which contains hydrogen is used instead of using hydrogen which is dangerous and difficult to handle.

Results and Observation

- The copper oxide glows.
- While the copper oxide is black the copper oxide inside the tube starts to turn brown/metallic red as it is reduced from copper oxide into copper.
- When the experiment is completed copper element remains in the tube as residue.
- The colour change from black to brown/metallic red/orange shows that the remaining residue is copper as opposed with the copper oxide which is used at the start of the experiment.
- Hydrogen in the Bunsen gas reduces copper oxide into copper.
- The hydrogen combines with the oxygen in the copper oxide to form water vapour (steam).

Conclusion

- Hydrogen acts as a reducing agent.
- It reduces copper oxide into copper and gives steam as a by product

The word equation can be written as:

Copper oxide + hydrogen $\rightarrow$ copper + water.

Qn: Write down the chemical reaction. (2)

Experiment 2

To reduce zinc oxide with carbon

Method

- ✚ Mix carbon and zinc oxide together in a crucible lid.
- ✚ Using tongs, hold the lid over flame and heat strongly.
- ✚ Observe the changes.

Observations

Carbon is able to remove oxygen from zinc oxide. The process is called reduction.

When zinc oxide is heated with coke (carbon), zinc and carbon dioxide are obtained.

Word equation: carbon + zinc oxide $\rightarrow$ zinc + carbon dioxide.

Qn: State the chemical equation. (2)

Conclusion

Metals are obtained by reducing their oxides by carbon as a reducing agent.

Experiment 3

Reducing copper with carbon

Aim: To reduce copper oxide with carbon.

Materials: black copper oxide, tongs, crucible lid, powdered carbon, burner

Method

1. Mix the copper oxide and powdered carbon together in the crucible lid.
2. Using tongs, hold the lid over the hottest part of the flame and heat strongly until you see some change in the mixture

Observation and Results

- The mixture smokes and burns.
- The copper oxide is reduced while carbon dioxide is reduced.

Conclusion

Copper can be reduced by carbon to give carbon dioxide and copper.

Copper oxide + carbon $\rightarrow$ copper + carbon dioxide.

Qn: Write down the chemical equation. (2)

Experiment 4

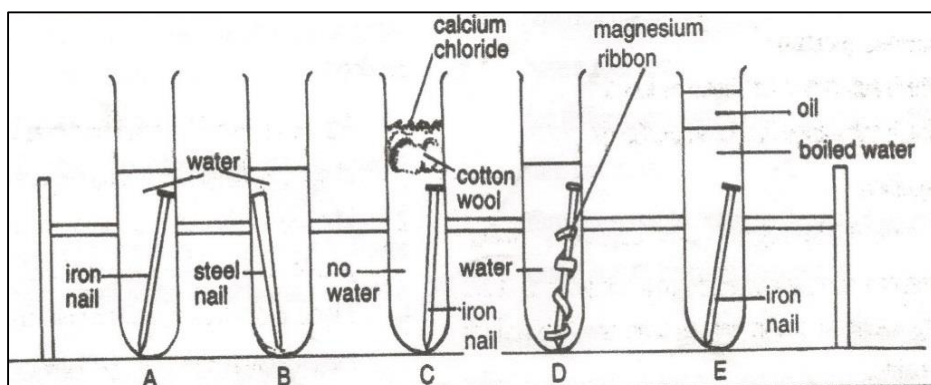
Oxidation of iron (Rusting)

- The oxidation of iron is called rusting. Oxygen is added to the iron and iron objects look unsightly and eventually start to disintegrate.

Experiment

Rusting of Iron

Method: see focus on science bk 3 pg 119



Observations

The iron nail in water rusts the most

The steel nail is alloyed with other elements hence resist corrosion.

The iron nail in dry air does not rust.

The nail with magnesium ribbon is protected from rusting; magnesium corrodes instead of the nail. Boiled water contains no oxygen so nail does not rust.

Conclusion

Oxygen and water must be present for iron to rust.

Reversible reactions

Have a forward reaction and a backward reaction happening at the same time e.g. water boils to steam, solid iodine (grey) turns into a purple vapour when heated and reforms crystals solid when cooled.

Physical changes are reversible reactions or processes.

Metals – Iron

Occurrence

Occurs as haematite (iron iii oxide), found at Redcliff and Buchwa. Main use is to make steel.

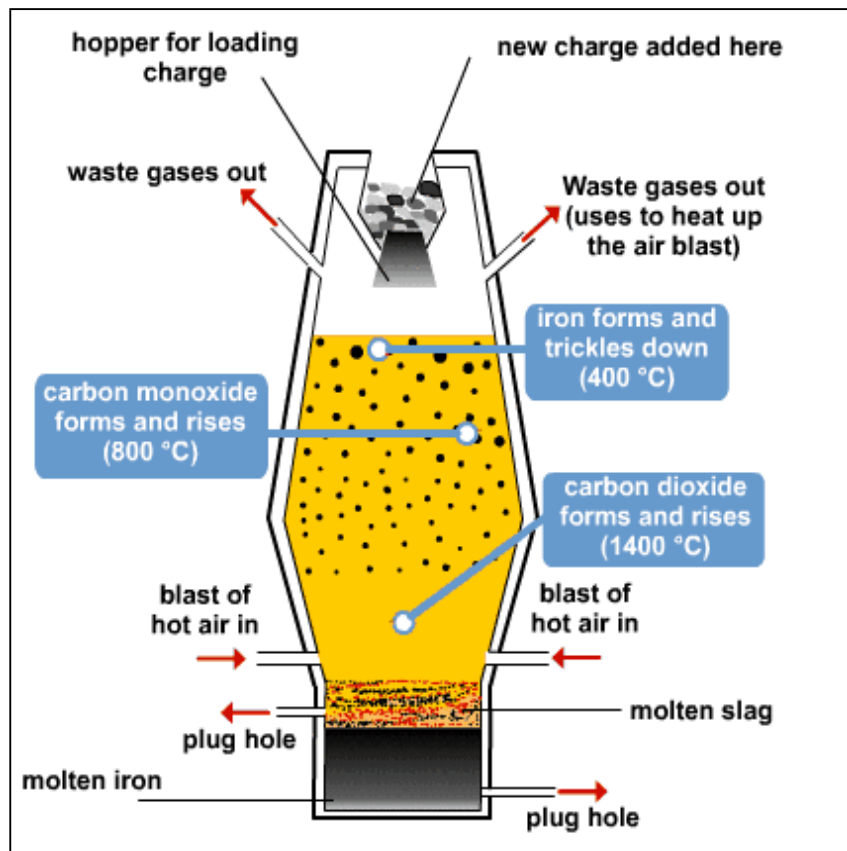
Extraction of Iron

Done in the blast furnace, iron iii oxide is reduced to iron using reducing agents carbon and carbon monoxide.

Raw materials

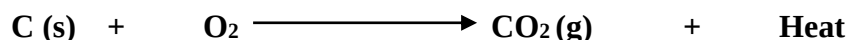
Raw materials	Source	Uses
Limestone/ CaCO_3	Redcliff	Removes impurities in the form of slag.
Haematite/Iron ore/ Fe_2O_3	Redcliff/Buchwa/Ripple creek	The source of iron.
Coke/ Carbon	Hwange	Burns to produce heat for the smelting purpose. It reducing agent,
Hot air/Oxygen	Sable Chemicals	Hot air- contains oxygen for burning coke i.e. Causes coke to burn.

The Blast Furnace

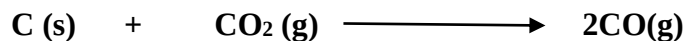


Describe reactions in the blast furnace

1. Near the bottom of the furnace, the carbon in coke combines with oxygen to give carbon dioxide.



2. In the middle of the furnace, more carbon reacts with carbon dioxide to form carbon monoxide.



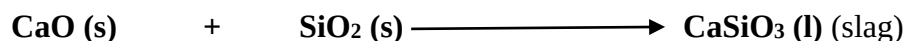
3. In the top, carbon monoxide reacts with iron(III) oxide to produce molten iron, which runs down to the bottom of the furnace



- The molten iron collects at the bottom of the furnace
4. The limestone is decomposed by heat to give calcium oxide



5. Iron ore contains many impurities (silicon, sulphur, phosphorus, etc.) Sand (silica), SiO_2 , reacts with calcium oxide to produce slag (calcium silicate). Slag runs down to the bottom of the furnace, floating on top of molten iron

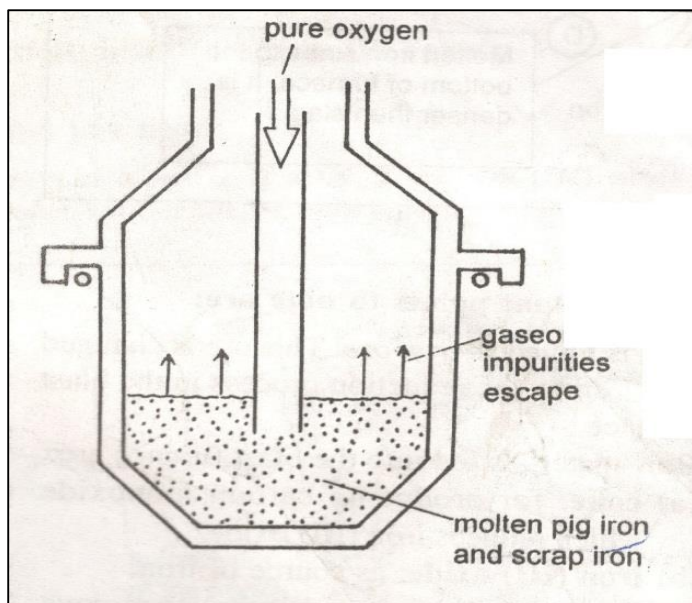


- Molten iron & slag tapped off separately in furnace. Slag is for road construction.

Waste gases

- These are carbon dioxide and nitrogen. They come out at the top of the furnace.
- **Explanation:** The carbon dioxide is from the reduction reaction in stage 3. The nitrogen is from the air blast. It has not taken part in the reactions so has not been changed.

Oxygen lance process



Is a batch process i.e. the products are manufactured at a given time before adding more raw materials (like pig iron) in the process.

- Oxygen is blown into molten iron (pig iron) through a tube called a lance.
- Temperature increases, all impurities are oxidised.
- Impurities like carbon, sulphur and phosphorus burn into gaseous oxides which quickly escape.
- A pure iron called pig wrought iron is obtained (99% iron).
- Wrought iron is then mixed with other metals to form ALLOYS.

Alloy is a mixture of one metal with one or more other metals and or carbon.

Alloys of iron are pig iron (cast iron), mild steel and stainless steel.

Question 1

State 2 properties and 2 uses of each of the above alloys of iron. (12)

ORGANIC CHEMISTRY

Organic chemistry is a branch of chemistry connected with compounds of hydrogen and carbon (hydrocarbons).

Hydrocarbons

- Are compounds of hydrogen and carbon only e.g. Alkanes, alkenes, alkyne and alcohols.

Name the members of the homologous series with 3 carbon atoms

Homologous Series:

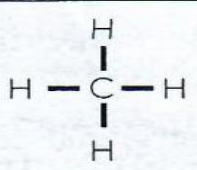
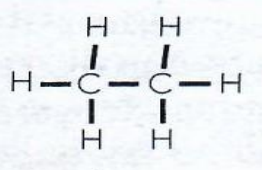
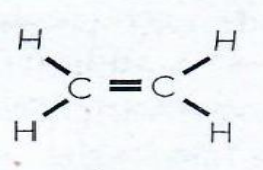
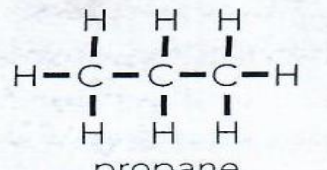
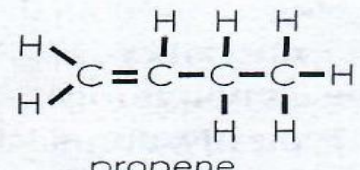
- These are families of compounds that differ only in the length of their carbon chain.

Alkanes

- Are hydrocarbons that have only a single C – C or C – H bond.
- They have a general formula C_nH_{2n+2} e.g. Methane (CH_4), Ethane (C_2H_6) and Propane (C_3H_8).
- Alkanes have only single carbon – carbon bonds, so are **saturated**.)

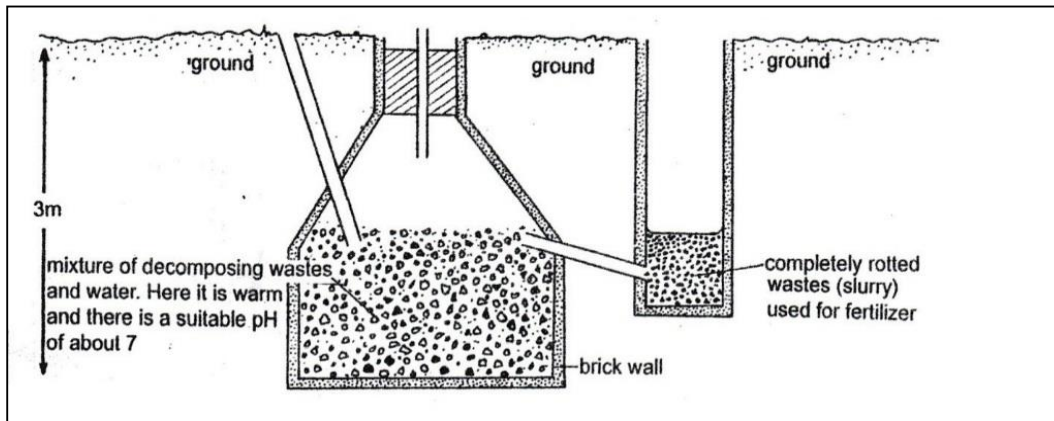
Alkenes

- Are hydrocarbons with a double C = C bond.
 - They have a general formula C_nH_{2n} e.g. ethene (C_2H_4) and Propene (C_3H_6).
 - They contain C=C double bonds, they are called **unsaturated**.
- Draw the displayed structures of methane, ethane, propane, ethene and propene

Alkanes	Alkenes
 methane	none
 ethane	 ethene
 propane	 propene

Biogas Production

The Biogas Digester



Cow dung is collected while fresh and mixed with water to keep it in a semi – solid state which enables chemical reactions to effectively take place and to ensure the pH of about 7 is kept. The bacteria that digest cow dung are anaerobic bacteria. The decomposition of organ matter without oxygen is fermentation. Biogas is a mixture of methane, some carbon dioxide, hydrogen and hydrogen sulphide. The products of fermentation are biogas and sludge used as farm manure. Manure, human waste, plant matter can be fed into the digester.

Factors Affecting Biogas Production

Type of waste, the temperature (optimum are between 35 and 55 degrees Celsius), pH not too acidic or alkaline, metals such as zinc will stop fermentation of waste by bacteria and time that the waste is in the digester.

QN What are the uses of biogas? [4]

1. Cooking
2. Lighting
3. Operating a refrigerator

END

FORM 4

BIOLOGY

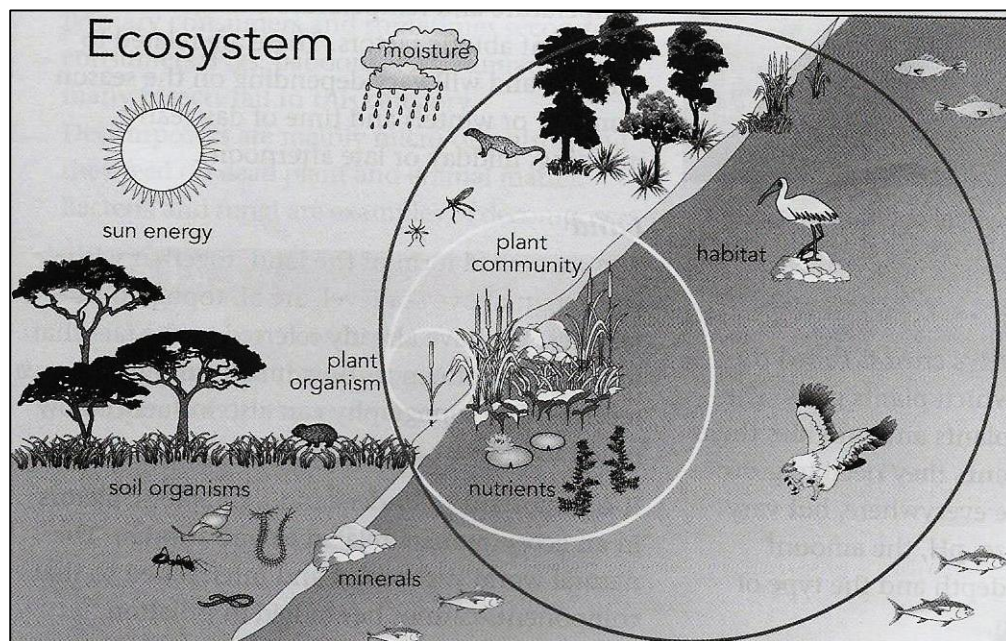
Cells and Levels of Organisation

ECOLOGY

Is the study of the interaction of living things with their environment.

ECOSYSTEMS

An ecosystem is a community of living things together with their non living environment with which they interact. e. g a pond, garden, soil e.t.c. the earth being a large ecosystem with plant and animal species and their physical surroundings put together.



Components of an Ecosystem

Abiotic and Biotic

Abiotic / Physical components

- a) Soil-soils support plants .Soils provide plants with nutrients and water. However, not all ecosystems contain soil. Lakes and ponds contain water.
- b) Air-oxygen and carbon dioxide are important for respiration and photosynthesis (define these terms).
- c) Light-is the ultimate source of energy for all living organisms and vital for photosynthesis to take place.
- d) Temperature- it varies from season to season and is related to sunlight and its changes is very important abiotic factor.
- e) Water- the quantity and quality of water varies from one ecosystem to another. Water affects the type of plants and animals found in an ecosystem

f) Humidity- areas of low and high humidity has an influence on ecosystems. For example deserts and coastal areas with low and high humidity respectively.

g) Land- the topography can influence rainfall type, amount and sunlight and area receives.

Biological / living / biotic components

- Producers and consumers are two kinds of living things in any ecosystem.
- The plants that make their own food and the animals that feed on them. The plants are the producers which supply energy to the consumers in the system.
- Consumers are grouped into primary (leaf eater or herbivores) , secondary (carnivores & omnivores) and tertiary consumers (decomposers).
- When plants and animals die ,their bodies still contain energy and complex chemicals. These chemicals are broken down and decomposed by micro-organisms, mostly bacteria and fungi. These organisms are called decomposers.

NB: 1. The distribution of living components in an ecosystem is influenced by physical components like shelter and food availability.

2. Energy from the sun flows into an ecosystem and is lost in various ways and stages.

Role of biological components in the soil

-Micro-organisms - Include bacteria and fungi. They improve soil fertility by decomposing plant and animal matter

-Earthworms-they improve drainage and aeration by making tunnels in soil hence water penetrates easily. They increase organic matter by pulling plants and grass into the soil. Recycling of nutrients and mixing topsoil and subsoil thereby increasing soil depth.

-Termites – help to stick soil particles together as they make fertile mounds. They secrete an alkaline substance which reduces soil acidity.

-Nematodes – they digest plant sticks, stems and grass helping in humus formation.

Feeding Relationships in a Natural Ecosystem

Plants are producers of food and animals are consumers. Organisms in an ecosystem depend on each other for living. The relationships amongst organisms are shown by food chains and food webs.

Types of Ecosystems

Two types are the natural and artificial ecosystem.

Natural Ecosystems

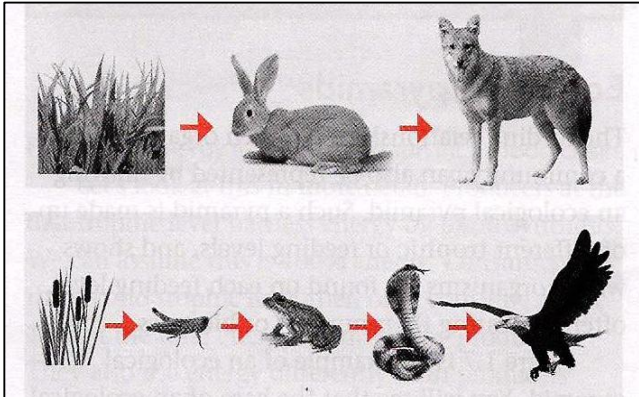
- There are no human interferences.
- Pests and diseases are part of the ecosystem and no control measures taken.
- Animals graze freely and no fertilizer application.
- There is natural balance of nature.

Feeding relationships

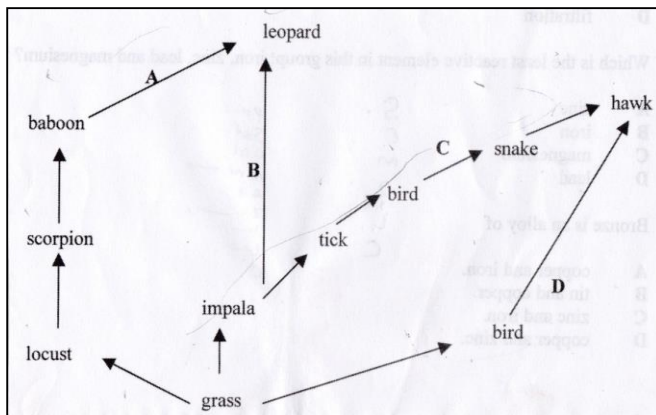
Food Chains

A food chain is a feeding relationship in which energy in producers is transferred to consumers at different trophic levels. Food chains start from the producer to consumers. e.g.

Grass → locusts → lizards → birds.



- Grass is the producer. It is eaten by locusts as primary consumers. Locusts are eaten by lizards (secondary) and lizards are eaten by birds.



- Short food chains are advantageous than long food chains as they influence the amount of energy that gets to the last link in the chain.
- If one organism is removed, the chain is affected to an extent a new source of food is needed or face extinction.

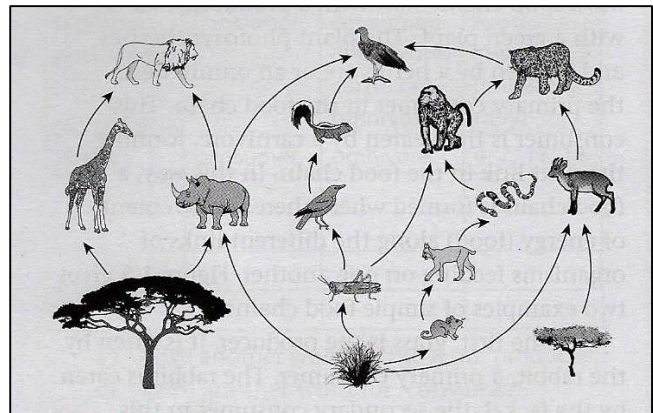
Each stage of the food chain is called a **trophic level**.

Trophic level is the position of an organism in the food chain. The four trophic levels are:

- Producers
- Primary consumers
- Secondary consumers and
- Tertiary consumers

Food Webs

Food webs are interconnections of food chains or show a complex feeding relationship between organisms. An organism may feed on more than one organism and in turn it may be eaten by others. However, plants are the primary food source and the sun is the only energy input.



Exercise

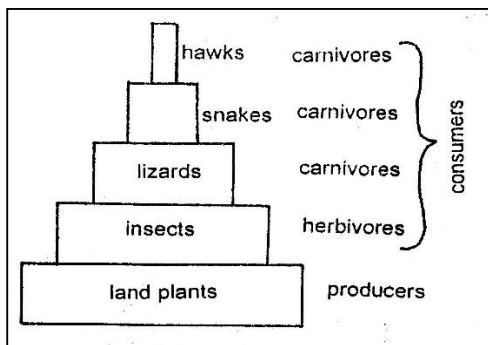
QN3. Draw a food web and extract two food chains from the food web. (8)

Ecological pyramids

The feeding relationships between organisms in a community is represented in pyramid of numbers and biomass.

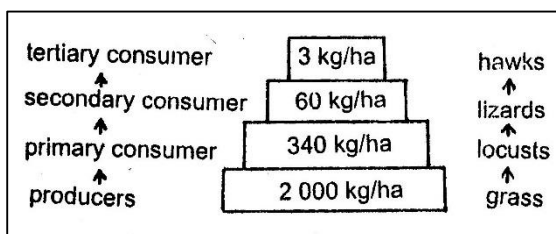
Pyramid of Numbers

A pyramid of numbers illustrates the number of organisms at each trophic level in a food chain. Here, organisms interact with each other through energy transfers and are linked by their feeding habits. On the higher levels of the pyramid of numbers can be a tertiary consumer or a decomposer for example hawk. The number of organisms decreases as you go up the pyramid. Again, energy is lost at each level. However, some pyramid of numbers may be inverted if one producer supports many consumers.



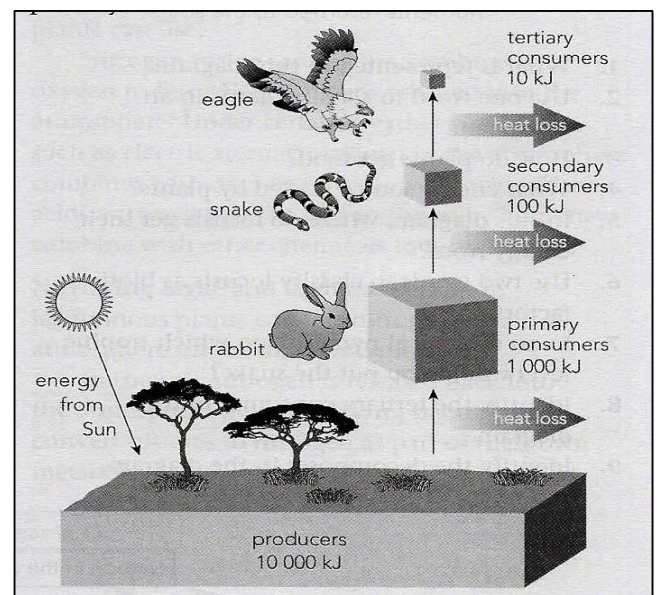
Pyramid of Biomass

It shows the mass of organic material at each trophic level. It decreases because of energy which is lost at each level. Producers have the largest total biomass than those at high levels. Thus, high trophic levels have very small numbers of large organisms with low biomass per unit area of the ecosystem.



Energy Flow

- The sun is the source of energy.
- It is converted to chemical energy during photosynthesis.
- As energy moves from one level to another, some is lost.
- From the 100 % energy we will have, about 10% of the energy can be consumed.
- Energy lost through the process of respiration (as heat).
- Energy used up for movement (to search for food, find a mate, escape from predators...).
- Warm-blood animals (birds and mammals) maintain a standard blood temperature – they lose heat to the environment.
- Warm-blood animals lose heat energy in faeces and urine.
- Some of the material in the organism being eaten is not used by the consumer, for example a locust does not eat the roots of maize, and some of the parts eaten are not digestible.
- So, we can say that consumers can pass about 10% of the total energy they receive to the next trophic level.

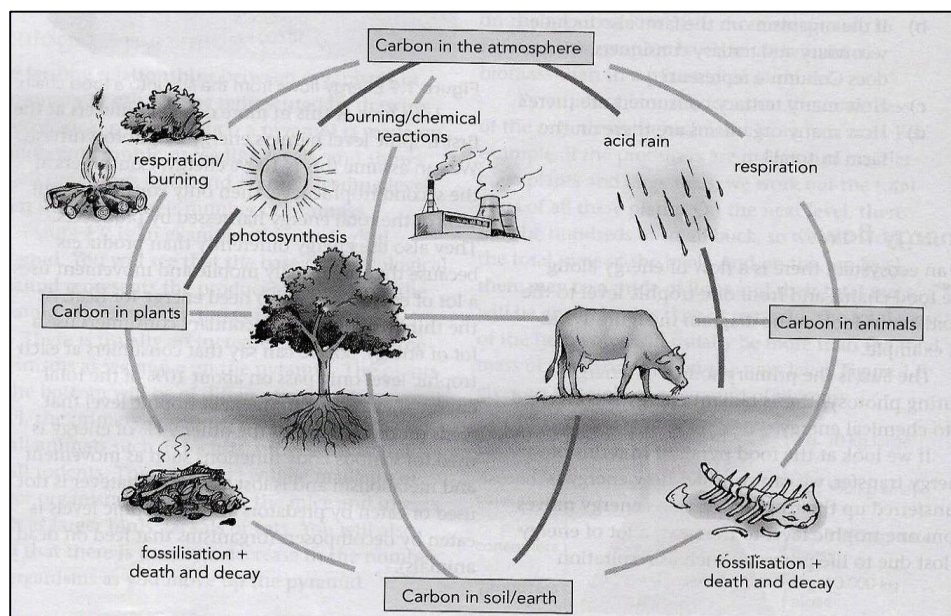


Question 4. What are the main causes of energy loss at each trophic level in pyramids of biomass?(3)

Nutrient Cycles in Ecosystems

Living organisms need substances such as oxygen, water, nitrogen, carbon and phosphorus. These chemicals are recycled through ecosystems and will be available time and again.

Carbon Cycle

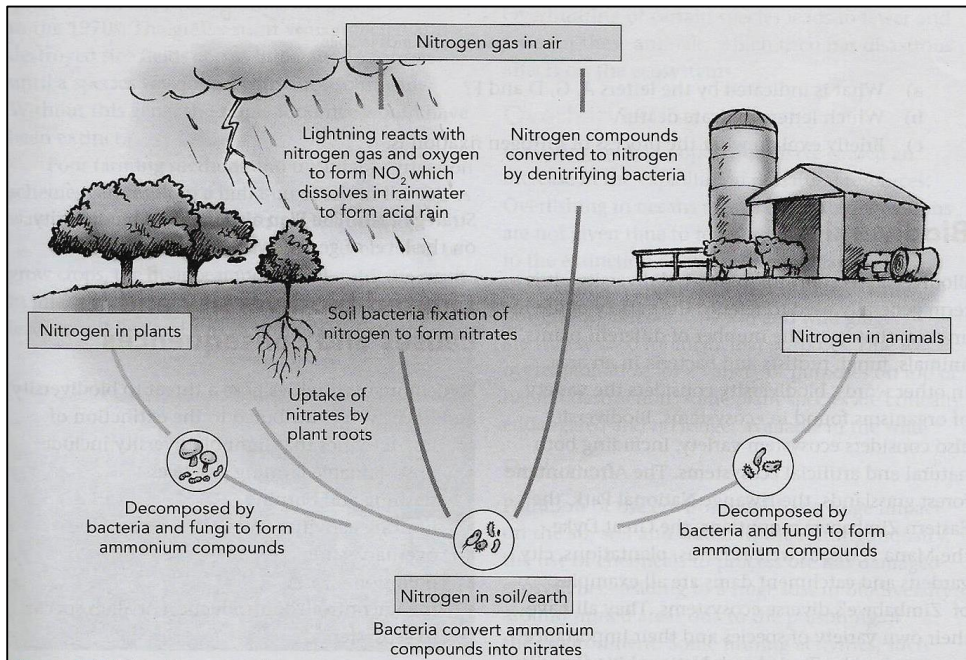


- Carbon in ecosystems comes from carbon dioxide in the atmosphere.
- Plants remove carbon from the air during photosynthesis and build it into food compounds.
- Animals obtain carbon by eating plants.
- Decomposers get carbon from remains of dead plants and animals.
- During respiration, carbon is released in the atmosphere by micro-organisms, plants and animals.
- Is also accelerated by human activities such as burning fuels and deforestation.
- Calcium carbonates in shells react with acid rain to release carbondioxide back into the atmosphere.

Green House Effect

Carbon dioxide is a gas which acts as an insulator to keep the earth warm. It traps the sun's energy in the atmosphere. As carbon dioxide increases, so does the temperature of the earth (Global warming).

Nitrogen Cycle



- Nitrogen is present in all proteins of animals and plants.
- When animals and plants die, proteins are decomposed by bacteria and fungi.
- Plants reabsorb nitrates and are transferred to animals through ingestion.
- Soil organisms such as bacteria, millipedes, worms and fungi act on plant and animal matter to change material to ammonium compounds.
- Ammonium compounds are further decomposed by nitrifying bacteria (Nitrosomonas and Nitrobacter).
- Nitrosomonas converts ammonium compounds into nitrites and nitrobacter converts nitrites to nitrates.
- Nitrates are in form of nitrogen which is mainly absorbed by plants.
- About 79% of the atmosphere is nitrogen and a small amount is changed by lightning into ammonia then to ammonium compounds which can be absorbed by plants.
- A small fraction of nitrogen in the air is converted into nitrates by nitrogen -fixing bacteria (Rhizobium –live in the nodules of legumes).
- If soil aeration is poor, bacteria uses oxygen contained in the nitrates and in the process, nitrogen gas is released into the air (denitrification).

Artificial ecosystems

- It is a natural ecosystem that has been interfered with and altered by people. It is one reflecting human interference and change e.g. fish pond, garden, orchard etc.
- Artificial ecosystems are usually characterized by low species diversity e.g. mono-culture.
- The natural balance that allows an ecosystem to be self-sustaining is disrupted in an artificial one. The system may be forced to produce more than it would naturally.
- Crops and animals produced are removed. To maintain self-sustaining artificial ecosystem energy, water and fertility must be supplied.

- Seeds which are supposed to be dispersed naturally are introduced from elsewhere, planted in prepared soil close together.
- Natural predators and control mechanisms are removed in an artificial system and where monocultivation is practiced; pest numbers build up alarmingly leading to the introduction of outside control.

Biodiversity

Refers to a variety of life in an area (plants and animal species). It includes a variety of plants and animals in artificial and natural ecosystems for examples in grasslands, national parks, farms and plantations.

Causes of limited species diversity

- Unsustainable farming practices. List examples of these practices.
- Slash and burning.
- Hunting and poaching.
- Overharvesting.
- Pollution.
- Invasion by alien species (Kariba weed, water hyacinth and cherry pie).

Consequences of limited species diversity in artificial ecosystem

- Soil erosion.
- Soil infertility.
- Need for the use of artificial fertilisers.
- Plants and animals pests and diseases built up.
- Extinction of some species.
- Loss of ecosystems and land degradation.
- Much human and fuel energy and water is spent in maintaining an artificial productivity level which is intended to produce more biomass than a natural ecosystem would.

Home Work

QN 2.What are the advantages and disadvantages of artificial and natural ecosystems? (8).

Management of Ecosystems

Humans play a major role in managing ecosystems if they are aware of the importance of conserving ecosystems.

Ways

- Ground cover- people can help an ecosystem by maintaining ground cover.
- Making use of marginal lands. This can be achieved by improving methods of farming, manuring, mulching and afforestation.

Carrying capacity- the maximum number of organisms the area can support without deterioration.

Human Activities on Ecosystems

- Poor farming methods.
- Burning fuels (pollution).
- Deforestation.

Question

1. a) What is limited species diversity? (2)
b) What are the problems caused by limited species diversity? (4)
c) State the advantages of biodiversity. (4)

NUTRITION

Diet

A good diet on its own will not make you skillful or fit as a performer or a sportsperson, but will help you make the most of your abilities through nutrition; hence, nutrition is very important. Your body has the ability to maintain a constant body temperature and a lot of energy is needed to do so. Thus it is important that you should have a good balanced diet to get all the seven components beneficial for nutrition.

Balanced Diet

- Is a diet that contains all the **main nutrients** in the correct **amounts** and **proportions** to maintain good health. Everyone, whether involved in sport or not should try to eat a healthy balanced diet.

Components of a Balanced Diet and Functions

Carbohydrates

They are needed for energy to do all types of work and are required in large quantities. Sugars and starch are important carbohydrates in a diet. Starch includes maize, rice and other cereals. Sugars are in fruits such as mangoes and apples and biscuit and sweet drinks.

Proteins

These are for building of tissue (growth) and repair of worn out tissues or cells. Are also used to speed up chemical reactions in the body. Proteins are obtained from lean meat, fish, milk, eggs, round nuts, peanut butter, cow peas and e.tc. When proteins are digested, they are broken down into amino acids. These acids are absorbed into blood stream and build up the cytoplasm of cells and tissues. The unused amino acids are changed to glycogen and then stored or oxidized to provide energy.

Fats

Fats help the body from losing heat therefore, one feels warmer. Fats also provide energy, but fats only contain carbon, hydrogen and oxygen in different proportions as to the carbohydrates. Sources of fats are peanut butter, cooking oil, fat meat and margarine.

Mineral salts

They are sometimes referred to as fast salts or minerals. Protein, carbohydrates and fats provide the body with carbon, hydrogen, sulphur, oxygen and phosphorus. However, there are more elements found in food we eat e.g. calcium, sodium, iron and others. Calcium is for building bones and teeth and other salts maintain the body in shape and function and iron makes up blood (forms a part of haemoglobin molecule found in red blood cell). Iodine produces a hormone called thyroxin that is needed for growth. Food such as fish, eggs, dried vegetables, liver meat, ishwa contain mineral salts and dairy products.

Vitamins

They are for health of the body and protection against diseases. Vitamins do not form body structures or provide energy but are needed in the body for chemical reactions in cells. Once a diet lacks vitamins people may suffer from vitamin- deficiency diseases. Vitamin A is needed for good eye sight, health skin, growing bones and immune system. Vitamin C is an antioxidant that protects the body from harmful

radicals and help in absorption of iron. Vitamin D or califerol helps absorption of calcium They are from plants and animals for example from cabbages, butter, cheese, tomatoes okra, milk, fruits, eggs and carrots.

Water

It is an important part of cytoplasm. Substances are carried around the body in watery solution in blood e.g nutrients, oxygen and wastes. It is also important in the digestion of food by softening it and moving it. Soft tissues and joints rely on the cushioning that water provides for them. In addition, water is obtained from fruits such as oranges and cucumbers and melons.

Roughage

It is the indigestible parts of plant foods food in cereals such as oats and bran. It absorbs water, providing bulk to faeces. It binds with cholestrol to reduce cholestrol levels in blood. Helps in digestion i.e. help movement of waste matter in the colon and the rectum (making them softer) so that constipation is prevented and also absorbs poisonous substances and are ejected as faeces. Define the term constipation. (2)

Malnutrition

is the result of not eating a balanced diet. There may be:

- Wrong amount of food: too little or too much.
- Incorrect proportion of main nutrients.
- Lacking in one or more key nutrients.

Effects of malnutrition

- a) Obesity - Too much food (carbohydrate, fat or protein).
- b) Coronary heart disease
 - Too much saturated/animal fat in the diet results in high cholesterol levels.
 - Cholesterol can stick to the walls of arteries, gradually blocking them.
 - If coronary arteries become blocked, the results can be angina and coronary heart disease.
- c) Starvation
 - Too little food can result in starvation.
 - Extreme slimming diets, such as those that avoid carbohydrate foods, can result in the disease anorexia nervosa.
- d) Childhood protein-energy malnutrition (Kwashiorkor)
 - Wrong proportion of nutrients e.g. too much carbohydrates (starchy foods) and a lack of protein can lead to kwashiorkor in young children.

Deficiency Diseases – define.

Kwashiorkor

Is due to lack of protein. The symptoms include underweight, enlarged abdomen and the children is inactive.

Rickets

Are skeletal disorder as a result of lack of calcium and vitamin D which is made in animal cells with the help of sunlight It results in weak bones hence the bones bend under the body weight, resulting in rounded legs and deformities of the pelvis, skull and ribcage. What are the corrective measures to this disease? (3)

Bleeding gums

This occurs when one lacks vitamin A. The symptoms are weak gums and bleeding gums especially when one is brushing the teeth. Eating of fruit such as oranges help to prevent bleeding of gums.

Anemia – lack of iron in the body results in lack of haemoglobin formation hence low oxygen is transported to cells hence less energy is made in cells, leading to tiredness and weakness. How can it be controlled? (4)

Goiter

Is due to lack of iodine since iodine is needed by the thyroid gland in the neck. The swelling of the thyroid gland in the neck is goiter. In children, mental and physical development is retarded. People must include iodine in their diets and use iodine salt in food.

Night blindness

It is caused by lack of vitamin A rich foods. A person cannot see proper during the night. One should eat liver, green vegetables and carrots.

Scurvy

Vitamin C deficiency causes scurvy. A person with scurvy bruises easily, has joint and muscle pains. It is treated by taking vitamin tablets and eating a lot of fruits.

Food Tests

Foods can be tested so that we find food groups they contain. This is crucial in the sense that people are able to plan their diets and plan diets for other people.

Experiments

NUTRIENT	REAGENT(S)	PROCEDURE	RESULTS WHEN NUTRIENT PRESENT	WHEN NUTRIENT IS ABSENT
Starch	-Iodine Solution	-Add iodine to food sample -Observe colour changes	-Brown to blue black	-Brown
Simple/reducing sugar(glucose)	a)Benedict's Solution	-add benedict to the food sample. - heat over a water bath. - shake the test tube and examine colour.	a) Blue to orange/brick red	a) Blue
	b)Clinistix	- State the mtd.	b) Pink to purple	b) pink

Protein	a) <u>Biuret Test</u> copper sulphate solution and sodium hydroxide solution (equal quantities) b) Albustix	- prepare food sample. -add sodium hydroxide into the test tube. -add copper sulphate not so rapidly. -shake at each drop. - State the mtd.	a) Blue to purple/violet b) Green to Dark green	a) Blue b) Green
Fats	a) Ethanol/Emulsion Test b) Transparent Test	- add ethanol to the food sample and shake well. -add water and shake. - State the mtd	a) milky white emulsion b) Transparent Mark	clear

RESPIRATORY SYSTEMS

Respiration

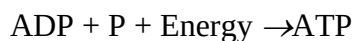
Is the breaking down of glucose to release energy.

Aerobic Respiration

It is the complete breakdown of food (glucose) to release a relatively large amount of energy. The food is oxidized or burnt using oxygen to release energy.

Energy carrier molecules

1. ATP (Adenosine triphosphate)- is formed when a phosphate group is joined to ADP (Adenosine diphosphate) with two phosphate groups.



In ATP molecule, energy is stored as chemical energy. One glucose molecule yields 38 ATP molecules. ATP MOLECULE will release energy when required and releases a phosphate group to form ADP.

$\text{ATP} \rightarrow \text{ADP} + \text{P} + \text{energy}$.

Oxygen is needed for a glucose molecule to be broken down.

Word equation

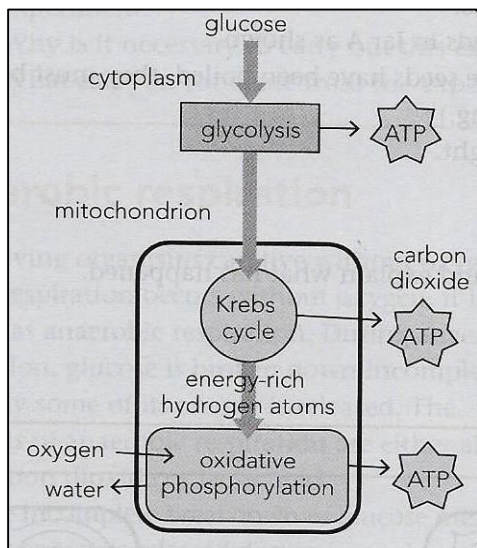
enzymes

Glucose + oxygen $\rightarrow$ carbon dioxide + water + energy.

Energy is released when glucose is burnt. Glucose and oxygen are supplied to body cells in blood. Water and carbon dioxide are released as wastes.

Aerobic respiration process

It occurs in three stages: glycolysis, the krebs cycle, Oxidative phosphorylation.



1. Glycolysis- splitting up of glucose without oxygen and produces ATP molecules.

2. The krebs cycle- it starts in the mitochondria. Pyruvic acid is broken into energy rich hydrogen atoms and carbon dioxide which will be released into the air.

3. Oxidative phosphorylation- occurs in the mitochondria. In the first phases, rich hydrogen atoms are given off but in this stage, they are processed into 36 ATP molecules. Once energy is stored, the low-energy or tired out hydrogen atoms combine with oxygen to form water molecules.

Question 1

What are the differences between photosynthesis and respiration? (5)

Anaerobic Respiration

Respiration takes place without oxygen. The food is not completely broken down and small amount of energy is released.

Yeast, a fungus, respire anaerobically. Human muscle is able to respire anaerobically when the demand for oxygen exceeds the supply.

Alcoholic fermentation

enzymes

Glucose $\rightarrow$ carbon dioxide + alcohol + energy.

Lactic acid fermentation

❖ Occurs in animals. During strenuous exercise, circulatory system cannot supply enough oxygen to muscles hence get extra energy by respiring anaerobically. The glucose is broken down into lactic

acid. After the exercise, lactic acid is broken into carbon dioxide and water. In order to do this, a lot of oxygen is needed and thus why people continue to be quite hard after exercise for some time. The advantage of being able to do this is that the muscles can still function and obtain some energy even though there isn't much oxygen available.
Lactic acid causes stiffness in muscle (muscle fatigue).

enzymes

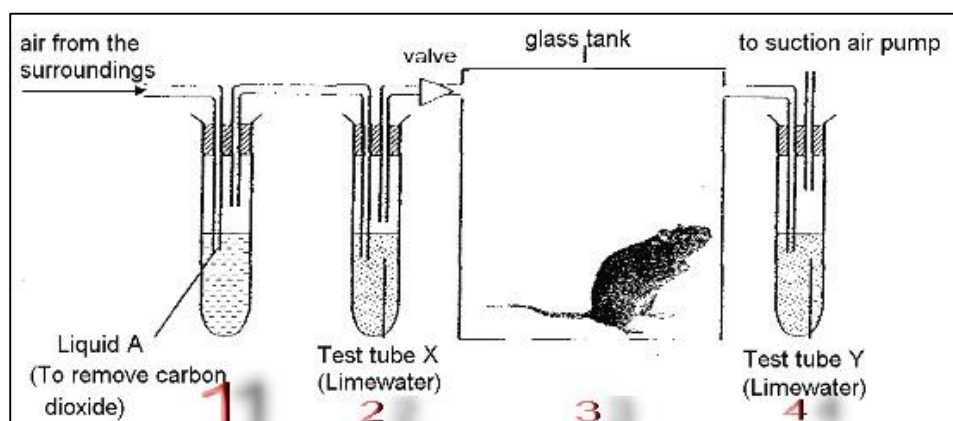
Glucose → lactic acid + energy.

Experiment 1

Which gas is released by respiring organisms/animals?

Apparatus: potassium hydroxide, lime water or bi-carbonate indicator, gas jar, a frog/ rat, test tubes e.t.c

Method



- ✚ Arrange the apparatus as shown:
- ✚ Put an animal in a jar or flask No 3.
- ✚ Put equal amounts of indicator in test tubes 1 (A), 2 (X) and 4 (Y).

Observations

At 1 carbon dioxide is removed from the air coming in.

At 2, the indicator is a check to make sure there is no carbon dioxide.

The respiring organism is at 3.

At 4, there is the carbon dioxide which turns milky.

Results

Lime water in 4 or D turns milky.

Conclusion

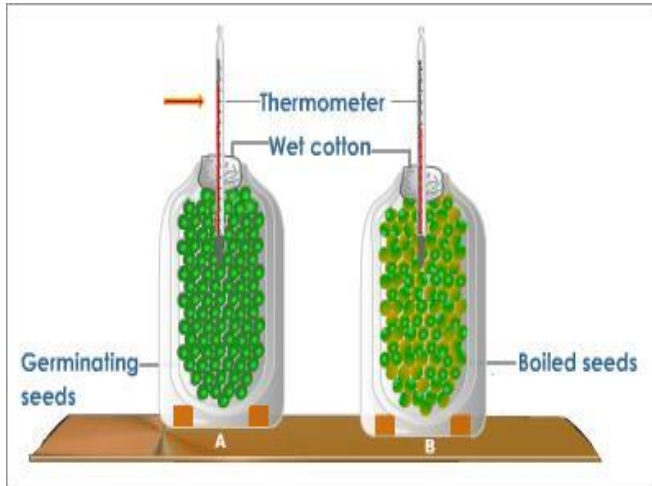
Respiring organisms release carbon dioxide.

Experiment 2

To prove that germinating seeds produce heat energy

Materials: 2 vacuum flasks, thermometers, cotton wool, 2 stands, soaked seeds, disinfectant, boiled seeds,

Method



- ✚ Soak the seeds to soften them.
- ✚ Boil half of the soaked seeds to kill them. Allow them to cool, wash them in disinfectant.
- ✚ Place them in an insulated container and set thermometer in unboiled seeds and another in boiled seeds.

- ✚ Record the temperature of each container and leave for few days.
- ✚ Record the temperature of each day.

Changes in temperature

Day	Flask A	Flask B
1	25	25
2	26	25
3	27	25

Results

- Temperature in flask A rises by 2 while temperature in flask B remain unchanged.
- Identify the error in the experiment.

Conclusion

Germinating seeds produce heat energy.

TRANSPORT SYSTEMS

Transpiration

Adaptations of a leaf for transpiration

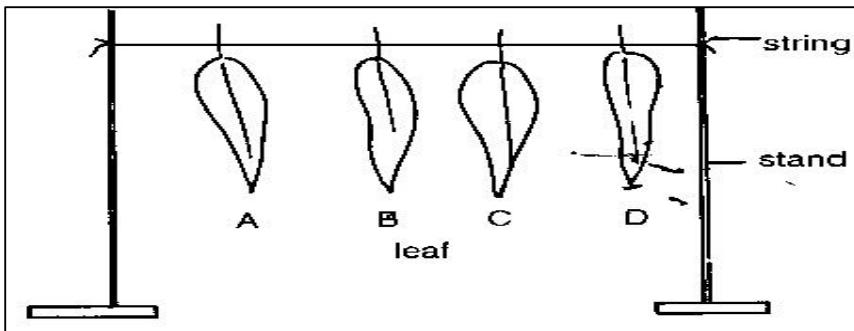
- ❖ Leaf size and surface- some trees have leaves with small surface area so they can transpire less.
- ❖ Have waxy leaf cuticle which reduces excessive water loss. Water cannot escape waxy layer. e.g. pumpkin leaves.
- ❖ Presence of hairs on the leaf – hairs trap water vapour and air and thus lessen the loss of more water.
- ❖ Stomata- opens during the day and closes at night. during drought, stoma will close to save water
- ❖ They develop few stomata on the upper surface. This ensures that direct rays from the sun are prevented from striking most of the stomata as they are on the lower surface / underside thus there will be less water loss.
- ❖ Leaf arrangement- some leaves will be butterfly shaped, some will move closer and hang vertically for example mopane leaves.

Experiment 1

To determine which surface of a leaf loses more water vapour

Method

1. Take 4 leaves of similar size.
2. Carefully smear both surfaces of leaf A with petroleum jelly / vaseline.
3. Cover the ventral surface (underside) of leaf B with petroleum jelly.
4. Cover the dorsal surface or (upper side) of leaf C with petroleum jelly.
5. Leave the one remaining leaf (D) / both surfaces uncovered .
6. Weigh each of the leaves on the balance.
7. Place a little vaseline on the cut end of the leaf stalk and suspend a piece of string between two retort stands and attach each of the 4 leaves to the string. Leave until next lesson.
8. Observe the appearance of the leaves and reweigh them.
9. Calculate the loss in mass of each of the leaves.



Observations

- Leaf A does not lose water at all.
- Leaf B loses water.
- Leaf C loses little water.
- Leaf D dries out most or fast.

Results

- More water is lost at D because more stomata are on the ventral surface of the leaf.

Conclusion

- The ventral surface loses more water than the dorsal surface of the leaf.

Interpretation

- The vaseline prevents evaporation.
- The untreated leaf and the leaf with its dorsal surface sealed show the greatest degree of shrivelling, so it is from the ventral surface that leaves lose most water by evaporation.
- More rapid results can be obtained by sticking small squares of blue cobalt chloride paper to the dorsal and ventral surface of the same leaf using transparent adhesive tape.
- Cobalt chloride paper changes from blue to pink as it takes up moisture. By comparing the time taken for each square to go pink, the relative rates of evaporation from each surface can be compared.

Blood

Blood is a circulating tissue which consists of blood cells and blood plasma. Blood cells include red blood cells, white blood cells and platelets.

Question 1

- a) What are the function of the following blood cells?
 - i. Red blood cells.
 - ii. White blood cells.
 - iii. Platelets.
- b) What is blood plasma? (6)

Blood Plasma

Plasma is a yellowish fluid consisting of water, nutrients, wastes, gases and plasma proteins. It is the medium in which blood cells are moved to all parts of the body. Plasma transports digested food substances, absorbed from the ileum to all parts of the body. It moves excretory products from tissues to the excretory organs e.g. urea to the kidneys and carbon dioxide to the lungs. It also distributes heat of the body.

Red Blood Cells

Red blood cells are small, round biconcave discs without a nucleus. They transport oxygen. Red blood cells have haemoglobin which combines with oxygen to form oxyhaemoglobin which is taken to all tissue where it releases the oxygen.

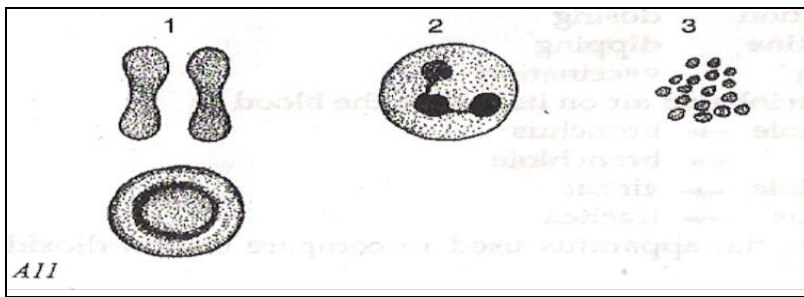
White Blood Cells

These are larger than the red blood cells, have no definite shape and have a nucleus. White blood cell protect the body from infection and disease. They attack invading foreign particles and germs such as bacteria and viruses in two ways. 1. Some are phagocytic – they engulf and destroy invading bacteria and viruses, thus preventing them from spreading to the rest of the body. 2. By secretion of antibodies by some white blood cells. Antibodies attack and neutralize disease agents.

Blood Platelets

Are the smallest numerous bodies found in blood. They have no nuclei and are oval shaped. They are responsible for clotting of blood. They secrete enzymes which assist in conversion of special plasma proteins which form thread like structures on the wound. The thread form clot which prevent loss of blood and further entry of germs.

Structures of Red blood cells, white blood cells and blood platelets



Functions of blood

- ✓ Is a medium that carries nutrients, oxygen and plasma proteins to where they are needed.
- ✓ Carries waste substances produced by the body to organs of secretion such as lungs and kidneys.
- ✓ Has a homeostatic function i.e. maintaining of temperature and water, salt and glucose levels at constant optimum conditions.
- ✓ Distributes heat and protects the body against infection.

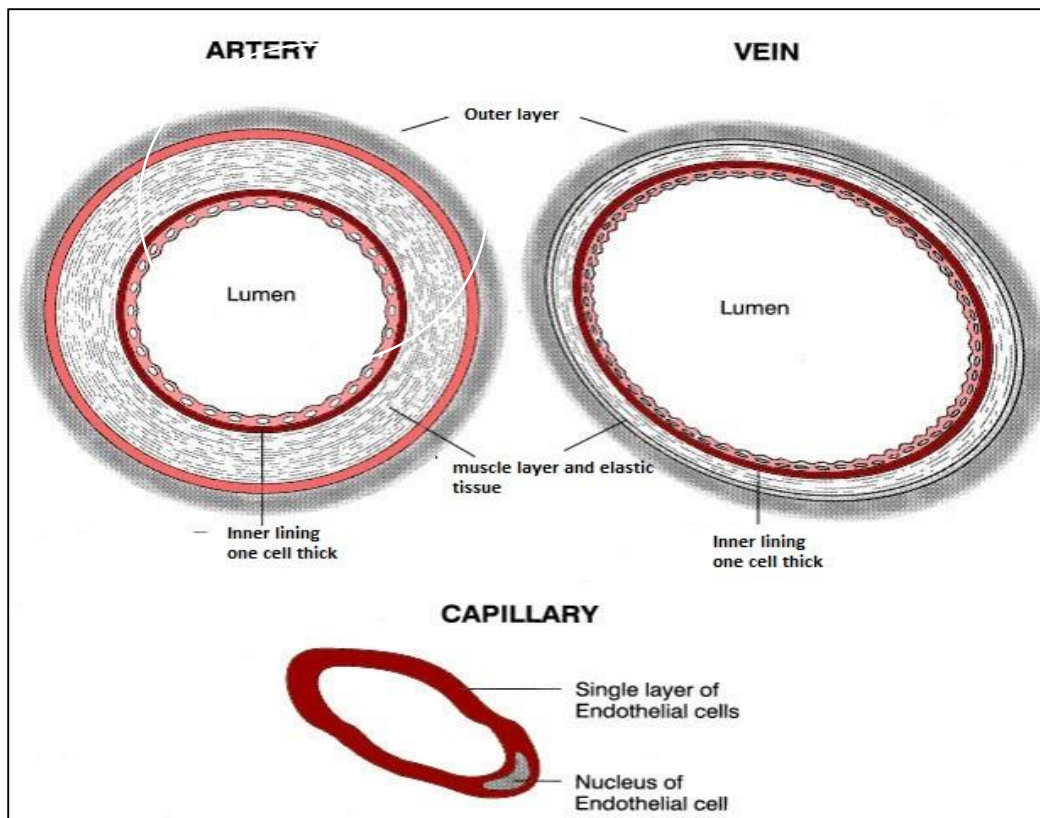
Blood Vessels

is a tube with a space in the centre, called the lumen

The Arteries.

Arteries transport blood from the heart to all parts of the body except the pulmonary artery.

They are thick walled muscular and elastic in order to resist surges of high pressure caused by the pumping action of the heart. They have a narrow passage (lumen). Carry blood of high pressure. Have no valves.



The Veins

Veins transport blood from all parts of the body to the heart except the pulmonary vein. They are thin-walled, less muscular and have wider lumen than arteries. Have valves to prevent backflow of blood. They transport blood under very little pressure.

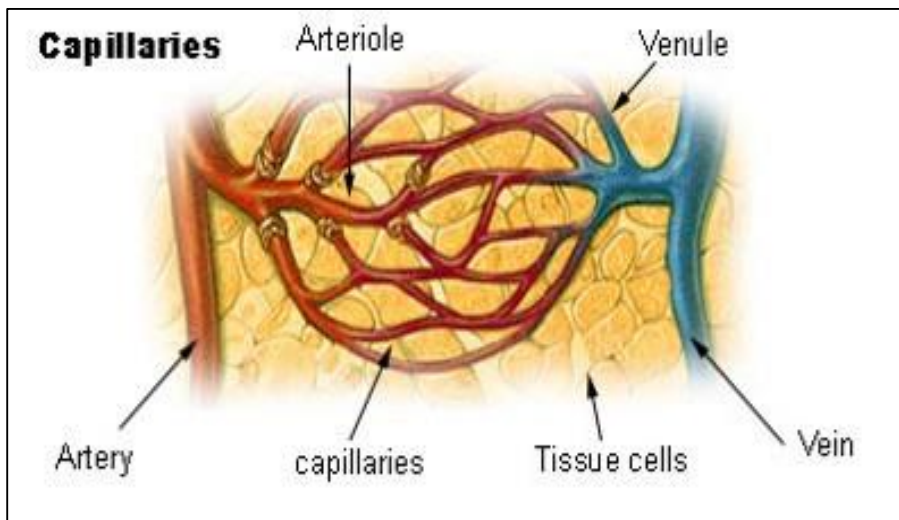
Home Work

2. a) What are capillaries? (2)

b) State two functions of capillaries. (2)

The Capillaries

These are small / microscopic fine tubes connecting the arteries to the veins. Their walls are one cell thick. They are permeable, allowing exchange of materials. Water and dissolved substances are allowed to pass in and out of the capillaries while oxygen, carbon dioxide, dissolved food and excretory substances are exchanged within tissues round the capillary. Transport blood with low pressure and have valves.



REPRODUCTIVE SYSTEM

Vegetative Reproduction in Plants

- Is also called propagation or reproduction without seeds.
- It is the production of new plants by methods other than seed formation.
- This is a non-sexual form of reproduction. Only one parent is involved to produce other plants by use of stem buds with or without leaves or roots.

Methods of vegetative reproduction

Natural vegetative reproduction

- Some plants develop specialized structures that can be used as organs of nature vegetative reproduction.
- Shoots develop from these structures and grow till they develop their own adventitious root system and later become independent parent plant.

Examples are:-

- Rhizomes –is an underground stem that grows parallel to the ground and develop shoots from buds at nodes on underground stems. The stem become swollen with stored food. e.g. ginger, sweet potatoes, grass / lawn , canna lilies etc.
- Tubers- are underground stems that are swollen with food e.g.
 - Stem tubers – these are the swollen ends of stem structures that develop shoots which grow horizontally at first and grow down into the soil heaped around the plant e.g. potatoes, yams.
 - Root tubers – are lateral root that has been modified to function as a storage organ. The middle part that store nutrients is enlarged and can involve the whole root. The food reserves are used to produce new tubers e.g cassava and sweet potatoes.

Artificial vegetative propagation

- Buds - can be induced to produce new plants from cuttings by layering and bud grafting e.g. sugar cane , cassava etc.
- Cuttings -are small parts of plants. These are cut off and used for producing other plants e.g. sugar cane, sweet potato, cassava etc.
- Suckers - these are outgrowth which grow from the stem below ground level. The terminal bud of a sucker grows upwards through the soil and form leaves above the surface. Roots form in the underground portion of the sucker and an independent plant is formed which loses its connection with the parent. e.g. coconuts, and bananas.

Experiment:

Growing plants using vegetative means.

Materials: trays of soil, plant cuttings of sugarcane, mulberry, sweet potatoes, roses

Method



- ✓ Cut pieces into sections and leave the bud/eye.
- ✓ Remove most of the leaves.
- ✓ Plant sections into the soil and make sure the sweet potato sugarcane cutting is covered with soil.
- ✓ Record number of cuttings planted.
- ✓ Irrigate the trays or beds.

Results

- ✓ The cuttings develop roots and grow after some weeks.

- ✓ Removing leaves reduces transpiration.
- ✓ Plants grow from the buds hence more plants grow than cuttings planted.

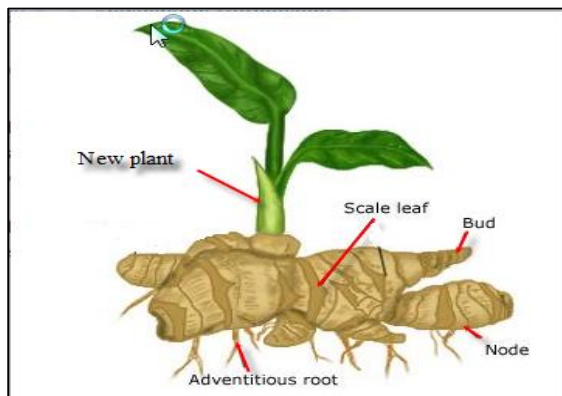
Advantages of vegetative reproduction

- Plants are identical to the parent plant
- Grow quickly as long conditions are favourable.
- Good chances of survival due to availability of stored food.
- Maturing more quickly than seed planted at the same time.
- Don't rely upon pollination, fertilization and seed dispersal.

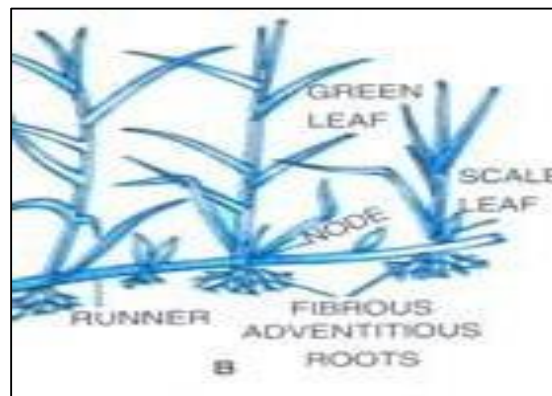
Disadvantages

- No variety (not showing any genetic variation for parent plant).
- Vulnerable to the same pests and diseases as parent plant.
- Cannot reproduce in large numbers as in seeds.
- Cannot be dispersed away from the parent plant thereby causing overcrowding.

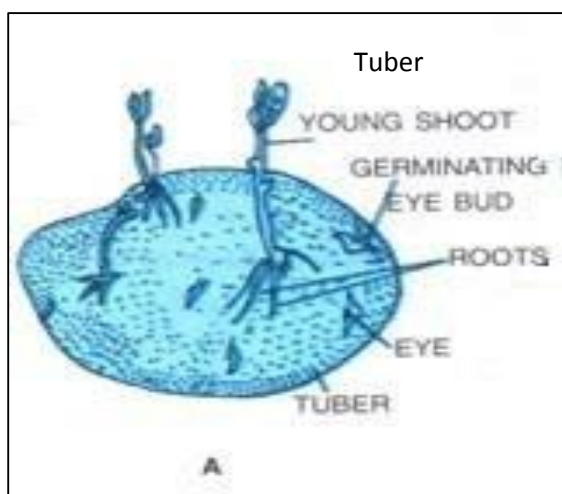
Rhizomes



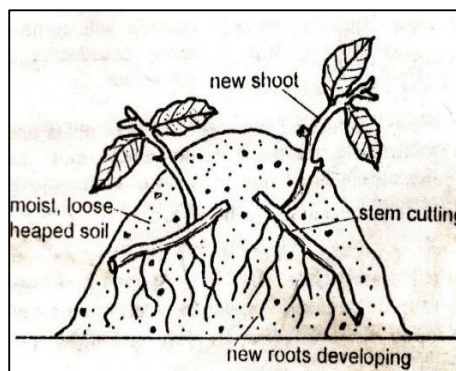
Rhizomes



Tuber



Cuttings



Question 3

- State the different between stem tubers and root tubers. (4)
- Outline the differences between sexual and asexual reproduction. [6]

Solution b)

Sexual	Asexual
Two plants	One plant
Variation occurs in offsprings	Offsprings are genetically identical to each other and their parent
Fertilisation takes place	No fertilisation takes place
Gametes are involved	No gametes involved
Mixing of hereditary material	No mixing of hereditary material.

Methods of Human Contraception

There are a number of contraception methods for preventing fertilisation or conception. Some methods are very effective and safe while others are less reliable.

Natural methods

	Abstinence	Woman avoids sexual intercourse completely
How they work	Rhythm method	Woman keeps track of her body temperature (that rises at ovulation) and doesn't have sex during this period
	Withdrawal	Woman checks if the mucus produced in her vagina has become slippery or not Man withdraws his penis before ejaculation.
Advantages	Useful for couples who don't want to use other measures of birth control for religious or other reasons No side effects	
Disadvantages	It is never possible to be 100% certain about ovulation period High risk of STIs	

Hormonal Methods

How they work	Contraceptive Pills	Contraceptive pills containing sex hormones can be taken by the woman; this stops the production of egg cells in the ovaries
		Sometimes contains copper.
Advantages	A very effective method only when contraceptive pills are taken at the right time.	
Disadvantages	It is important to have regular checkups by a doctor as some women do experience side effects of contraceptive pills	

- Contraceptive pills play a vicious part in **river pollution** and cause some male fish to change gender as well!

Mechanical/ Barrier Methods

How they work	Condom	• A condom is a piece of rubber sheath • A condom is placed upon the erect penis and acts as a barrier between the sperm and the vagina.
	Femidom	• A femidom is a female version of a condom and is used similarly
	Diaphragm (or cap)	• A diaphragm is a circular and slightly domed piece of rubber • Inserted into the vagina on the top of the cervix • Diaphragms are often used with spermicide for the best results
	IUD(Intra-Uterine-Device)	A device called as an IUD can be used. • An IUD releases hormones that prevent implantation and development of any fertilized egg cell.

Condom is a very safe method of contraception only if it is used correctly.

Condom helps in the prevention of HIV and gonorrhoea

Spermicides - Spermicides can be used to kill sperm that enter the vagina

Diaphragm is also a very safe and reliable method IF used with spermicide

Advantages Suggests advantages of some barrier methods above. (8)

Disadvantages Care must be taken when using a **condom** or a femidom; no sperm should escape through it.

Surgical Methods

	Vasectomy	• In a man the sperm ducts are cut and tied, thus preventing the passage of any sperms produced.
How it works	Tubal ligation	• In a woman the oviducts are cut or tied, stopping egg cells from travelling down the oviduct.

Advantages

- Extremely reliable and sure method of contraception.
- Have no side effects.

Disadvantages

- The tubes cannot be often opened.
- Not suitable for young couples who may wish to have children later.

HEALTH AND DISEASES

Immunity - is the ability of the body to resist infection.

1. Acquired Immunity/Active Immunity

-When one suffers from a disease like small pox or measles, they develop immunity. One will not suffer from the disease again. This is because antibodies are produced against the pathogen and they remain active in the blood ready for new attack.

2. Natural Passive Immunity

-Passed on through the placenta or from mother to child during breast feeding.
-Antibodies are transferred from the mother to the child to gain immunity.

3. Artificial Immunity

-Antibodies are produced after vaccination. A vaccine containing dead and weakened pathogen of a disease. When they are injected into the body, the body produces antibodies that destroy the weak pathogens.

Infants acquire immunity through breast feeding and immunization.

Immunisation

-Is to offer artificial immunity.

In Zimbabwe children are vaccinated to:

- Eliminate diseases e.g. to eliminate measles.
- Protect children against the killer diseases such as Polio, Diphtheria, Measles, whooping cough, Tetanus, T.B. Hepatitis B.
- Introduce a vaccine of certain diseases for the first time e.g. rubella vaccine.

N.B. B.C.G is a vaccine against T.B.

Vitamin A supplements are provided to prevent vitamin A deficiency in children (6 months to 59 months).

The programmes are carried out for free.

NB: H.IV. Causes a destruction of the immune system (bodies defence system) that is why a person suffers from many diseases. It interferes or destroys the white blood cell that produces antibodies.

Advantages of Breast Feeding

It is hygienic, it has all food nutrients, more of a balanced diet and it contains antibodies.

Question 4

Describe passive and active and artificial immunity. (9)

HIV / AIDS

HIV and Its effects

➤ HIV stands for Human Immunodeficiency Virus.

- It lives in the body fluids including blood, saliva, semen, tears etc
- It enters the body in the following ways:
 - a) During unprotected sex with an infected person.
 - b) Through sharing contaminated (by HIV) needles when administering drugs.
 - c) During blood transfusion if the blood is contaminated by the virus.
 - d) From infected mother to child during pregnancy, at birth and during breastfeeding.
- The virus reduces the body's ability to fight infection and disease by destroying the body's white blood cells responsible for producing the antibodies.
- A person is deemed to be HIV positive if that person's blood is tested for the HIV virus and it is found to be present in his/her blood.
- A person can look healthy and yet be HIV positive

AIDS

- Stands for Acquired Immuno Deficiency Syndrome.
- It is a stage during the HIV infection when the HIV has destroyed the body's defence system, causing the individual to become susceptible (vulnerable) to multiple infections and failing to recover.
- This stage can (and will likely) lead to death
- There is no cure for AIDS
- .Exercising a healthy diet, maintaining a positive attitude, seeking early treatment when an opportunistic infection surfaces and
- Taking Antiretroviral drugs (ARVs).
- All significantly prolong a person's life.

FORM 4
CHEMISTRY SECTION

SEPARATION

Paper chromatography

Paper Chromatography

- Is a method of separating and identifying mixtures.

Application of paper chromatography

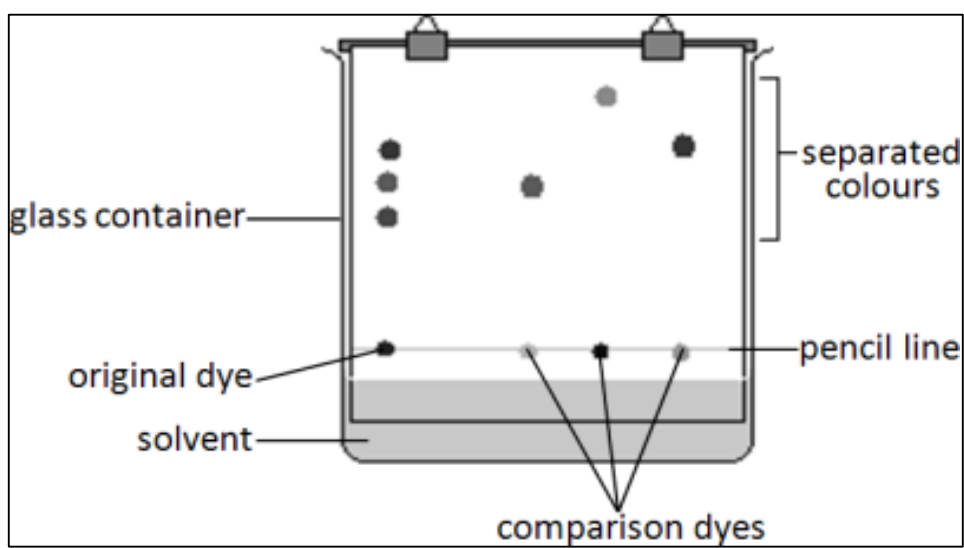
- Separates and identify mixtures of coloured substances in dyes.
- Separates substances in urine, drugs & blood for medicinal uses.
- To find out whether athletes have been using banned drugs.

Separating Mixtures of Coloured Substances

1. Obtain a dye sample then put a drop of the sample on a pencil line drawn on the filter paper then dip the paper into a solvent with the level below the spot.
2. The dye will dissolve in solvent and travel up the paper at different speed. Hence they are separated.

Identifying Mixtures of Coloured Substances

- In the diagram on the right, drop of sample dye is placed on pencil line. The result shows that:
 - a) The sample dye is made of 3 colours.
 - b) Comparison dyes are of one of the compositions of the original dye as the spots are of same colour and distance.
 - c) A comparison dye isn't part of sample.



MATTER

Trends in the periodic table

The elements in each numbered group shows trends in their properties. For example as you go down group 1, the elements become more reactive or as you go down group 7 the elements become less reactive and so on.

Group 1: *The alkali metals*

Their physical properties:

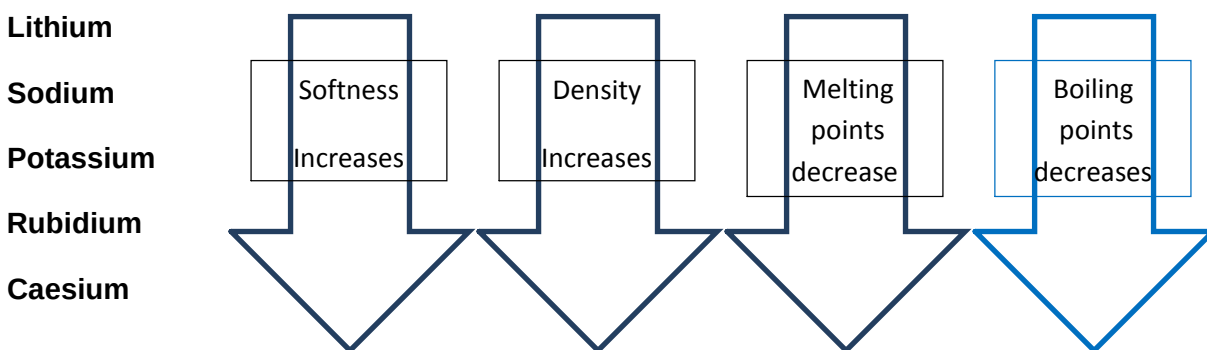
1. Like all metals, they are good conductors of heat and electricity.
2. They are softer than most other metals and they have low density.
3. They have low melting and boiling points, compared to most metals.

Their chemical properties:

1. All alkali metals react vigorously with water, releasing hydrogen gas and forming hydroxides. The hydroxides give alkaline solutions.
2. They react with non-metals. With chlorine they react to make chlorides and with oxygen they make oxides.

They form ionic compounds in which the metal ion has a charge of 1+. The compounds are white solids, which dissolve in water to give a colorless solution.

The trend in physical properties



Why they have similar properties?

Because atoms with the same number of valency electrons react in a similar way.

As you go down the group reactivity increase.

Why?

Because the atoms get larger down the group because they add electron shells.

Group 7: *The halogens*

A non-metal group.

- Form colored gases.
- Are poisonous
- Are brittle and crumbly in their solid form, and do not conduct electricity.
- Form diatomic molecules (means they exist as 2 atoms)

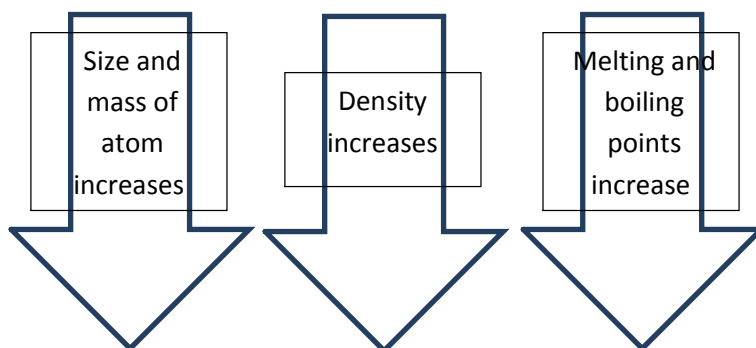
Trends in their physical properties

Fluorine

Chlorine

Bromine

Iodine



Trends in their chemical properties

Reactivity increases as you go up group 7.

Why?

Because the smaller the atom, the easier it is to attract the electron – so the more reactive the element will be.

Why are they so reactive?

Because their atoms are only one electron short of a full shell.

Group 0: *The noble gases*

A non-metal group

- Contains colorless gases, which occur naturally in air
- Monatomic – they exist as single atoms
- Unreactive because they have a full outer shell.

Trends in their physical properties

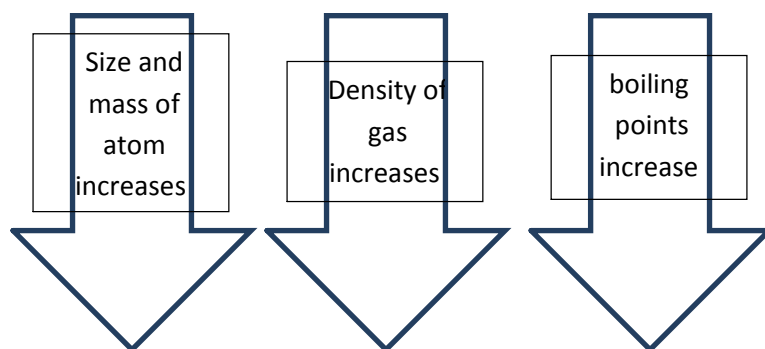
Helium

Neon

Argon

Krypton

Xenon



Uses of noble gases

Noble gases are unreactive, making them safe to use. They also glow when current is passed through them at low pressure.

<u>Gas</u>	<u>Use</u>
Helium	-Used to fill balloons and airships, because it is much lighter than air and will not catch fire
Neon	-Used in advertising signs. It glows red, but the color can be changed by mixing it with other gases.
Argon	-Used as a filler in ordinary tungsten light bulbs. (oxygen would make the tungsten filament burn away) -Used to protect metals that are being welded. It won't react with the hot metals (unlike oxygen)
Krypton	-Used in lasers. For example for eye surgery and in car headlamps
Xenon	-Used in lighthouse lamps, lights for hospitals operating theatres, and car headlamps.

The transition elements

The transition elements are the block of 30 elements in the middle of the periodic table. They are all metals.

Their physical properties

- Hard, tough and strong
- High melting points (mercury is an exception)
- Malleable and ductile
- Good conductors of heat and electricity
- High density

Their chemical properties

1. They are much less reactive than the metals of group 1.
2. They show no clear trend in reactivity, unlike the metals of group 1.
3. Most transition metals form colored compounds
4. Most can form ions with different charges (they have variable valency)
5. They can form more than one compound with another element
6. Most transition metals can form complex ions

Uses of transition metals

- The hard strong transition metals are used in structure such as bridges, buildings, cars etc.
- Many transition metals are used in making alloys.
- Transition metals are used as conductors of heat and electricity.
- Many transition metals and their compounds act as catalysts

Group II – the alkaline earth metals

Group II consists of the five metals beryllium, magnesium, calcium, strontium and barium, and the radioactive element radium. Magnesium and calcium are generally available for use in school.

These metals have the following physical properties:

- They are harder than those in Group I.
- Have low melting and boiling points.
- Have low densities, but not low as those in group 1.
- They are silvery-grey in colour when pure and clean.

Chemical properties

- Magnesium reacts extremely slowly with cold water, but easily with steam.
- Calcium reacts with water to produce calcium hydroxide.
- They tarnish quickly, however, when left in air due to the formation of a metal oxide on their surfaces.

Experiment 1

Aim: Heating a mixture of iron filings and sulphur.

Materials: iron filings, sulphur powder, evaporating dish, burner, burner stands

Method:

- Add equal amounts of sulphur and iron into an evaporating dish.
- Bring a magnet near the mixture and observe what happens.
- Heat the mixture and when the mixture starts to glow stop heating.
- Leave the apparatus to cool and bring a magnet to the solid after it cooled.

Observations

On the first instance, the iron filings are picked / attracted to the magnet. Iron and sulphur retained their original colours.

When heated a new substance is formed which is red. The substance is not attracted to the magnet.

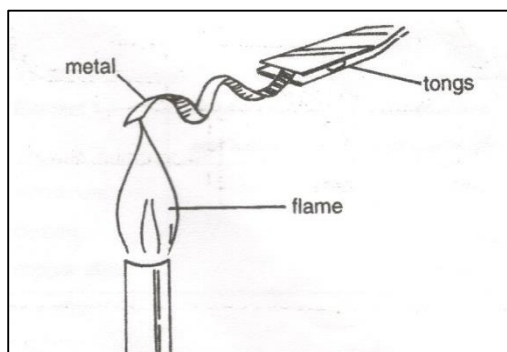
Iron + sulphur + heat → iron sulphide (compound).

Experiment 2

Aim: Burning magnesium ribbon.

Materials: magnesium ribbon, burner, sand paper, tongs

Method:



- Clean a strip of magnesium with sand paper.
- Examine and describe the physical properties of the cleaned magnesium.
- Heat the mag strip holding it with the tong until it catches fire.
- Remove it from the flame. It will continue burn for some minutes.
- Examine the new compound after the magnesium has finished burning.

Observation

When cleaned the magnesium strip is silver in colour. When it burns, a white brilliant dazzling flame is produced resulting in a white powder being formed (magnesium oxide).

Magnesium + oxygen $\rightarrow$ magnesium oxide.

Metals and Non- Metals

Elements are categorized into metals and non-metals.

Examples of metals include aluminium, silver, iron, tin zinc etc and non-metals are hydrogen, chlorine, carbon etc.

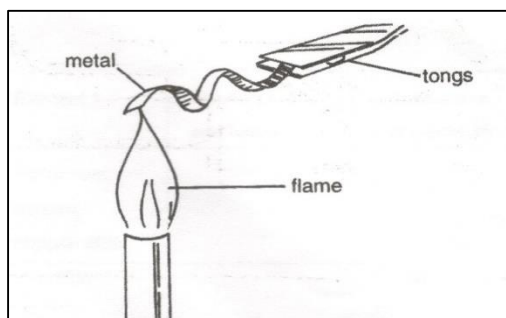
Heating metals in air

Other metals react when heated to form oxides; however some will not react and are said to be noble metals.

Experiment 1:

Heating magnesium in air.

Methods: See Science Today Bk 3 Pg 92



Results

element	Appearance before heating	reaction
magnesium	Shiny silver solid	Burns with a bright flame and leave a white residue.

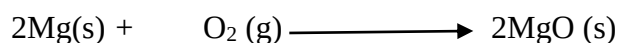
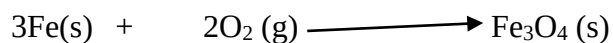
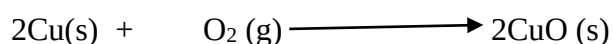
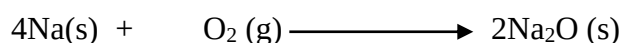
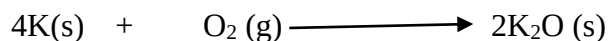
Conclusion

Magnesium produces a bright flame, when hot is yellow and leave white residue.

Word equation is: Magnesium + Oxygen $\rightarrow$ magnesium oxide.

Summary

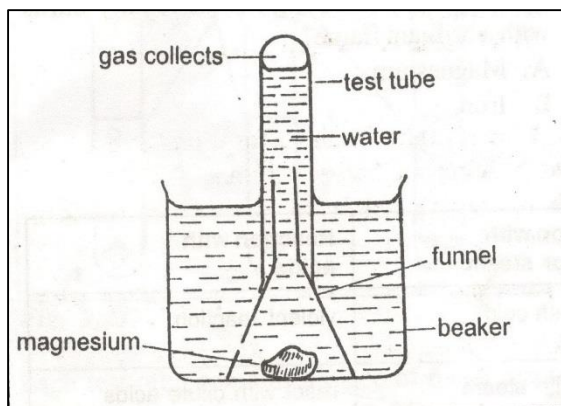
- Potassium tarnishes in the presence of oxygen
- Sodium burns with a yellow flame
- Copper powder burns with dull red glow
- Iron powder burns with bright yellow flame
- Magnesium burns with a bright white flame to produce white solid



Reactions of Metals with Water

Experiment 2

Method: see Science Today Bk 3 pg93

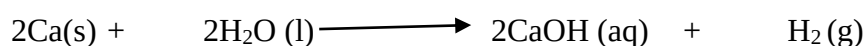
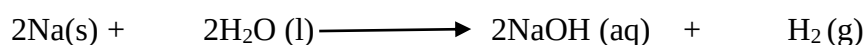
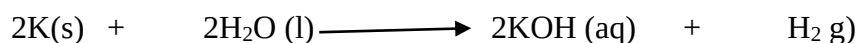


- One of the products of the reaction with water is hydrogen gas
- hydrogen produces a pop sound with a lighted glowing splint.
- magnesium is less reactive than calcium.

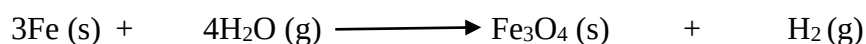
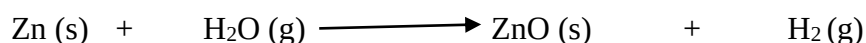
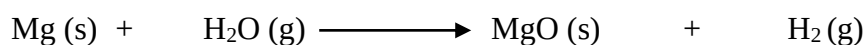
Word equation: metal + water → metal oxide + hydrogen i.e. magnesium + water → magnesium oxide + hydrogen.

Summary

- Potassium, Sodium and Calcium reacts vigorously with cold water



- Magnesium reacts very slowly with cold water but vigorously with steam
- Zinc do not react with cold water but reacts slowly with steam
- Iron reacts with steam but does not react with cold water. Rusting occurs very slowly in the presence of oxygen.



- Copper have no reaction with water and steam

Reactivity Series

- Metals can be arranged in order of their reactivity.
- The reactivity series is based on the reaction of metals with water or dilute hydrochloric acid.
- When metal reacts, the atom loses electrons to become an ion. The more readily a metal gives up electrons to form ions, the more reactive it is.

Potassium	most reactive
Sodium	
Calcium	
<u>M</u> agnesium	most reactive
<u>A</u> luminium	↓
<u>Z</u> inc	decreasing reactivity

Iron



Lead

Copper

least reactive

For the reactivity series, use the word MAZILC by noting the underlined letters of each element.

Metals Reacting With Acids

When metals react with a dilute acid, hydrogen is released. The metal dissolves and disappears and a salt is also formed .e.g. metal+ acid→ salt + hydrogen.

Hydrochloric acid →chlorides.

Sulphuric acid → sulphates.

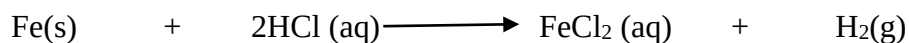
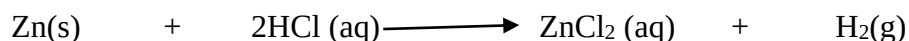
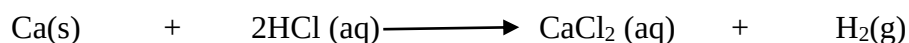
Examples

Magnesium + hydrochloric acid →magnesium chloride + hydrogen gas.

Zinc + hydrochloric acid → zinc chloride + hydrogen gas.

*Use other metals to come up with the above equation. (8)

- Potassium and Sodium reacts explosively to with acids.
- Calcium reacts vigorously with dilute acids.
- Magnesium reacts very fast with dilute acids.
- Zinc reacts moderately with dilute acids
- Iron reacts slowly with dilute acids



- Lead reacts with warm hydrochloric acid slowly
- Copper has no reaction with dilute hydrochloric acid

Oxidation and Reduction

Oxidation is the addition of oxygen and removal of hydrogen. Oxygen is an oxidising agent when added to an element during combustion or burning. E.g. Carbon + oxygen →carbon- dioxide; Sulphur +oxygen →sulphur dioxide.

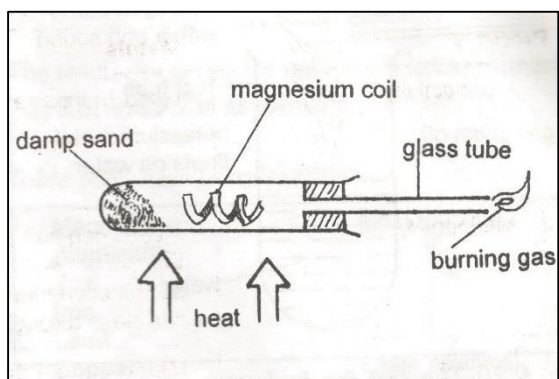
Experiment 3

Reaction of Magnesium with Steam

The metal reacts to form an oxide. The oxidising agent supplies oxygen.

Materials: magnesium ribbon, hard glass, test tube, water, sand, 2 burners, cork, clamp stand, splint, glass tube.

Method



- ✚ Set up apparatus as shown.
- ✚ Heat damp sand and magnesium ribbon.
- ✚ As soon as the ribbon starts to glow, place a lighted splint close to the open end of the glass tube.
- ✚ Continue heating until reaction is complete.

Observations and results

Magnesium reacts with water (steam) releasing hydrogen.

When magnesium is heated strongly and hot steam is passed over it catches fire and burns brightly releasing heat and hot hydrogen which burns when it meets the air.

The residue is magnesium oxide (white powder in appearance)

Equation

Magnesium + water vapour (steam) $\rightarrow$ magnesium oxide + hydrogen.

The test tube may swell and break due to high heat involved.

Conclusion

Magnesium reacts with steam to form magnesium oxide and hydrogen.

ACIDS, SALTS AND BASES

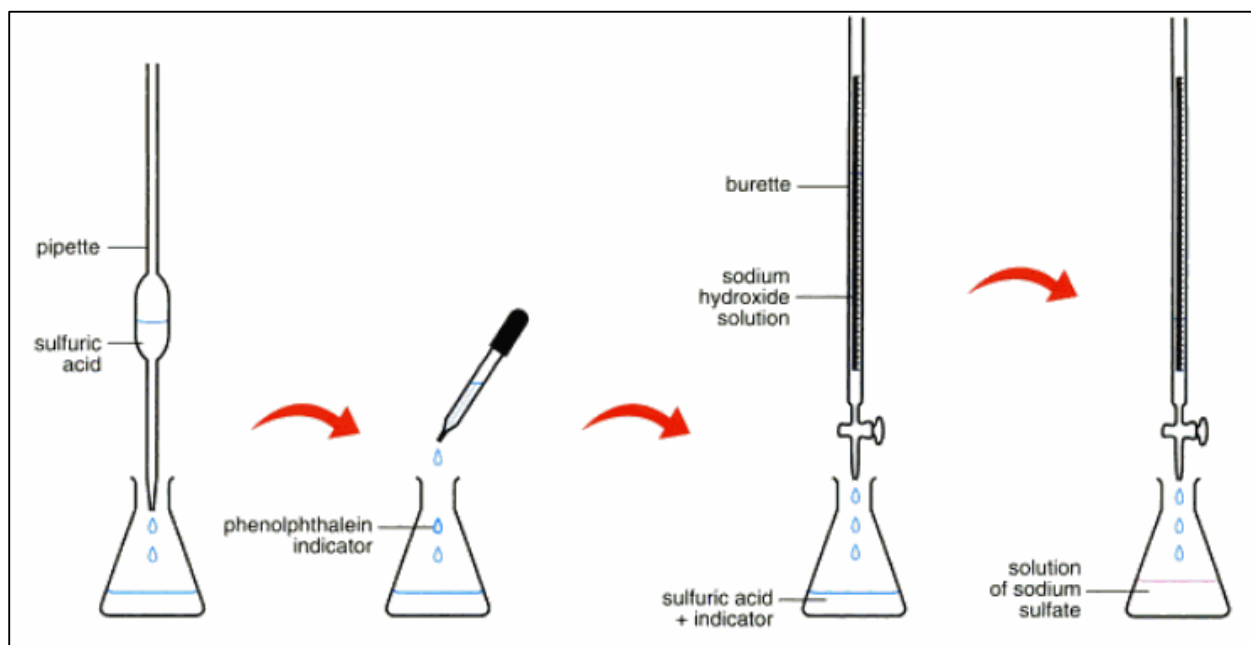
TITRATION

- Technique used to find the concentration of acids and to make soluble salts from an acid and an alkaline e.g. finding the concentration of ethanoic acid in a sample of vinegar.

Identify apparatus used in a titration

- A pipette
- Phenolphthalein.
- A burette

Describe an acid-base titration procedure



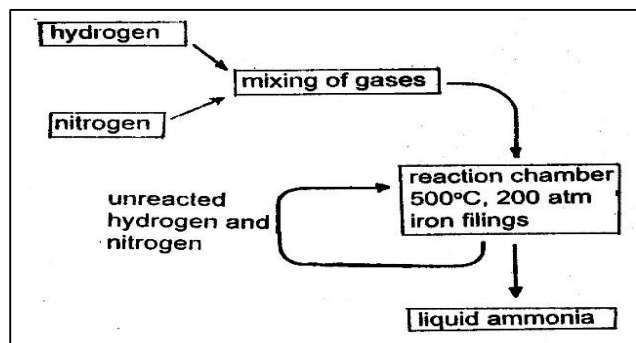
- Using a pipette, 25cm³ of Hydrochloric acid is placed in a conical flask. A few drops of indicator e.g. Phenolphthalein are added.
- Dilute Hydrochloric acid is placed in a burette and then released into conical flask until the indicator changes colour which means all the acid has just reacted. The volume of alkali added from the burette to acid is measured.
- The experiment is now repeated. 25.0cm³ of Hydrochloric acid is placed in a titration flask as before, but no indicator is added.
- The NaOH is placed in a burette and the same volume of this alkali is added to the flask as before.
- The flask then contains a solution of sodium chloride without excess acid or alkali.
- The sodium chloride is obtained by evaporating most of the water and crystallizing the salt.

INDUSTRIAL PROCESSES

Ammonia (NH₃)

- It is a poisonous, colourless gas that has a pungent smell.
- It burns in oxygen to produce nitrogen dioxide and water.

Manufacture of Ammonia- Haber Process



Step 1

- Nitrogen and hydrogen are scrubbed to remove impurities like carbon dioxide.
- This means washing a stream of gas with an alkali to dissolve all carbon dioxide in the alkali.

Step 2

- The gas mixture of 1: 3 is compressed under a pressure of around 200 atmospheres.

Step 3

- The compressed gas is pumped into a converter chamber where the reaction occurs.
- The gas flows over beds of fine iron which acts as catalyst.
- Temperature of 450-500 degrees Celsius helps the reaction on.
- The reaction is reversible. Nitrogen + hydrogen $\leftrightarrow$ ammonia. Deduce the chemical equation. (2)

Step 4

- Not all the mixture reacts, but only 15% is converted to ammonia.
- Ammonia is cooled so it is collected.
- Unreacted nitrogen and hydrogen are simply recycled through the converter chamber.

Step 5

- The liquid ammonia is allowed to flow into tanks and stored under pressure until use.

NB: It is not wise to increase temperature beyond 500 degrees Celsius because ammonia can decompose back to nitrogen and hydrogen.

Uses of Ammonia

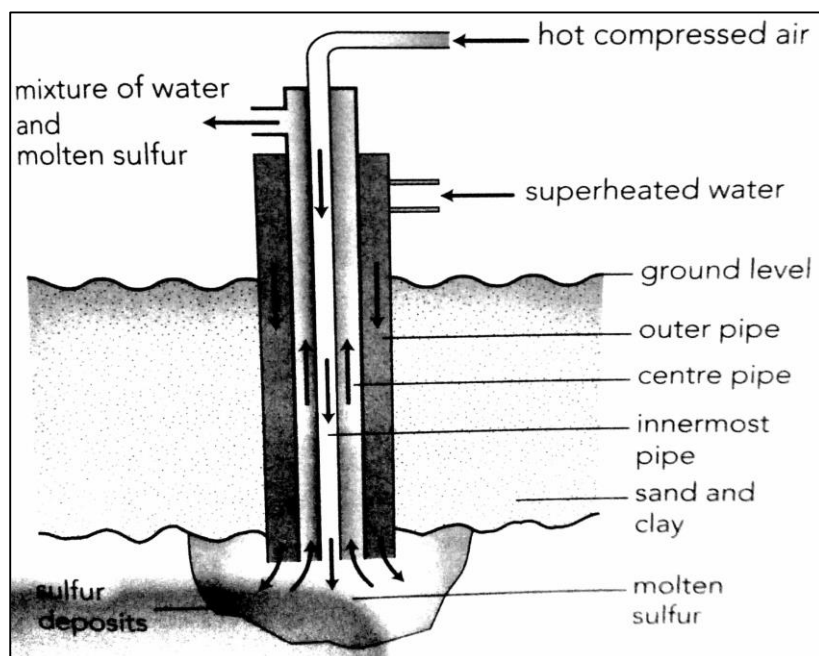
- ✚ Manufacture of fertilizers.
- ✚ Nitric acid production.

- ✚ Used as refrigerant.
- ✚ Making of dyes, drugs, explosives and fibres.
- ✚ Water purification and drugs.

Sulphuric Acid production

The extraction of sulphur

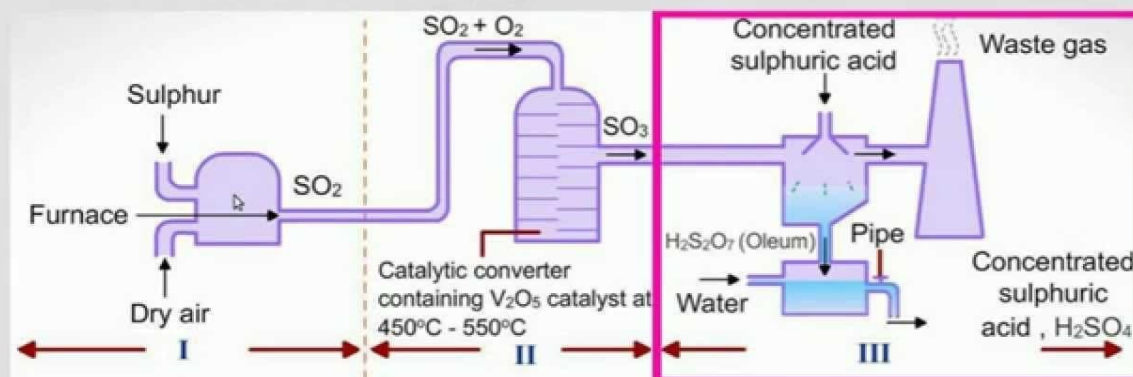
- The Frasch process is used.
- The process uses superheated water in the extraction. Superheated water is water that has been heated beyond its boiling point, but kept in the liquid phase by allowing a great pressure to it.
- The superheated water is pumped underground where it melts the sulfur but not the surrounding rocks.
- The liquid sulfur is then pumped to the ground level where it cools and sets.



Contact Process

Manufacture of sulphuric acid in industry

►► Contact process



Stage 1

Production of sulphur-dioxide

- Sulphur is further oxidised to yield sulphur-dioxide or strongly heating sulfide ores in air.
- $\text{S} + \text{O}_2 \rightarrow \text{SO}_2$

Stage 2

Production of sulphur-trioxide

- Sulphur-dioxide is further oxidized to produce sulphur-trioxide.
- Catalyst- vanadium (v)oxide (V_2O_5) It speeds up the reaction.
- $2\text{SO}_2 + \text{O}_2 \leftrightarrow 2\text{SO}_3$
- To ensure production, a temperature of 450-500 degrees Celsius, pressure of 1 atmosphere and catalyst are needed.

Question 1

Write down the word equation for production of sulphur-trioxide.(2)

Stage 3

To make oleum (Absorption)

- Sulphur-trioxide is absorbed in concentrated sulphuric acid to obtain oleum.
- Sulphur trioxide + sulphuric acid $\rightarrow$ oleum.
- $\text{SO}_3 + \text{H}_2\text{SO}_4 \rightarrow \text{H}_2\text{S}_2\text{O}_7$

Stage 4

To make sulphuri acid (Dilution)

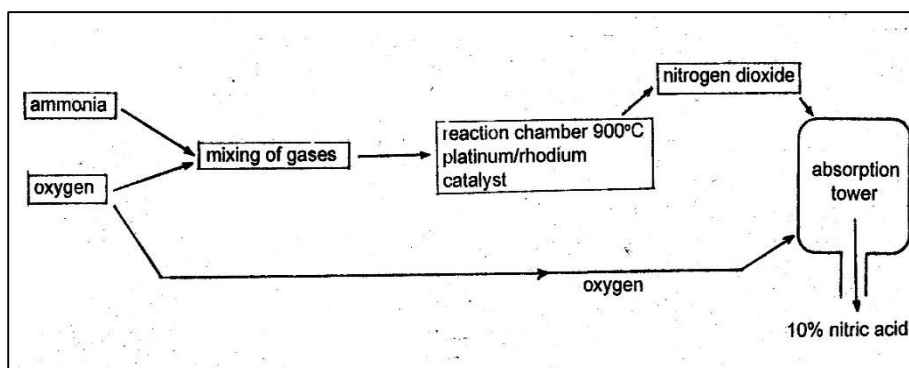
- Oleum is converted into twice as much concentrated sulphuric acid by adding water carefully.
- 98% concentrated sulphuric acid is obtained.
- $H_2S_2O_7 + H_2O \rightarrow 2H_2SO_4$
- NB: Not all sulphur dioxide is converted to sulphur trioxide in stage 2. This unreacted gas can pollute the environment if not disposed off properly by reacting with water to form acid rain.

Question 2

State the uses of sulphuric acid. (10)

- 1) The acid in a car battery.
- 2) Making detergents.
- 3) Metal treatment and anodising.
- 4) A catalyst
- 5) A dehydrating agent.
- 6) Making fertiliser.
- 7) Paints and dyes.
- 8) Mineral processing
- 9) Oil refining
- 10) Wastewater processing
- 11) Paper making
- 12) Manufacture of plastics.

Manufacture of Nitric Acid



This is catalytic oxidation of ammonia.

Stage 1

- Ammonia reacts with oxygen to obtain nitrogen dioxide and water, platinum rhodium catalyst, temp of abt 900 degrees Celsius and pressure of 10-15 atmospheres.
- $4NH_3 + 5O_2 \rightarrow 6H_2O + 4NO$

Stage 2

- On cooling, nitrogen monoxide is further oxidised to nitrogen dioxide.
- $2NO + O_2 \rightarrow 2NO_2$

Stage 3

- Nitrogen dioxide absorbed in water in the presence of oxygen to give nitric acid.
- Nitrogen dioxide + oxygen + water → nitric acid (dilute).
- $2H_2O + HNO_3 \rightarrow NH_4NO_3$

Uses of Nitric Acid

- ✚ Manufacture of fertilizers.
- ✚ Manufacture of explosives.
- ✚ Making dyes, drugs, nylon and plastics.

Manufacture of Ammonium Nitrate Fertilizer

- Nitric acid + ammonia → ammonium nitrate
- (Acid) (Base) (Salt)
- $NH_3 + HNO_3 \rightarrow NH_4NO_3$
- The two substances combine chemically in a process called neutralisation.

Industrial Gases and Their Uses

Gas	Source	Uses	Test
hydrogen	Electrolysis of water	Making ammonia, margarine and is a fuel	Produces a pop sound
oxygen	Fractional distillation of liquid air, electrolysis of water	Sulphuric and nitric acid manufacturing, medical use, welding and oxygen lance process	Relights a glowing splint
nitrogen	Fractional distillation of air	Manufacture of ammonia	
ammonia	Haber process	manufacture of nitric acid, fertilizers, in refrigerant	Turns damp red litmus paper blue
Carbon dioxide	Fermentation of glucose, fractional distillation of air	Fire extinguishers, fizzy drinks, refrigerants	Turns lime water milky

OXIDATION AND REDUCTION

Alloy is a mixture of one metal with one or more other metals and or carbon.

Alloys of iron are pig iron (cast iron), mild steel and stainless steel.

Question 1

State 2 properties and 2 uses of each of the above alloys of iron. (12)

Alloys of iron

Steel

Iron made from blast furnace is not good as:

- It contains impurities which makes it brittle (can break easily).
- It cannot be bent or stretched.
- Most iron is converted into steel which is an alloy of iron and carbon with small amounts of other elements.

Advantages of steel

- it is strong and tough.
- it can be bent and stretched without shattering

Making Steel

- Impurities of iron is removed by blowing oxygen into molten iron to change the impurities into oxides. They are then combined with CaO and removed as slag.
- Carbon and other metals are added in certain amount to make steel.

Different Types of Steel:

Mild steel – is a low carbon steel with 0.25% carbon.

- It is strong and quite malleable. It is used for car bodies, ships, railway lines and steel rods to reinforce concrete.

Hard steel – is a high-carbon steel with about 1% carbon.

- It is harder than mild steel and less malleable. It is used to make tools

Stainless steel – is iron with large amounts of chromium and nickel.

- It is hard, shiny and doesn't rust. It is used to make cutleries, medical instrument and pipes in chemical industries.

ORGANIC CHEMISTRY

Alcohols

- The **alcohols** are the family of organic compounds that contain the **OH** group.
- Are homologous series with general formula $C_nH_{2n+1}OH$
- They have $-OH$ functional group (**hydroxyl group**) meaning they will all react in a similar way.
- All alcohols end with suffix **-ol**
- This tells you that the OH group is attached to a carbon atom at one end of the chain.
- For alcohol, the $-OH$ is not of hydroxide ion, OH^- , but is covalent bond between oxygen and hydrogen, $O-H$

First three members of the series

- Methanol, CH_3OH
- Ethanol, C_2H_5OH or CH_3CH_2OH
- Propanol, C_3H_7OH or $CH_3CH_2CH_2OH$

Ethanol, an important alcohol

- Ethanol is the alcohol in alcoholic drinks.
- It is a good solvent. It dissolves many substances that do not dissolve in water.
- It evaporates easily – it is **volatile**. That makes it a suitable solvent to use in glues, printing inks, perfumes, and aftershave

Making Ethanol

- Fermentation of sugars with yeast
- Reacting ethene with steam

Fermenting glucose

- Fermentation is breakdown of sugars into smaller molecules by microorganisms.
- $C_6H_{12}O_6 + C_6H_{12}O_6(aq) \rightarrow 2C_2H_5OH(aq) + 2CO_2(g)$
- Temperature is kept constant at $37^\circ C$ to prevent destruction of yeast at higher temperatures.
- Oxygen is removed by limewater and carbon dioxide is produced during fermentation.
- Alcohol is separated from solution by fractional distillation.

Reacting Ethene with Steam

- Ethene and steam are passed over phosphoric acid H_3PO_4 (as a catalyst) under high temperature of $300^\circ C$ and pressure of 65 atm.
- $C_2H_4(g) + H_2O(g) \rightleftharpoons C_2H_5OH(aq)$
- Since this is reversible reaction, both ethene and water are produced aside from ethanol. The ethanol is separated by fractional distillation.

Uses of Alcohol

As organic solvent; alcoholic drink; preservatives; vehicle fuel

Experiment 1

Investigating fermentation

Apparatus: 2 test tubes, bicarbonate indicator solution, litmus paper, yeast and starch solution.

Method

-  Arrange apparatus as shown below.
-  Note colour changes after 30 minutes.
-  Dip red litmus paper into the solution A, observe what happens.

Observations and Results

Brick – red bicarbonate indicator turns yellow to show presence of carbon dioxide. Thus, the gas produced during fermentation is carbon dioxide.

Red litmus paper turns to blue to show the presence of alcohol, an alkaline substance.

If one uses lime water, it turns to milky from clear indicating the presence of carbon dioxide.

Uses of ethanol

- a. It is used as a fuel.
- b. It is a solvent for organic compounds.
- c. It is used in alcoholic drinks.
- d. It is mixed with methanol to get methylated spirits used in laboratory.
- e. It is used to produce drugs, cosmetics, detergents, plastics and lubricants.

Global warming

Greenhouse effect is the trapping of heat from sun by greenhouse gases to regulate Earth temperature so that not all heat is reradiated back to space.

However, increased industrialization releases more greenhouse gases to atmosphere, contributing to Global Warming (increase in temperature of Earth's atmosphere due to trapping of heat by greenhouse gases).

Examples of Greenhouse Gases are:

1. Carbon dioxide which is naturally occurring or by combustion of hydrocarbons.
2. Methane which occur naturally or emitted during production of fuels or from decaying vegetable matter.
3. Nitrous Oxide is produced by industrial and agricultural activities, and by incomplete combustion of hydrocarbons.
4. Water vapour
5. Ozone

Causes of Global Warming

State any four causes of global warming. [4]

Effects of Global Warming

- Rising sea Levels.
- Loss of biodiversity.
- Desertification.
- Increased tropical storms.

- Heat waves and forest fires
- Droughts and famines (crop failure).
- Skin cancer (diseases).
- It melts polar icebergs.
- Floods low lying areas and coastal regions,
- Alter the climatic conditions of certain places.

Tackling Global Warming

- Reduce the use of fossil fuels.
- Use alternative forms of energy such as wind, tidal and hydroelectric power.
- Use more electric vehicles.
- Reduce number of cars on road.
- Create efficient engines in cars to ensure complete hydrocarbon combustion.
- Reduce, Reuse, Recycle.
- Afforestation and Reforestation.

PHYSICS

DATA PRESENTATION

Pie Charts - display proportions of multiple data.

Advantages

- Simpler to understand and interpret.
- Summarise large amount of data in visual form.

Disadvantages

- Do not show exact values of quantities.
- They do not show changes of quantities over time.
- They can not show relationships between variables.

MEASUREMENTS

S. I Unit for base quantities

Quantity measured	S.I Unit	Symbol
Length	Metre	m
Mass	Kilogram	kg
Time	Second	s
Temperature	Kelvin	k
force	Newton	N
power	watt	w
Current	Ampere	a
Multiples of the quantity	Number of moles	N

Derived SI units

Quantity measured	Formula	Symbol
Density	Mass / Volume	kg / m ³
Area	LXW	m ²
Velocity	Length / time	m /s
Acceleration	Velocity / time	m/s ²
Work	Force x distance	J
Potential difference	Work / charge	V

- A constant 20N force pulls a cart over a distance of 6m. Calculate the work done.
- Work done = Force X displacement
= 20N X 6m
= 120 Nm or 120 Joules.

FORCE

Pressure in fluids

Pressure is force acting over an area.

It is the ratio of weight/force to the area supporting it.

The bigger the surface area, the smaller the forces or weight each unit area supports.

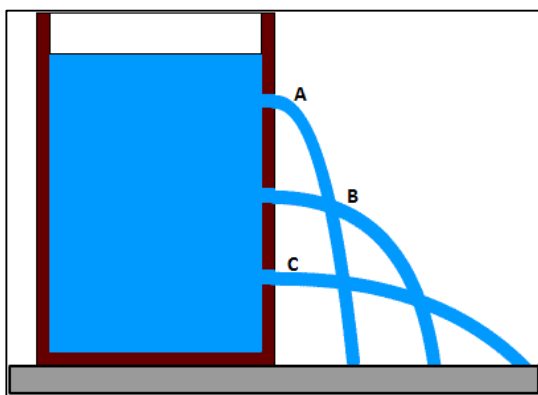
$$\text{Pressure} = \frac{\text{force (N)}}{\text{area (m}^2\text{)}} = \text{N/m}^2 = (\text{Nm}^2)$$

Pressure in fluids

- A fluid/liquid cannot be compressed.
- Gravity pulls a liquid into a container and the liquid exerts a pressure on the other container.

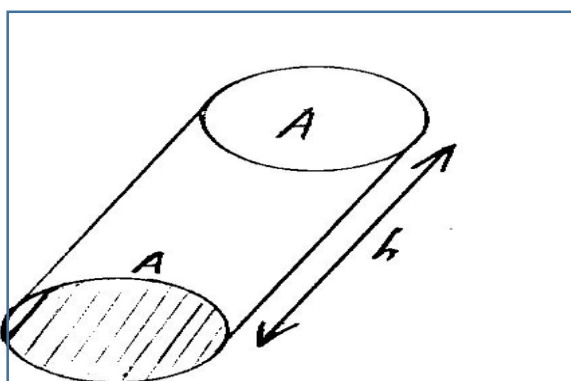
Principles in pressure in fluids

- Pressure in a liquid at a certain depth acts equally in all directions.
- Pressure increases with depth because of the weight above. In the above diagram pressure at A is smaller than at B and C.



- Pressure increases with density. A heavier (denser) liquid e.g. mercury exerts more pressure than water of the same volume.
- Similarly if a tin with three holes is filled with water comes out with greatest force from the hole at the bottom (C).
- Dam walls are thicker at the bottom to withstand great pressure of water at the bottom.

Pressure of Liquid



Mass of liquid = mass $\times$ density
 $= A \times h \times p$

Weight of liquid = $A \times h \times p \times g$

Pressure of liquid = $\frac{A \times h \times p \times g}{A}$

P = hpg where;

h = height of the liquid

g = force of gravity

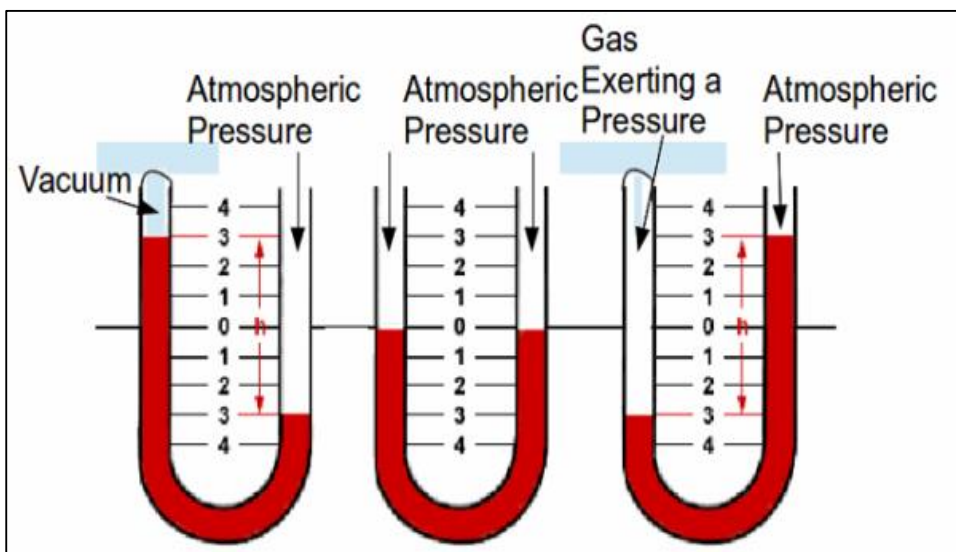
p = density of liquid.

-We use a manometer to measure pressure.

-It is a U-tube filled with liquid e.g. water or mercury.

-When both arms are open to atmosphere pressure, the levels of the liquid is the same.

A manometer



- Gas/liquid pressure can be measured using a manometer.
- It is a U-tube partly filled with a liquid such as mercury or water.
- When both arms are open to atmospheric pressure the levels of the liquid are the same in both arms.
- However, if one arm is connected to a gas supply the levels change.
- The liquid in the left arm is forced downwards by the gas pressure.
- The pressure of the gas is then balanced by atmospheric pressure plus the column of liquid with height h .
- The pressure of the gas is equal to the atmospheric pressure at the time of measurement plus the pressure of the water in column h .

Summary

Pressure of gas = atmosphere + (h)

Pressure = $P + pgh$

Where

p = pressure

P = atmospheric pressure

$P(\rho)$ = density

g = gravity

h = height difference in pressure

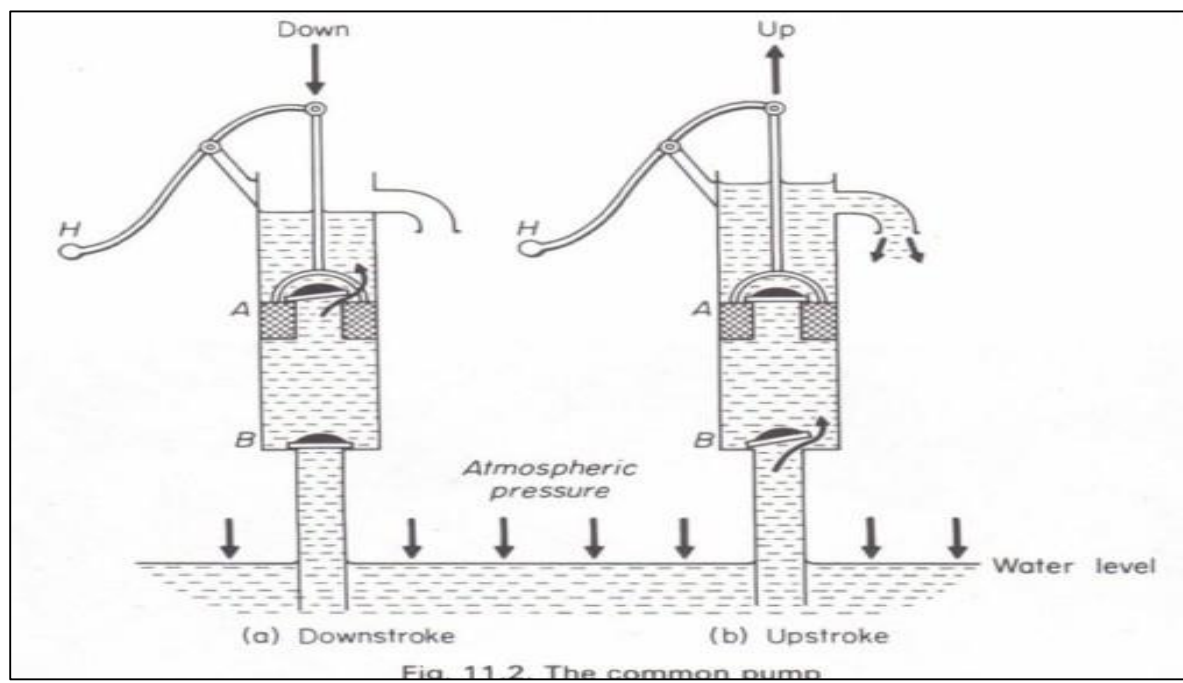
level .

Pumps

-Pumps are used to lift liquids and gases using atmospheric pressure.

Types of pumps

Lift pump



– Has two valves. One valve is the position, the other near the base of the cylinder. When one valve is open, the other is closed. The pump is primed (filled with water) first before it can be used so as to prevent air leaking past it.

Upstroke

-Is when piston moves upwards valve A (in the piston) is closed and pressure in the cylinder fall.

-Atmospheric pressure then pushes water up the pipe through valve which is at the bottom which would have open. Water above the piston is forced out of the water spout.

Down stroke

-Piston moves downwards. Valve B closed due to pressure of water on it and valve A open allowing water to pas upwards into the cylinder above the piston.

Disadvantages of a lift pump.

-Needs priming before use (laborious)

-It only delivers water on the upstroke also making it laborious.

-Only raise water through height of 9 meters.

2. Force pump

-They force water to more than 10 metres

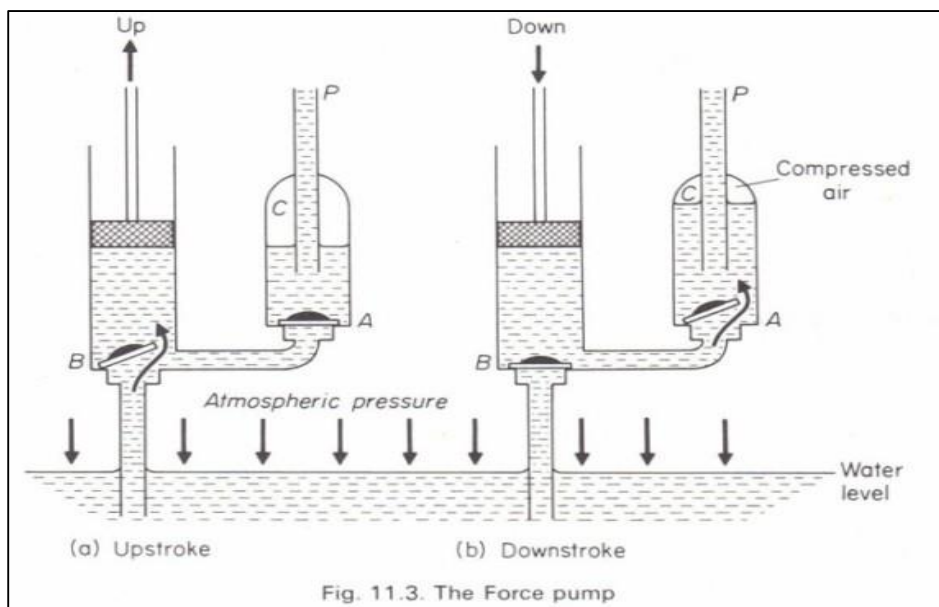
-They do not need priming before use.

Upstroke

-Piston moves up, the outlet valve closes.

-Pressure in the barrel decreases

-Inlet valve opens and water enters the barrel



Down stroke

➤ Piston moves down and inlet valve closes.

➤ Outlet valve opens forcing water into the reservoir and out at the spout.

➤ Water flows continuously from the spout because of the pressure of air in the reservoir.

3. Blair pump

-It is a combination of a force and a lift pump.

-It is used in a covered well.

-It lifts water to a height of about of 10 meters.

-It gives a steady supply of water for small communities.

- It has got a handle connected to a piston which moves in a cylinder that is fixed below the water level in a well.

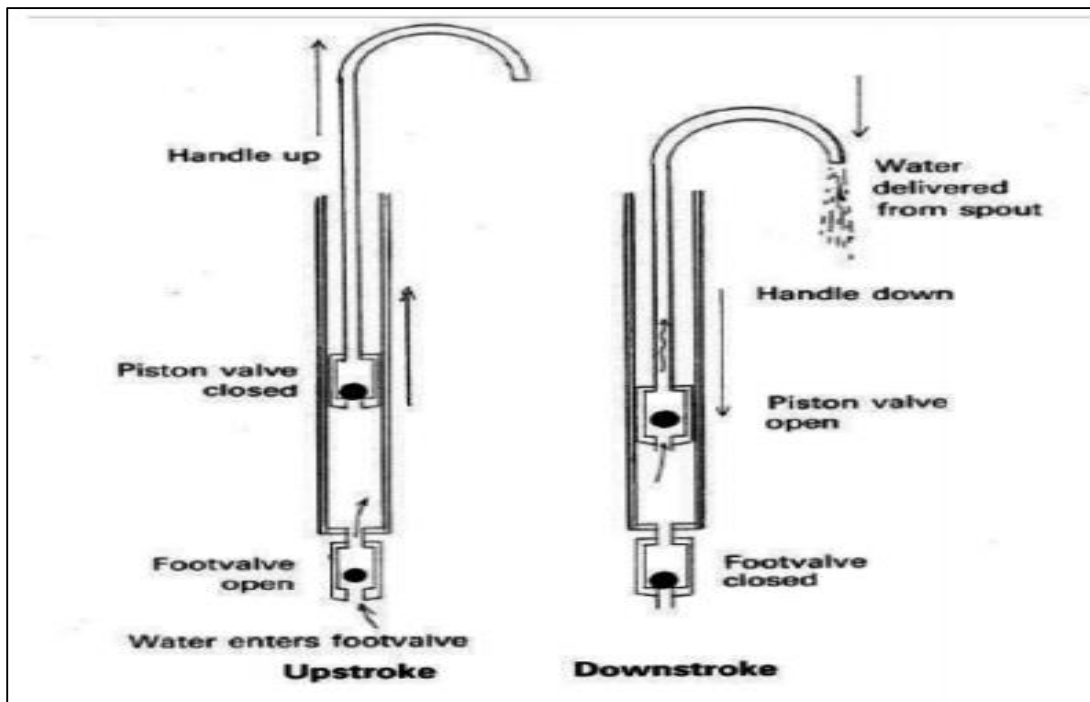
-It has two valves which both open only upwards.

Up stroke

Piston moves up; atmospheric pressure of water in the well pushes water against valve 1 (foot valve), which opens and water enters the cylinder. Piston valve will be closed.

Down stroke

Piston moves down, valve 1 (foot valve) closes pressure of water in cylinder forces valve 2 (piston valve) to open and water enters the handle and flows out of the spout.

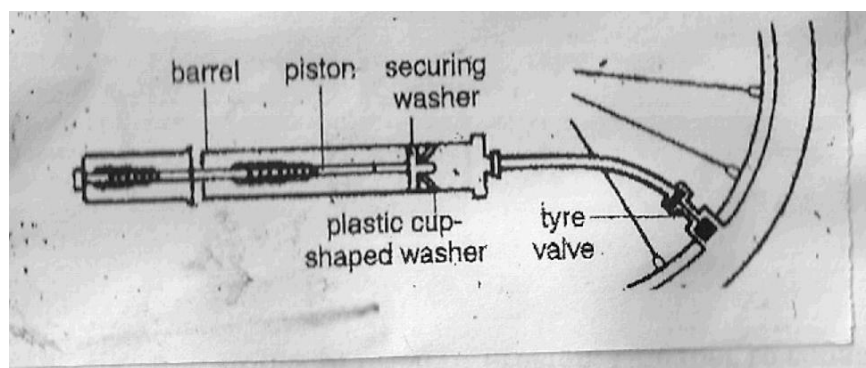


Advantages

- The Blair pump is made of plastic making it light, cheap and easy to assemble, operate and repair.
- No lubricating oil or grease is used (water is colourless and odourless).

Bicycle pump

-Air is compressed in the cylinder by the piston which is fitted with leather or rubber washer to give an air tight fit. When piston is pushed inwards, the air in the cylinder is forced through the rubber valve in the rubber tube. This prevents air from escaping.



Withdrew piston

- Rubber washer move away from the barrel
- Air pressure at front of barrel reduces and air moves from the back barrel.

Pushing piston

- Washer moves back to the wall of the barrel. Air in front is compressed and tyre is forced to open. Air enters the tyre.

Fluid systems

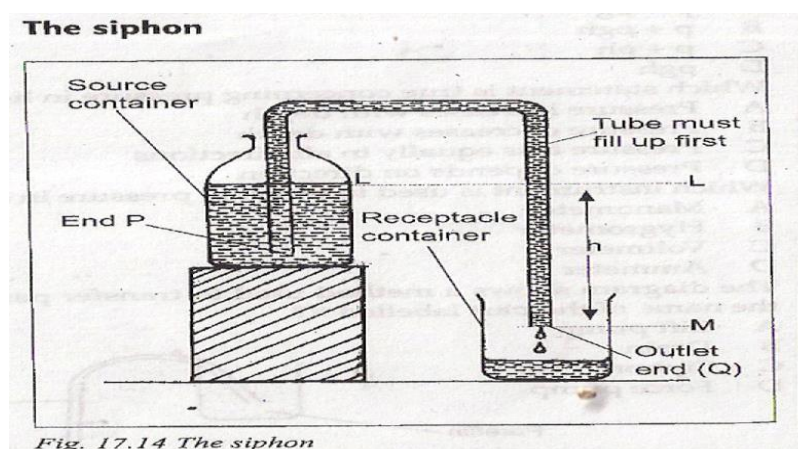
Works on the principle that;

- ✚ Liquids transmit equally in all directions.
- ✚ Liquids cannot be exposed or compressed.

Examples of fluid systems includes:

The siphon

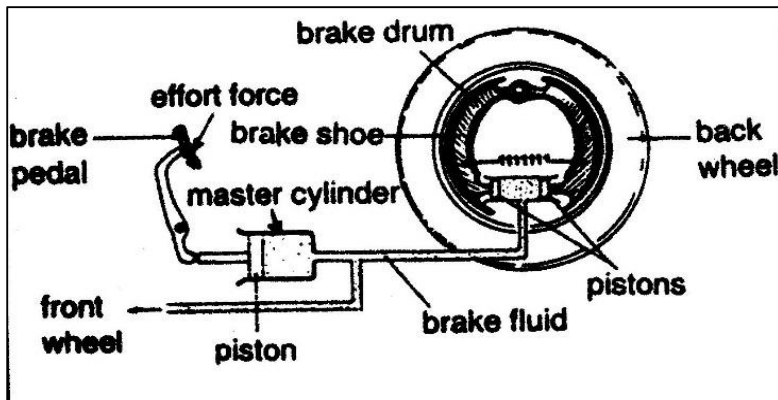
- is a tube used to empty liquid from a tank that is not fitted with an outlet.
- When outlet Q is open, pressure inside the tube will be greater than outside.
- Liquid in the tube will flow out. This reduces the pressure in the tube to air pressure on the surface at P which pushes more liquid.
- The tube must be full of water and one end of the tube must be below level of liquid in the tank.
- The liquid to be siphoned should be at a higher level than where it is being taken.



- A fountain of water is produced because pressure of water at P is higher than at Q.
- The higher pressure causes water to come at the mouth of the tube.

Hydraulic breaks

Car breaking system



-Pressure of foot on the brake pedal causes the piston in master cylinder to exert pressure on the brake fluid.

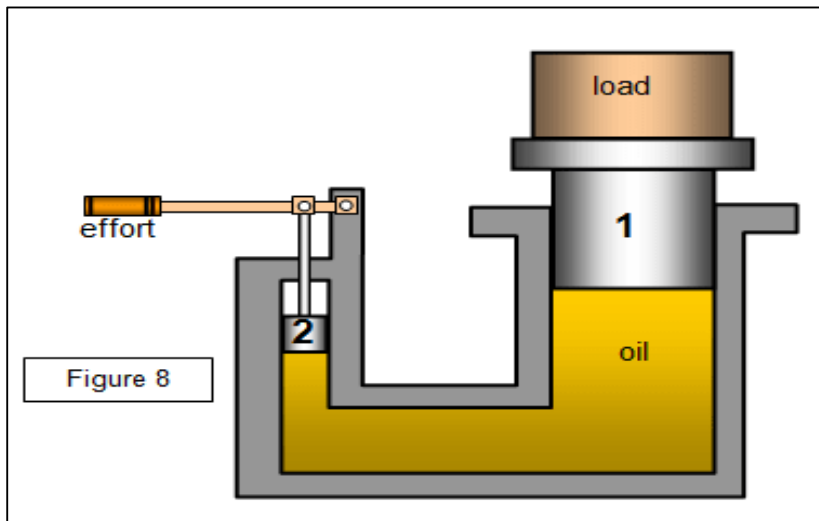
-The pressure is passed on equally to the piston in the wheel cylinders

-This pressure forces the brake pads against the brake drums.

NB: Air bubbles must not be found in the brake fluid as air is compressible and thus reducing the effort of the braking system. The principles used in hydraulic systems are that liquids cannot be compressed and an applied pressure is transmitted.

Hydraulic jack

These are used to lift cars.



To work the hydraulic jack :-

These use the facts that-;

- Liquids cannot be squeezed.
- Liquids pass on pressure applied to them.
- Effort is exerted to push down the small piston.
- Pressure is transmitted through the liquid and acts on the large piston
- The large piston is lifted up a small amount, compared with how far the small piston has been moved down. Force is applied at Area 2 to lift a load at 1. The pressure exerted at piston 2 is

transmitted equally to the larger piston 1 which is carrying the load. Piston 1 moves up raising the load.

$$\text{Formular: } \frac{\text{Force}}{\text{Area}} = \frac{\text{Force}}{\text{Area}}$$

For example;

If a downward force of 20N act on small piston (piston 2) of area 10 cm². The pressure applied to liquid would be –

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = \frac{20\text{N}}{10\text{cm}^2} = 2 \text{ N/cm}^2$$

The pressure is passed through the liquid and acts on the large piston (piston 1) of Area 100cm². Upward force on piston 1 would be -:

$$\text{Pressure} = \frac{\text{Force}}{\text{Area}} = 2\text{pa} \times 100\text{cm}^2 = 200 \text{ N.}$$

NB; Thus a load of 200N can be raised by an effort of 20N.

NB; In machines such as hydraulic brakes and hydraulic jacks, work input equals work output, ignoring energy losses.

$$1. \text{ M.A} = \frac{\text{Load}}{\text{Effort}}$$

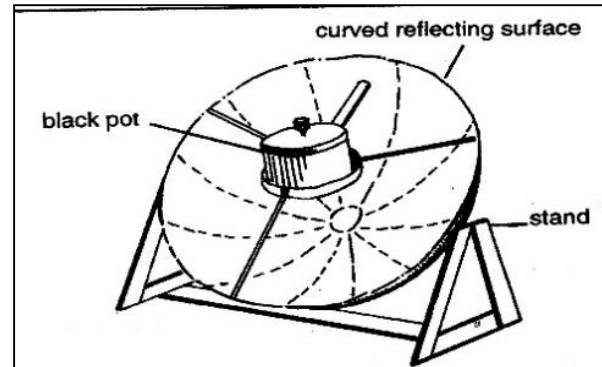
$$2. \text{ V.R} = \text{Ratios of Areas i.e } \frac{100\text{cm}^2 \text{ (load area)}}{10\text{cm}^2 \text{ (effort area)}}$$

$$3. \text{ EFFICIENCY} = \frac{\text{M.A}}{\text{V.R}} \times 100\%$$

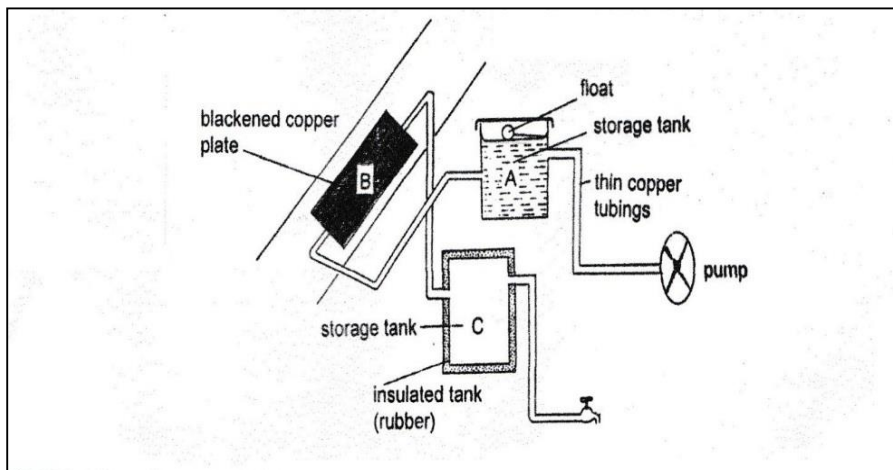
ENERGY

Solar Cooker

The surface of the solar cooker is shiny so as to reflect heat as possible. The sun's rays are reflected by the curved surface and come to a focus at the pressure cooker. The solar cooker uses a renewable source of energy. It is fast to cook food with a pressure cooker. The method is environmental friendly and is easy to use once installation done.



The Solar Water Heater



- ✓ Solar water heating system is mounted on roof of building so as to energy from the sun to be used to produce warm water.
- ✓ The solar panel is painted black to absorb radiant heat.
- ✓ Cold water is pumped into a tank or from a tap system.
- ✓ The water is taken from a tank through copper tubes.
- ✓ These thin copper tubes pass through the solar panel and the heat absorbed by the solar panel is quickly absorbed by the passing water at B where there is Copper Plate.
- ✓ The heated water moves to a storage tank through the copper tubes.

A vacuum or Thermos flask

- ✓ Keeps hot liquids hot or cold liquids cold.
- ✓ It is very difficult for heat to travel into or out of the flask.
- ✓ Transfer by conduction and convection is minimised by making the flask a double-walled glass vessel with a vacuum between the walls (Figure 24.4).
- ✓ Radiation is reduced by silvering both walls on the vacuum side.
- ✓ Then if, for example, a hot liquid is stored, the small amount of radiation from the hot inside wall is reflected back across the vacuum by the silvering on the outer wall.
- ✓ The slight heat loss that does occur is by conduction up the thin glass walls and through the stopper.

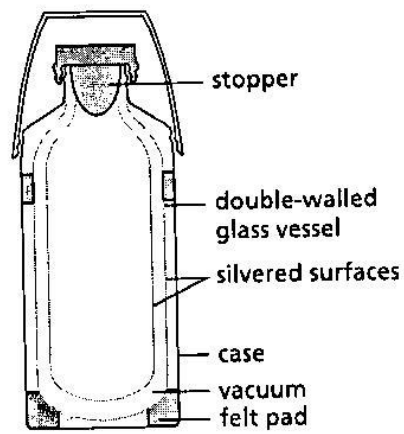


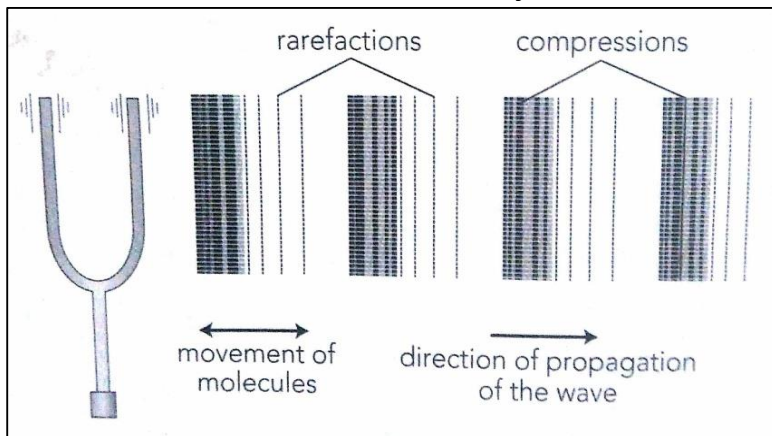
Figure 24.4 A vacuum flask

TELECOMMUNICATION

Define the term (2).

Sound transmission

- Sound waves are longitudinal waves which are transmitted when particles of a medium vibrate back and forth in the same direction as the direction of movement of wave.
- For example, when a loud speaker vibrates outwards, it brings air particles closer to each other forming a compression.
- When it vibrates inwards, it pulls air particles apart to form a rarefaction.
- A series of compression and rarefactions move sound forwards but particles of the medium only move forward and backwards and thus why sound needs a medium to be propagated.



Characteristics of sound

1. The wave length (L) – is the distance from the start of one compression to the start of the next compression. It can be measured from the centre of one rarefaction to the centre of the next rarefaction.
2. Frequency (f) – is the rate at which complete wave lengths pass a given point in one second and is measured hertz (Hz). Humans can only hear sounds of frequencies between 20 Hz and 20kHz. Thus the audible range of frequency.
3. Amplitudes of a sound wave (a) – the maximum distance that particles move from their rest positions. As amplitude increases, particles move further from their positions hence sound increases loudness as more sound energy is produced.
4. The velocity (v) – the distance per unit time that a wave-front travels in one second. Sound travels faster in solids than liquids and slowest in gases.
5. The pitch of a sound – is a measure of how high or low a note is. A sound of high pitch has high frequency and vice versa.

How a telephone works

Three important steps are:

- When you speak into the mouthpiece, your voice makes a diaphragm in the microphone inside vibrate. Sound energy is converted to electrical energy.
- The electrical energy travels from the phone, through exchanges, to another phone. These exchanges are computer terminals that make electrical connections between the transmitting and receiving phone.

- A diaphragm in the loudspeaker inside the earpiece of a receiving phone converts incoming electrical energy back into sound energy.

Thus, making a telephone call, is all about converting sound to electrical energy, the electrical energy is carried down a very long wire and finally electrical energy is turned back in sound.

The cellphone

- A microchip inside a cellphone takes vibrations of your voice into a tiny microphone and turns it to strings of numbers that are transmitted as radio waves by means of the phone's antenna of the phone to the nearest mast.
- The mast works as the antenna in the phone, transmitting the radio waves across the next phone mast and the next, until it reaches the one closest to the person you are calling.
- From that mast it is transmitted to the person's phone where the encoded digital message is decoded back into electrical pulses that get the speaker of the receiver vibrating to produce sound waves.

Email

- The internet converts information into a string of numbers and transmitting these through a combination of radio waves through antennas, electrical pulse through phone lines and pulses of lights through fibre optics.
- The computer chips in the computer breaks up information into smaller, addressed pieces called "packets".

Telecommunication Signals

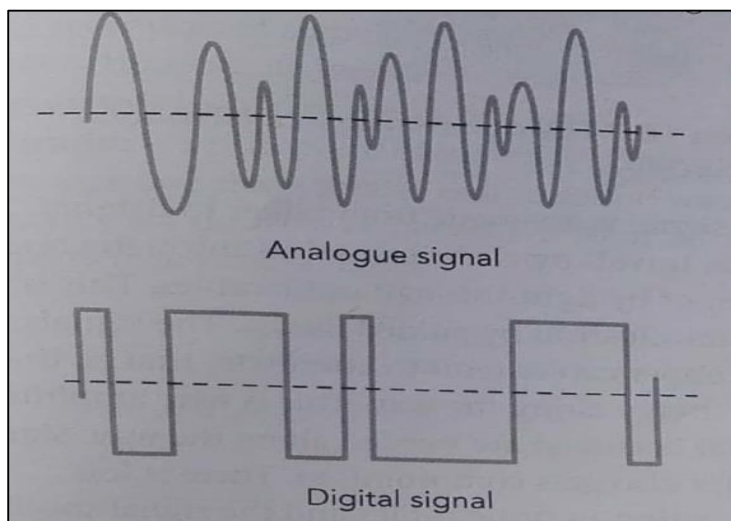
- Two types are analogue and digital.

Analogue

- Is made up of continuous wave that varies over time.
- The signal varies in amplitude and frequency and takes many values.
- The wave contains the message that is being transmitted.

Digital

- Is an electrical signal that has been chopped into bits. They look like square waves of different sizes.
- The bits are in the form of pulses that can take two values, either a 1 or 0. Therefore, a digital signal is made up of a series of ones and zeros, either "on" (1) or "off" (0).



Media for signal transmission

- Modern communication includes radio, television, fixed line phones, cell phones and other hybrid forms. There is need to transmit signals in all these systems.
- Signals carry messages and are transmitted through guided and wireless media.

Guided media

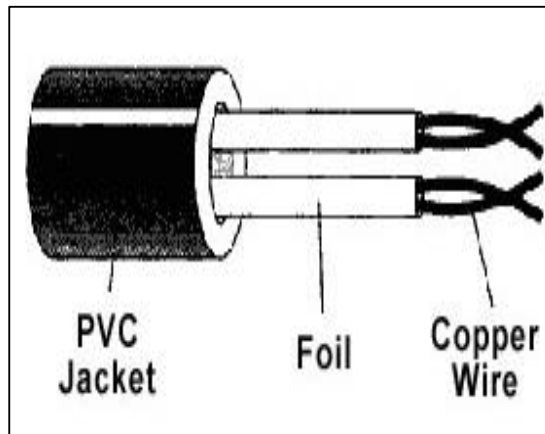
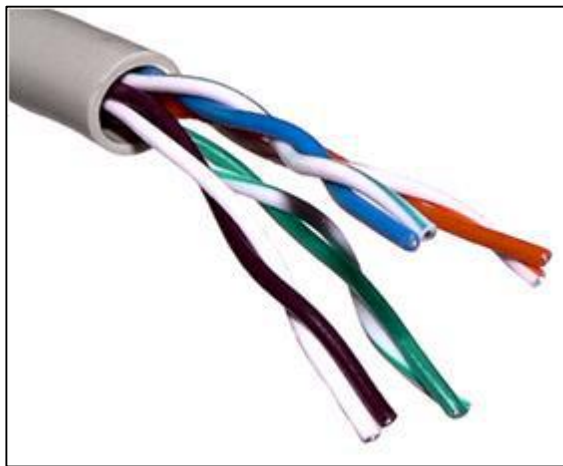
- In guided media, transmission is restricted within boundaries of a cable. 1. Twisted cables, sheathed pair cables, 2. coaxial and 3. optical fibre.

Signal transmission in cables

- The twisted, braided and coaxial cables have copper wire through which electrical signal travels.
- The optic fibre has a thin glass threads through which light travels carrying signals with out leaving the glass cable until it reaches its destination.

Twisted Pair Cable

- Most common form of wiring in data communication.
- It consists of two identical wires wrapped together in a double helix.
- A special type of twisted pair cable known as **CAT5** or **CAT6** is mostly used in a specific type of LAN namely **Ethernet**, hence it is also known as **Ethernet cable**.



Why the wires are twisted in TWISTED PAIR CABLE?

- The twisting of wires reduces **Crosstalk**. The bleeding of a signal from one wire to another and which can corrupt signal and cause network errors. This form of signal interference is called **Crosstalk**.

Advantages

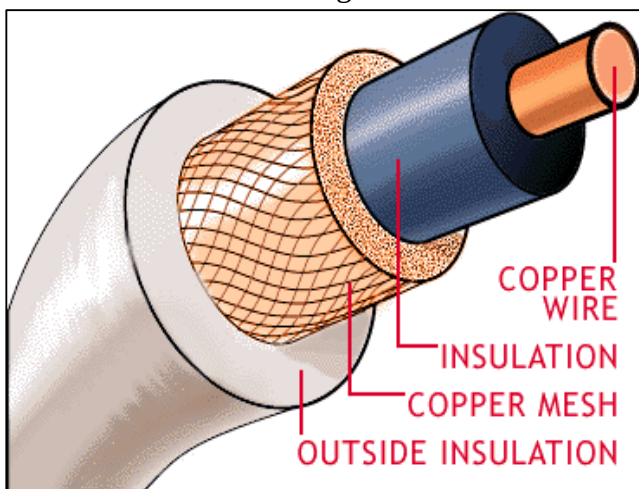
1. It has low weight.
2. It is very inexpensive.
3. It is easy to install and maintain.

Disadvantages

1. Because of high attenuation (weakening of signal), it is incapable carrying a signal over long distances without the use of repeaters.
2. It has low bandwidth (data carrying capacity) capabilities

Coaxial cable

- This type of cable consists of a solid wire core surrounded by one or more wire shields, each separated by some kind of plastic insulator. The inner core carries of the cable carries the signal.
- This cable has high electrical properties and is suitable for high speed communication. It is widely used for television signals.



Advantages

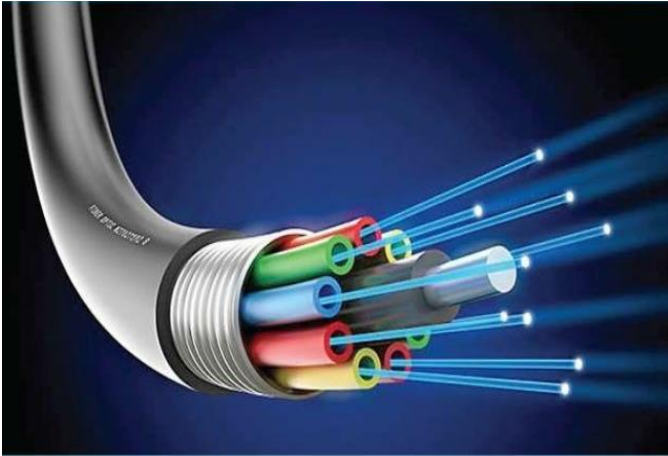
1. It provides higher bandwidths as compared to twisted pair cable.
2. The data transmission characteristics of coaxial cables are considerably better than those of twisted pair cables.
3. The coaxial cables can be used for broadband transmission.

Disadvantages

1. Expensive compared to twisted pair cables.
2. The coaxial cables are not compatible with twisted pair cables.

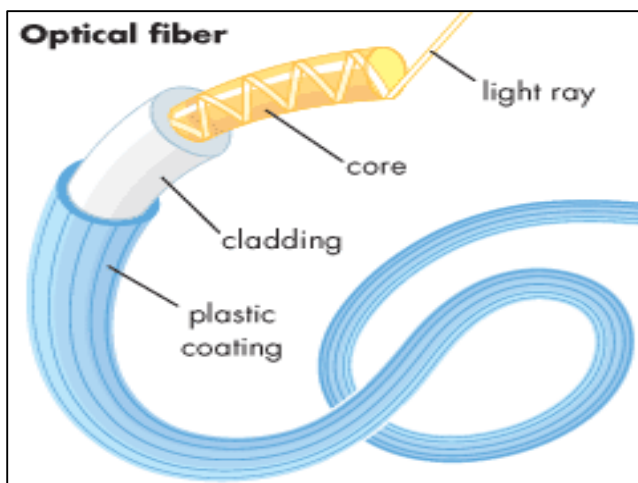
Optical fibers

- It consists of thin strands of glass which are used to carry light from a source at one end of the fiber to a detector at the other end. The bandwidth of this medium is very high.



The fiber cable consists of three pieces:

1. **Core** : the glass strand through which the light travels.
2. **Cladding** : which is a covering of the core that reflects light back to the core.
3. **Protective coating** : which protects the fiber cable from hostile environment.



Advantages

1. It is immune to electrical and magnetic interference i.e., noise in any form because the information is travelling on a modulated light beam.
2. It guarantees secure transmission and has a very high transmission capacity.
3. It is highly suitable for harsh industrial environments.

Disadvantages

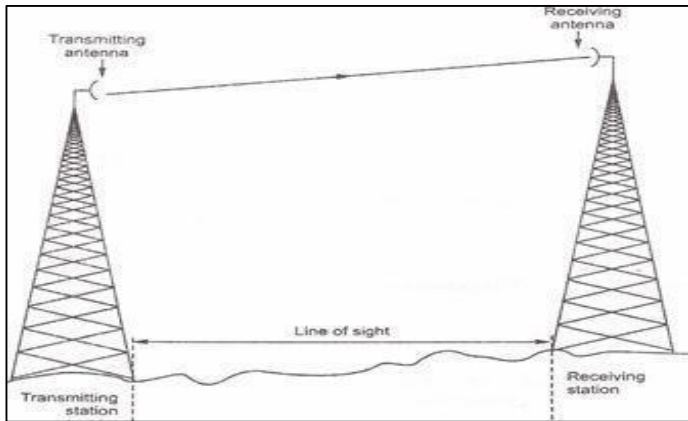
1. Optical fibers are quite fragile and may need special care in installation.
2. Because of noise immunity, optical fibers are virtually impossible to tap.
3. They are the most expensive of all the cables.

Unguided media / Wireless media

1. Microwave
2. Radio wave
3. Satellite
4. Infrared
5. Laser
6. Bluetooth

Microwave

- Microwave signals are wireless transmission media and are used for long distance transmission. The microwave transmission consists of a transmitter, receiver and the atmosphere.
- In microwave communication, parabolic antennas are mounted on towers to send a beam to other antennas tens of kilometers away. The higher the tower, the greater the range. With a 100-meter high tower, distances of 100 km between towers are feasible.
- The microwave transmission is line of sight transmission.



Advantages

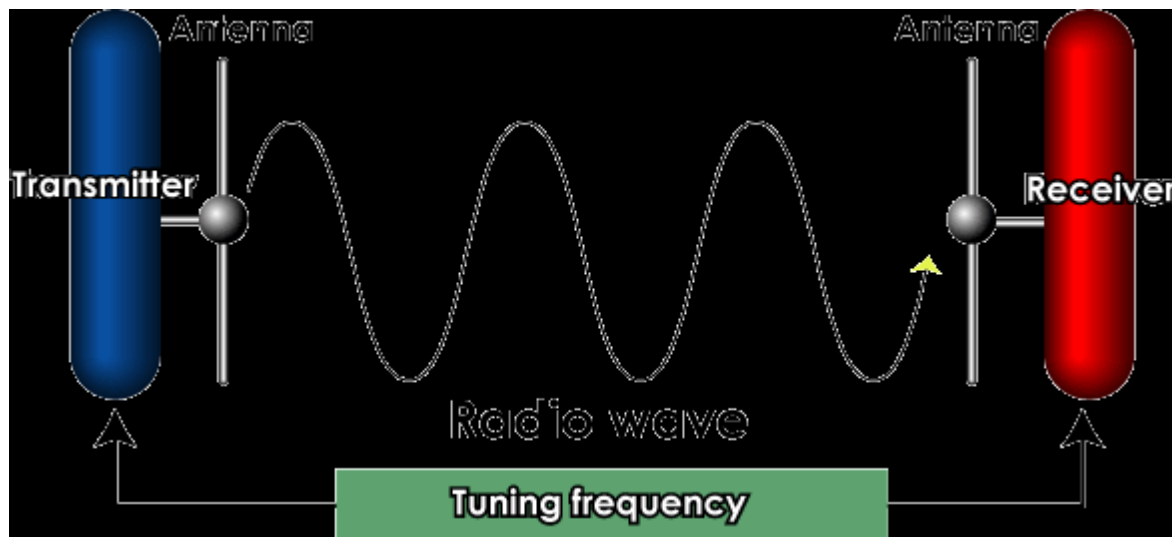
1. This transmission media proves cheaper than digging trenches for laying cables.
2. It offers ease of communication over difficult terrain.

Disadvantages

1. Microwave communication is an insecure communication.
2. Microwave signals are susceptible to weather effects like rains, thunderstorms etc.

Radio wave

- This transmission media make use of radio frequencies. In this transmission continuous sine waves are used to transmit information.
- Transmission through radio wave consists of two parts: (i) transmitter (ii) receiver.
- The transmitter takes some sort of message (it could be the sound of someone's voice, pictures for a TV set etc.), encodes/changes it onto a sine wave and transmits it with radio waves. The receiver receives the radio waves and decodes the message from the sine wave it receives. Both the transmitter and receiver use antennas to radiate and capture the radio signal.

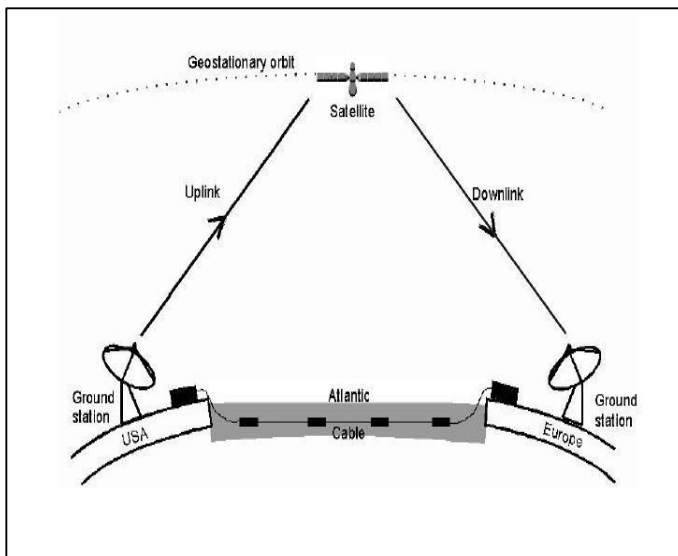


Advantages

1. This transmission media proves cheaper than digging trenches for laying cables.
2. It offers ease of communication over difficult terrain.

Satellite

- Satellite communication is special case of microwave relay system. In satellite communication the earth station consists of a satellite dish that functions as an antenna and communication equipment to transmit and receive data from satellites passing overhead.
- The satellites accept data/signals transmitted from an earth station, amplify them, and retransmit them to another earth station.



Disadvantages

1. Radio wave communication is an insecure communication.
2. Radio wave signals are susceptible to weather effects like rains, thunderstorms etc.

Advantages

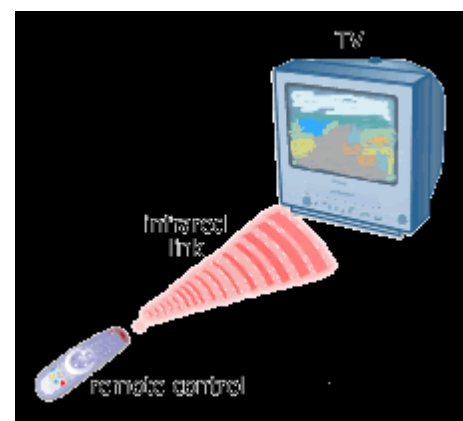
1. The area coverage through satellite transmission is quite large.
2. The laying and maintenance of intercontinental cable is difficult and expensive and this is where the satellite proves to be the best alternative.

Disadvantages

1. The costs associated with satellites are significantly high.

Infrared

- This type of transmission uses infrared light to send data. TV remotes, automatic garage doors, wireless speakers etc., all make use of infrared as transmission media.
- The infrared light transmits data through the air and can propagate throughout a room, but will not penetrate walls.

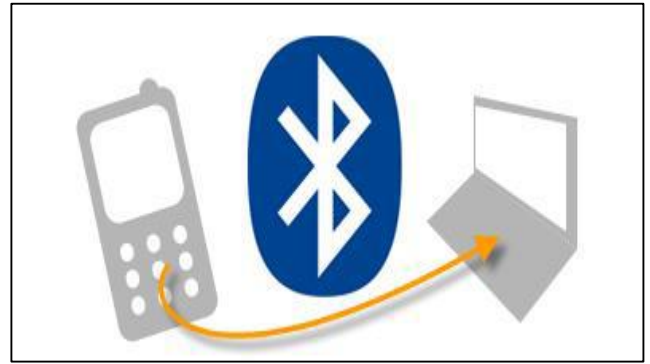


Laser

- The laser transmission requires direct line of sight. It is unidirectional like microwave, but has much higher speed than microwaves.

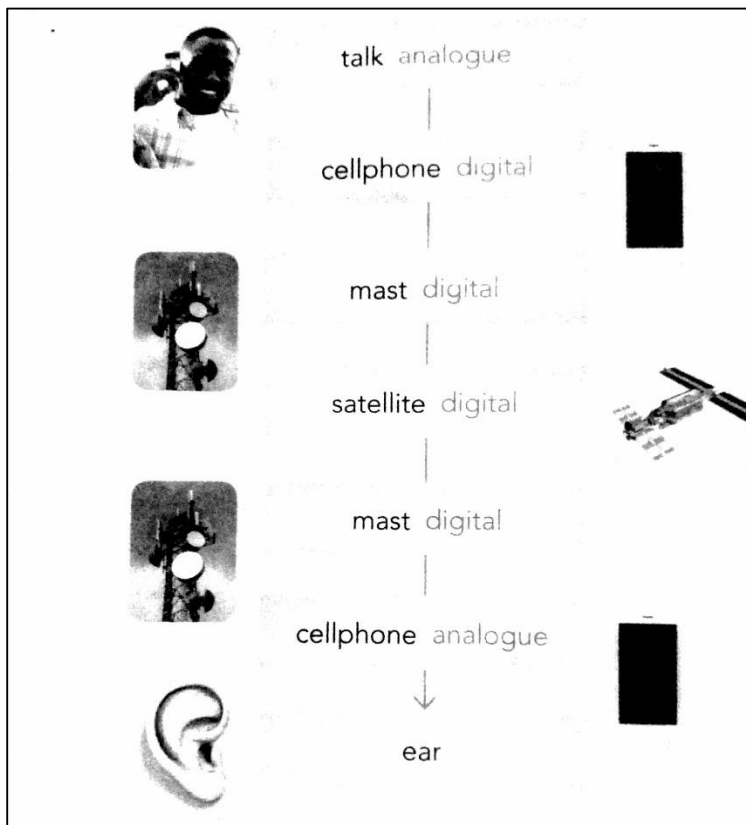
Bluetooth

- Bluetooth is a telecommunication industry specification that describes how mobile phones and computers can be easily interconnected using a short range wireless connection.
- The maximum range of connection is 10 meters. Data can be exchanged at a rate of 1 megabit per second.

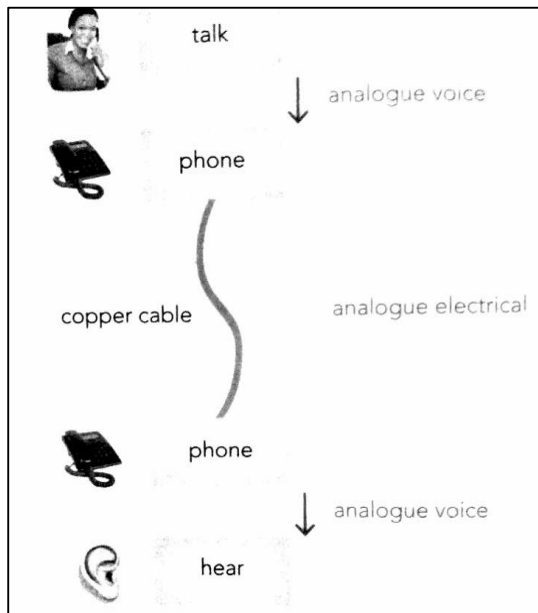


Communicating by cell phone

- When communicating by cell phone, signal is travelling through unguided media.
- At first, its analogue, the cell phone digitalises it before sending it off to mast.
- The signal is converted back to analogue by another cell phone and into sound waves for the other person to hear.
- Signals arrive within a fraction of a second despite the large distance between the two.



Communicating by telephone



- There are wires all the way from the start to the end.
- The signal is analogue from talker to listener.
- Signal travels by electrical pulses through copper wires, or by light through optic cables (Guided media).
- Signal boosters or amplifiers are needed along the way so that cables do not lose energy.
- Hence signal speed is fast.

Magnetism and electricity

The generation of electricity

- Electricity is also produced in power stations using generators.
- Their names are derived from the source of energy used.
- In power stations, the turbine connected to the generator is driven by the movement of either water or wind, steam etc.

Thermal Power

- Fuel such as coal, oil and natural gas can be used to heat water in huge boilers to produce steam. Chemical energy → Heat energy → Kinetic → Electrical energy.
- The steam at high pressure turns turbines.
- Turbines turn generators to produce electricity.
- E.g. Hwange and Munyati power station.

Advantages

1. Energy production using coal can be increased or decreased according to demand
2. The technology to burn coal to generate electricity already exists

Disadvantages

1. Coal is finite so will eventually run out.
2. Many existing reserves are becoming harder to extract or are in environmentally sensitive areas.
3. Fossils releases large amounts of greenhouses gases when burnt.
4. Mining deep underground fossils is very dangerous.
5. Coal is very bulky and expensive to transport around the world.

Solar Power

- Photovoltaic cells use sunlight to produce electrons between two films of silicon.
- The electron flow can be channelled in order to complete an electrical current.

Advantages

1. It is a clean form of energy.
2. It is an infinite resource.

3. Panels can be used locally e.g. on top of someone's house.
4. It can be used to heat water and generate electricity.

Disadvantages

1. It is expensive to make solar panels.
2. The sun does not shine all the time.
3. Not every country gets adequate levels of sun.
4. They can't be used at night.
5. It is hard to store surplus energy.
6. Supply does not always equal demand.

Wind Power

- Using the energy of the wind to drive a turbine to generate electricity.

Advantages

1. It is a clean form of energy.
2. It is an infinite resource.
3. It can be used on a local scale e.g. in your back garden.
4. Technology is proven.

5. They can be placed at sea or on mountains away from settlements

Disadvantages

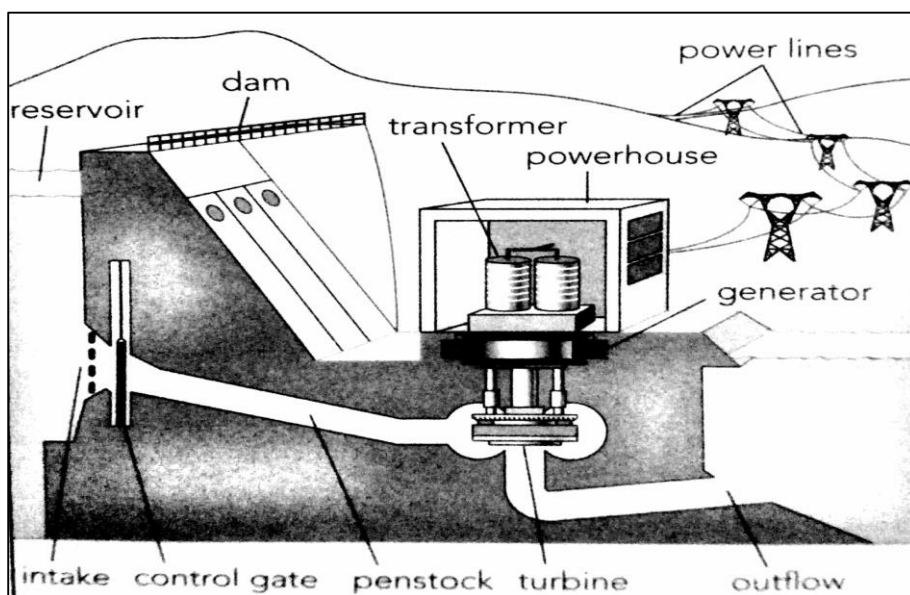
1. Visual pollution.
2. Noise pollution.
3. Wind is unreliable.
4. They are expensive to install, especially offshore.
5. It is hard to store surplus energy.
6. They have to be turned off in very strong winds.

HEP (Hydroelectric power)

- Using the energy of falling water in rivers to drive generators.
- Potential energy → Kinetic energy → Electrical energy.
- Kinetic energy is converted into electrical by **a.c** generator.

Hydro – electric power generation at Kariba

- Water leaves the lake (head of water) at high pressure and goes through the penstocks (concrete and steel pipes).
- The larger the depth of water, the more the pressure hence high electricity generation.
- High speed water hits cups of turbines.
- Turbines start spinning and rotating.
- They in turn generators to which they are attached with pistons.
- Generators produce electricity.
- The water passes through surge chambers and return back to the river through tailrace pipes.
- Electricity is transmitted to huge transformer which regulates the power.
- Electricity is transmitted through cables to homes for use.



Advantages

1. It is a clean form of energy
2. It is finite as long as rivers are managed properly.
3. The built dam can also prevent flooding.
4. The reservoir behind the dam can be a store of water.

Disadvantages

1. Only a limited number of suitable rivers.
2. Can hamper navigation up and down river.
3. Reservoirs may force resettlement.
4. Migration patterns of animals may be disrupted.
5. Dams can flood large areas of land.

Electricity

Electricity Transmission

- The system of power stations and transmission lines is called the **national electricity grid**.
- The alternating current (A.C) produced by huge A.C generators at Hwange and Kariba is stepped up by transformers.
- The electricity is transmitted as alternating current (A.C) of up to 500 KV (500 000 volts).
- Substations then transform (step down), the high voltage to 230 volts for domestic use.
- 15% of electrical energy is lost in the transmission process.

Electrical Hazards

- Damaged insulation.
- Overheating cables.
- Damp conditions.

Safety components in homes

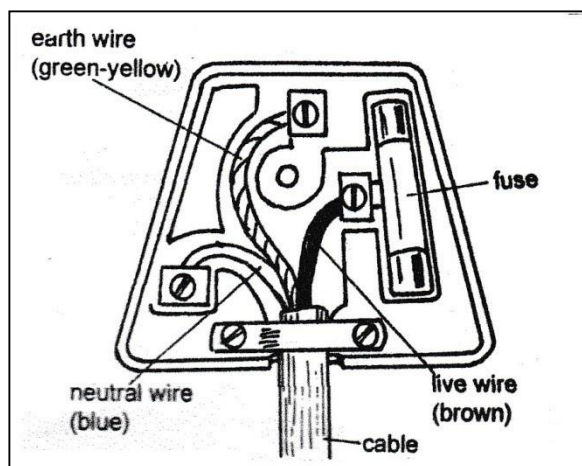
- Circuit breaker- an electric switch that interrupts current flow.

- Fuse- a component from a metal with a low melting point that melts when strong current flows over. This prevents overheating and damage of wiring and appliances.
- Earth leakages- will break the current if anything connects the current with the earth.
- Switches –are placed on the live wire of a circuit so as to break the circuit when current is not required.

Safety Precautions

- Damp conditions – ensure that all gadgets are away from water or your hands are not wet before touching electric gadgets or appliances.
- Do not carry out repairs; only hire the electrician with the knowledge and should disconnect main circuit breakers before.
- Avoid overloading circuits.
- Earthing – reduces electric shock and being electrocuted in that appliances are earthed using an earth wire which would carry current when there is leakage of electricity due to a fault.
- Use insulated cables and wear rubber footwear when using appliances.
- Use three pin plugs on all permanent appliances.

The 3 – pin plug



- Live wire – brown, neutral wire – blue and earth wire – green – yellow. The live wire carries electric current and is considered alive. The earth wire carries current only when there is a leakage due to a fault. The live wire is at full potential of 250 volts while the neutral carries current away and has a potential of -250 volts. At the back of the case of the 3 – pin plug, are letters E, N, L indicating Earth, Neutral and Live respectively. The **insulator** (rubber or plastic) prevents short circuiting and prevent the user from **electrocution**.

The 2 – pin plug

- A 2 – pin plug is connected only to the live and neutral wires and has no earth wire. Appliances such as steam irons, radio and TV sets use the 2 – pin plug.

Experiment 1

Aim: To demonstrate the effect of a short circuit/overloading of a circuit on a fuse.

Materials: 6 or 12 V battery, copper wires, fuse wire/steel wool strand, switch, car headlamp, torch bulb, switch, 2 nails, wooden block

Method

- Set up with circuit with a torch bulb in the circuit.
- Close the switch and observe the fuse wire.
- Replace the bulb in the circuit with a car hand-lamp which draws more current than the torch bulb.
- Close the switch and observe the fuse.
- Open the switch and replace the fuse wire.
- Take a wire and replace the bulb with it.
- Close the switch and observe the fuse

Results and Observations

- The fuse wire remains intact when a torch bulb is used.
- The fuse blows when the torch bulb is replaced with a car head lamp
- The fuse also blows when a wire is used to short circuit the circuit.

Conclusion

- Fuses can protect a circuit from overloading/short circuits by blowing and breaking the circuit.

Uses of electricity in the home

- a) Heating.
- b) Lighting.
- c) Powering electrical devices.

Buying electricity

- Electrical energy from the National Grid used in household is measured on a meter in each house. We buy electricity at a fixed price per unit of energy. In Zimbabwe, it costs \$ 0.90 per kWh.
- We use the Kilowatt – Hour to measure the amount of electricity used.
- One kilowatt – hour is the energy used in one hour by electrical appliances.
- Number of kWh units = $\frac{\text{watts}}{1000} \times \text{hours}$.

Or $1 \text{ kWh} = 1000 \text{ J/s} \times 3600 \text{ s}$
 $= 3\,600\,000 \text{ J} = 3.6 \text{ MJ}$

- A 3 kW electric fire working for 2 hours uses 6 kWh of electrical energy – usually called 6 ‘units’. Electricity meters, which are joulemeters, are marked in kWh: the latest have digital readouts.

Example 1: How much electricity is used by 60w lamp running for 20 hours?

Solution: $= \frac{60}{1000} \times 20$
 $= \frac{1200}{1000}$
 $= 1.2 \text{ kW}$

Therefore if 1 kilowatt costs \$ 0.90, the cost of 1.2 kW is \$ 1.08

Methods of saving electricity

- a) Use of energy saving bulbs and low power rated appliances.
- b) Finding other alternative sources of energy.
- c) Use of solar panels.
- d) Switching off gadgets and appliances not in use.

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